

Exhibit

5

United States Patent [19]
Herbert

[11] **Patent Number: Des. 316,397**

[45] **Date of Patent: ** Apr. 23, 1991**

[54] **PORTABLE, RECHARGEABLE STORAGE BATTERY**

[75] **Inventor: Raymond F. Herbert, Yorkshire, United Kingdom**

[73] **Assignee: Powerbank Limited, United Kingdom**

[**] **Term: 14 Years**

[21] **Appl. No.: 232,895**

[22] **Filed: Aug. 16, 1988**

[52] **U.S. Cl. D13/106; D13/103**

[58] **Field of Search D13/8, 4, 5, 9, 10, D13/103-107, 119; 320/2-5, 27; 429/48, 54, 121, 122, 96, 97, 99, 100, 160, 163, 176, 178, 187**

[56] **References Cited**

U.S. PATENT DOCUMENTS

D. 235,454 6/1975 Lindow, Jr. D13/10

D. 285,191 8/1986 Humphreys et al. D13/8
D. 291,300 11/1987 Chaney D13/8
3,963,972 6/1976 Todd 429/97 X
4,300,087 10/1981 Meisner 320/2
4,392,661 7/1983 Langenstein 429/100 X

Primary Examiner—Susan J. Lucas

Assistant Examiner—Joel Sincavage

Attorney, Agent, or Firm—Kinney & Lange

[57] **CLAIM**

The ornamental design for a portable, rechargeable storage battery, as shown and described.

DESCRIPTION

FIG. 1 is a side elevational view of a portable, rechargeable storage battery showing my new design, the opposite side elevational view being a mirror image;

FIG. 2 is a rear elevational view thereof;

FIG. 3 is a front elevational view thereof;

FIG. 4 is a top plan view thereof;

FIG. 5 is a bottom plan view thereof.

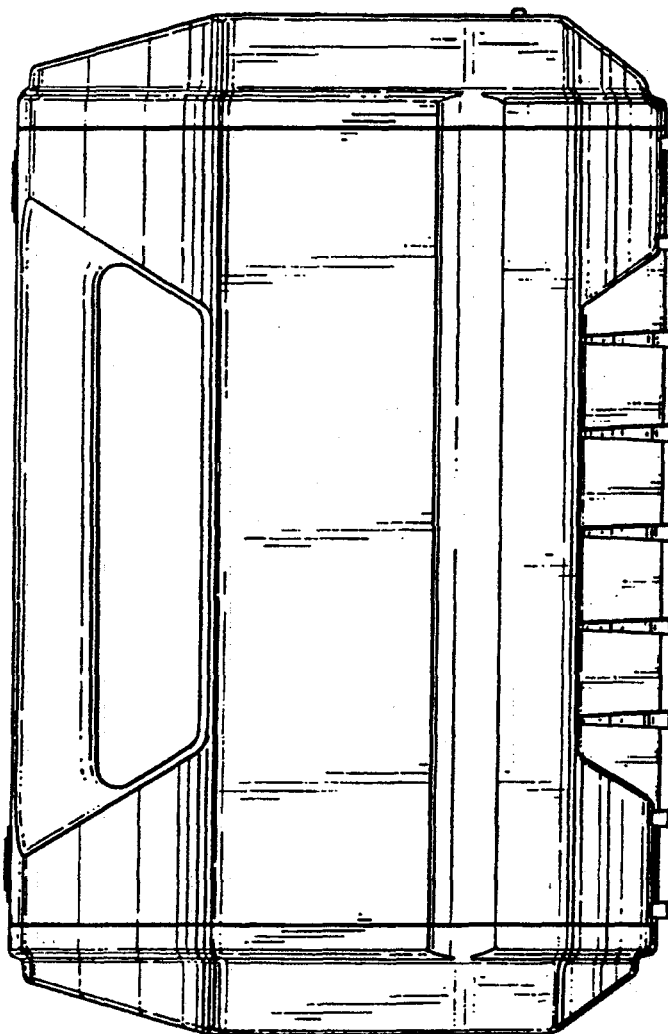


FIG. 1.

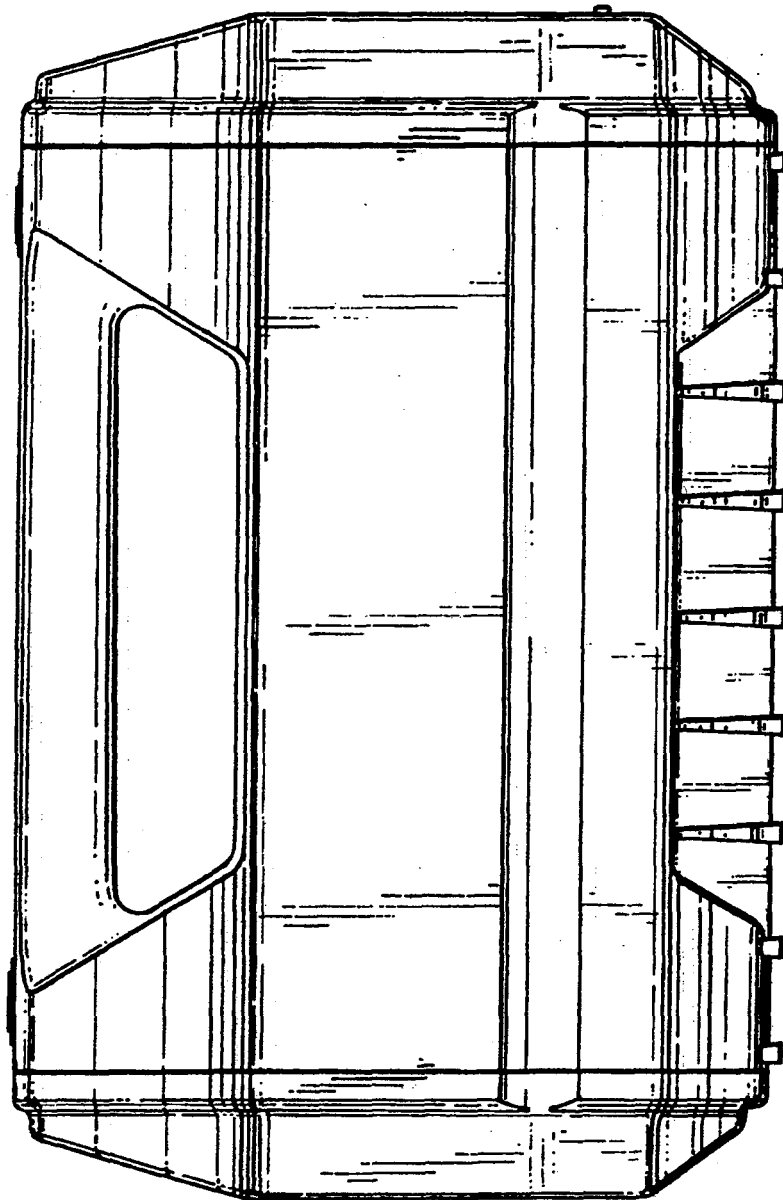


FIG. 2.

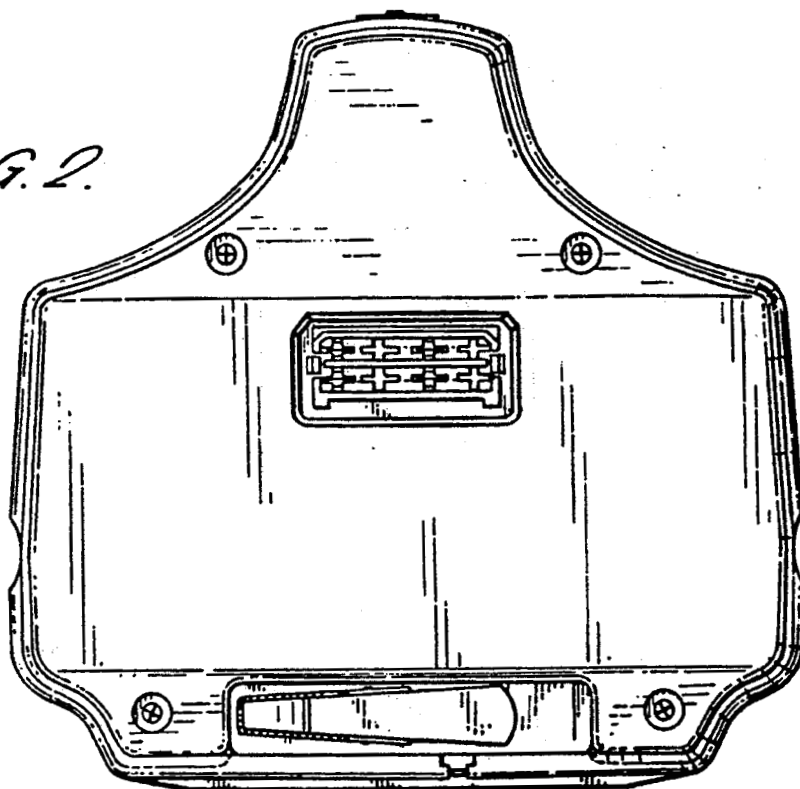


FIG. 3.

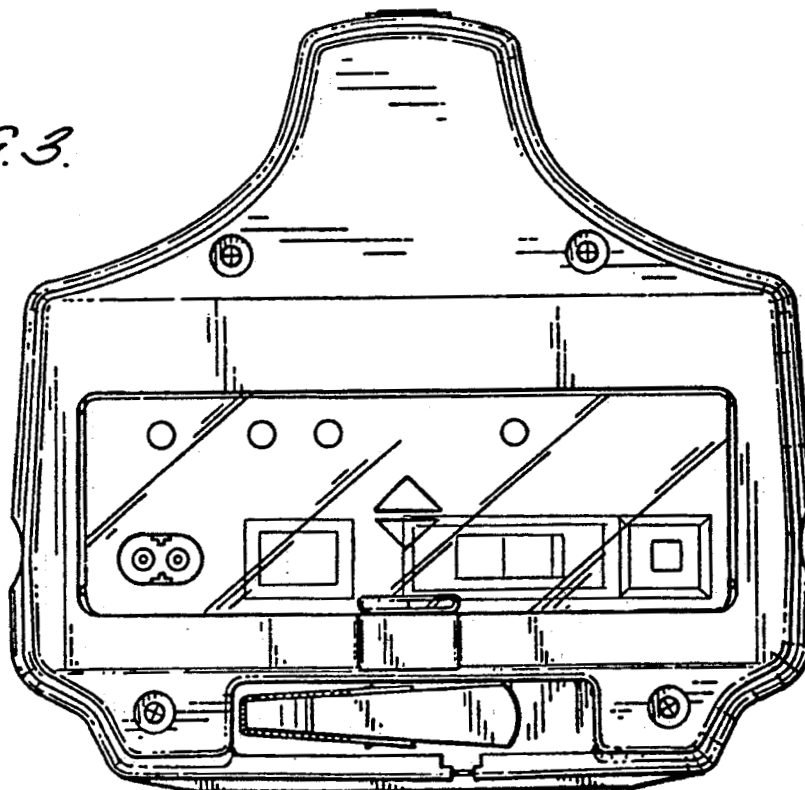


FIG. 4.

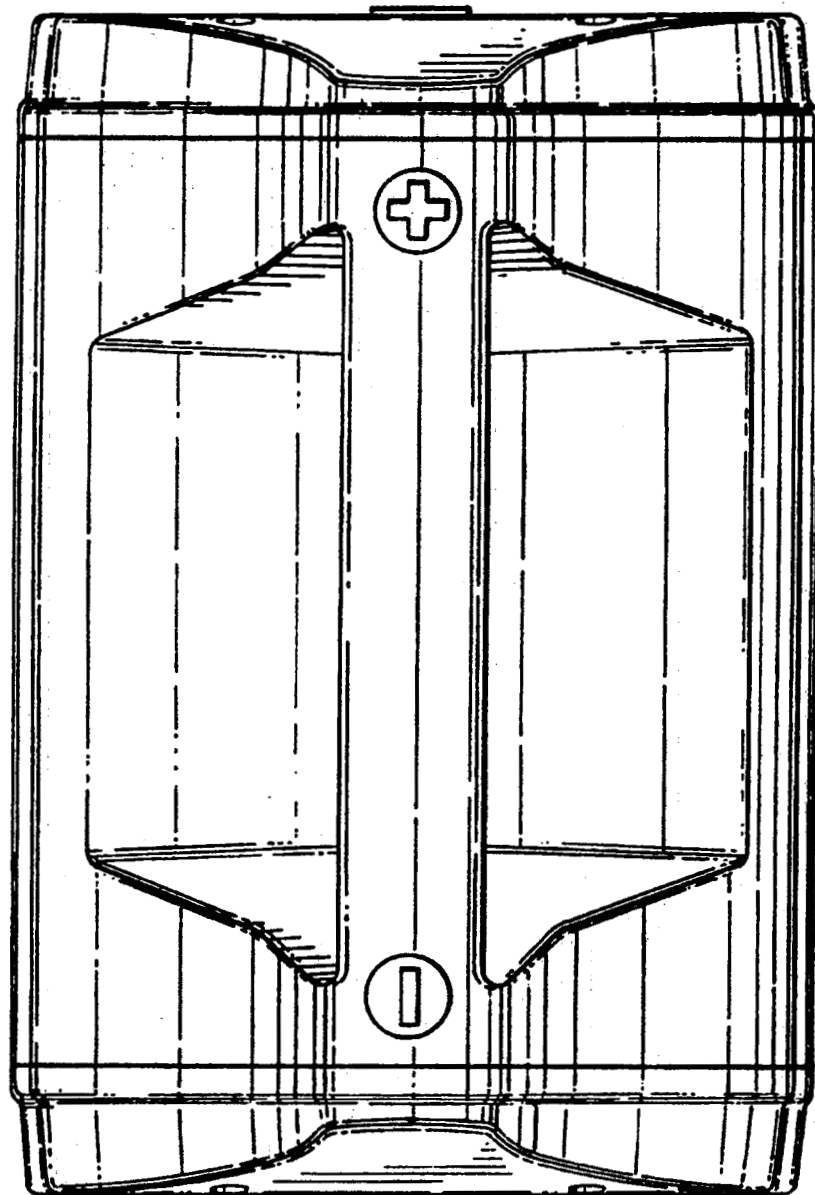
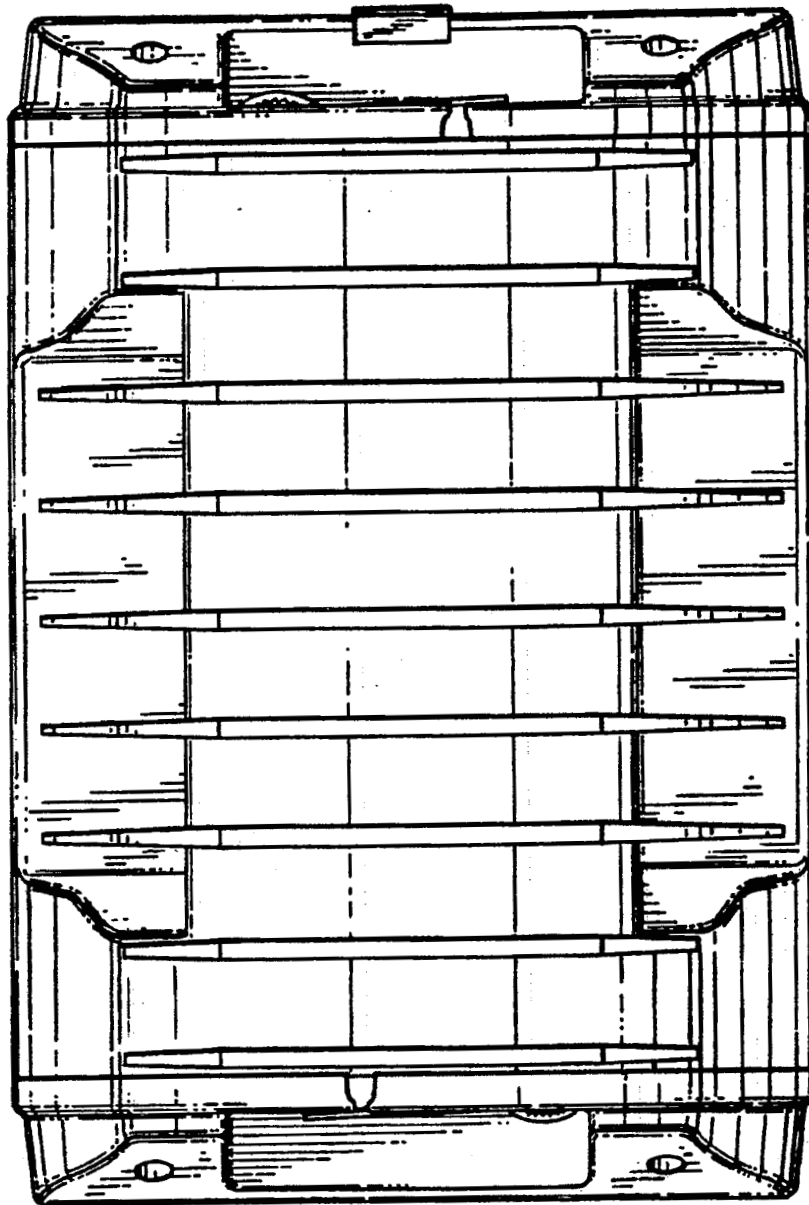


FIG. 5.





US00D334560S

United States Patent [19]

Wilson

[11] Patent Number: Des. 334,560

[45] Date of Patent: ** Apr. 6, 1993

[54] **RECHARGEABLE POWER PACK FOR VARIOUS MODULAR DEVICES**

[75] Inventor: Michael Wilson, Mission Viejo, Calif.

[73] Assignee: Innova Electronics Corporation, Fountain Valley, Calif.

[**] Term: 14 Years

[21] Appl. No.: 769,674

[22] Filed: Sep. 30, 1991

[52] U.S. Cl. D13/106; D13/110

[58] Field of Search D13/103, 110, 106; 320/2; 307/150; 429/97, 187

[56] References Cited

U.S. PATENT DOCUMENTS

D. 300,920	5/1989	Gierke		D13/103
D. 302,971	8/1989	Gierke		D13/103
D. 310,509	9/1990	Fushiya et al.		D13/103
D. 316,397	4/1991	Herbert		D13/103 X
D. 321,680	11/1991	Blount et al.		D13/103

1,984,398	12/1934	Dame		429/97 X
2,978,596	4/1961	Robirds		307/150
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3,963,972	6/1976	Todd		320/2
4,300,087	11/1981	Meisner		307/150 X

OTHER PUBLICATIONS

Radio Shack Catalog—1990—p. 10, "Cellular Telephone CT 201".

Primary Examiner—Wallace R. Burke

Assistant Examiner—Joel Sincavage

Attorney, Agent, or Firm—Stetina and Brunda

[57] CLAIM

The ornamental design for a rechargeable power pack for various modular devices, as shown and described.

DESCRIPTION

FIG. 1 is a front perspective view of the rechargeable power pack for various modular devices of the present invention showing my new design; and, FIG. 2 is a rear perspective view thereof.

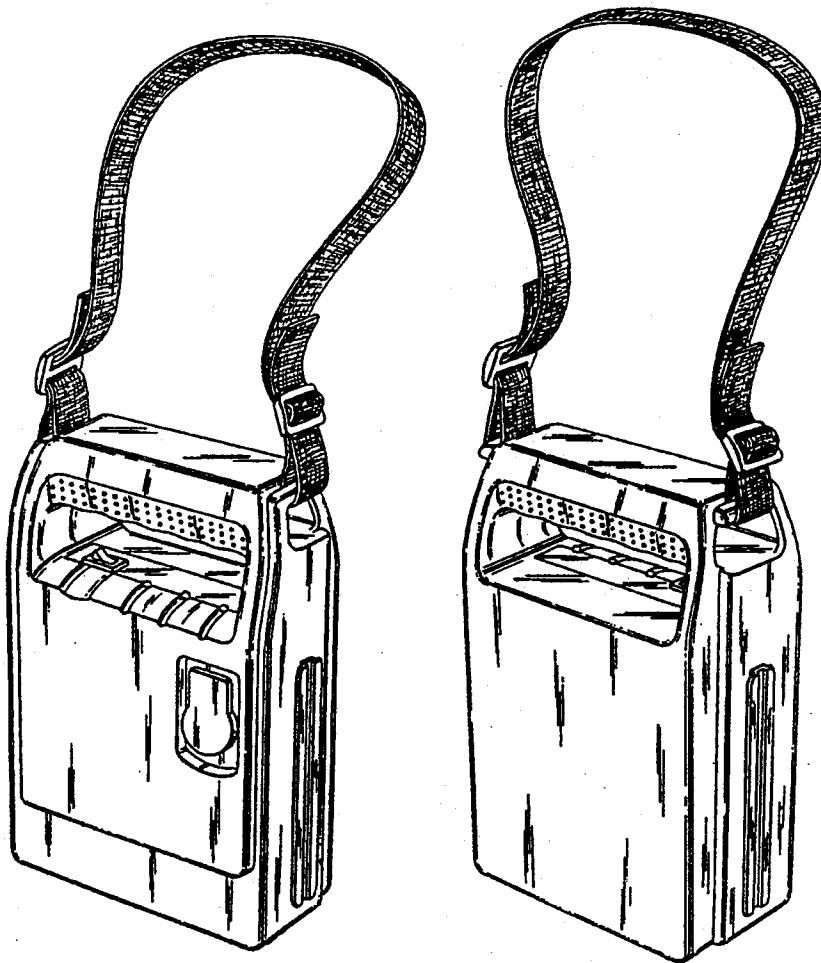


Fig. 1

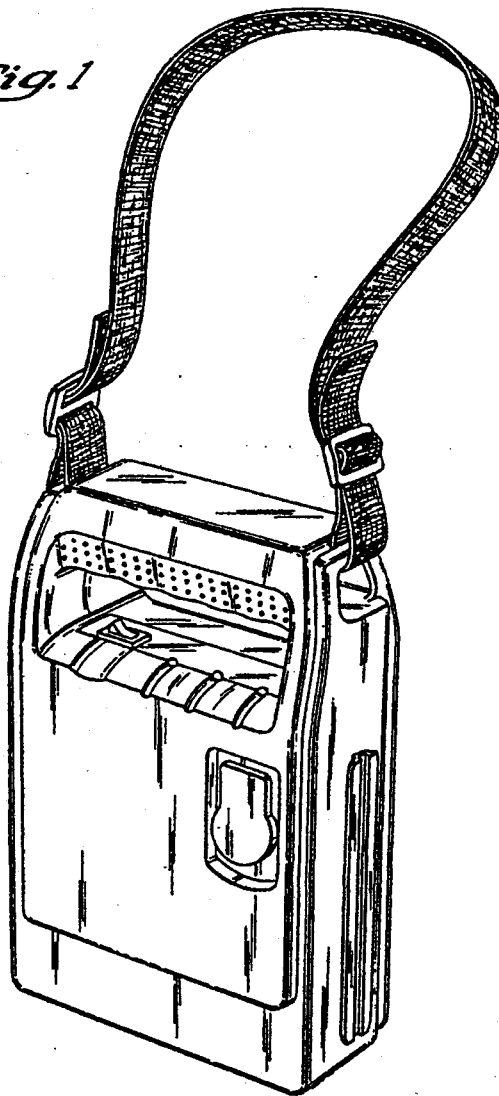
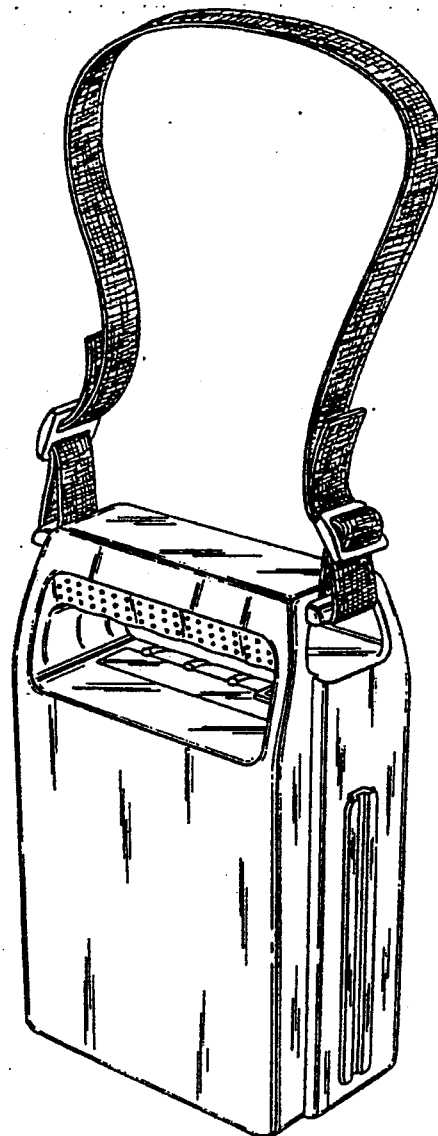


Fig. 2





US00D374859S

United States Patent [19]

Guay et al.

[11] Patent Number: Des. 374,859

[45] Date of Patent: **Oct. 22, 1996

[54] PORTABLE POWER SUPPLY

[75] Inventors: Etienne Guay; André Côté, both of
Montreal, Canada[73] Assignee: Booster PAC International
Corporation, St-Jean-sur-Richelieu,
Canada

[**] Term: 14 Years

[21] Appl. No.: 18,788

[22] Filed: Feb. 15, 1994

[52] U.S. CL. D13/110

[58] Field of Search D13/103, 106,
D13/112; 307/64, 65, 66, 150; 206/333;
429/96, 97, 98, 99, 100, 121, 163, 176,
185, 187; 320/2

[56] References Cited

U.S. PATENT DOCUMENTS

D. 235,454	6/1975	Lindow, Jr.		D13/103 X
D. 316,397	4/1991	Herbert		D13/106
D. 334,560	4/1993	Wilson		D13/106

2,610,987	9/1952	Betz		429/96
2,718,540	9/1955	Betz		429/96
5,214,368	5/1993	Wells		320/2

Primary Examiner—Joel Sincavage

Attorney, Agent, or Firm—McGlew and Tuttle, P.C.

[57] CLAIM

The ornamental design for a portable power supply, as
shown and described.

DESCRIPTION

FIG. 1 is a front view of a portable power supply showing
my new design;

FIG. 2 is a front and right side perspective view thereof;

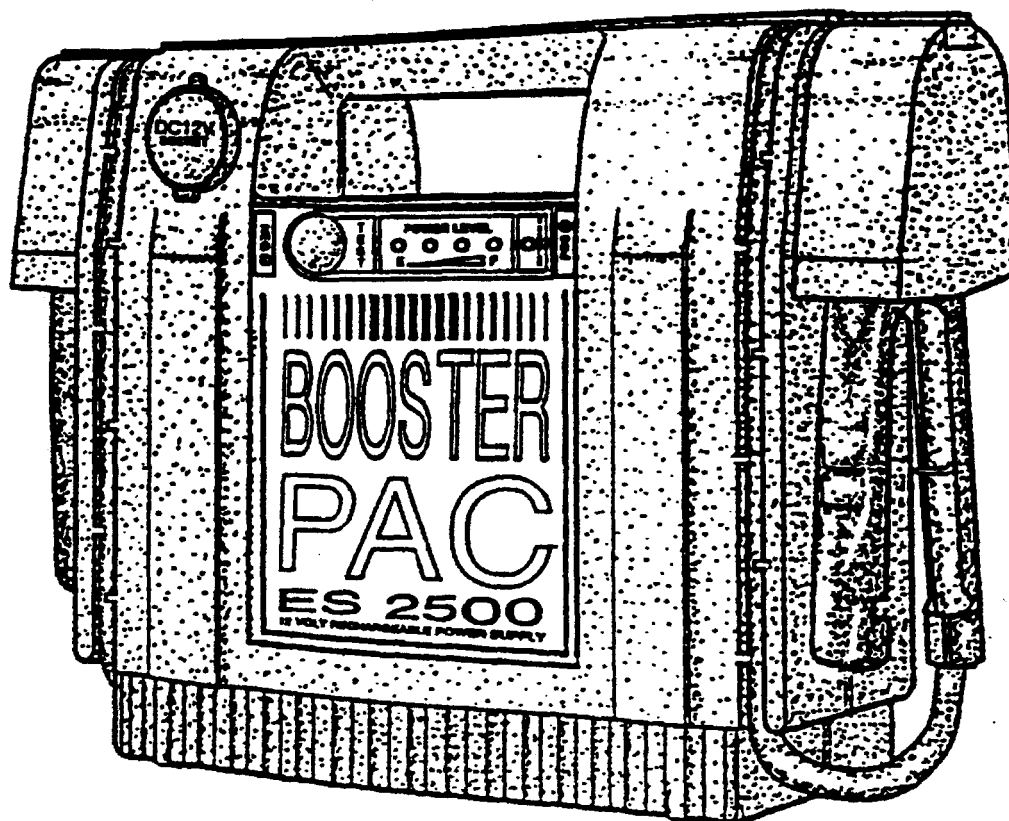
FIG. 3 is a top view thereof;

FIG. 4 is a right side view thereof, the left side view thereof
being a mirror image except for the color of the cable and
clamp;FIG. 5 is a front view thereof showing one cable clamp
removed from its storage position;

FIG. 6 is a rear view thereof; and,

FIG. 7 is a bottom view thereof.

1 Claim, 4 Drawing Sheets



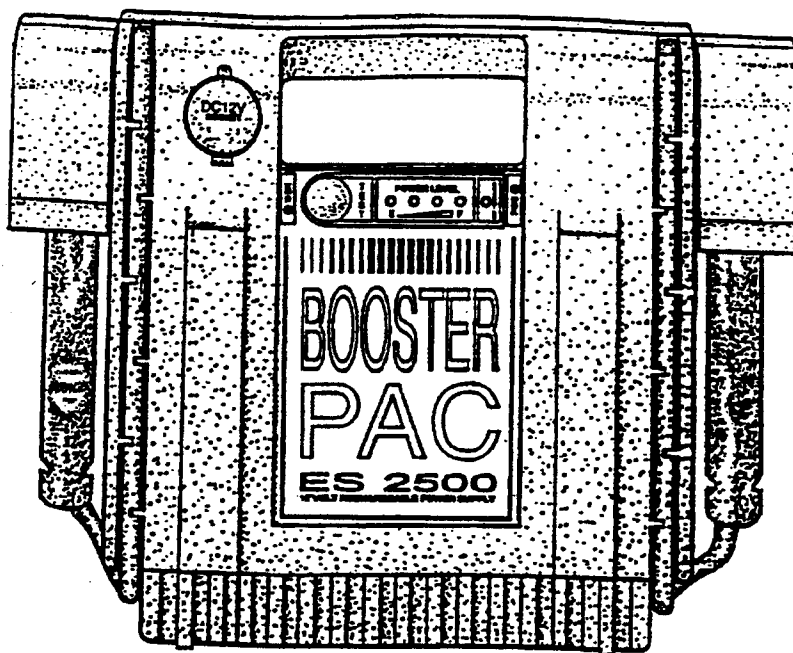


FIG. 1

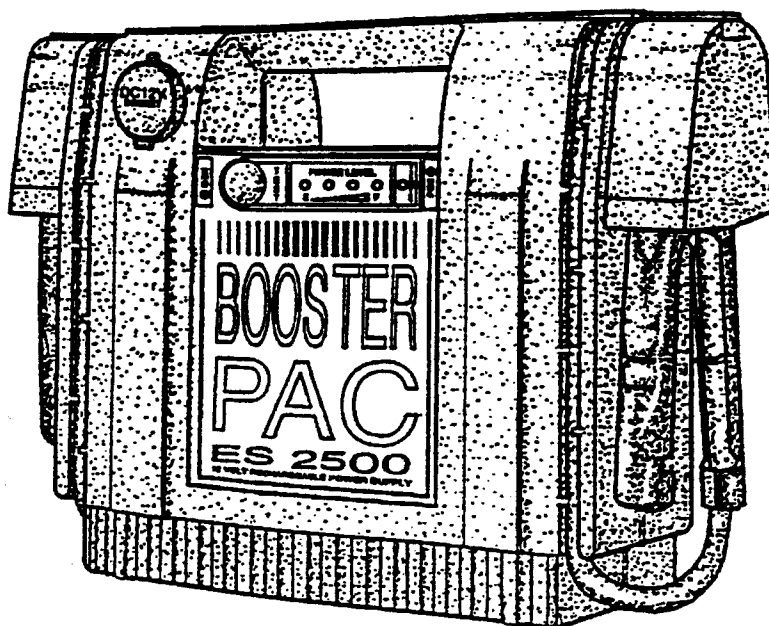


FIG. 2

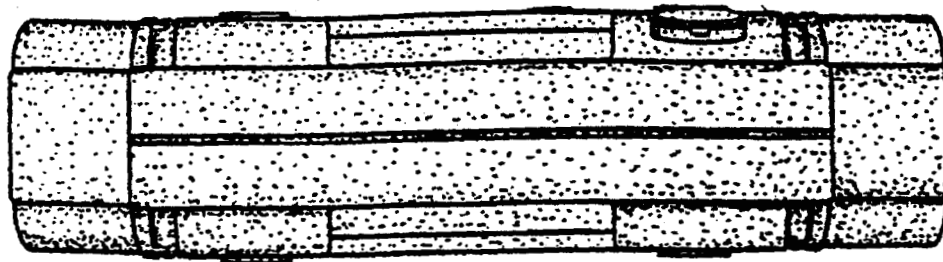


FIG. 3

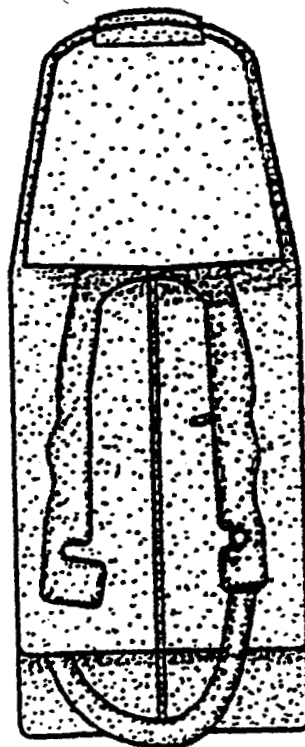


FIG. 4

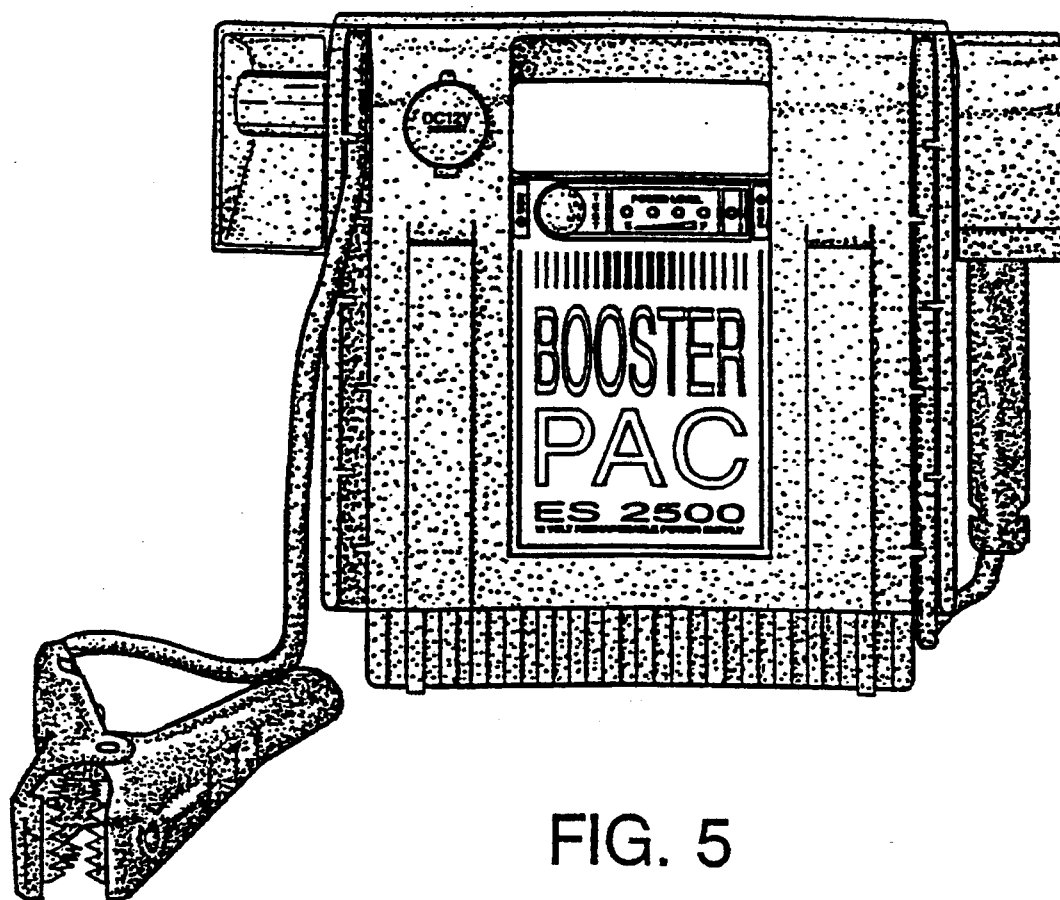


FIG. 5

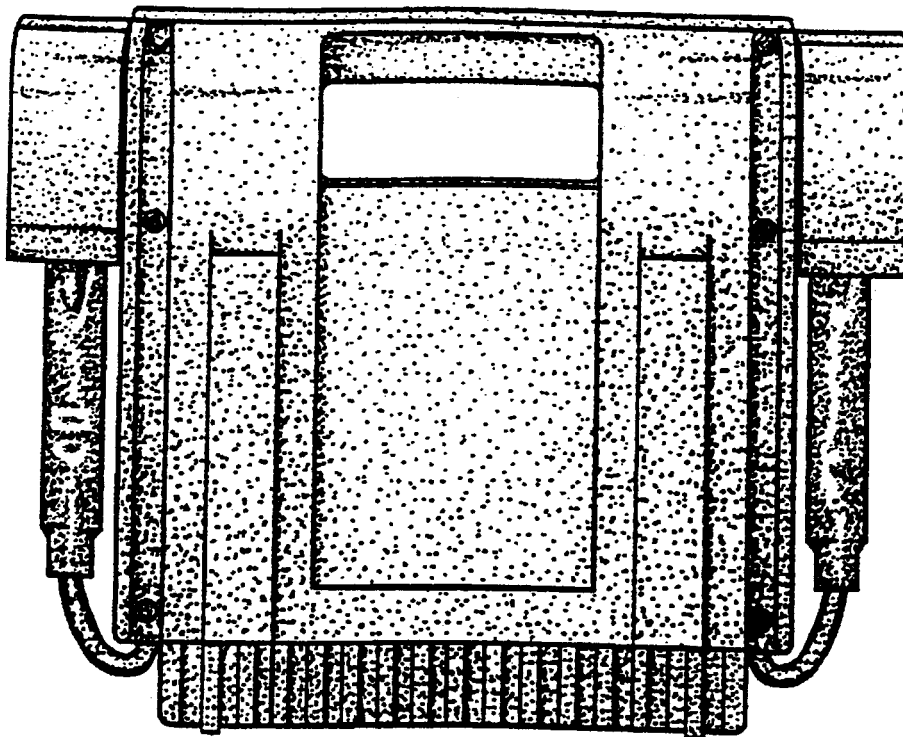


FIG. 6

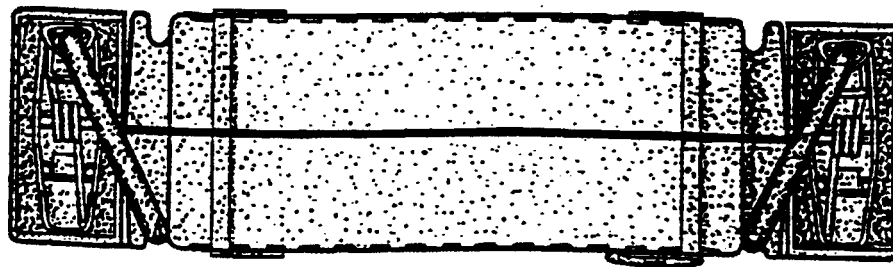


FIG. 7



US00D386148S

United States Patent [19]

Katooka et al.

[11] Patent Number: Des. 386,148

[45] Date of Patent: ***Nov. 11, 1997

[54] **PORTABLE POWER SUPPLY UNIT WITH AC-DC CONVERTER**[75] Inventors: **Massao Katooka**, Kawanishi; **Toru Arai**, Shimochoja-machi; **Shigeru Okamoto**, Hirakata; **Kenzo Danjo**, Kyoto-fu; **Masaharu Tanaka**, Osaka, all of Japan[73] Assignee: **Sansha Electric Manufacturing Co., Ltd.**, Higashiyodogawa-ku, Japan

[*] Notice: The portion of the term of this patent subsequent to Jan. 14, 2011, has been disclaimed.

[**] Term: 14 Years

[21] Appl. No.: 55,295

[22] Filed: May 31, 1996

Related U.S. Application Data

[63] Continuation-in-part of Ser. No. 39,633, Jan. 1, 1995.

[30] **Foreign Application Priority Data**

Dec. 6, 1994 [JP] Japan 6-37410

[51] LOC (6) CL 13-02

[52] U.S. Cl. D13/110

[58] Field of Search D13/110; 307/64, 307/65, 150; 363/15

[56] **References Cited****U.S. PATENT DOCUMENTS**

D. 264,962 6/1982 Fellmann D13/110
D. 276,495 11/1984 Sylvia D13/110 X
D. 280,318 8/1985 Barone D13/110
D. 311,899 11/1990 Lenzing et al. D13/110 X

D. 334,560 4/1993 Wilson D13/110 X
4,295,179 10/1981 Read 174/52.1 X
4,300,087 11/1981 Meisner 37/150 X

OTHER PUBLICATIONS

SanRex Arc Friend (4 pg. brochure) (date prior to Jun. 6, 1994).

Primary Examiner—Joel Sincavage

Attorney, Agent, or Firm—William H. Murray; Robert E. Rosenthal

[57] **CLAIM**

The ornamental design for a portable power supply unit with AC-DC converter, as shown and described.

DESCRIPTION

FIG. 1 is a front and right side perspective view of a portable power supply unit with AC-DC converter showing our new design;

FIG. 2 is a rear and left side perspective view thereof;

FIG. 3 is a front elevational view thereof;

FIG. 4 is a rear elevational view thereof;

FIG. 5 is a left side elevational view thereof;

FIG. 6 is a right side elevational view thereof;

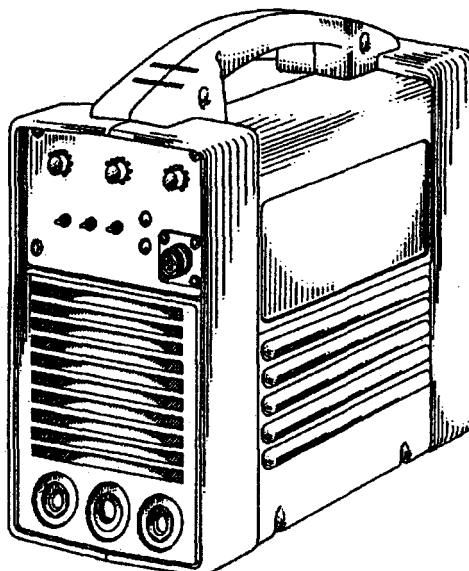
FIG. 7 is a top plan view thereof;

FIG. 8 is a bottom plan view thereof; and,

FIG. 9 is a cross-sectional view thereof along the line A—A in FIG. 3.

The cross-sectional view is included for the purpose of showing external surface contours which may not be clear from the remaining illustrations. Any features of internal construction which are incidentally shown form no part of the claimed design.

The portable power supply unit with AC-DC converter of our new design is a portable power supply unit for converting AC power to DC power and supplying DC power to, for example, an arc welder or a plasma arc cutter.

1 Claim, 9 Drawing Sheets

U.S. Patent

Nov. 11, 1997

Sheet 1 of 9

Des. 386,148

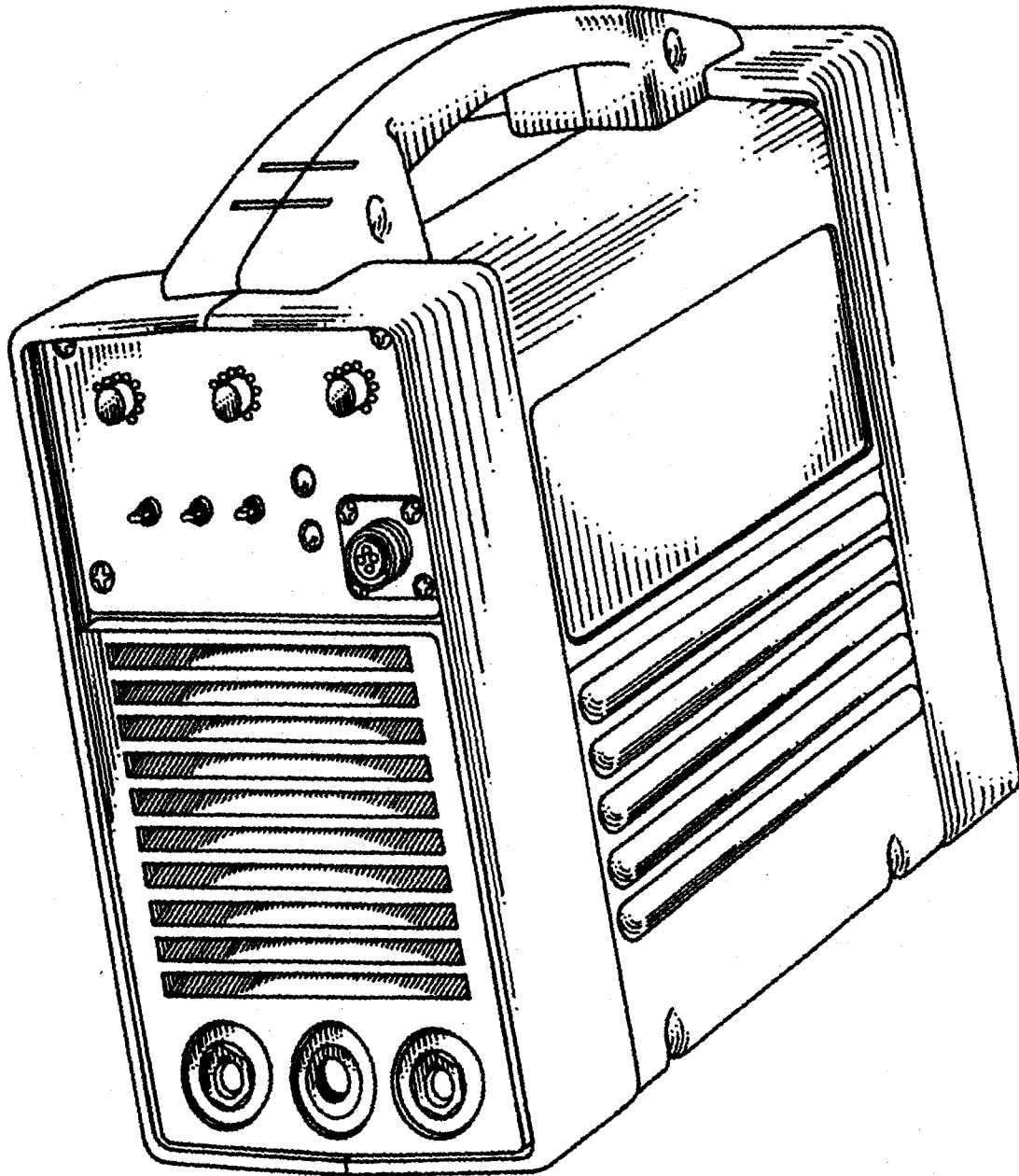
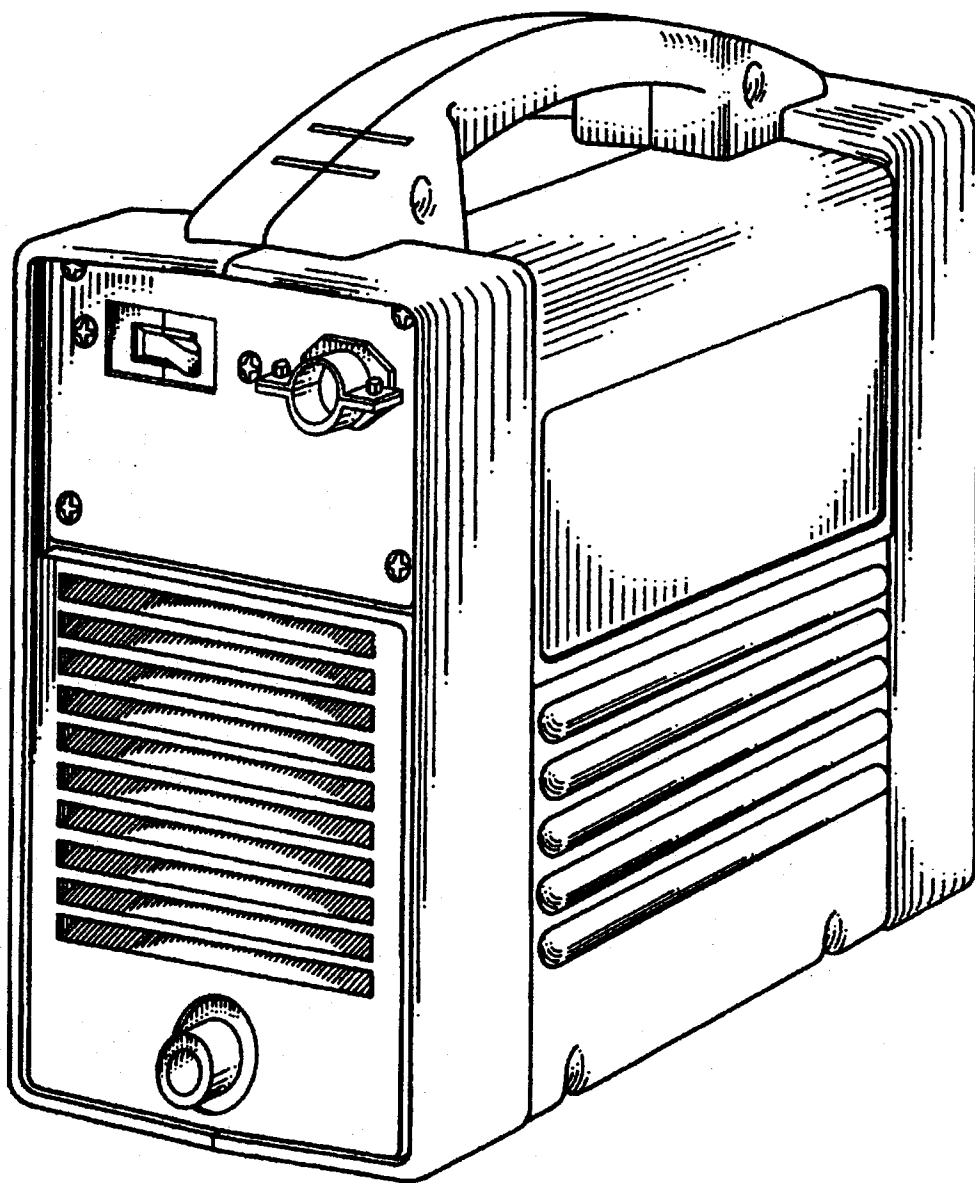
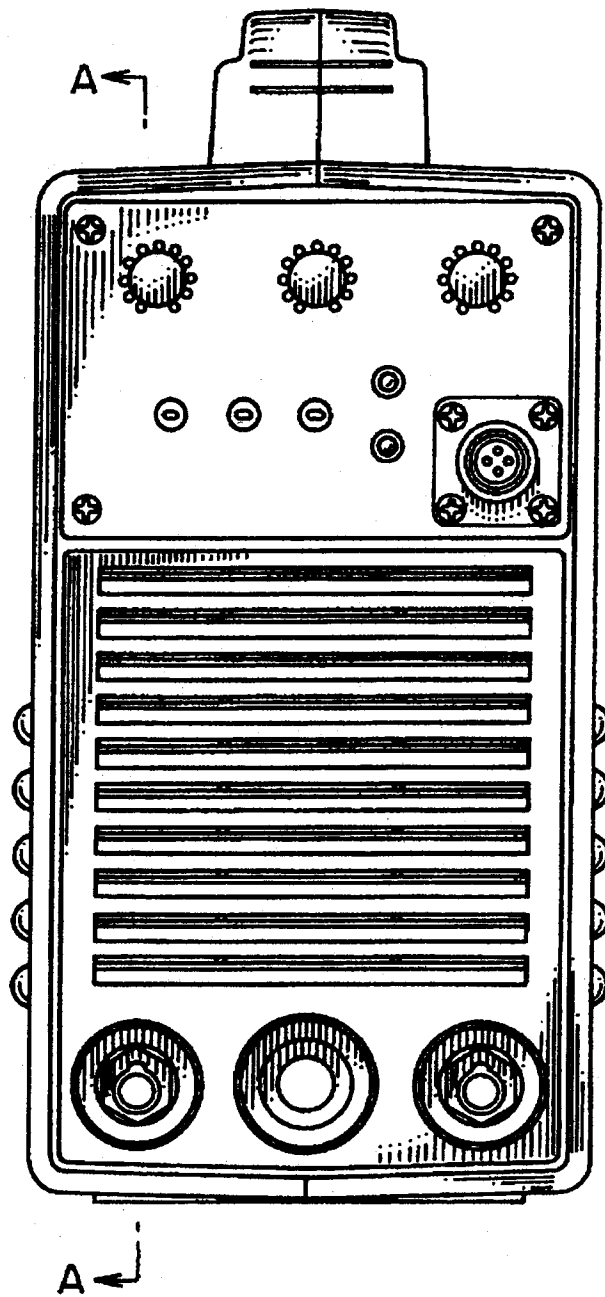


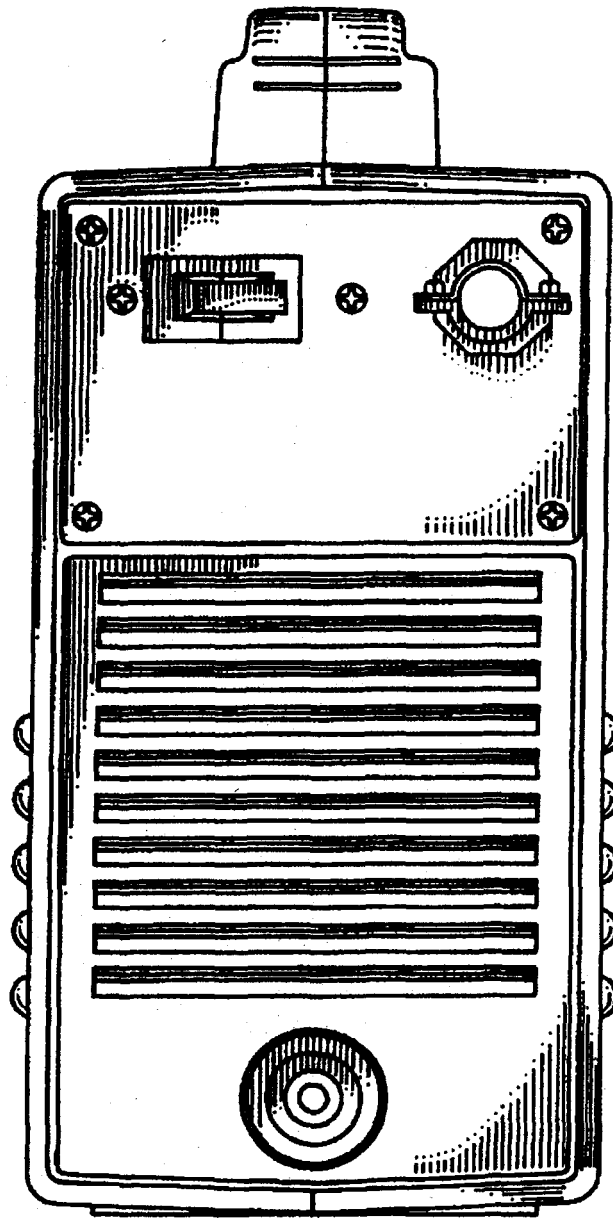
FIG. 1



F I G . 2



F I G . 3



F I G . 4

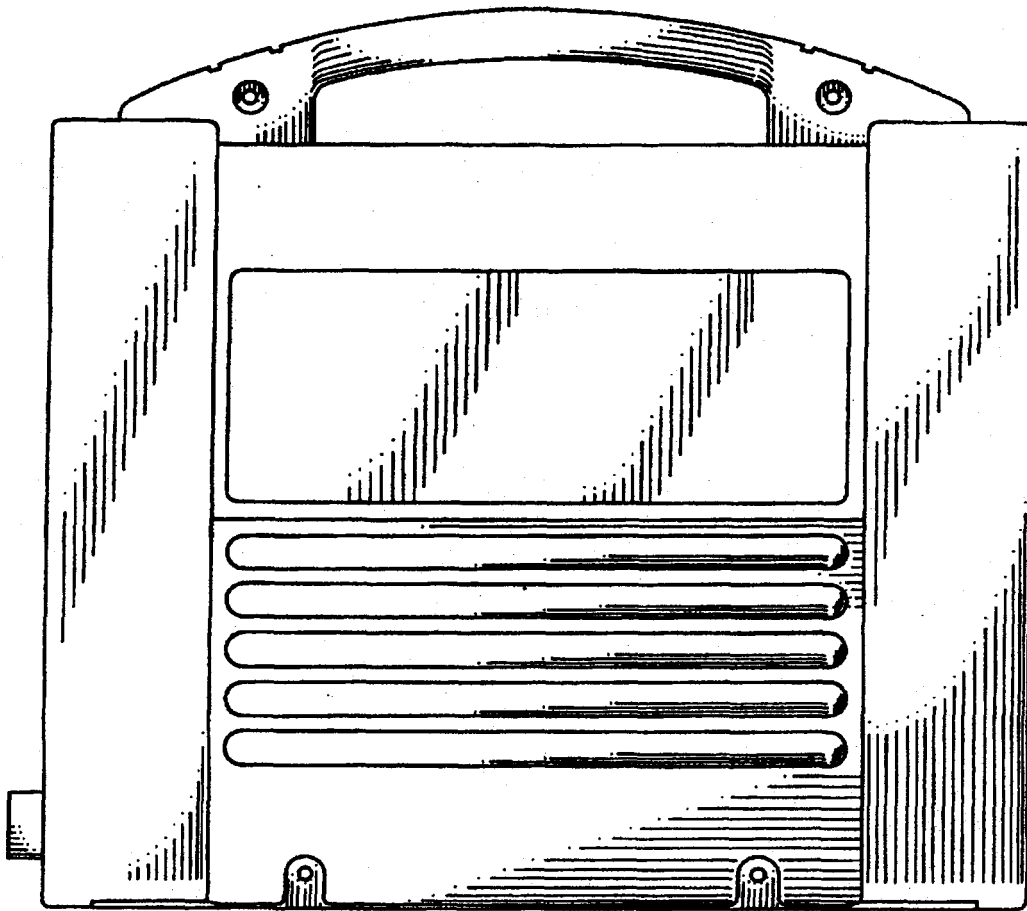
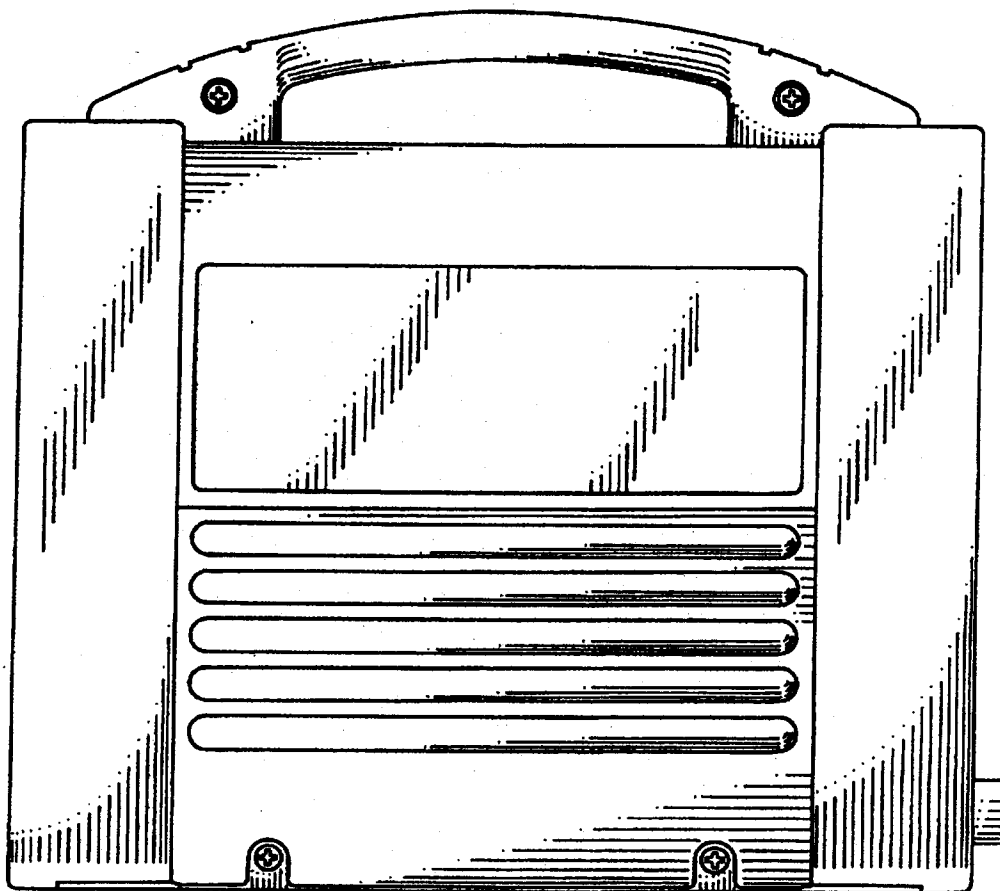
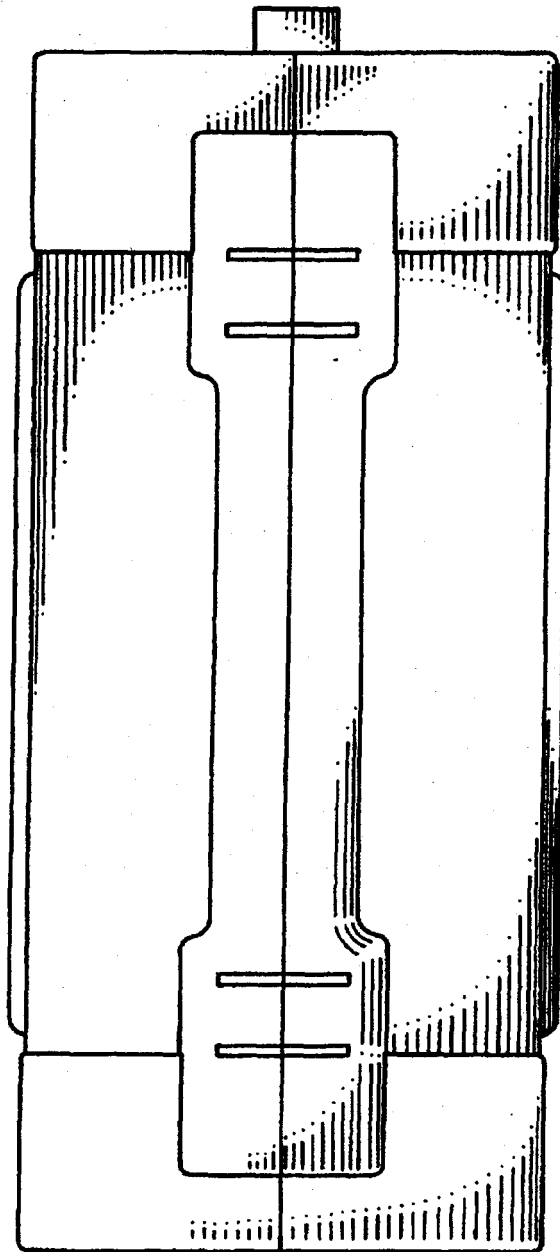


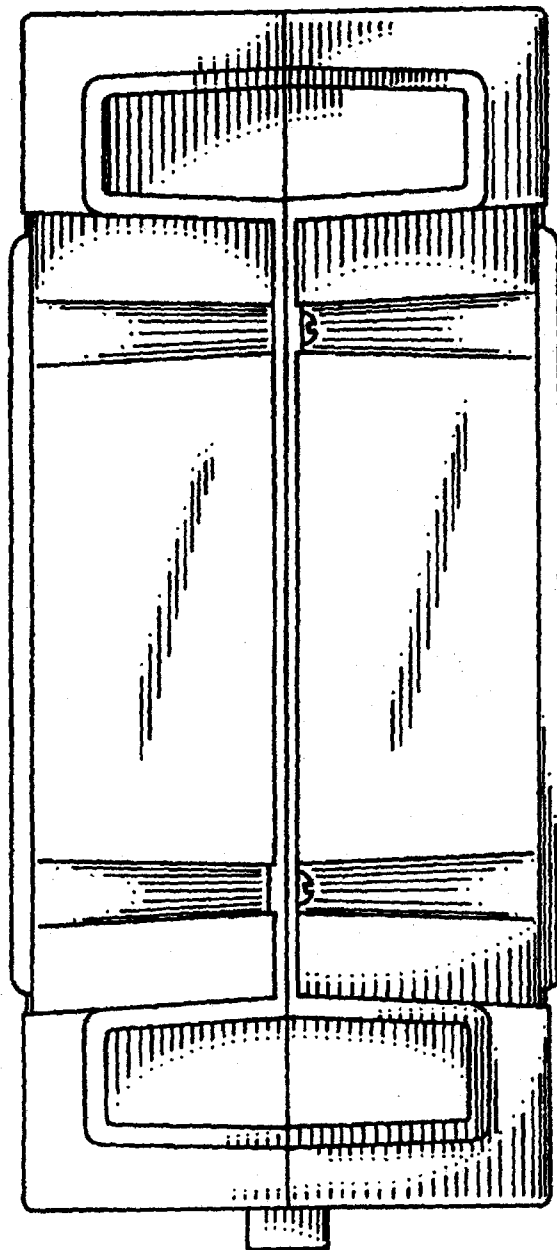
FIG. 5



F I G . 6



F I G . 7



F I G . 8

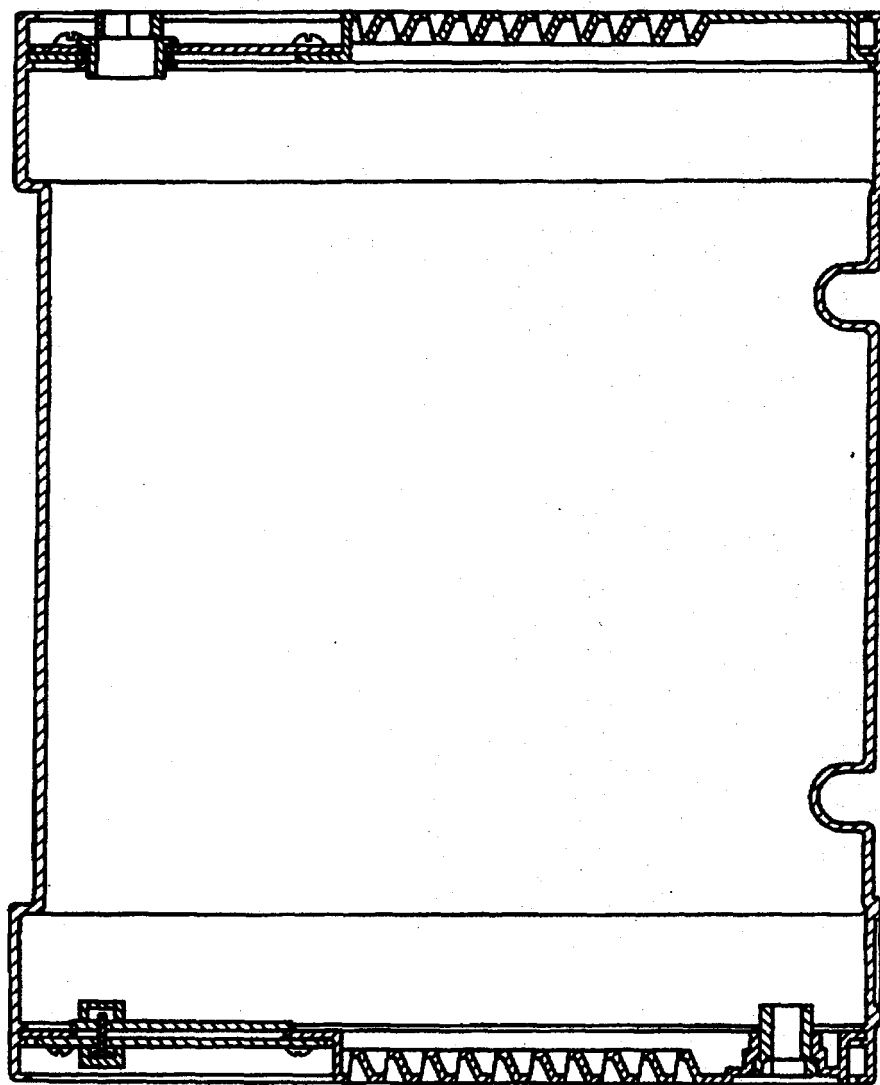


FIG. 9

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : Des. 386,148
DATED : November 11, 1997
INVENTOR(S) : Masao Katooka et al.

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

~~Item~~ [73] should read: --Assignee: Sansha Electric Manufacturing
Company, Limited, Osaka, Japan--.

Signed and Sealed this
Twenty-fourth Day of March, 1998

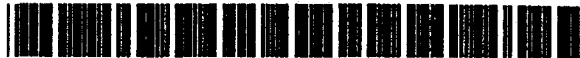
Attest:



BRUCE LEHMAN

Attesting Officer

Commissioner of Patents and Trademarks



US00D393446S

United States Patent [19]

Bellofatto et al.

[11] Patent Number: Des. 393,446

[45] Date of Patent: **Apr. 14, 1998

[54] **BATTERY PACK**

[75] Inventors: Steven Bellofatto, Closter, N.J.;
Matthew Froehlich, Kings Park, N.Y.;
Arthur Gurtman, Great Neck, N.Y.;
Bernard Gurtman, North Hills, N.Y.;
Scott Salmon, Jersey City, N.J.

[73] Assignee: Autostart Marketing Incorporated,
Great Neck, N.Y.

[**] Term: 14 Years

[21] Appl. No.: 62,434

[22] Filed: Nov. 14, 1996

[51] LOC (6) CL. 13-02

[52] U.S. Cl. D13/106

[58] Field of Search D13/103, 106;
429/96-100, 176, 187

[56] **References Cited****U.S. PATENT DOCUMENTS**

D. 302,971	8/1989	Gierke		D13/106
D. 334,560	4/1993	Wilson		D13/106
D. 354,729	1/1995	Poon		D13/106
5,589,292	12/1996	Rozon		429/187 X

OTHER PUBLICATIONS

Owner's Manual, A Person's Companion, Anytime, Anywhere TITAN by HYUNDAI Rechargeable, Portable, Multi Purpose DC Power Supplier (DC 12V), published by Electronics of North American, Inc. in 1990, pp. 1-9.

Primary Examiner—Joel Sincavage

Attorney, Agent, or Firm—Baker & Botts

[57] **CLAIM**

The ornamental design for a battery pack, as shown and described.

DESCRIPTION

FIG. 1 is a three-dimensional perspective view of the battery pack of the present invention;

FIG. 2 is a top view of the battery pack of the present invention;

FIG. 3 is a front elevational view of the battery pack of the present invention;

FIG. 4 is a left side view of the battery pack of the present invention;

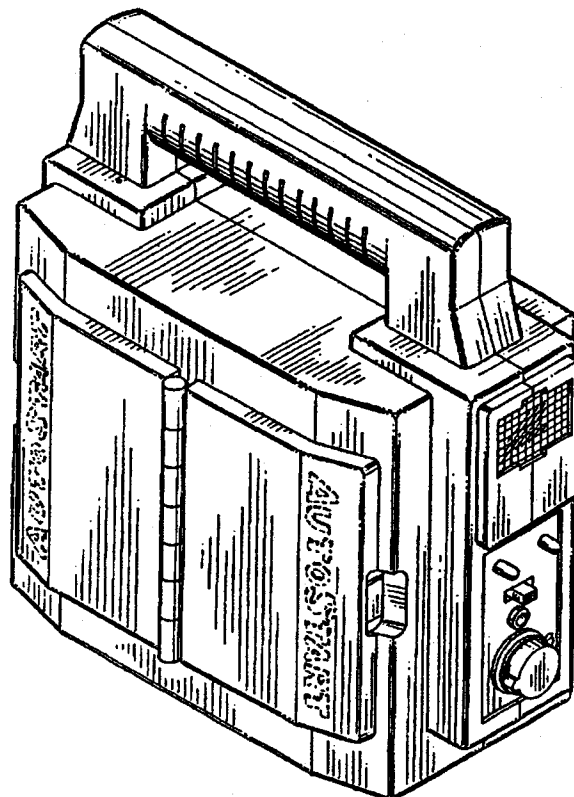
FIG. 5 is a right side view of the battery pack of the present invention;

FIG. 6 is a rear elevational view of the battery pack of the present invention; and

FIG. 7 is a bottom view of the battery pack of the present invention.

The broken line showing is included for the purpose of illustrating environmental structure only and forms no part of the claimed design.

1 Claim, 4 Drawing Sheets



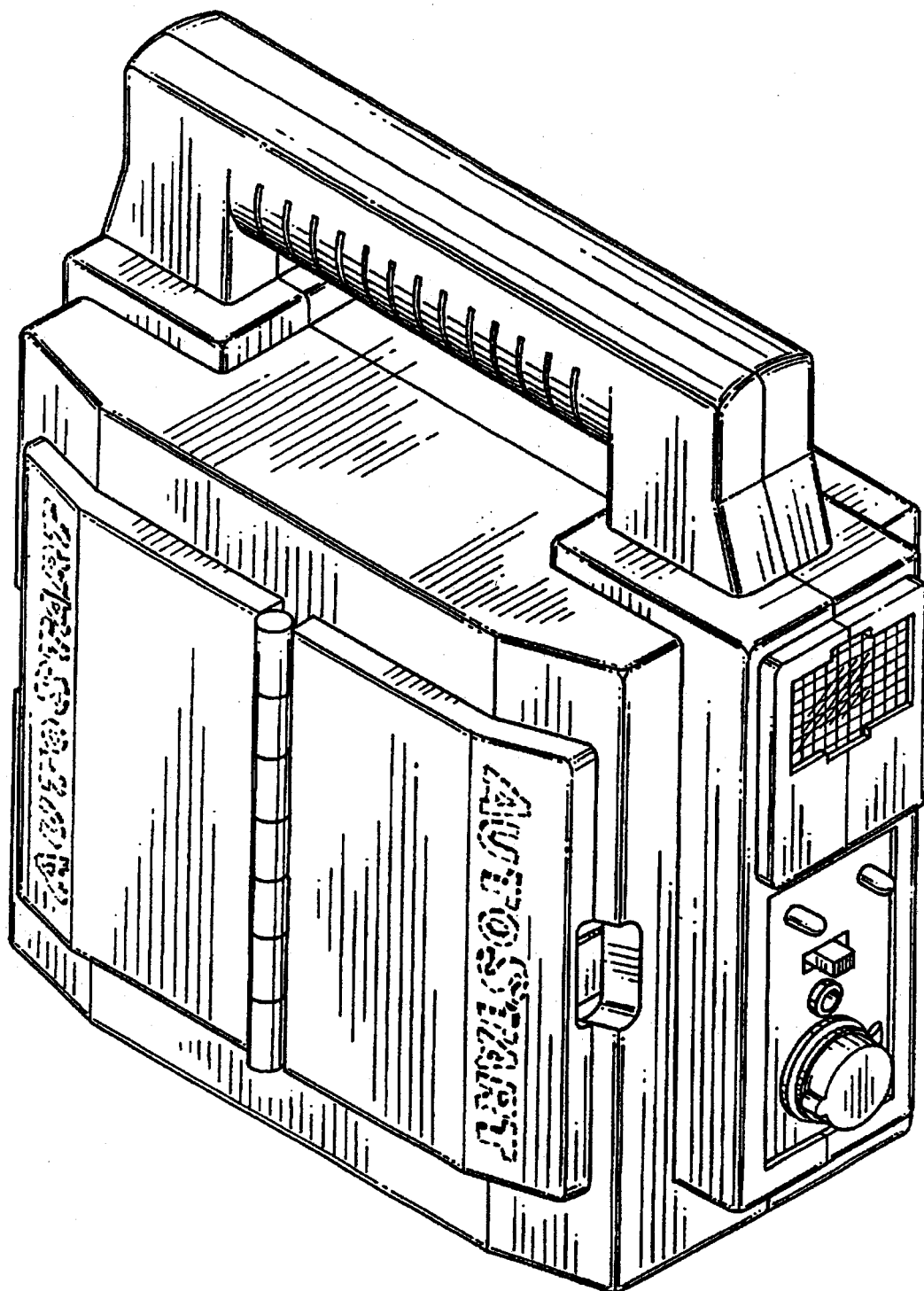


FIG. 1

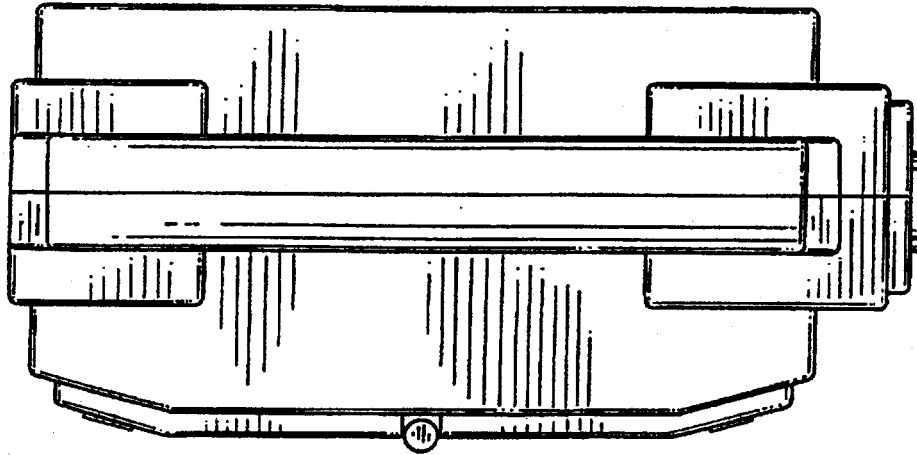


FIG. 2

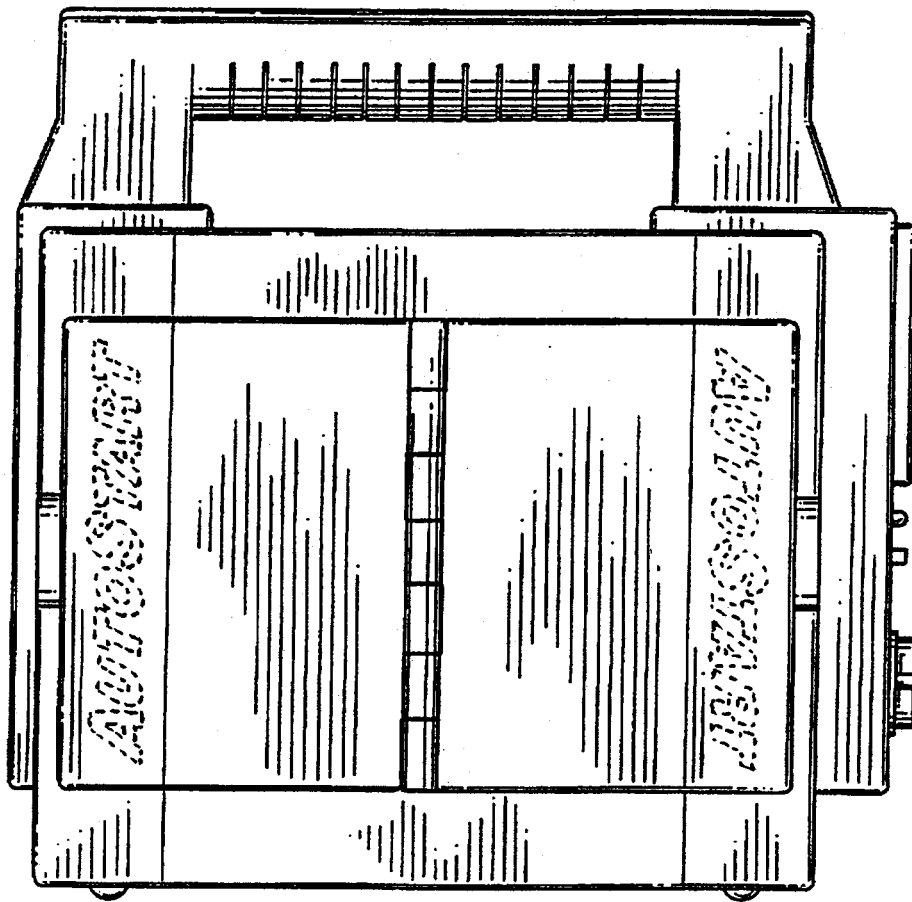


FIG. 3

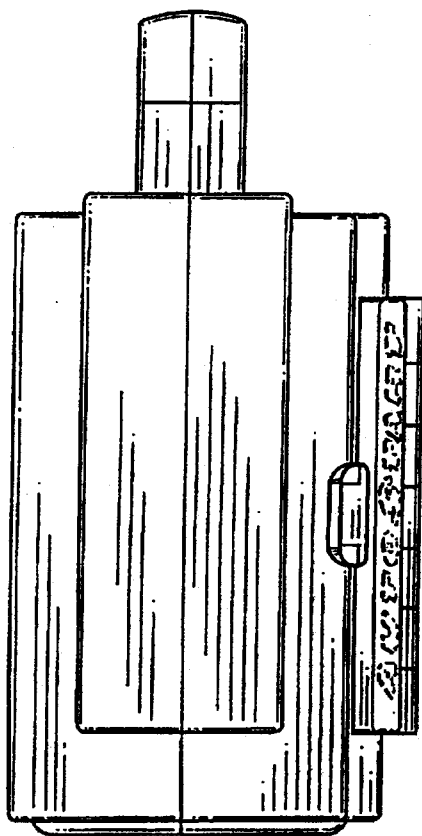


FIG. 4

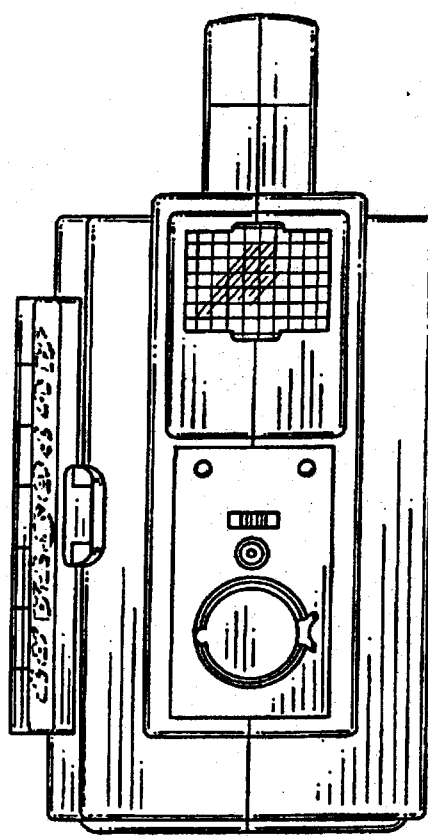


FIG. 5

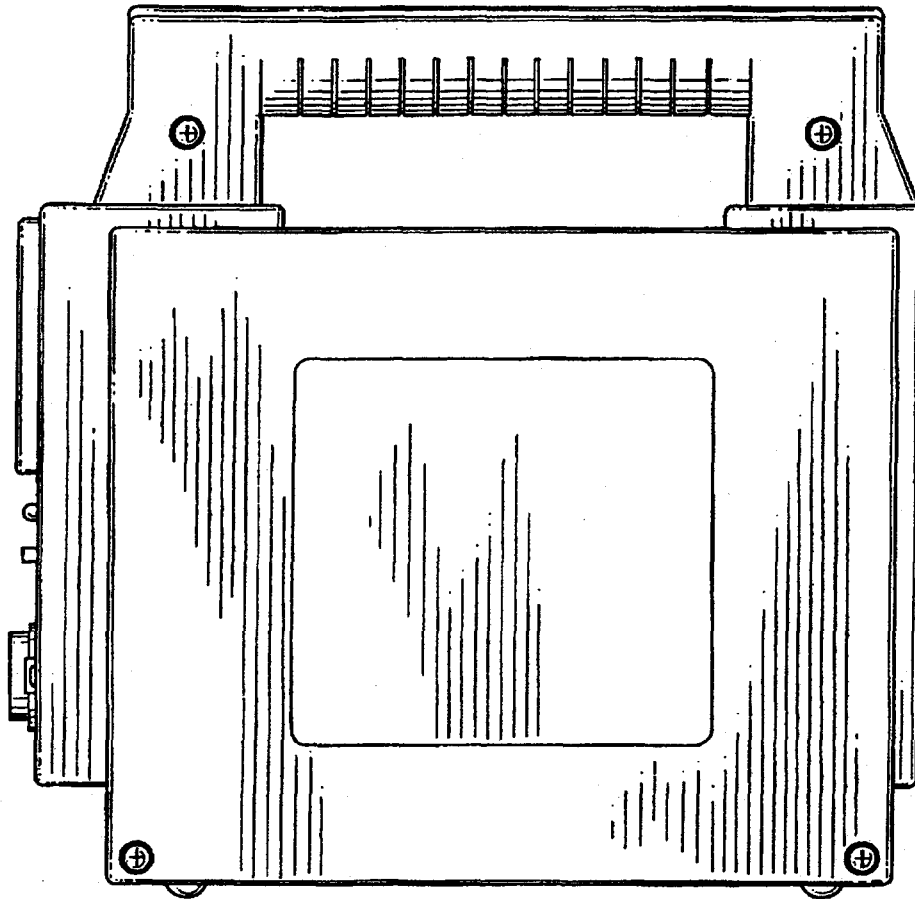


FIG. 6

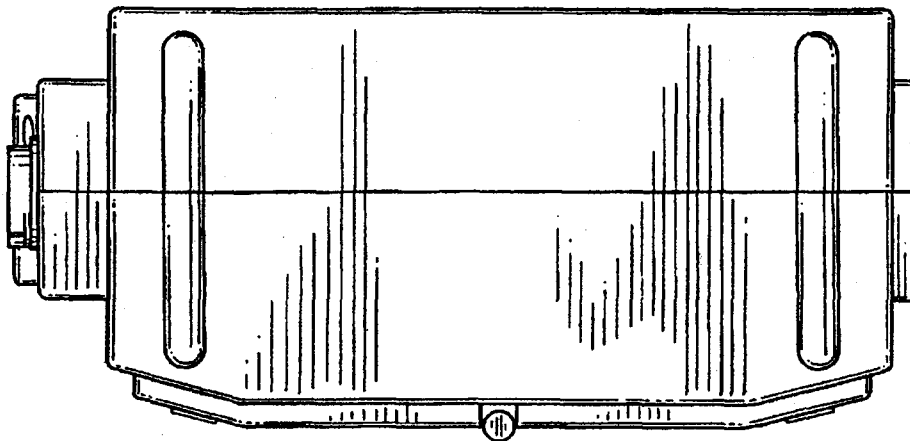


FIG. 7

[54] PORTABLE BATTERY CHARGER

[76] Inventors: Donald R. Manis, 7968 Camberly Dr.; Lon C. Hammock, 4915 W. Emory Rd., both of Powell, Tenn. 37849

[21] Appl. No.: 264,859

[22] Filed: Oct. 31, 1988

[51] Int. Cl.⁴ H02J 7/00; B65B 31/00

[52] U.S. Cl. 320/2; 141/38; 152/415; 439/504

[58] Field of Search 320/2, 25, 26; 439/501, 439/504; 152/415; 141/38

[56] References Cited

U.S. PATENT DOCUMENTS

2,966,248	12/1960	Armbruster	320/2 X
3,729,033	4/1973	Bayerkohler	141/38 X
3,967,133	6/1976	Bokern	320/2 X
4,161,682	7/1979	Corvette	320/2
4,443,751	4/1984	Humphrey	320/2
4,667,141	5/1987	Steele	320/2
4,776,766	10/1988	Brent	152/415 X
4,798,233	1/1989	Mooney	152/415 X

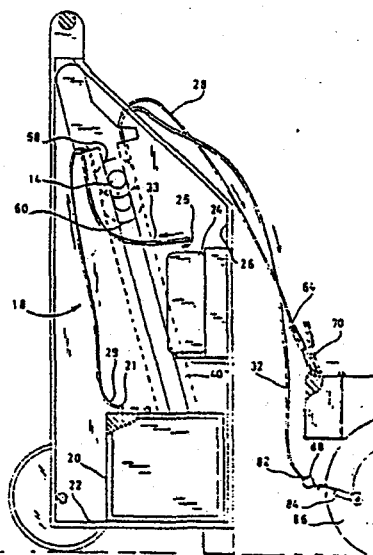
Primary Examiner—R. J. Hickey

Attorney, Agent, or Firm—Pitts and Brittan

[57] ABSTRACT

An improved portable battery charger (10) for boosting and charging automotive-type batteries. The battery charger (10) comprises rectifier means (34), an automotive-type battery (20), and an air compressor (24) powered by the battery (20) received within a charger housing (12). Positive and negative charging cables (30 and 28, respectively) are retractably received within first and second cable housings (14 and 16), respectively. The first and second cable housings are positioned adjacent oppositely disposed external walls of the charger housing (12) and are dimensioned such that the positive and negative charging clamps (76 and 70, respectively) may be slidably positioned over saddle notches (72) defined by the cable housings when the charging cables are fully retracted into the cable housings whereby the charging clamps (76 and 70) are physically isolated from each other by the charger housing 12, thereby preventing accidental contact between the clamps. When connected to a source of AC power, the rectifier means (34) provides a DC charging voltage to the battery (20). The charging cables are connected to the terminals of the battery (20) to permit the boosting or jump-starting of a vehicle with a dead battery when AC power is not conveniently available.

12 Claims, 5 Drawing Sheets



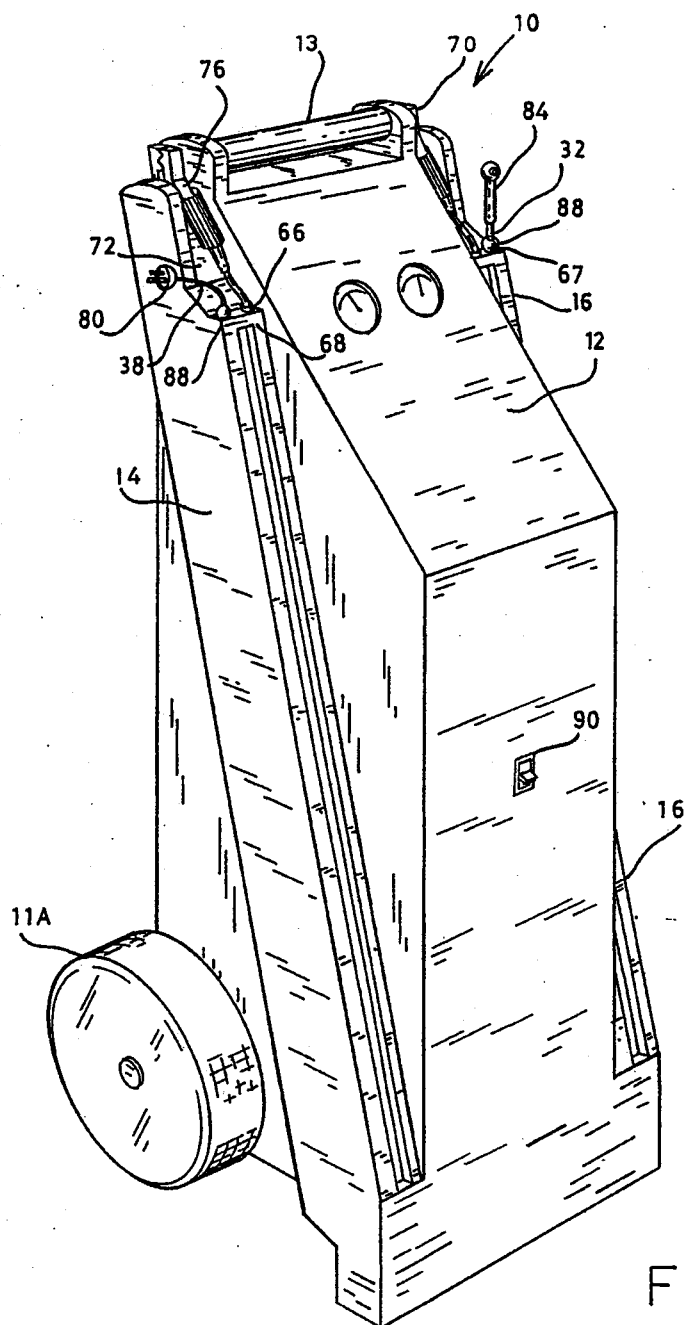


FIG. 1

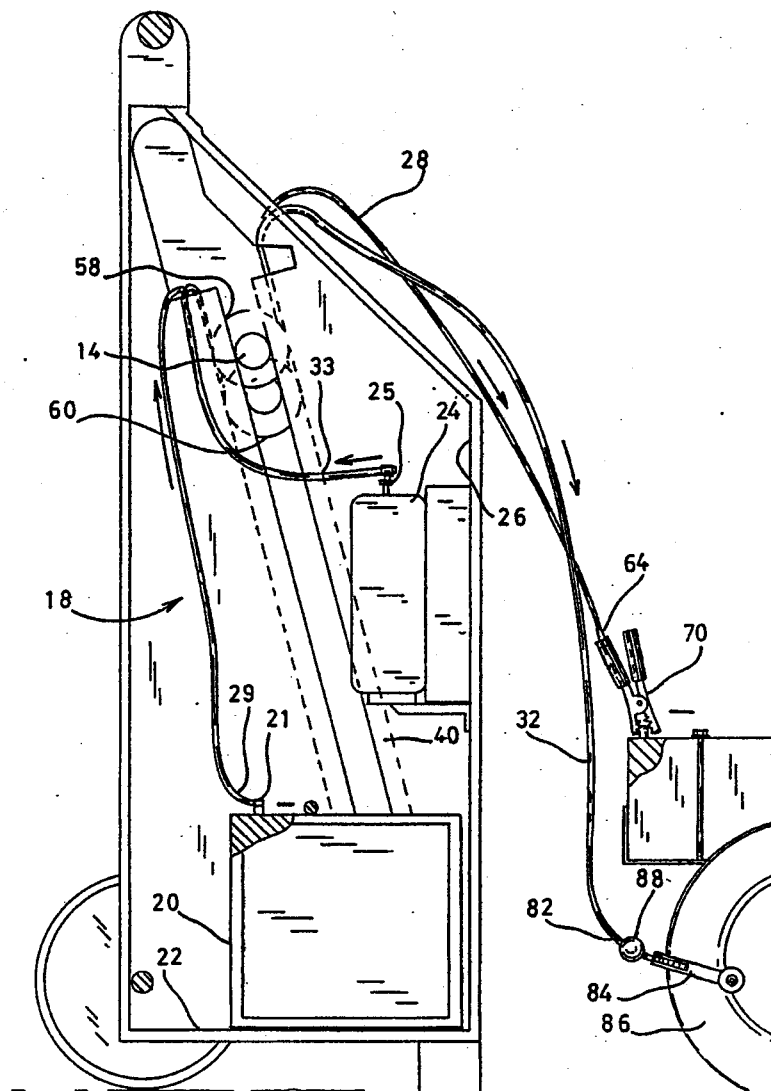


FIG. 2

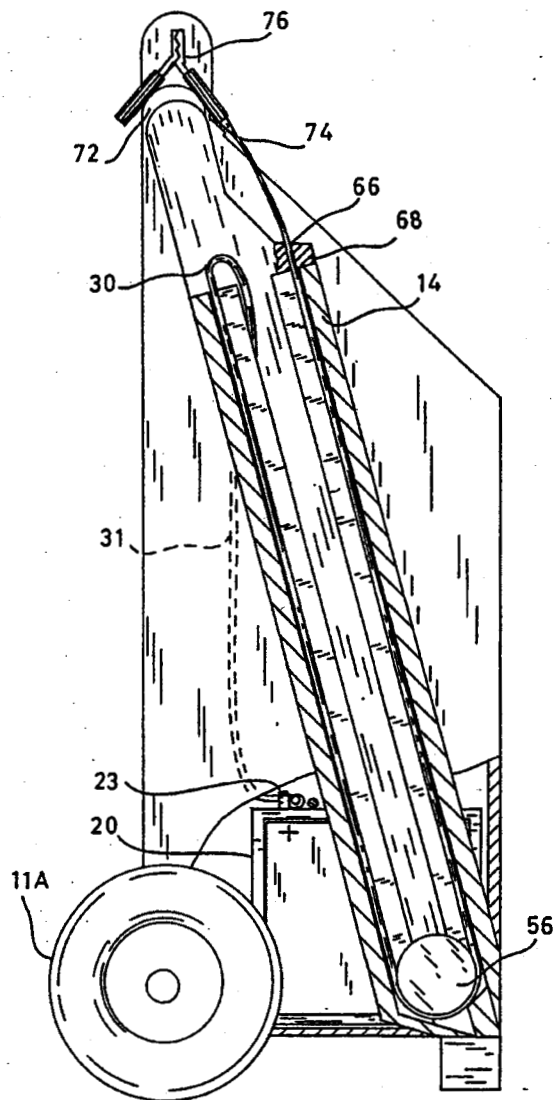


FIG. 3

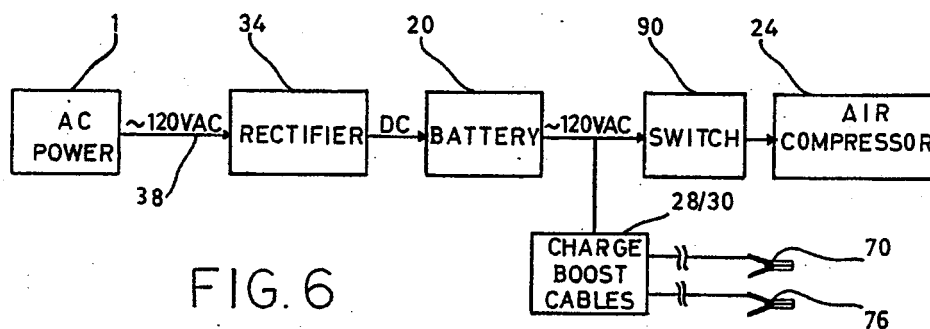


FIG. 6

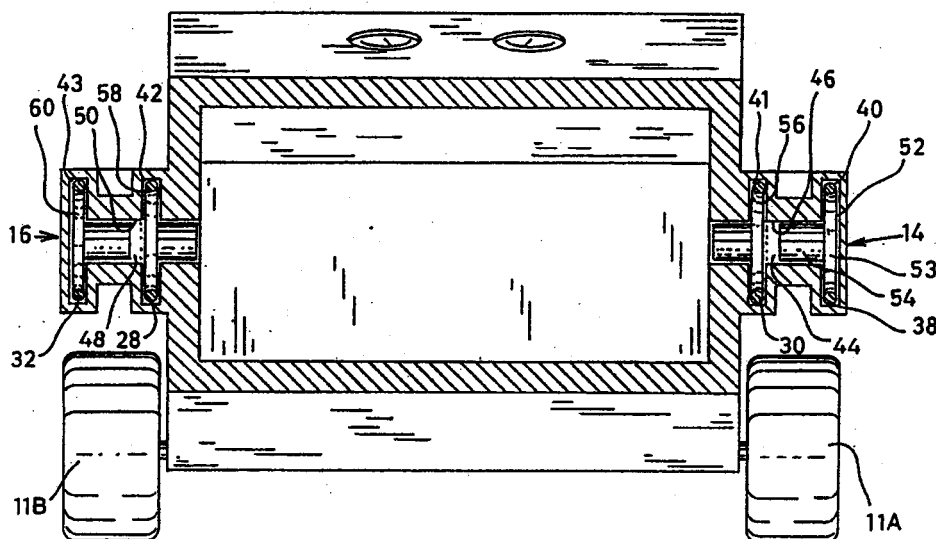


FIG. 5

PORTABLE BATTERY CHARGER

DESCRIPTION

1. Technical Field

This invention relates to a portable battery charger and more particularly to one used for boosting and charging automobile and other vehicle batteries.

2. Background Art

Battery chargers for charging automotive-type batteries are well known in the art. Most prior art chargers have been relatively simple devices which include rectifier means for converting conventional AC power to a DC charging source, and cables for connecting the DC charging source to the battery to be charged. The cables are usually provided with clamps for securing the cables to the terminals of the battery being charged. These prior devices are fairly adequate for use in garages or relatively limited areas where AC power is conveniently available. However, they are less than satisfactory for use in such areas as automobile sales lots, parking lots and garages, and other areas where AC power is not conveniently available. Even when the battery to be charged or boosted can be accessed with the use of an extension cord, such use is plagued with problems of broken and tangled extension cords, power loss in the extension cord, and tripping hazards. (As used herein, the terms boost and jump-start are synonymous and refer to the procedure of electrically connecting a charged battery in parallel with the discharged battery of a vehicle to permit the starting of the vehicle.)

In most prior art battery chargers, the charging cables are wrapped around cleats or lugs provided on the housing of the charger during storage. The cables can become tangled, twisted, frayed, and otherwise damaged because of the inadequacies of this storage method. Further, these prior art chargers have the disadvantage of being susceptible to dangerous sparking caused by accidental touching of the clamps when power is connected to the charger.

One prior art battery charger, that described in U. S. Pat. No. 4,161,682, has attempted to overcome the disadvantages of the previous chargers described above. The device described in the cited patent includes a battery housed within a common housing of the battery charger which permits jump-starting of a vehicle having an excessively discharged battery, even when AC power is not conveniently available. The device of the cited patent also includes means for retracting the charging cables into separate compartments of the charger's housing when they are not in use. However, this device does not adequately deal with the problem of accidental touching of the charging clamps when power is supplied to the cables. In the device of the cited patent, the charging clamps extend vertically from the uppermost portion of the charger's housing with no obstruction between the two clamps to prevent one from being pushed or pulled into contact with the other. Further, because there is no physical obstruction between the two clamps, they could easily be accidentally shorted by a metal object such as a screwdriver or wrench.

The applicant is unaware of any prior art battery charger which includes a self-contained source of compressed air for pumping up automobile tires and the like. Such a feature is particularly advantageous in a charger used in such places as automobile sales lots, parking lots

and garages, and other locations where neither AC power nor sources of compressed air are readily available.

Therefore, it is an object of the present invention to provide a portable battery charger for boosting and charging automotive-type batteries which is capable of jump-starting (i.e. providing sufficient power to crank the engine of) an automobile without the need of being connected to a source of AC power.

It is a further object of the present invention to provide a portable battery charger for boosting and charging automotive-type batteries which includes means for retractably storing the charging cables of such battery charger when they are not in use, thereby minimizing the damage to caused by the twisting, fraying, tangling, etc. of the cables.

It is another object of the present invention to provide a portable battery charger for boosting and charging automotive-type batteries which includes means for minimizing accidental contact between its charging clamps when power is applied to the cables.

It is yet another object of the present invention to provide a portable battery charger for boosting and charging automotive-type batteries which includes a self-contained source of compressed air for pumping up automobile tires and the like.

DISCLOSURE OF THE INVENTION

Other objects and advantages will be accomplished by the present invention which provides a portable battery charger for boosting and charging automobile and other vehicle batteries. The battery charger of the present invention comprises an upright housing containing a rectifier means, an automotive-type battery, and, in the preferred embodiment, a source of compressed air. The DC output of the rectifier means is electrically connected to the automotive-type battery. At least one wheel is secured to the housing proximate its lowermost portion for rollably engaging the ground or a floor.

Positive and negative charging/booster cables, each having proximal and distal end portions, are provided. An intermediate portion of the positive charging/booster cable is retractably received within a cable housing secured to a vertical exterior wall of the charger's housing; an intermediate portion of the negative charging/booster cable is retractably received within a second cable housing secured to an opposite vertical exterior wall of the charger's housing such that the first and second cable housings are separated by the charger's housing. The proximal end portions of the positive and negative charger/booster cables extend into the charger housing through cooperating openings in the charger housing and first and second cable housings, respectively, and are connected to the positive and negative terminals, respectively, of the automotive-type battery.

The distal end portions of the positive and negative charging/booster cables extend through openings defined in an upper portion of the first and second cable housings, respectively, and are connected to positive and negative charging clamps, respectively, exterior of the cable housings. The first and second cable housings are each provided with a saddle notch over which the positive and negative charging clamps, respectively, may be positioned when not in use. The cable housings are dimensioned such that the saddle notches are physically isolated from one another by the charger housing,

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thereby preventing the charging clamps from accidentally coming into contact when not in use. In the preferred embodiment, retraction means are contained within each cable housing whereby most of the intermediate portion of the charging cables can be deployed out through the opening in the upper portion of the cable housings for charging or boosting an external battery, and then retracted back into the cable housing when not in use. The charging/booster cables are dimensioned such that the retraction means bottoms out (i.e. stops retracting) just as the charging clamps are positioned over the saddle notches of the cable housings, thereby minimizing stress at the points where the cables are connected to the charging clamps.

In the preferred embodiment, each of the first and second cable housings are provided with dual cable channels and retraction means, thereby providing a spare in each. One of the spares is used to retract and house the AC supply cord of the rectifier means; the other is used to retract and house an air hose which is connected at a proximal end portion to the internal source of compressed air, and which has at its distal end portion a fitting suitable for inflating automobile tires and the like.

BRIEF DESCRIPTION OF THE DRAWINGS

The above-mentioned features of the invention will become more clearly understood from the following detailed description of the invention read together with drawings in which:

FIG. 1 illustrates a perspective view of a portable battery charger constructed in accordance with the various features of the present invention.

FIG. 2 illustrates a side elevation view, in section, of a portable battery charger constructed in accordance with the various features of the present invention.

FIG. 3 illustrates a side elevation view, partially in section, of a portable battery charger constructed in accordance with the various features of the present invention.

FIG. 4 illustrates a side elevation view, in section, of a portable battery charger constructed in accordance with the various features of the present invention.

FIG. 5 illustrates a top view, in section, of a portable battery charger constructed in accordance with the various features of the present invention.

FIG. 6 illustrates a block diagram of the circuitry of a portable battery charger constructed in accordance with the various features of the present invention.

BEST MODE FOR CARRYING OUT THE INVENTION

A portable battery charger incorporating various features of the invention is illustrated generally at 10 in the figures. Referring to FIG. 1, the battery charger 10 includes a charger housing 12 having wheels 11A and 11B secured to the housing 12 proximate its lowermost portion for rollably engaging the ground or a floor. In the preferred embodiment, a handle 13 is provided proximate the uppermost portion of the charger housing 12 to facilitate propelling the charger 10 to a desired location. The charger housing 12 is an elongated, substantially rectangular, upstanding box-like member having an internal cavity, indicated at 18 in FIG. 2. Substantially identical first and second cable housings 14 and 16, respectively, are positioned adjacent opposing exterior vertical walls 15 and 17, respectively, of the charger housing 12. It will be appreciated by those

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skilled in the art that the cable housings 14 and 16 may be separate housings secured to the charger housing 12, or, in the alternative, may be integrally constructed with the charger housing 12.

Referring now to FIG. 2, contained within the internal cavity 18 defined by the charger housing 12 is an automotive-type battery 20. Preferably, the battery 20 is positioned on a floor or shelf 22 proximate the lowermost portion of the housing 12 to provide a low center of gravity to the device 10 for stability. A negative charging cable 28, having a proximal end portion 29 connected to the negative terminal 21 of the battery 20, is provided. Similarly, as shown in FIGS. 3 and 4, a positive charging cable 30, having a proximal end portion 31 connected to the positive terminal 23 of the battery 20, is also provided. Also contained within the cavity 18 is a source of compressed air which in the preferred embodiment is a battery powered air compressor 24 mounted on an interior wall 26 of the housing 12. An air hose 32, having a proximal end portion 33 connected to the output fitting 25 of the air compressor 24, is provided. Referring to FIG. 4, rectifier means 34 are contained within the cavity 18 of the charger housing 12 and are secured to an interior wall 36 thereof. It will be appreciated by those skilled in the art that the rectifier means 34 can be any of a number of conventional battery charging circuits of various capacities which convert AC power to a DC charging voltage. An AC supply cord 38, having a proximal end portion 39 connected to the input of the rectifier means 34, is provided.

Referring now to FIGS. 5, a sectional top view of the portable battery charger 10 is illustrated. FIG. 5, when viewed with the other figures, illustrates retraction means contained within the cable housings 14 and 16. Channels 40 and 41 are defined by the first cable housing 14; similarly, channels 42 and 43 are defined by the second cable housing 16. The channels extend through substantially the length of the cable housings 14 and 16. A cooperating channel 44 which is perpendicular to the channels 40 and 41 is defined by the first cable housing 14 and the vertical wall section 46 of the charger housing 12; similarly, a cooperating channel 48 which is perpendicular to the channels 42 and 43 is defined by the second cable housing 16 and the vertical wall section 50 of the charger housing 12. A weighted member 52, having a wheel section 53 and a shaft section 54, is received within the void defined by the channel 40 and the cooperating channel 44; the wheel section 53 being closely received within the channel 40 with the shaft section extending into and being substantially received within the cooperating channel 44. Similarly, substantially identical weighted members 56, 58 and 60, each having a wheel section 53 and a shaft section 54, are received within the voids defined by the channel 41 and cooperating channel 44, the channel 42 and cooperating channel 48, and the channel 43 and cooperating channel 48, respectively. An intermediate portion of the positive charging cable 30 is looped underneath the wheel section 53 of the weighted member 56 such that the intermediate portion of the cable 30 is substantially received within the channel 41 when the weighted member is permitted to fall by gravity to the bottom of the channel 41. The proximal end portion 31 extends into the cavity 18 of the charger housing 12 through a void in the uppermost portion of the first cable housing 14 and a cooperating opening 62 in the vertical wall section 63 of the charger housing 12 where it terminates in an electri-

cal connection to the positive terminal 23 of the battery 20 as previously described. The distal end portion 74 of the positive charging cable 30 extends through a bore 66 in an upper end portion 68 of the first cable housing 14 and terminates in an electrical connection to a positive charging clamp 76. The positive charging cable is dimensioned such that when the intermediate portion of the cable is fully retracted within the channel 41 of the first cable housing 14, i.e. the weighted member 56 is at the bottom of the channel 41, there is just enough of the distal end portion 74 of the cable 30 extending through the bore 66 to permit the handles of positive charging clamp 76 to be slidably positioned over the recessed surface of a saddle notch 72 provided in the exterior of the cable housing 14 proximate its upper end without the electrical connection between the distal end portion 74 and the charging clamp 76 being stressed by the retraction force exerted on the cable 30 by the weighted member 56. Referring to FIGS. 1 and 2, it can be seen that an intermediate portion of the negative charging cable 28 is similarly received within the channel 42 of the second cable housing 16 with the distal end portion 64 of the negative charging cable 28 terminating in a negative charging clamp 70. Referring to FIGS. 1 and 4, it can be seen that an intermediate portion of the AC power supply cord 38 is similarly received within the channel 40 of the first cable housing 14 with the distal end portion 78 of the power cord 38 terminating in a conventional electrical plug 80. Referring to FIGS. 1 and 2, it can be seen that an intermediate portion of the air hose 32 is similarly received within the channel 43 of the second cable housing 16 with the distal end portion 82 of the air hose 32 terminating in a fitting 84 suitable for inflating a vehicle tire 86 and the like. Referring to FIG. 1, it can be seen that the AC power supply cord 38 and the air hose 32 are provided with means 88 proximate their distal end portions 78 and 82, respectively, for arresting the retraction of such end portions into the channels 40 and 43, respectively. As illustrated, the means 88 can be a spherically shaped member secured to the cord/hose which has a larger diameter than the second bore 67 in the cable housings 14 and 16 through which the end portions extend. Means 88 of the type illustrated reduce stress on the plug 80 and the fitting 84.

Referring to FIG. 1, it can be seen that the cable housings 14 and 16 are dimensioned such that, when the charging clamps 76 and 70 are positioned over the saddle notches 72 of the cable housings 14 and 16, respectively, the charging clamps 70 and 76 are physically isolated from one another by the charger housing 12. This is an important feature of the present invention, because it minimizes the danger of sparking caused by the clamps 70 and 76 accidentally coming into contact with one another when the cables are in the stored position. It is almost impossible for the clamps to be accidentally pushed or pulled into contact with each other because of the physical obstruction provided by the charger housing 12. The obstruction provided by the charger housing 12 also minimizes the possibility of accidental shorting of the clamps by a metal object such as a screwdriver or a wrench.

Referring to FIG. 6, a block diagram of the circuitry of the portable battery charger 10 is illustrated. AC power, indicated by the block 1, is supplied through a power supply cord 38, to a rectifier 34 comprising a battery charging circuit of conventional design, which converts the AC power to a DC charging voltage. The output of the rectifier 34 is connected to the automo-

tive-type battery 20, thereby charging the battery 20 when the power supply cord 38 is plugged into a conventional AC outlet. DC power is selectively supplied to the air compressor 24 through switch means 90, a preferred location of which is shown in FIG. 1, by the battery 20. Negative and positive charging cables 28 and 30, respectively, are connected to the negative and positive terminals 21 and 23, respectively, of the battery 20. The charging cables 28 and 30 terminate in charging clamps 70 and 76 at their distal end portions.

In light of the foregoing, it will be appreciated that the present invention provides a portable battery charger for boosting and charging automotive-type batteries. The portable battery charger 10 can be wheeled to a vehicle disabled by a discharged battery. The disabled vehicle can be started by boosting its battery by connecting the positive and negative charging clamps 76 and 70, respectively, to the positive and negative terminals of the vehicle's discharged battery. The battery charger 10 can be used to charge an automotive-type battery by connecting the positive and negative charging clamps 76 and 70, respectively, to the positive and negative terminals, respectively, of the battery to be charged and plugging the AC supply cord 38 into an AC power outlet. Further, the battery charger 10 can be wheeled to a location where neither AC power nor compressed air are readily available and used to inflate a vehicle tire 86 or the like by connecting the fitting 84 to the tire 86 and turning on the air compressor 24 which is powered by the internal battery 20 of the battery charger 10. When not in use, the charging cables 28 and 30 are retracted into the cable housings 14 and 16 and the charging clamps 70 and 76 are isolated one from the other by the charger housing 12 to prevent dangerous sparking caused by accidental contact between the charging clamps.

While a preferred embodiment has been shown and described, it will be understood that there is no intent to limit the invention to such disclosure, but rather it is intended to cover all modifications and alternate constructions falling within the spirit and scope of the invention as defined in the appended claims.

We claim:

1. A portable battery charger for boosting and charging automotive-type batteries comprising:
 - a charger housing;
 - an automotive-type battery received within said charger housing, said battery having a positive terminal and a negative terminal;
 - a first cable housing secured to said charger housing proximate a first vertical wall of said charger housing, said first cable housing having an upper portion defining a first bore and a first saddle notch disposed above said bore;
 - a second cable housing secured to said charger housing proximate a second vertical wall of said charger housing, said second vertical wall being oppositely disposed from said first vertical wall whereby said first and second cable housings are physically isolated one from the other by said charger housing, said second cable housing having an upper portion defining a second bore and a second saddle notch disposed above said second bore;
 - a positive charging cable having proximal and distal end portions, said proximal end portion being electrically connected to a positive terminal of said automotive-type battery, an intermediate portion of said positive charging cable being received

through said first bore and retractably received within said first cable housing, said positive charging cable terminating at its said distal end portion external to said first cable housing;

a negative charging cable having proximal and distal end portions, said proximal end portion being electrically connected to a negative terminal of said automotive-type battery, an intermediate portion of said negative charging cable being received through said second bore and retractably received within said second cable housing, said negative charging cable terminating at its said distal end portion external to said second cable housing; and positive and negative charging clamps electrically connected to said distal end portions of said positive and negative charging cables, respectively, said positive and negative clamps each having a pair of rearwardly disposed, diverging handles for being releasably received in said first and second saddle notches, respectively, whereby said clamps are held in a preselected position when said intermediate portions of said positive and negative charging cables are fully received within said first and second cable housings.

2. The portable battery charger of claim 1 wherein said first and second cable housings further comprise cable retraction means received therein.

3. The portable battery charger of claim 2 wherein said retraction means comprises said cable housings defining an elongated substantially vertical channel therein, a weighted member closely received within said elongated channel, and said intermediate portion of said charging cables being looped under said weighted member whereby said weighted member exerts a force on said charging cables tending to retract said intermediate portion into said cable housing.

4. The portable battery charger of claim 3 wherein said weighted member comprises a wheel-like portion and a shaft portion, said shaft portion having a common axis with said wheel-like portion and said shaft portion having a smaller diameter than said wheel-like portion.

5. The portable battery charger of claim 1 further comprising a source of compressed air received within said charger housing, an air hose having proximal and distal end portions, said proximal end portion of said air hose being connected to said source of compressed air, and said distal end portion of said air hose terminating in a fitting for facilitating the inflation of vehicle tires and the like externally of said charger housing.

6. The portable battery charger of claim 5 wherein said source of compressed air is an air compressor powered by said automotive-type battery.

7. The portable battery charger of claim 1 wherein said first and second cable housings further comprise cable retraction means received therein.

8. The portable battery charger of claim 7 wherein said cable retraction means comprises said cable housings defining an elongated substantially vertical channel therein, a weighted member closely received within said elongated channel, and said intermediate portion of said charging cables being looped under said weighted member whereby said weighted member exerts a force on said charging cables tending to retract said intermediate portion in said cable housing.

9. The portable battery charger of claim 8 wherein said weighted member comprises a wheel-like portion and a shaft portion, said shaft portion having a common

axis with said wheel-like portion and said shaft portion having a smaller diameter than said wheel-like portion.

10. A portable battery charger for boosting and charging automotive type batteries comprising:

- a charger housing;
- an automotive-type battery received within said charger housing, said battery having a positive terminal and a negative terminal;
- a positive charging cable having proximal and distal end portions, said proximal end portion being electrically connected to said positive terminal of said automotive-type battery;
- a negative charging cable having proximal and distal end portions, said proximal end portion being electrically connected to said negative terminal of said automotive-type battery;
- positive and negative charging clamps electrically connected to said distal end portions of said positive and negative charging cables, respectively; and
- an air compressor mounted within said charger housing and powered by said automotive-type battery, said compressor being provided with an air hose having a proximal end portion connected to said compressor and a distal end portion terminating in a fitting for facilitating the inflation of vehicle tires.

11. A portable battery charger for boosting and charging automotive-type batteries comprising:

- a charger housing;
- rectifier means received within said housing, said rectifier means having a DC output;
- an automotive-type battery received within said charger housing, said battery being electrically connected to said DC output of said rectifier means whereby said battery is supplied with a charging voltage when said rectifier means is approximately connected to an AC power source;
- a first cable housing secured to said charger housing proximate a first vertical wall of said charger housing;
- a second cable housing secured to said charger housing proximate a second vertical wall of said charger housing, said second vertical wall being oppositely disposed from said first vertical wall whereby said first and second cable housings are physically isolated one from the other by said charger housing;
- a positive charging cable having proximal and distal end portions, said proximal end portion being electrically connected to a positive terminal of said automotive-type battery, an intermediate portion of said positive charging cable being retractably received within said first cable housing, said positive charging cable terminating at its said distal end portion externally of said cable housing and said charger housing;
- a negative charging cable having proximal and distal end portions, said proximal end portion being electrically connected to a negative terminal of said automotive-type battery, an intermediate portion of said negative charging cable being retractably received within said second cable housing, said negative charging cable terminating at its said distal end portion externally of said cable housing and said charger housing;
- positive and negative charging clamps electrically connected to said distal end portions of said positive and negative charging cables, respectively;

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said first and second cable housings being dimensioned whereby said positive and negative charging clamps are physically isolated from each other by said charger housing when said intermediate portions of said positive and negative charging cables are fully received within said first and second cable housings, respectively, thereby preventing said charging clamps from accidentally coming into contact with one another while in a stored position; and
a source of compressor air including an air compressor powered by said automotive-type battery, and

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an air hose having proximal and distal end portions, said proximal end portion of said air hose being connected to said air compressor and said distal end portion of said air hose terminating in a fitting for facilitating the inflation of vehicle tires and the like externally of said charger housing.

12. The portable battery charger of claim 11 wherein said first and second cable housings further comprise notch-like portions over which said positive and negative charging clamps can be slidably positioned when not in use.

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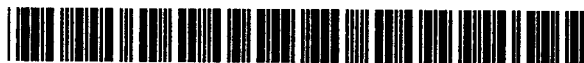
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US00D413098S

United States Patent [19]
Dayan

[11] **Patent Number: Des. 413,098**

[45] **Date of Patent: ** Aug. 24, 1999**

[54] **PORTABLE POWER STATION**

[75] **Inventor: Mervin A. Dayan, Eatontown, N.J.**

[73] **Assignee: Roadmaster (USA) Corp., Eatontown, N.J.**

[**] **Term: 14 Years**

[21] **Appl. No.: 29/091,383**

[22] **Filed: Jul. 29, 1998**

[51] **LOC (6) Cl. 13-02**

[52] **U.S. Cl. D13/110; D13/106**

[58] **Field of Search D13/106, 110;
307/150, 66; 429/96**

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*Primary Examiner—Joel Sincavage
Attorney, Agent, or Firm—Ezra Sutton*

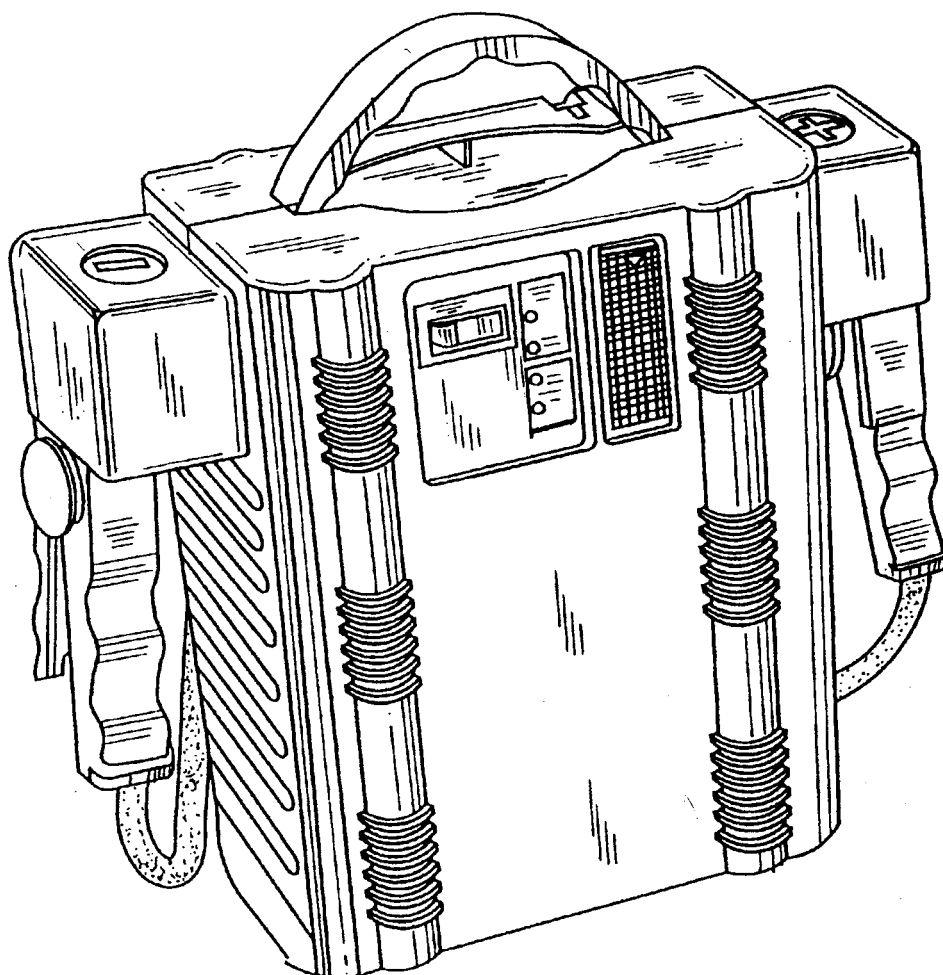
[57] **CLAIM**

The ornamental design for a portable power station, as shown and described.

DESCRIPTION

FIG. 1 is a perspective view of a portable power station embodying my new design;
FIG. 2 is a front view thereof;
FIG. 3 is a rear view thereof;
FIG. 4 is a side view thereof, the other side view being a mirror image thereof; and,
FIG. 5 is a top view thereof.

1 Claim, 5 Drawing Sheets



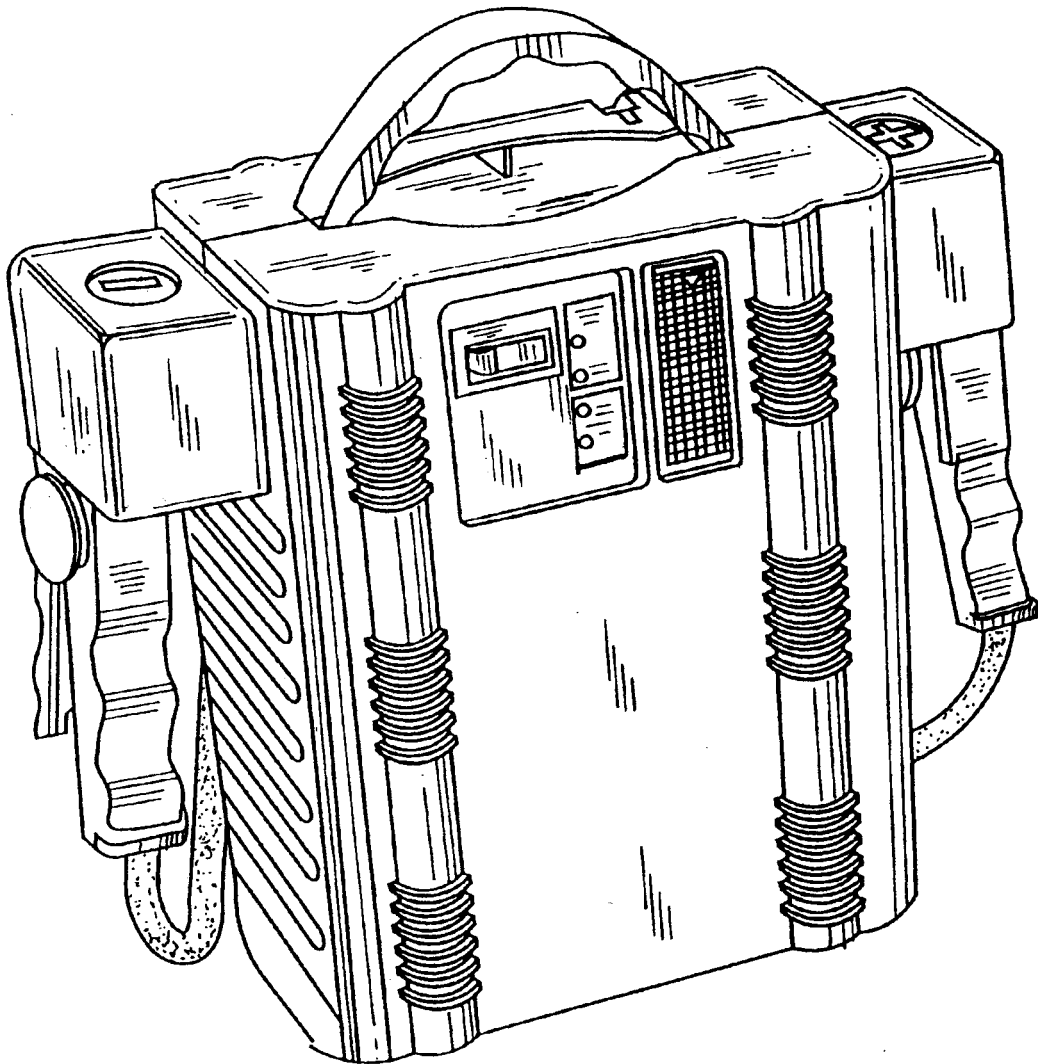


FIG. 1

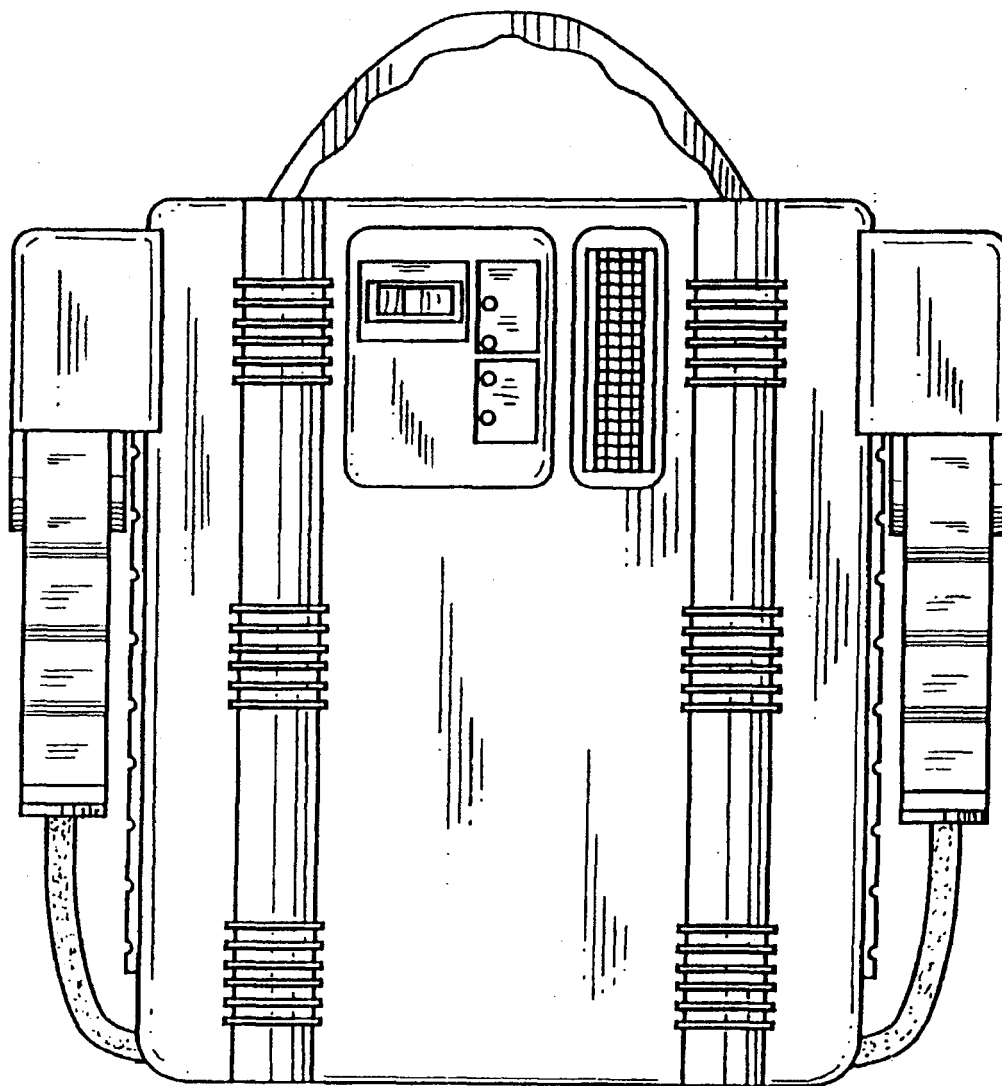


FIG. 2

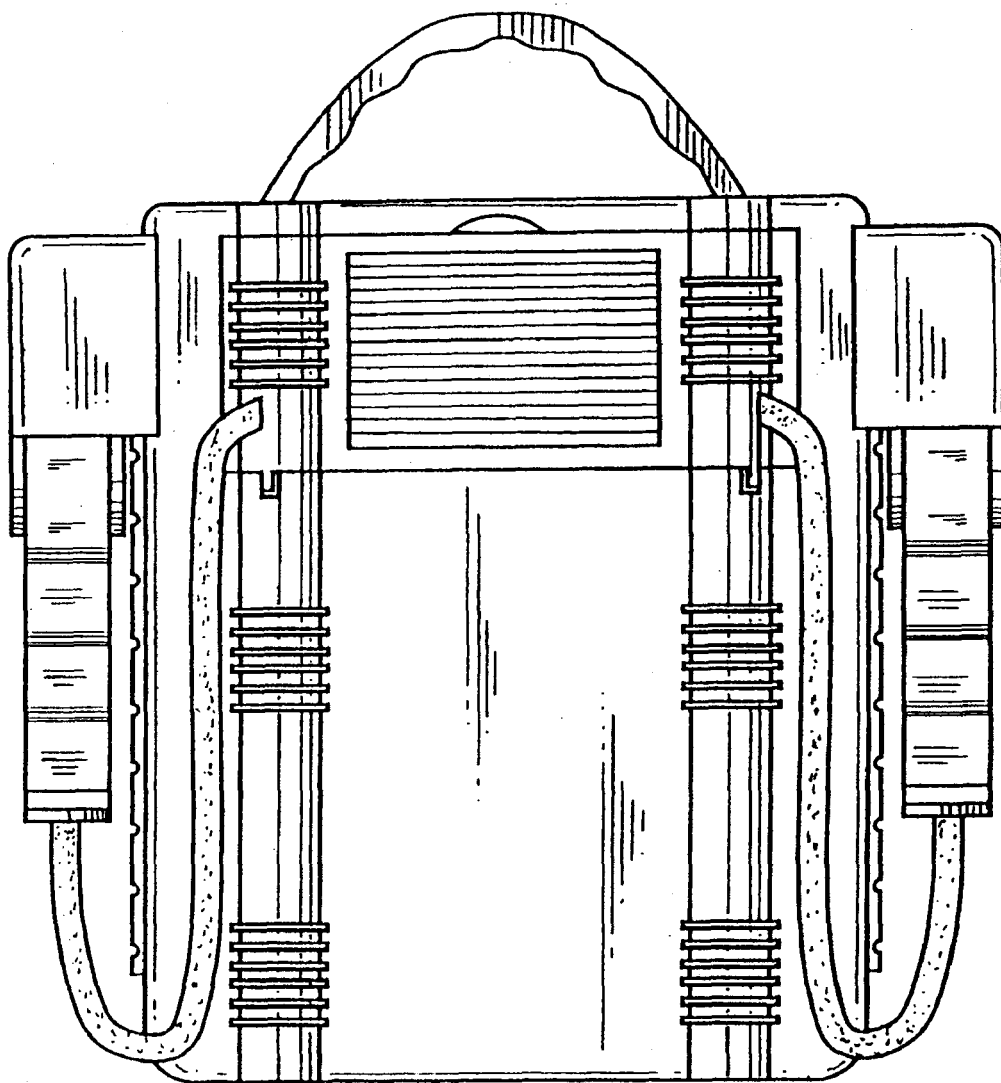


FIG. 3

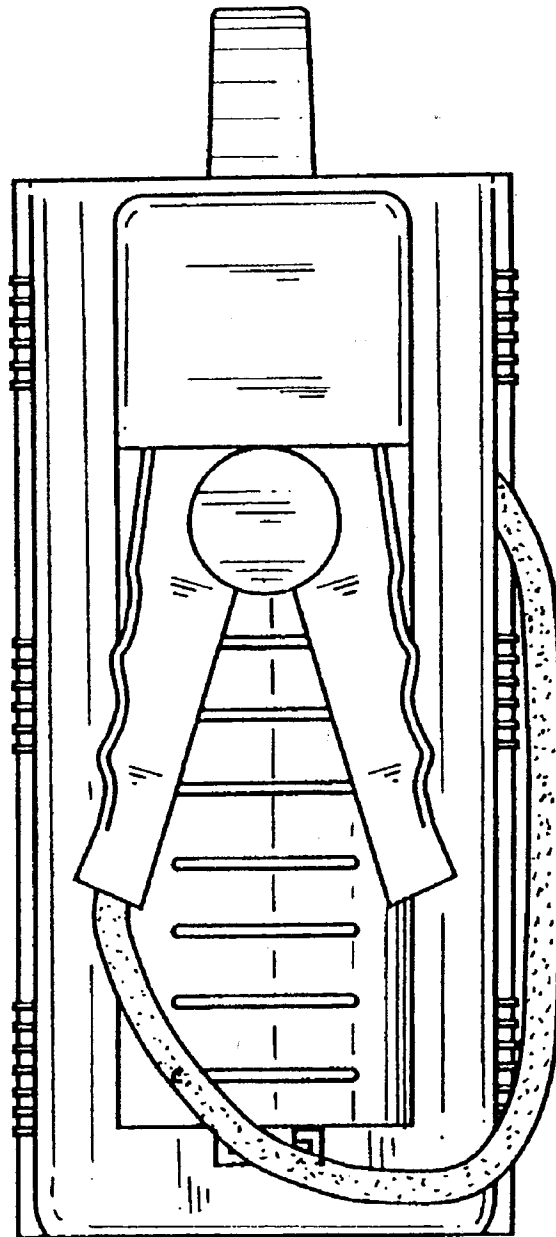


FIG. 4

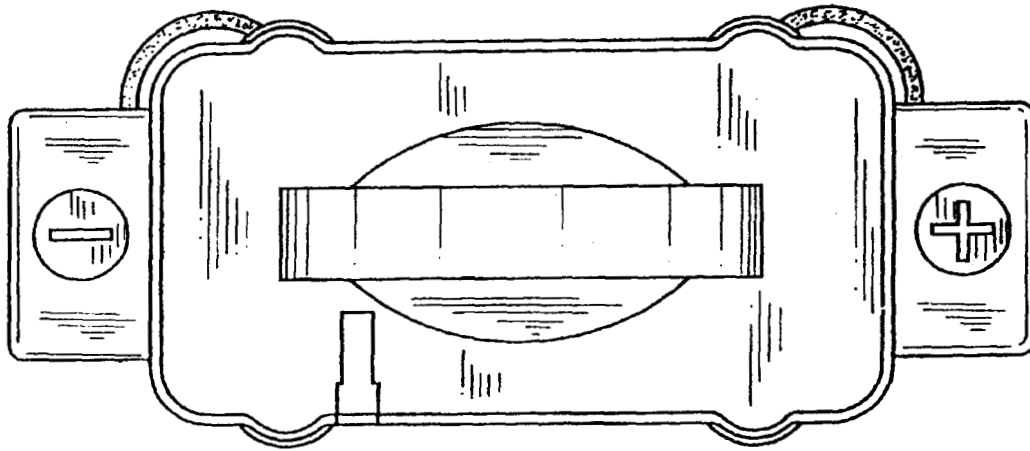


FIG. 5



US00RE37482B1

(19) **United States**
(12) **Reissued Patent**
Guay et al.

(10) **Patent Number:** **US RE37,482 E**
(45) **Date of Reissued Patent:** **Dec. 25, 2001**

(54) **PORTABLE POWER SUPPLY**
(75) Inventors: **Etienne Guay**, Oxford; **André Côté**,
Montreal, both of (CA)
(73) Assignee: **Century Mfg. Co.**, St. Paul, MN (US)
(21) Appl. No.: **29/095,253**
(22) Filed: **Oct. 20, 1998**

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Primary Examiner—Joel Sincavage

(74) *Attorney, Agent, or Firm*—Merchant & Gould P.C.

(57) **CLAIM**

The ornamental design for a portable power supply, as shown and described.

DESCRIPTION

FIG. 1 is a front view of a portable power supply showing my new design;

FIG. 2 is a front and right side perspective view thereof;

FIG. 3 is a top view thereof;

FIG. 4 is a right side view thereof, the left side view thereof being a mirror image [except for the color of the cable and clamp];

FIG. 5 is a front view thereof showing one cable clamp removed from its storage position;

FIG. 6 is a rear view thereof; and,

FIG. 7 is a bottom view thereof.

1 Claim, 4 Drawing Sheets

Related U.S. Patent Documents

Reissue of:

(64) Patent No.: **Des. 374,859**
Issued: **Oct. 22, 1996**
Appl. No.: **29/018,788**
Filed: **Feb. 15, 1994**

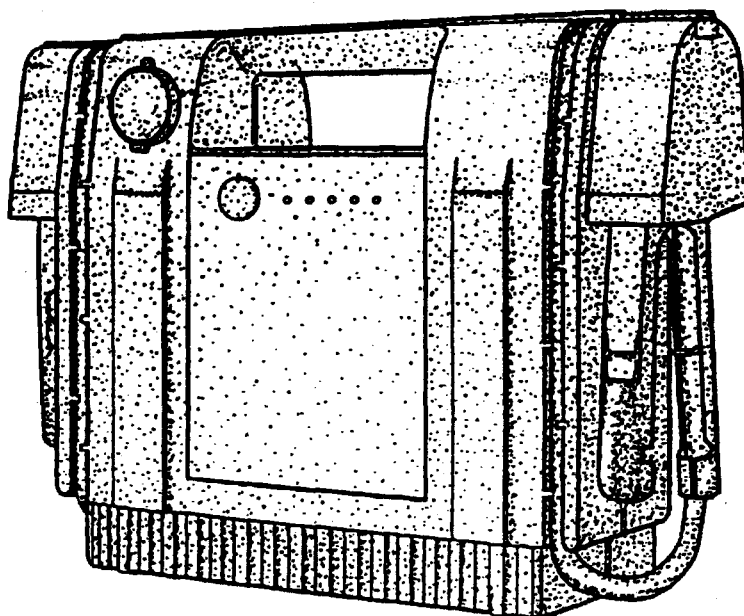
(51) **LOC (6) Cl.** **13- 02**
(52) **U.S. Cl.** **D13/110**
(58) **Field of Search** D13/103, 106,
D13/110; 307/64-66, 150; 429/96-100,
121, 163, 176, 185, 187; 320/104, 105

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Matter enclosed in heavy brackets [] appears in the original patent but forms no part of this reissue specification; matter printed in italics indicates the additions made by reissue.



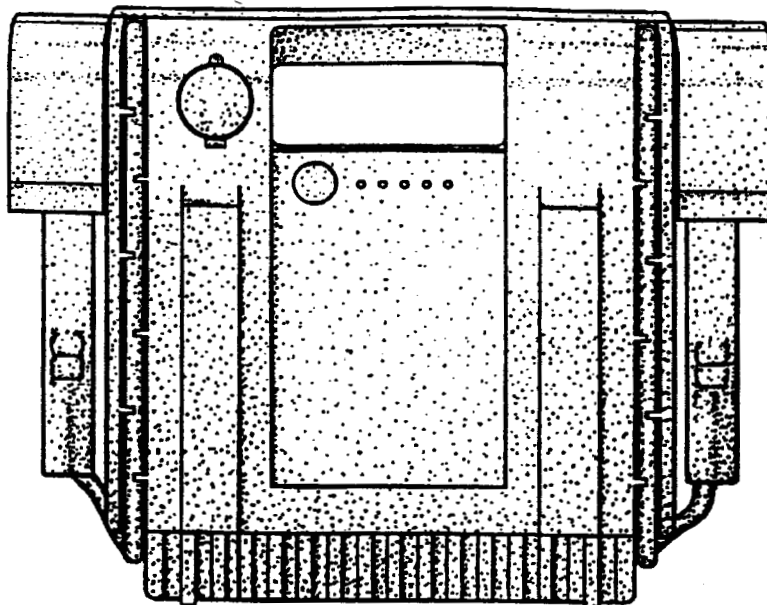


Fig. 1
(AMENDED)

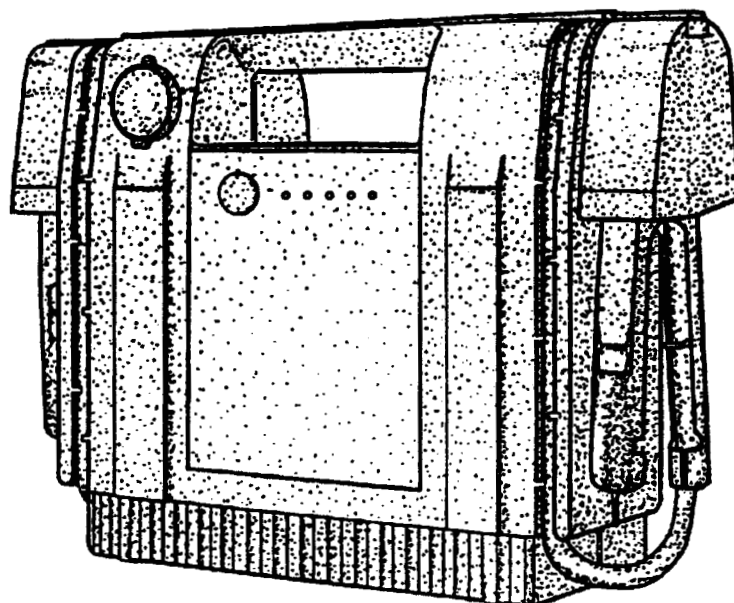


Fig. 2
(AMENDED)

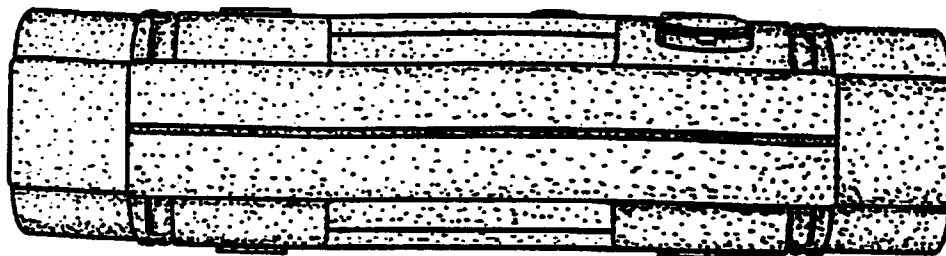


FIG. 3

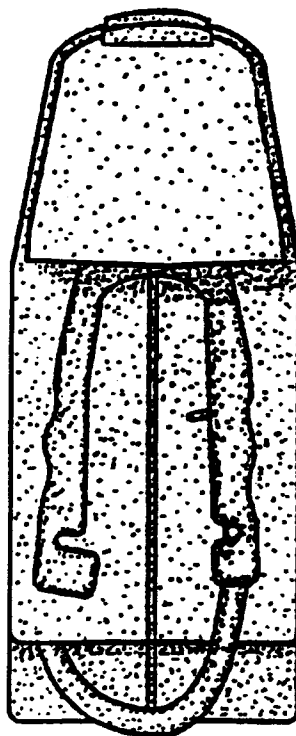


FIG. 4

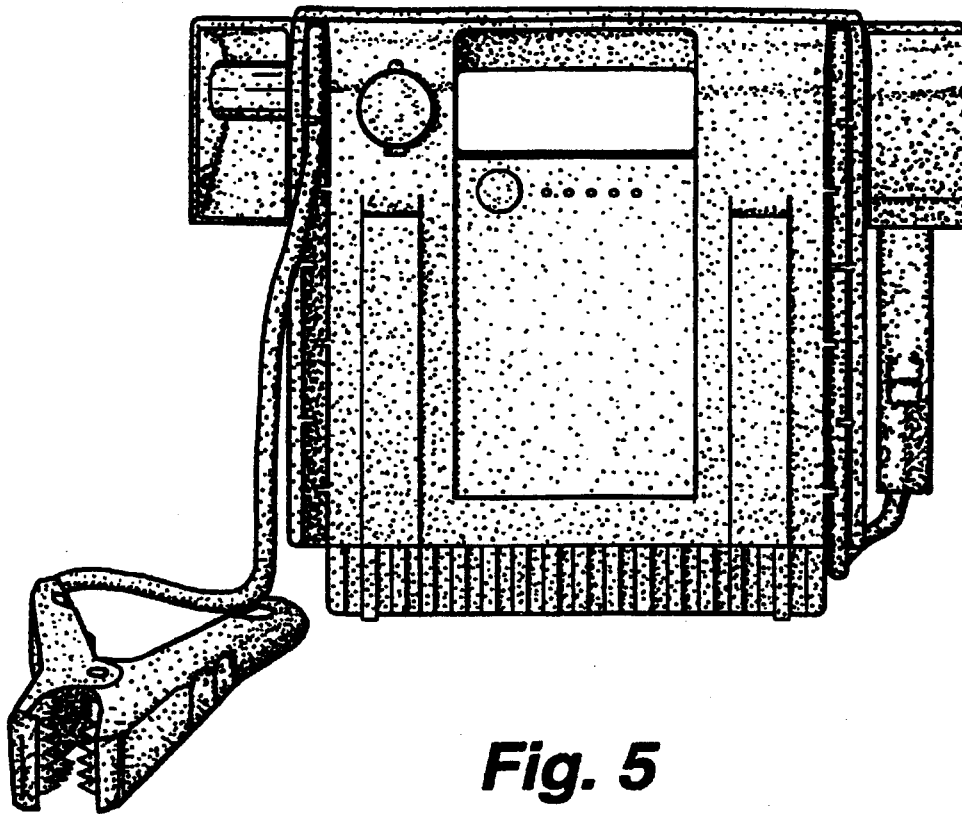


Fig. 5
(AMENDED)

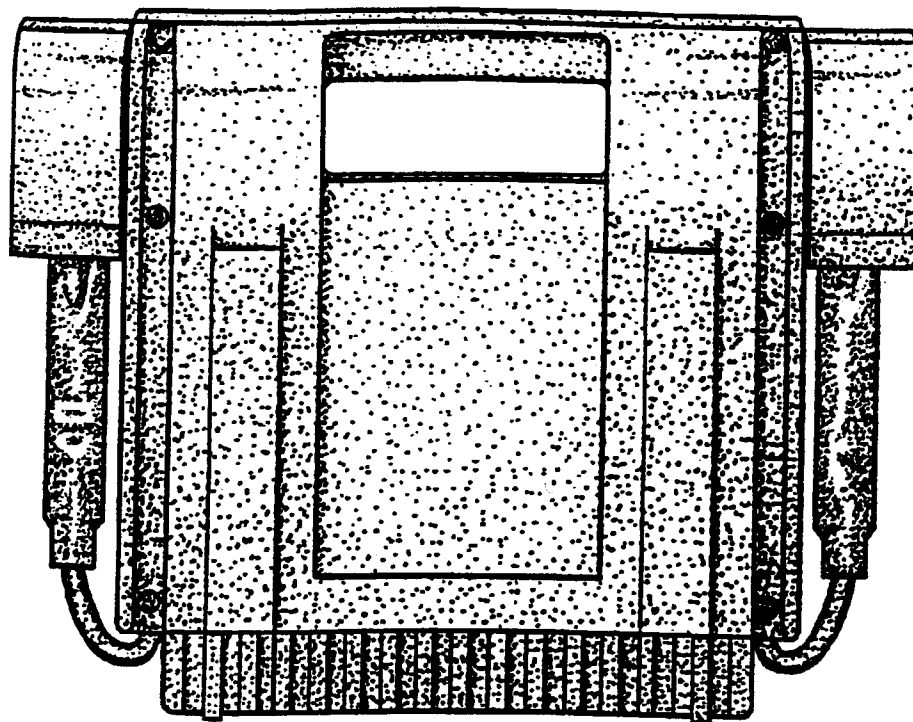


FIG. 6

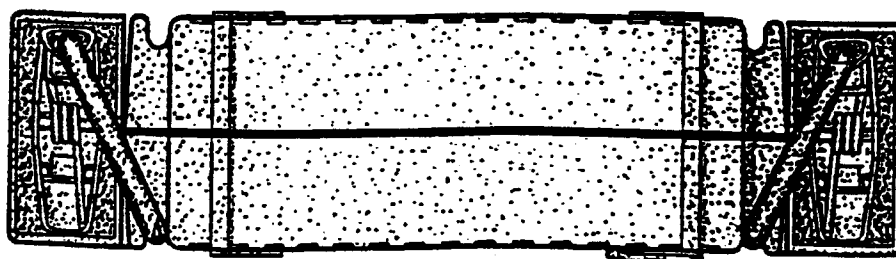
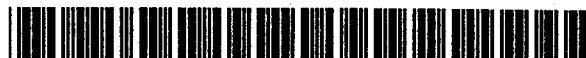


FIG. 7



US00D452474B1

(12) **United States Design Patent** (10) Patent No.: **US D452,474 S**
Casarella (45) Date of Patent: **** Dec. 25, 2001**

(54) **BATTERY BOOSTER**

(75) Inventor: **Tommaso Casarella, Bernex (CH)**

(73) Assignee: **Lemanía Credits & Finances
 Casarella Rosalba, Versoix (CH)**

(**) Term: **14 Years**

(21) Appl. No.: **29/122,453**

(22) Filed: **Apr. 28, 2000**

(30) **Foreign Application Priority Data**

Oct. 29, 1999 (XH) DMA/004 682

(51) **LOC (7) Cl.** **13-02**

(52) **U.S. Cl.** **D13/107**

(58) **Field of Search** **D13/106, 107,
 D13/108, 110; 320/105, 110-115**

(56) **References Cited**

U.S. PATENT DOCUMENTS

D. 374,859 • 10/1996 Guay et al. D13/110
 D. 413,098 • 8/1999 Dayan D13/110
 D. 425,019 • 5/2000 Zheng D13/110

D. 429,482 • 8/2000 Hussaini et al. D13/107
 D. 432,985 • 10/2000 Wan D13/107
 5,793,185 • 8/1998 Prelec et al. 320/104
 5,955,866 • 9/1999 Peng 320/105
 5,982,138 • 11/1999 Krieger 320/105
 6,002,235 • 12/1999 Clore 320/105
 6,130,519 • 10/2000 Whiting et al. 320/105
 6,155,870 • 12/2000 Valentine 439/504

* cited by examiner

Primary Examiner—Joel Sincavage

(74) *Attorney, Agent, or Firm*—Young & Thompson

(57) **CLAIM**

The ornamental design for said battery booster, as shown and described.

DESCRIPTION

FIG. 1 is a front view thereof.

FIG. 2 is a rear view thereof.

FIG. 3 is a top view thereof.

FIG. 4 is a side view from the right of FIG. 3; and,

FIG. 5 is a side view from the left of FIG. 3.

1 Claim, 2 Drawing Sheets

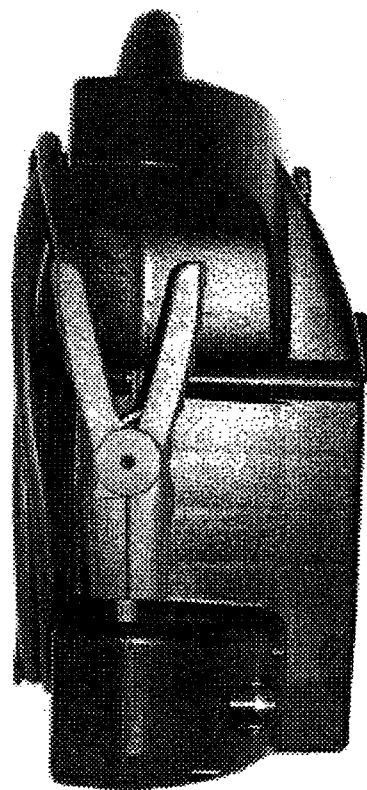
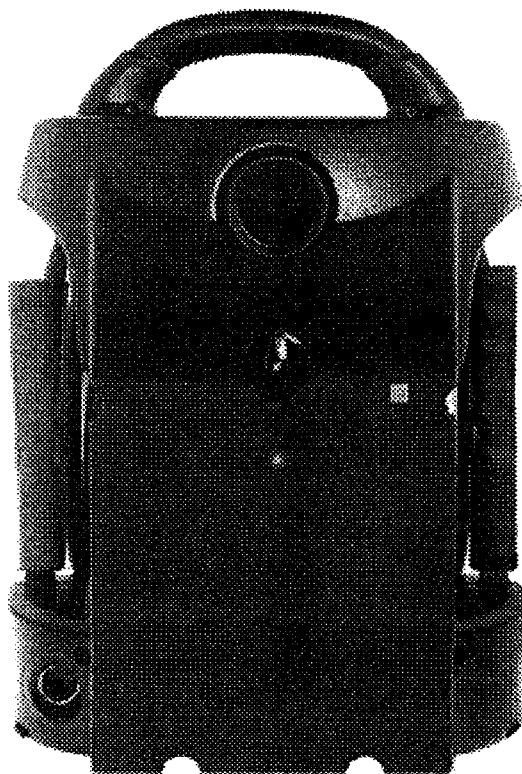


Fig.1

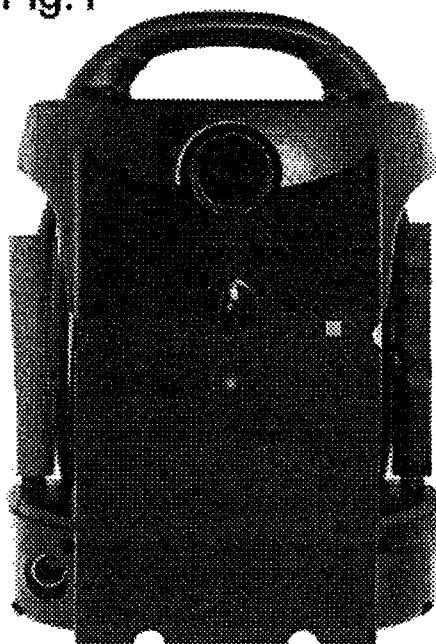
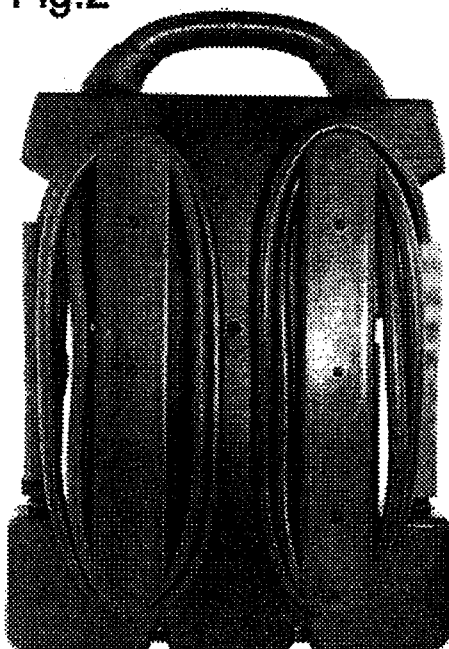
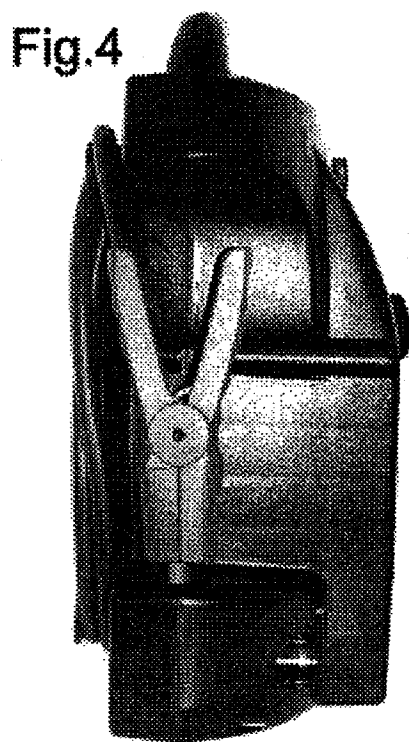
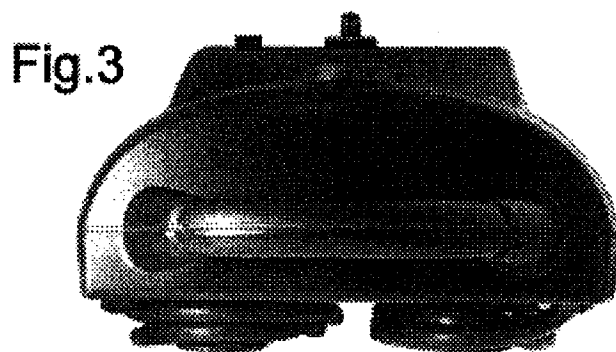


Fig.2







US00D448723B1

(12) **United States Design Patent** (10) Patent No.: **US D448,723 S**
Zhu (45) Date of Patent: **** Oct. 2, 2001**

(54) **PORTABLE AUTOMOTIVE BOOSTER**

(75) Inventor: **Yun Hua Zhu, Shanghai (CN)**

(73) Assignee: **Shanghai Autopower Resource Co., Ltd., Shanghai (CN)**

(**) Term: **14 Years**

(21) Appl. No.: **29/131,990**

(22) Filed: **Nov. 1, 2000**

(30) **Foreign Application Priority Data**

Jun. 27, 2000 (CN) 00 3 14995

(51) LOC (7) Cl. **13-02**

(52) U.S. Cl. **D13/107**

(58) Field of Search D13/106, 107,
D13/108, 118, 110; D14/149, 251, 253;
320/110-115, 105

(56) **References Cited**

U.S. PATENT DOCUMENTS

D. 374,859	* 10/1996	Guay et al.		D13/110
D. 413,098	* 8/1999	Dayan		D13/110
D. 425,019	* 5/2000	Zheng		D13/110
D. 429,482	* 8/2000	Hussaini et al.		D13/107
D. 432,985	* 10/2000	Wan		D13/107
5,589,292	12/1996	Rozon		429/170
5,793,185	* 8/1998	Prelec et al.		320/104
5,955,866	* 9/1999	Peng		320/105
5,982,138	* 11/1999	Krieger		320/105

6,002,235 * 12/1999 Clore 320/105
6,130,519 * 10/2000 Whiting et al. 320/105
6,155,870 * 12/2000 Valentine 439/504

* cited by examiner

Primary Examiner—Joel Sincavage

(74) Attorney, Agent, or Firm—BakerBotts LLP

(57) **CLAIM**

The ornamental design for a portable automotive booster, as shown and described.

DESCRIPTION

FIG. 1 is a perspective view of the portable automotive booster of the invention.

FIG. 2 is a front view of the portable automotive booster depicted in FIG. 1.

FIG. 3 is a rear view of the portable automotive booster depicted in FIG. 1.

FIG. 4 is a top view of the portable automotive booster depicted in FIG. 1.

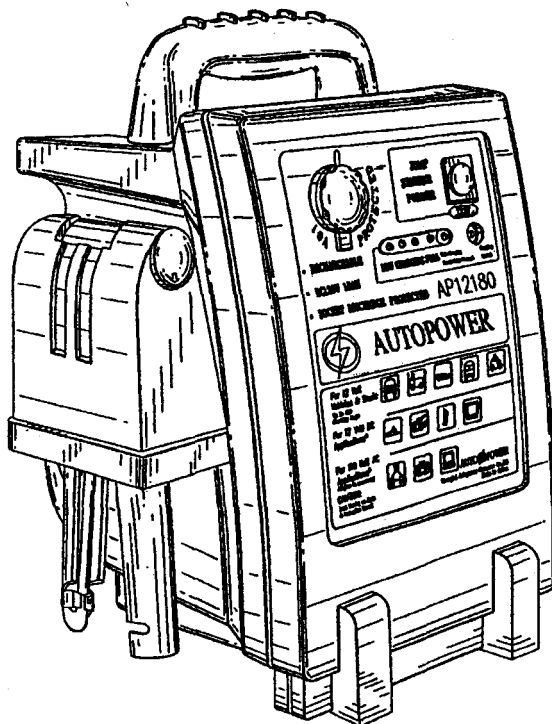
FIG. 5 is a bottom view of view of the portable automotive booster depicted in FIG. 1.

FIG. 6 is a left side view of view of the portable automotive booster depicted in FIG. 1.

FIG. 7 is a right side view of the portable automotive booster depicted in FIG. 1; and,

FIG. 8 is another perspective view of the portable automotive booster depicted in FIG. 1, in an alternate position.

1 Claim, 8 Drawing Sheets



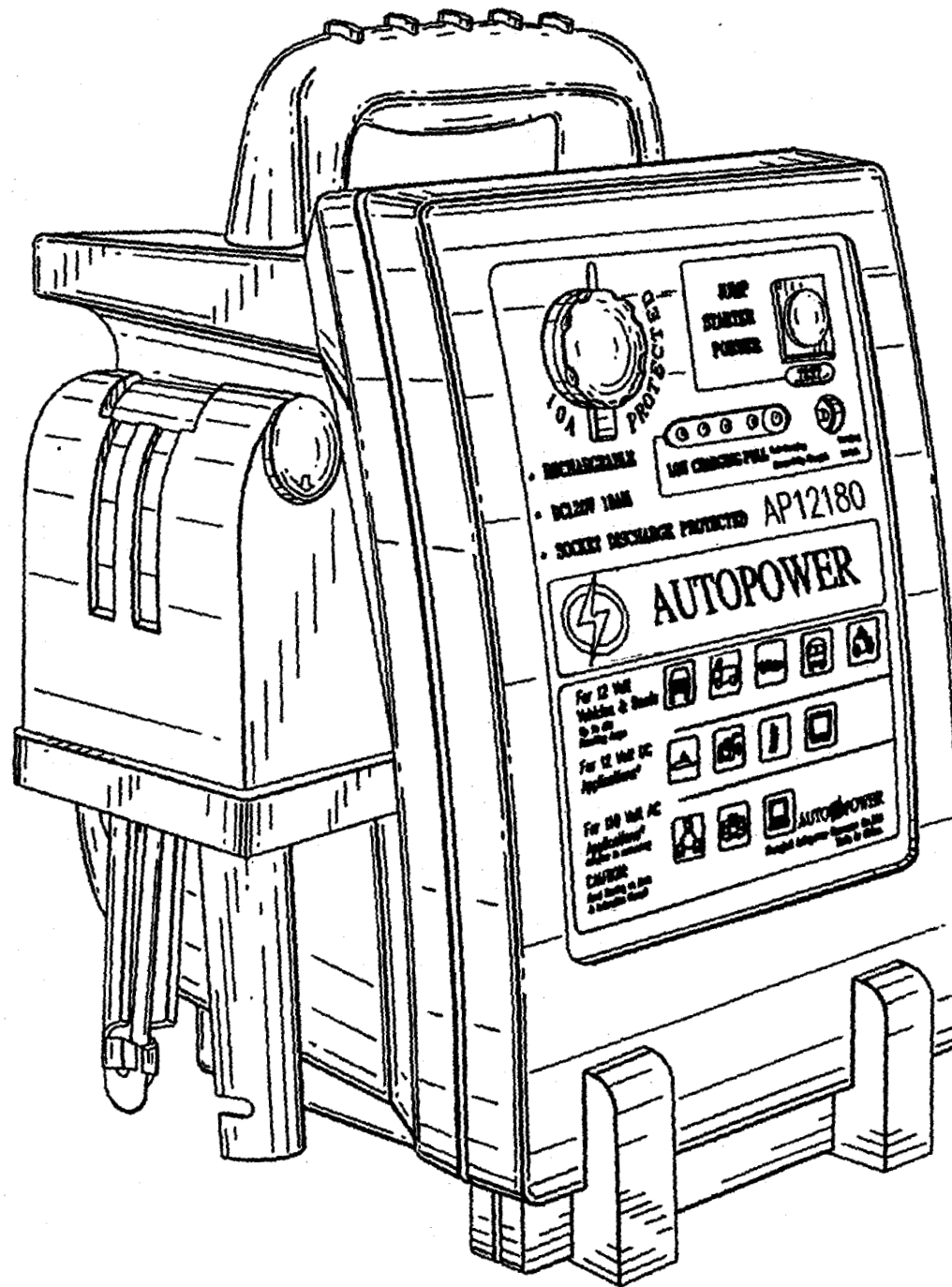


FIG. 1

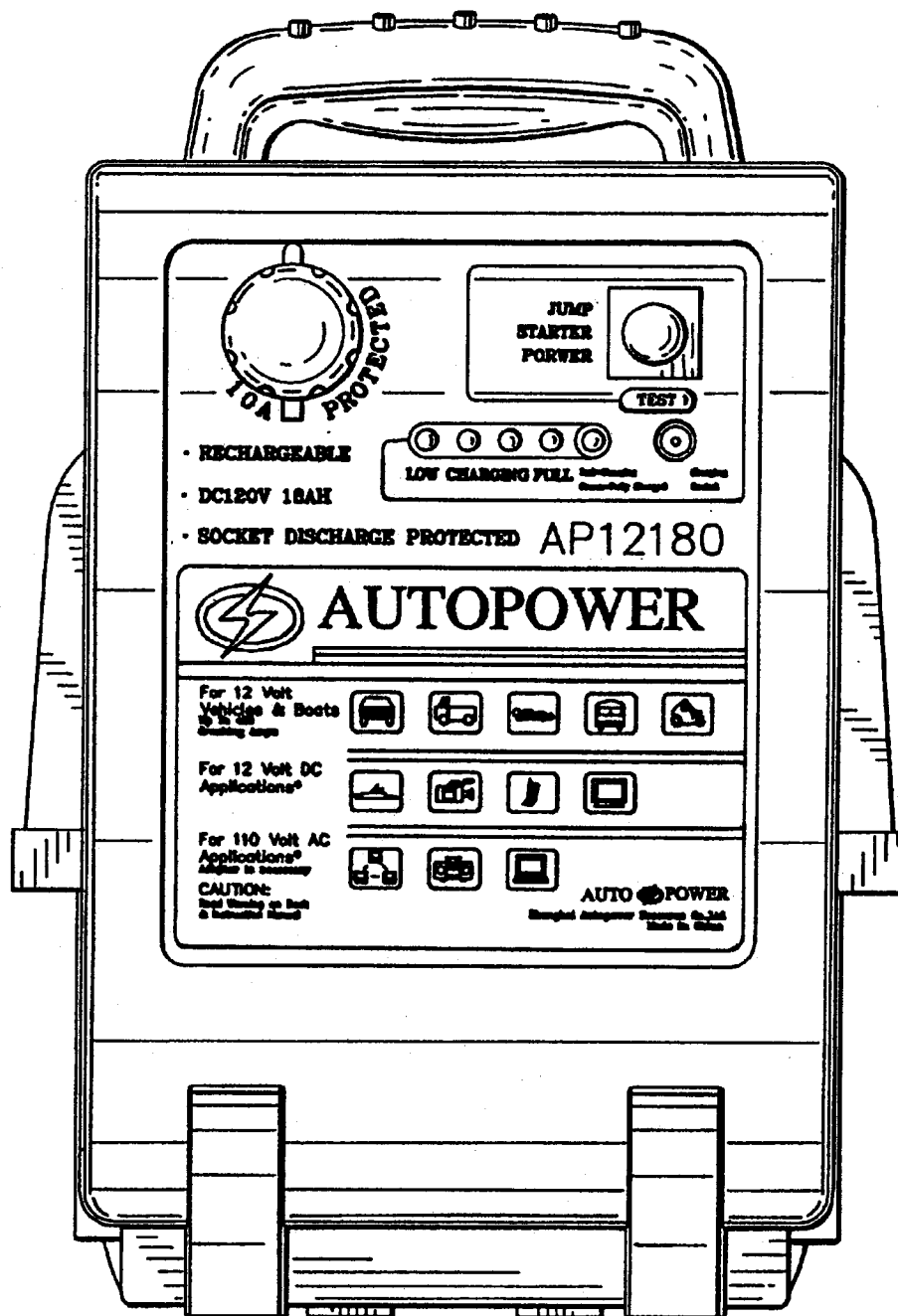


FIG. 2

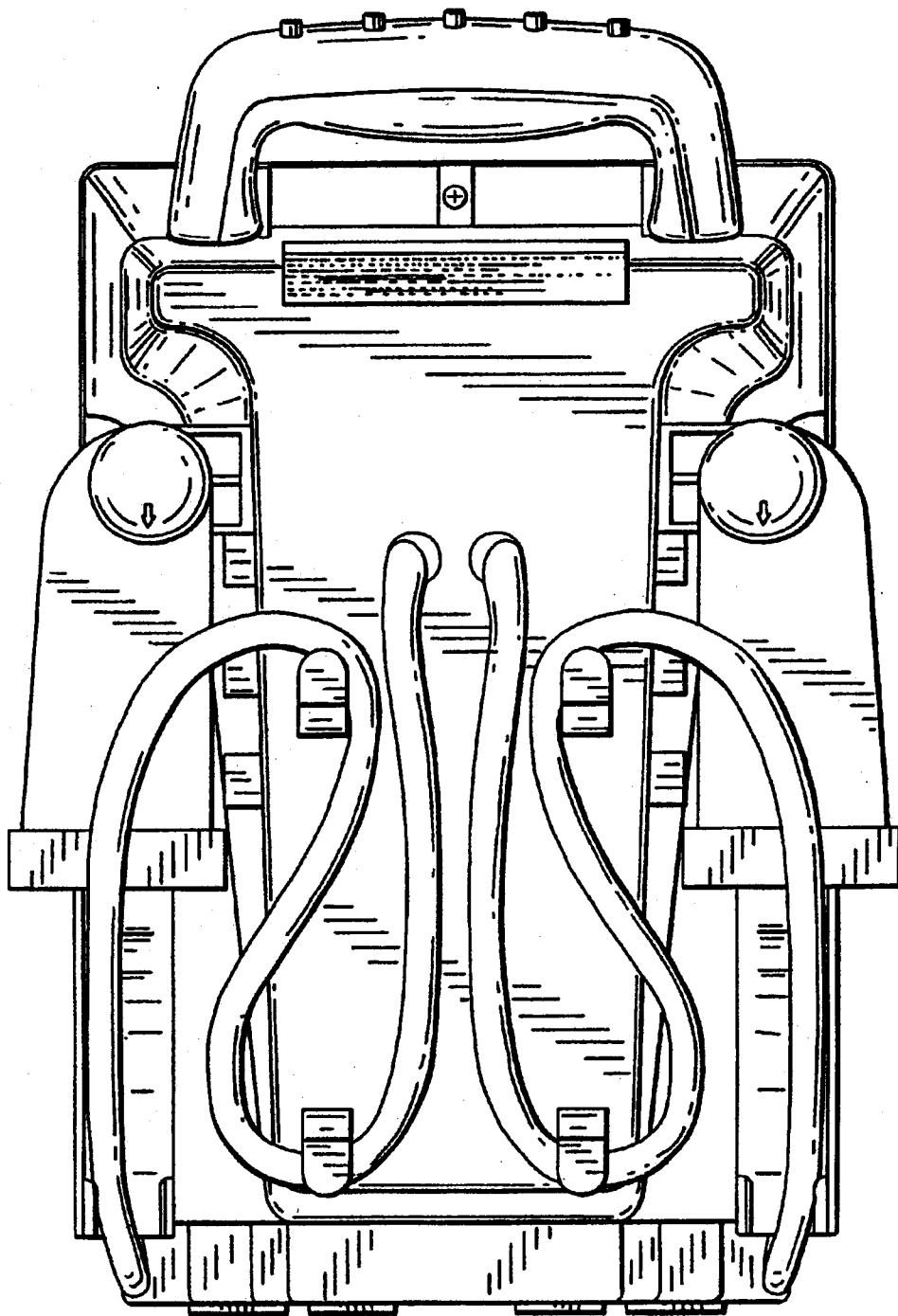


FIG. 3

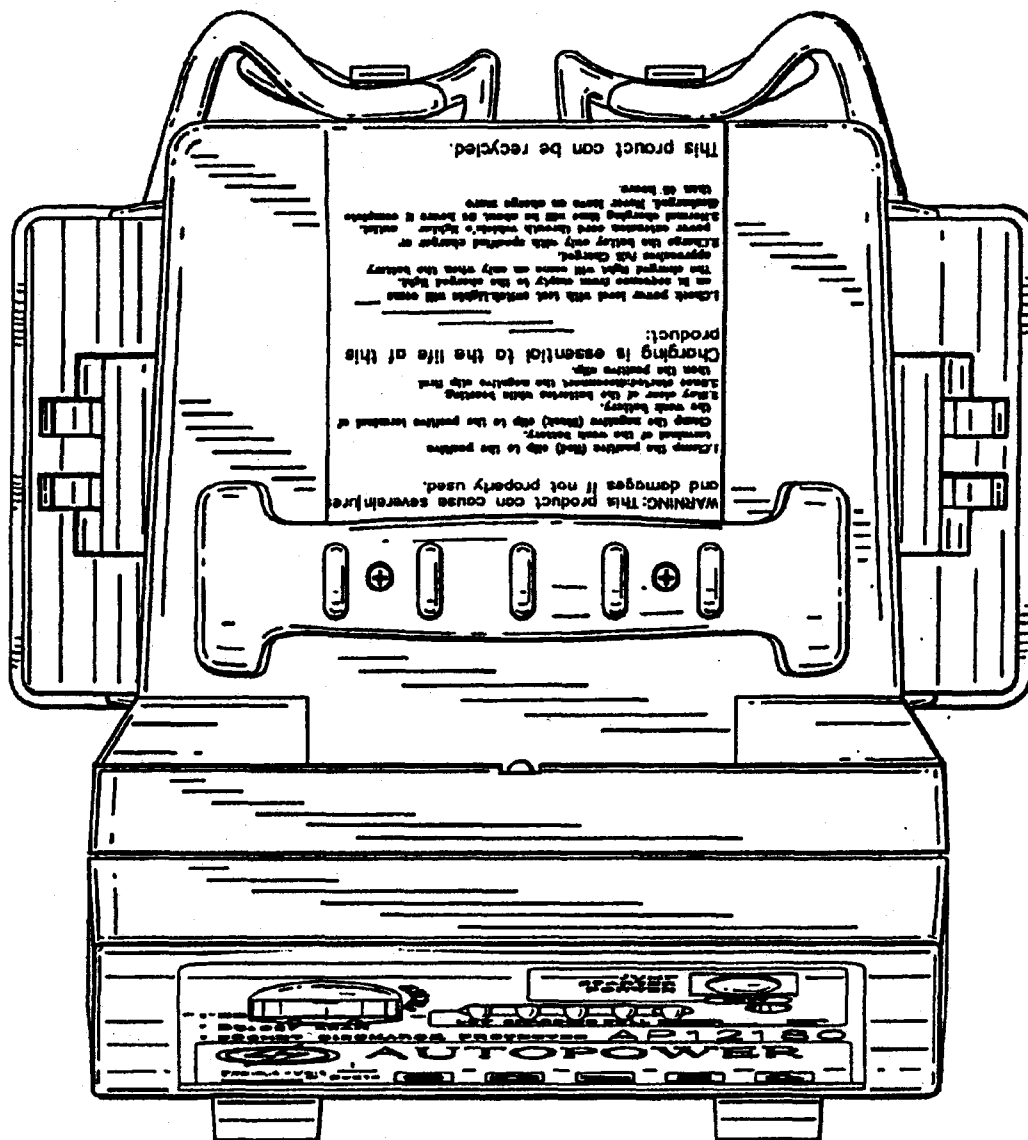


FIG. 4

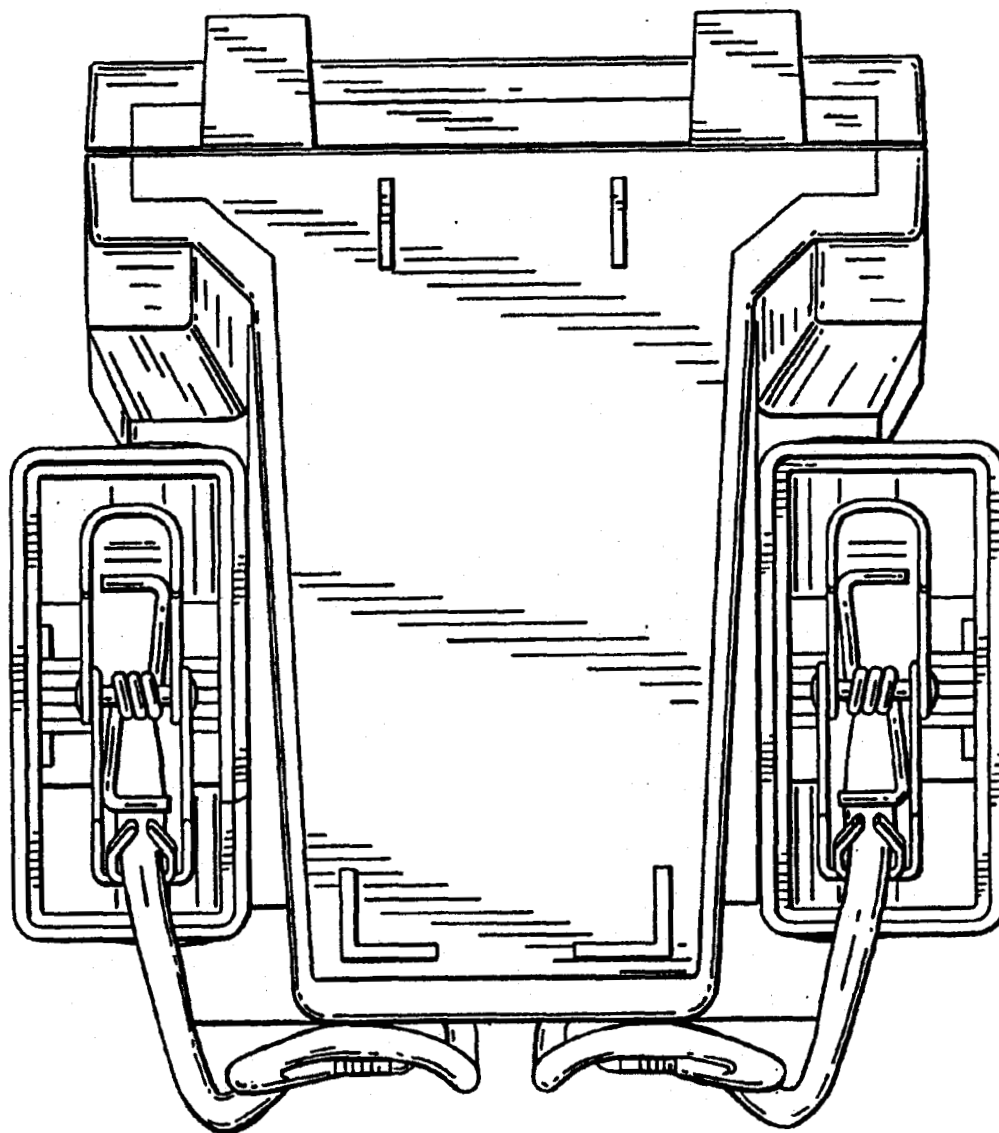


FIG. 5

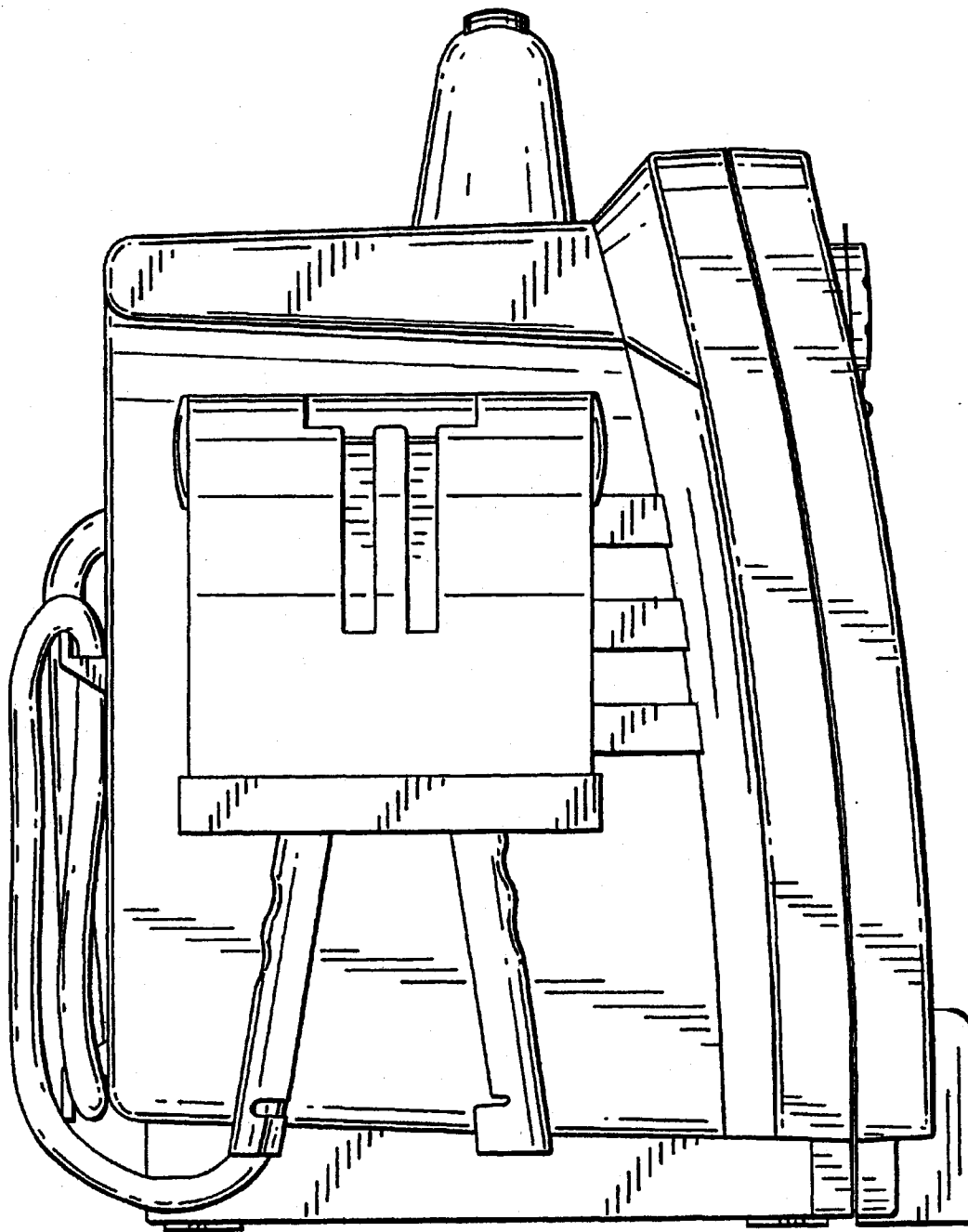


FIG. 6

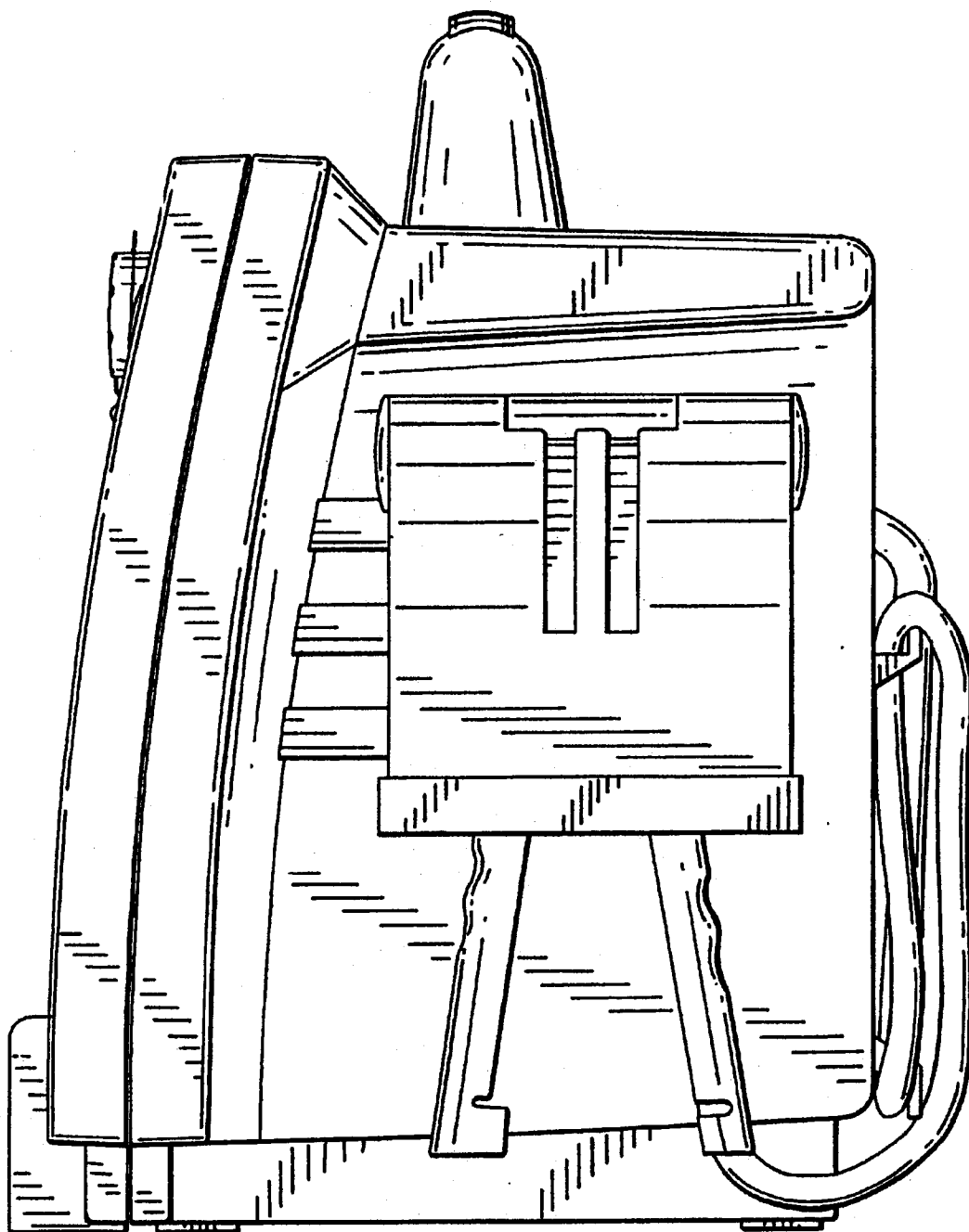


FIG. 7

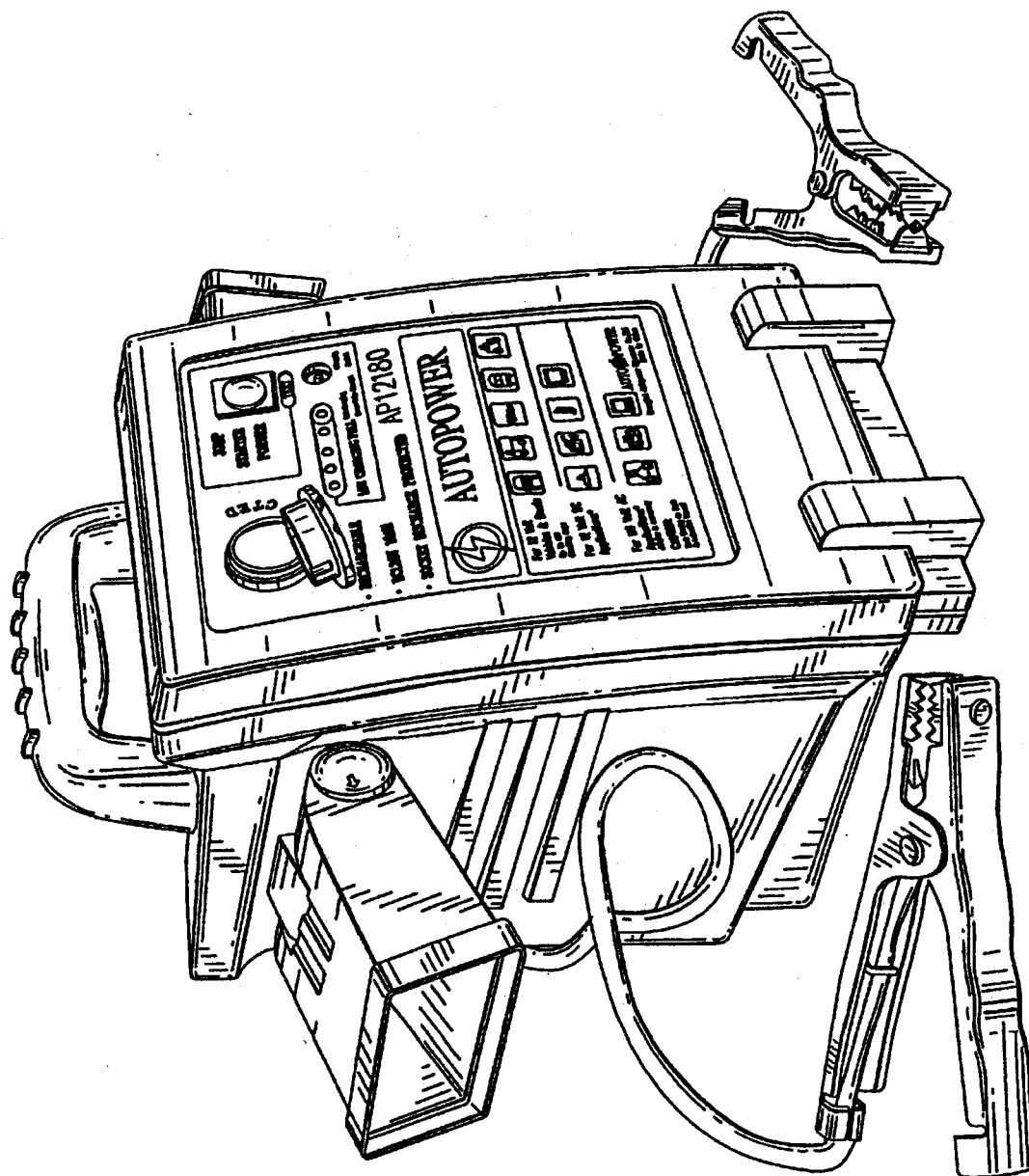


FIG. 8



US00D432985S

United States Patent [19]

Wan

[11] **Patent Number:** Des. 432,985[45] **Date of Patent:** ** Oct. 31, 2000[54] **ELECTRICITY CHARGING APPARATUS**[75] **Inventor:** Yiu Kwong Wan, Hong Kong, The Hong Kong Special Administrative Region of the People's Republic of China[73] **Assignee:** Fee Tat Holdings (H.K.) Limited, Hong Kong, The Hong Kong Special Administrative Region of the People's Republic of China[**] **Term:** 14 Years[21] **Appl. No.:** 29/115,967[22] **Filed:** Dec. 27, 1999[51] **LOC (7) Cl.** 13-02[52] **U.S. Cl.** D13/107[58] **Field of Search** D13/106, 107, D13/108, 110; 320/110-115[56] **References Cited****U.S. PATENT DOCUMENTS**

D. 374,859 10/1996 Guay et al. D13/110

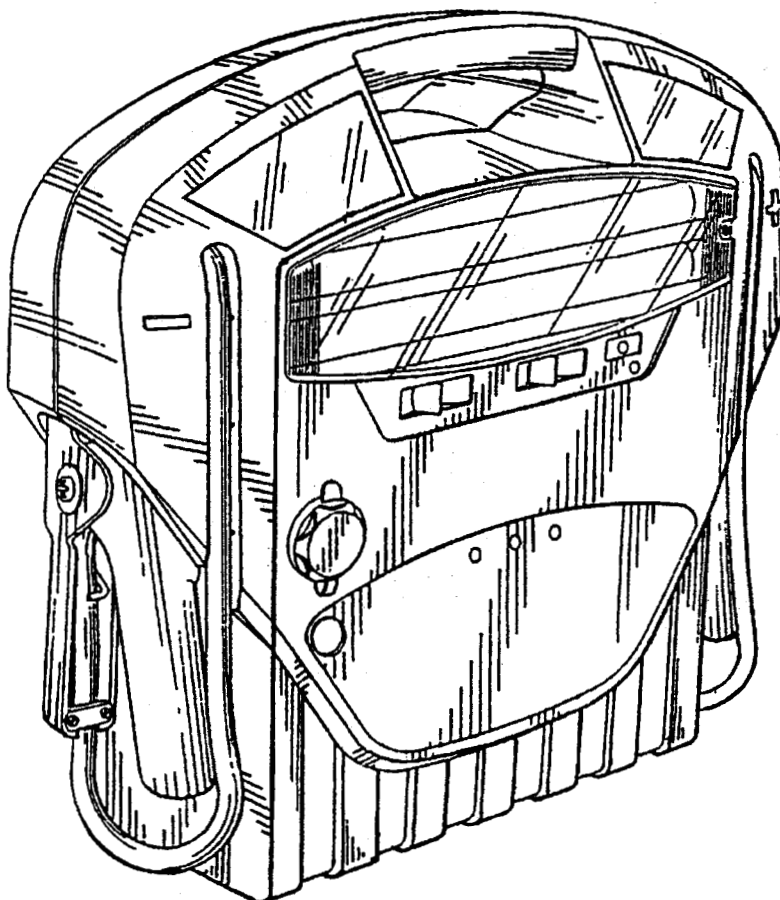
D. 407,065 3/1999 Chang D13/110
D. 408,782 4/1999 Poon D13/106
D. 413,098 8/1999 Dayan D13/110*Primary Examiner*—Joel Sincavage
Attorney, Agent, or Firm—Reese Taylor[57] **CLAIM**

I claim the ornamental design for an electricity charging apparatus, as shown and described.

DESCRIPTION

FIG. 1 is a front perspective view showing my new design for an electricity charging apparatus;
FIG. 2 is a front elevational view thereof;
FIG. 3 is a bottom plan view thereof;
FIG. 4 is a top plan view thereof;
FIG. 5 is a rear elevational view thereof;
FIG. 6 is a side elevational view taken from the right of FIG. 1;
FIG. 7 is a side elevational view taken from the left of FIG. 1; and,
FIG. 8 is a perspective view similar to FIG. 1 showing the electricity charging apparatus in a different mode.

1 Claim, 5 Drawing Sheets



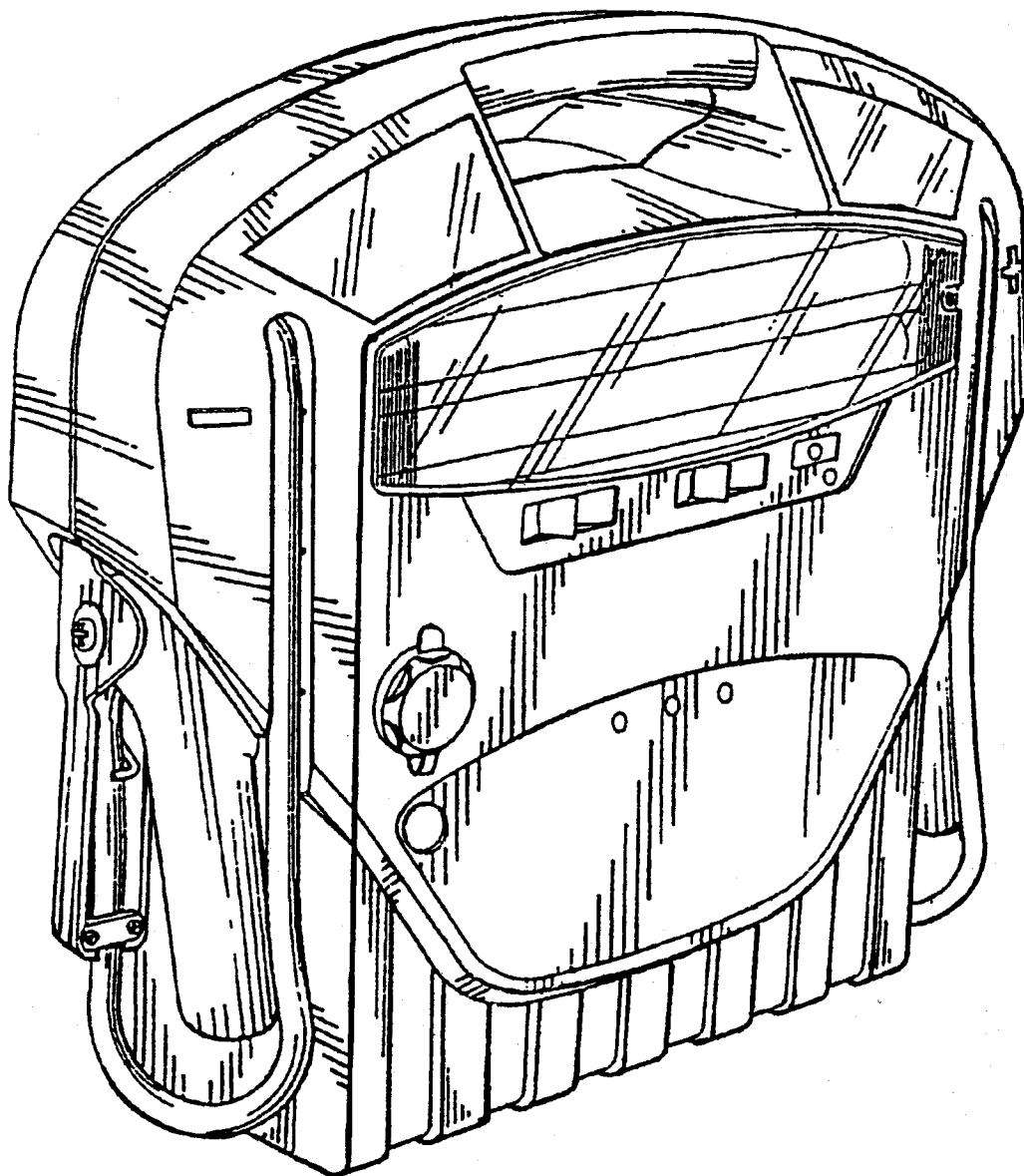


FIG. 1

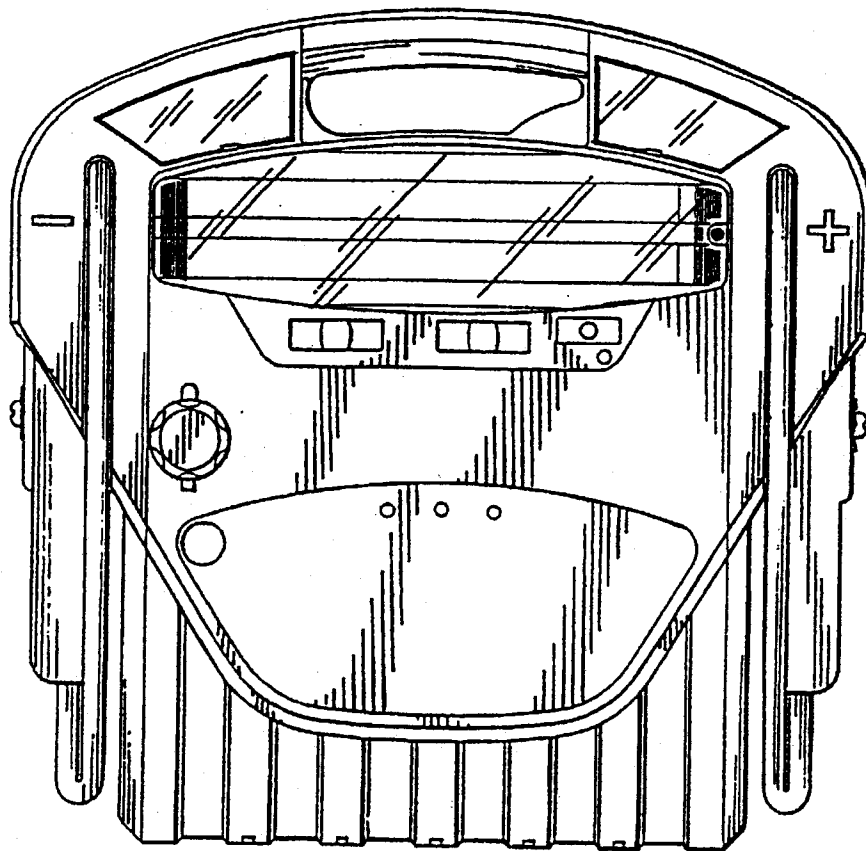


FIG. 2

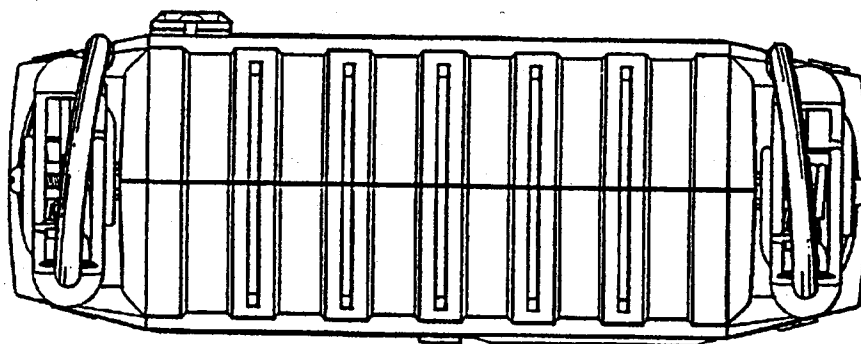


FIG. 3

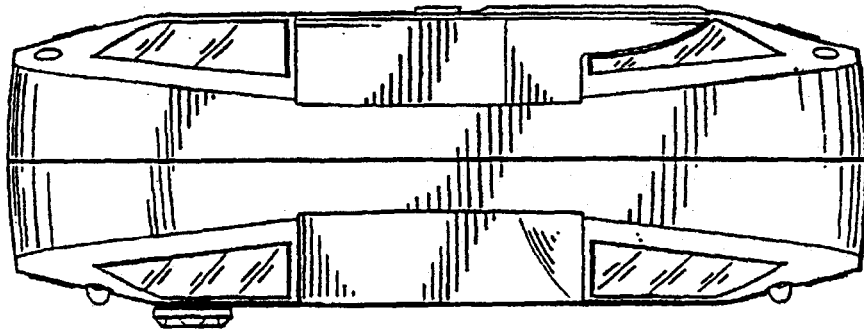


FIG. 4

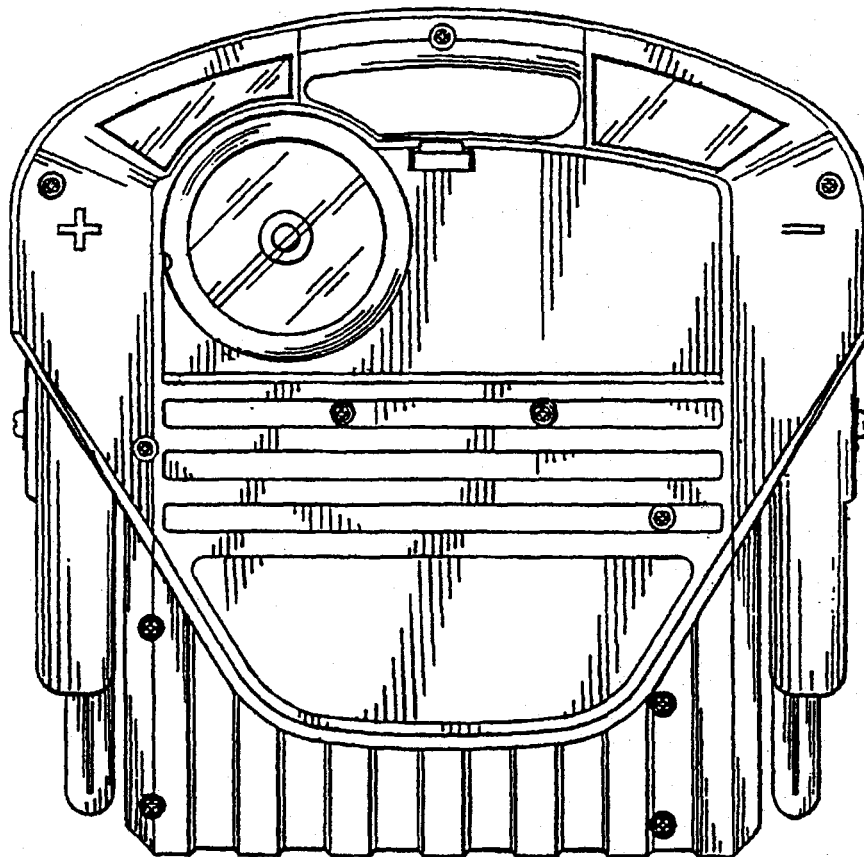


FIG. 5

FIG. 7

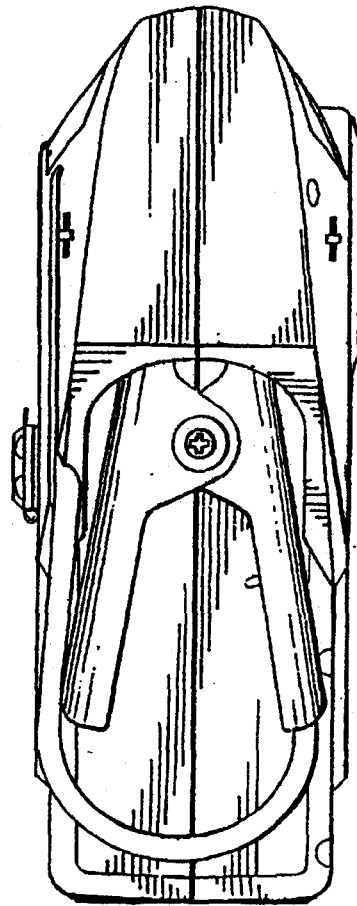
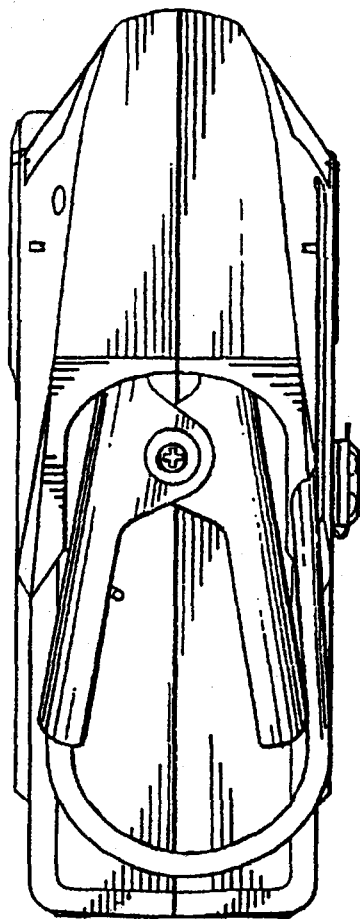


FIG. 6

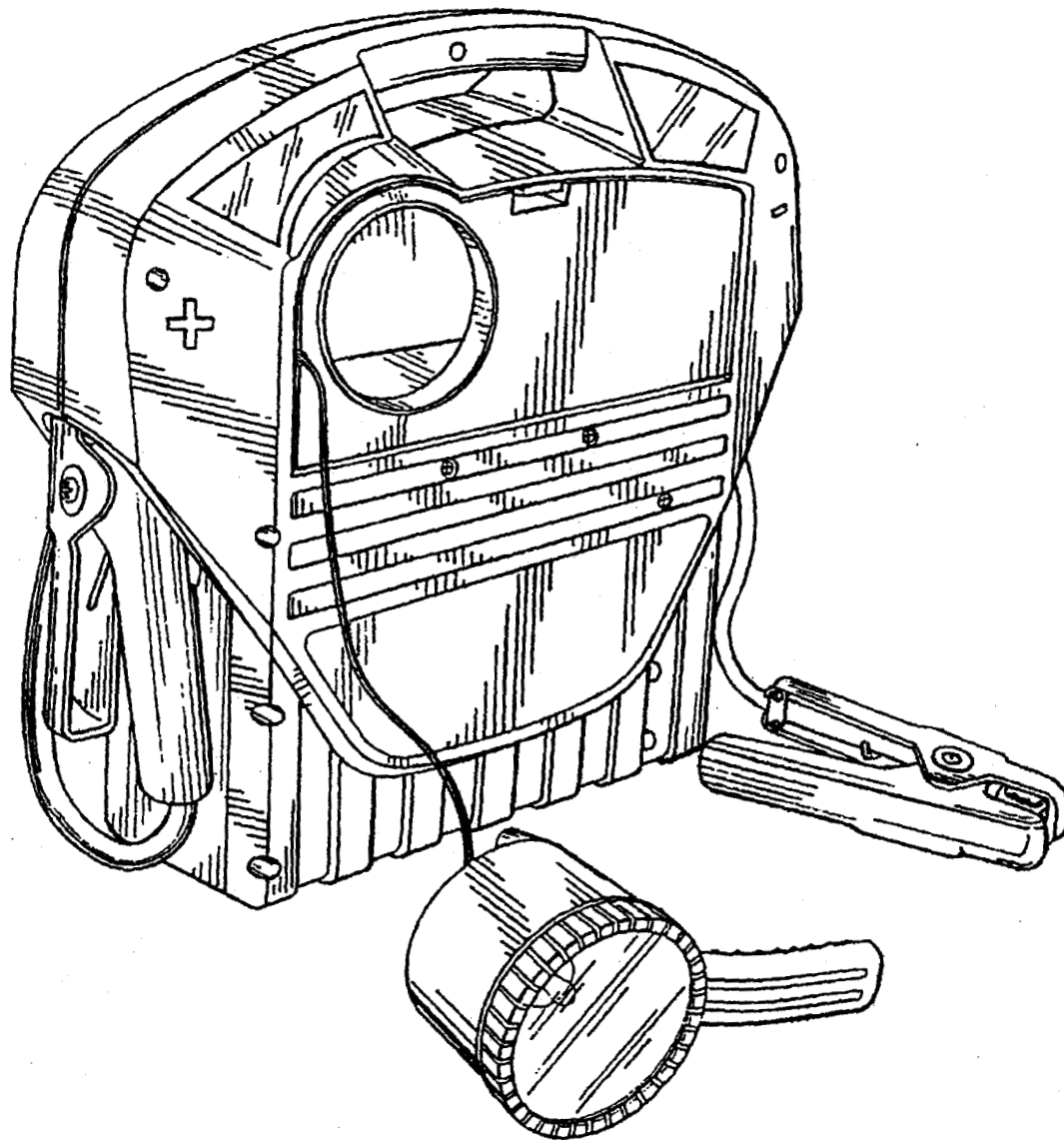


FIG. 8



US00D429482S

United States Patent [19]

Hussaini et al.

[11] Patent Number: Des. 429,482

[45] Date of Patent: ** Aug. 15, 2000

[54] BATTERY CHARGER

[76] Inventors: Saled Hussaini; Marc Iacovelli, both
of 5255 NW. 159th St., Miami, Fla.
33014

[**] Term: 14 Years

[21] Appl. No.: 29/110,308

[22] Filed: Sep. 3, 1999

[51] LOC (7) Cl. 13-02

[52] U.S. Cl. D13/107

[58] Field of Search D13/106, 107,
D13/108, 118, 110; 320/105, 110-115;
429/96

[56] References Cited

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D. 374,859	10/1996	Guay et al.	D13/110
D. 408,782	4/1999	Poon	D13/106
D. 413,098	8/1999	Dayan	D13/110
2,718,540	9/1955	Betz	429/96
5,589,292	12/1996	Rozon	429/170
5,793,185	8/1998	Prelec et al.	320/104
5,955,866	9/1999	Peng	320/105
6,002,235	12/1999	Clore	320/105

Primary Examiner—Joel Sincavage
Attorney, Agent, or Firm—Liniak, Berenato Longacre &
White

[57] CLAIM

The ornamental design for a battery charger, as shown and
described.

DESCRIPTION

FIG. 1 is a front view of the battery charger according to the
present invention.

FIG. 2 is a rear view of the battery charger of FIG. 1
according to the present invention.

FIG. 3 is a left side view of the battery charger of FIG. 1
according to the present invention.

FIG. 4 is a right side view of the battery charger of FIG. 1
according to the present invention.

FIG. 5 is a top view of the battery charger of FIG. 1
according to the present invention.

FIG. 6 is a bottom view of the battery charger of FIG. 1
according to the present invention; and,

FIG. 7 is a front perspective view of the battery charger of
FIG. 1 according to the present invention.

The broken line showing of the cables and clamps is
included for the purpose of illustrating environmental struc-
ture only and forms no part of the claimed design.

1 Claim, 6 Drawing Sheets

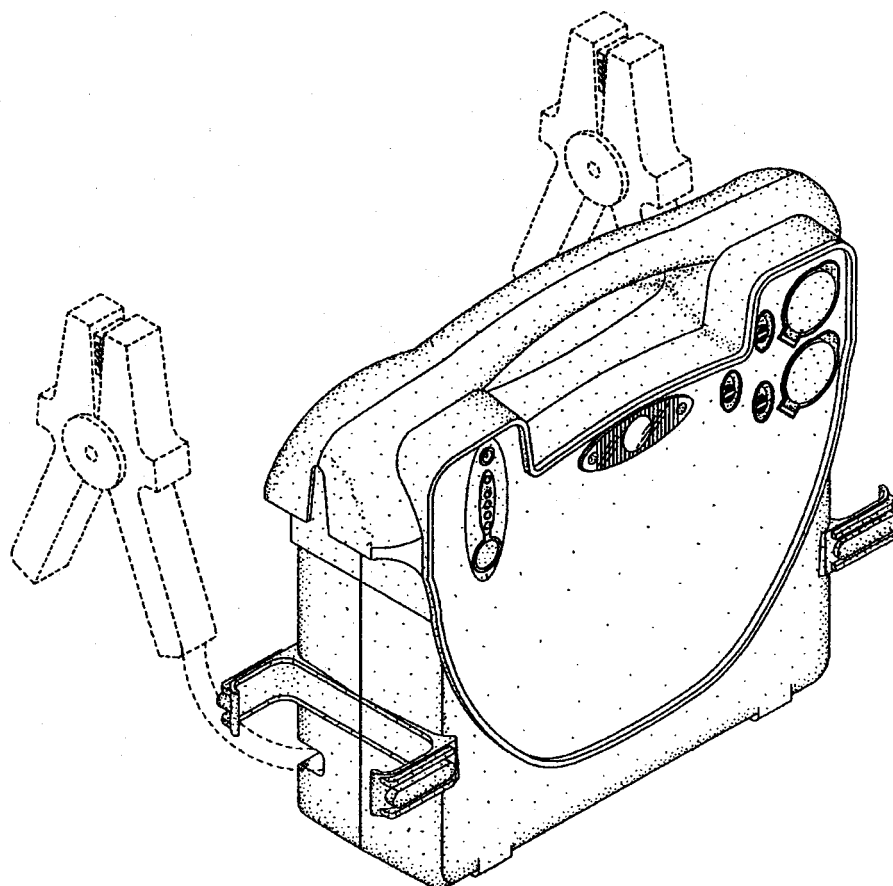


Fig. 1

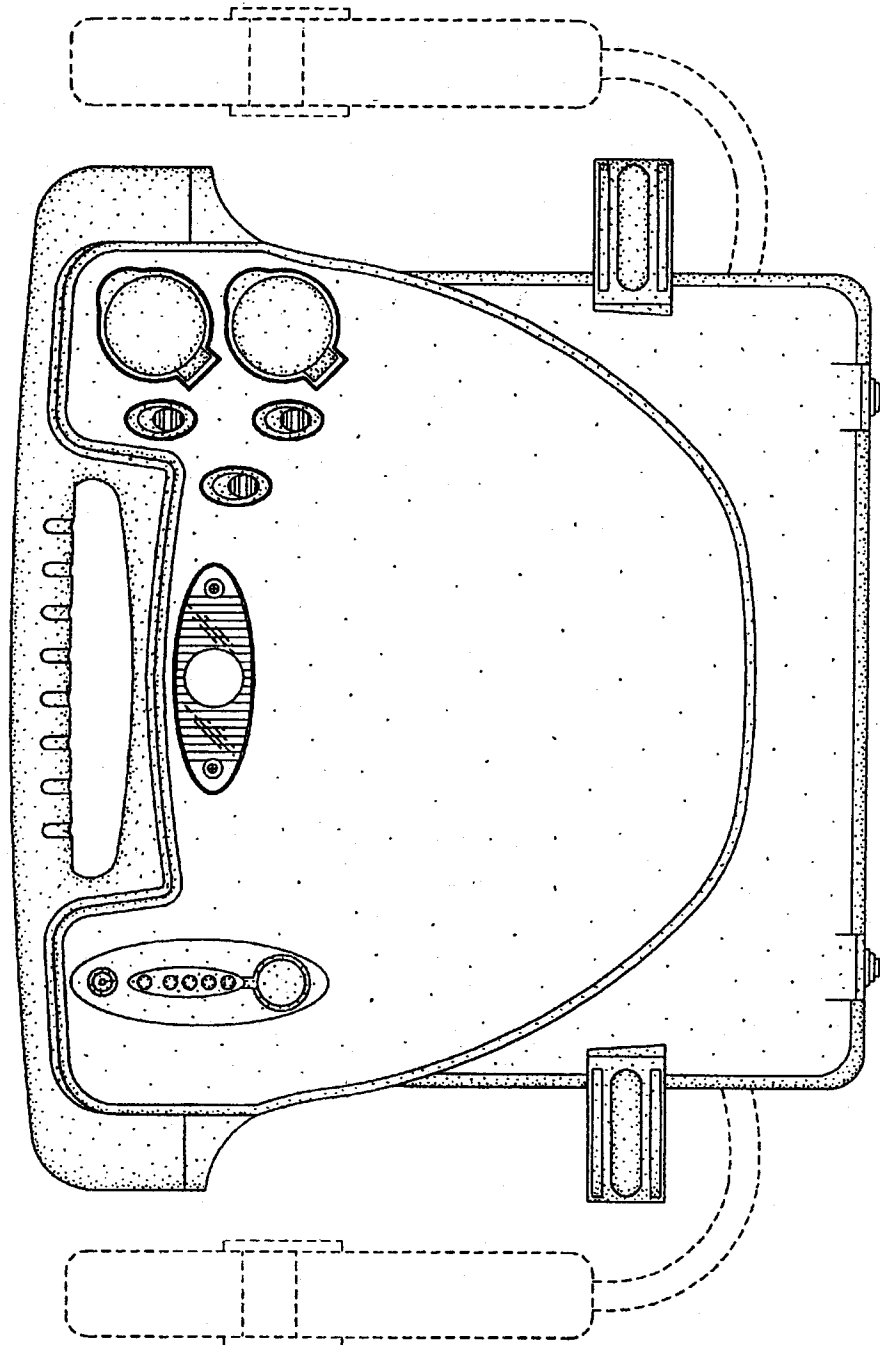


Fig. 2

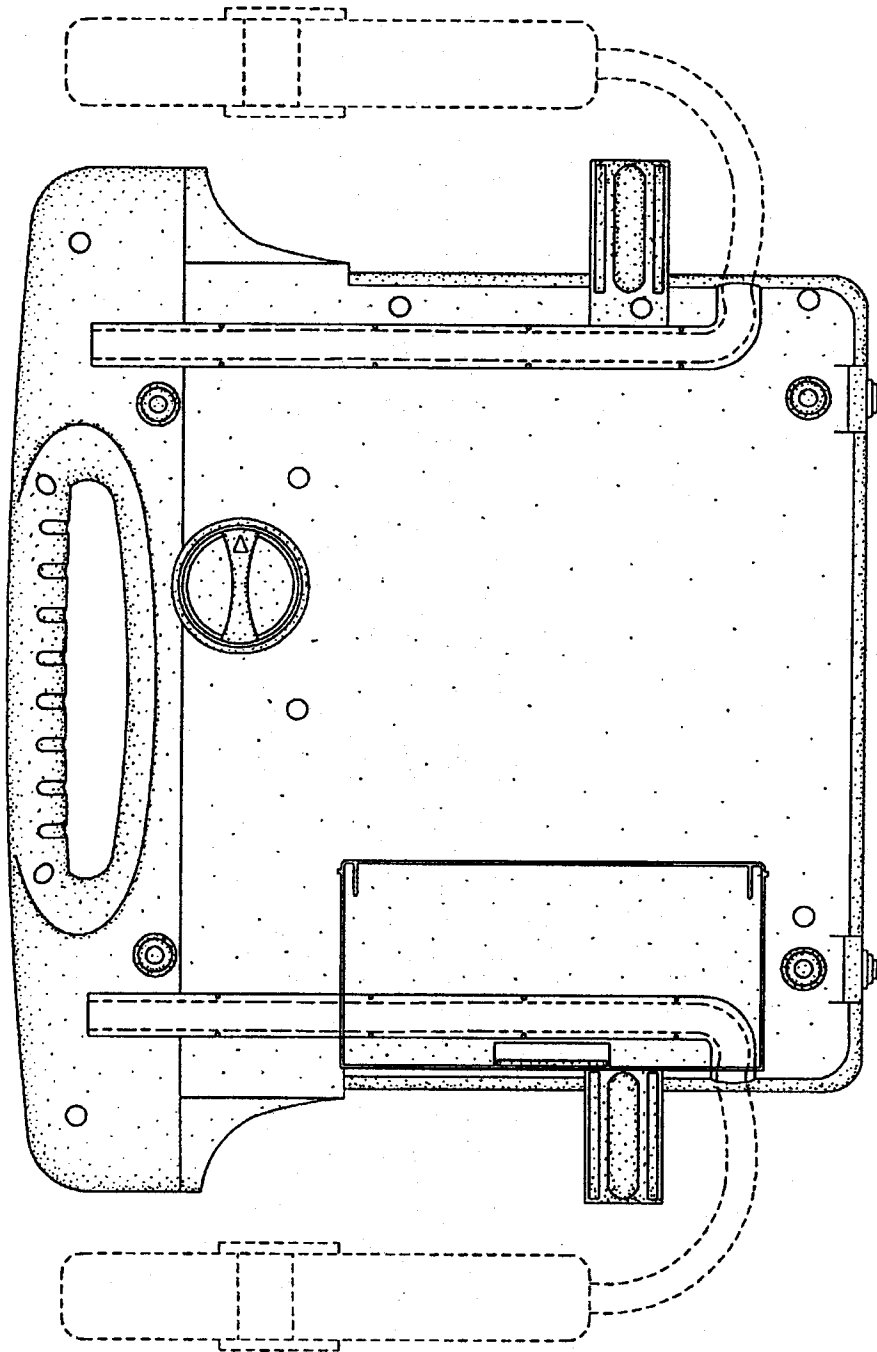


Fig. 4

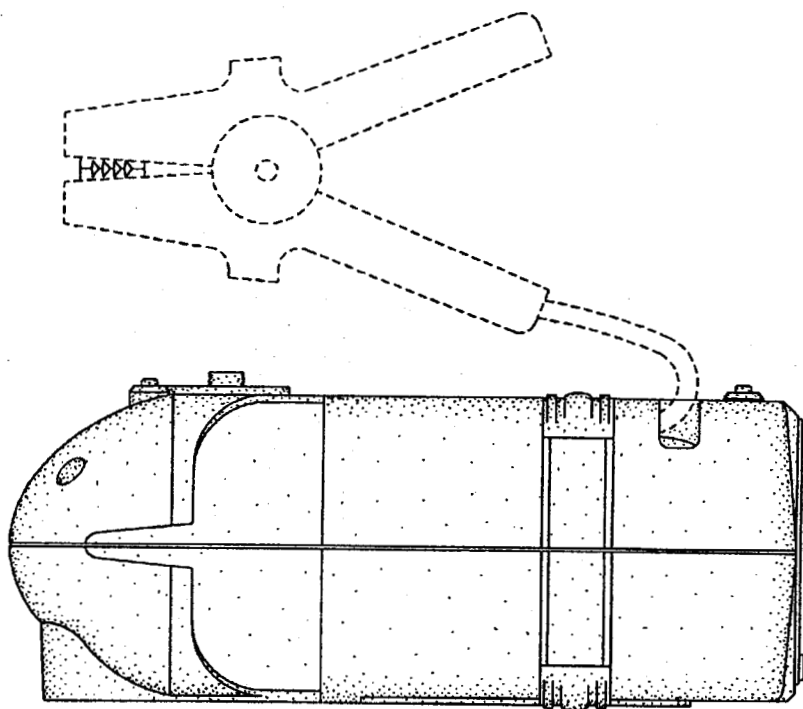


Fig. 3

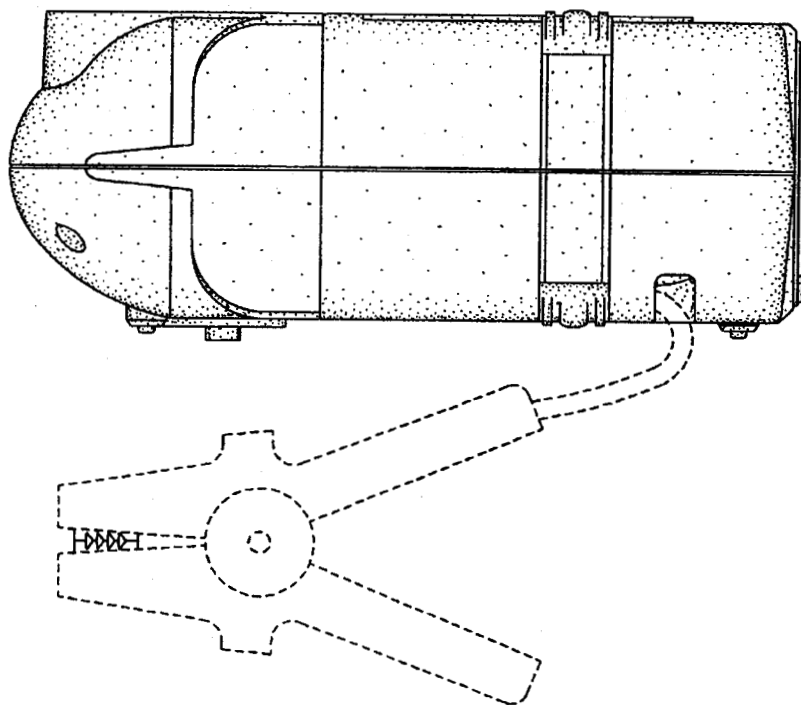


Fig. 5

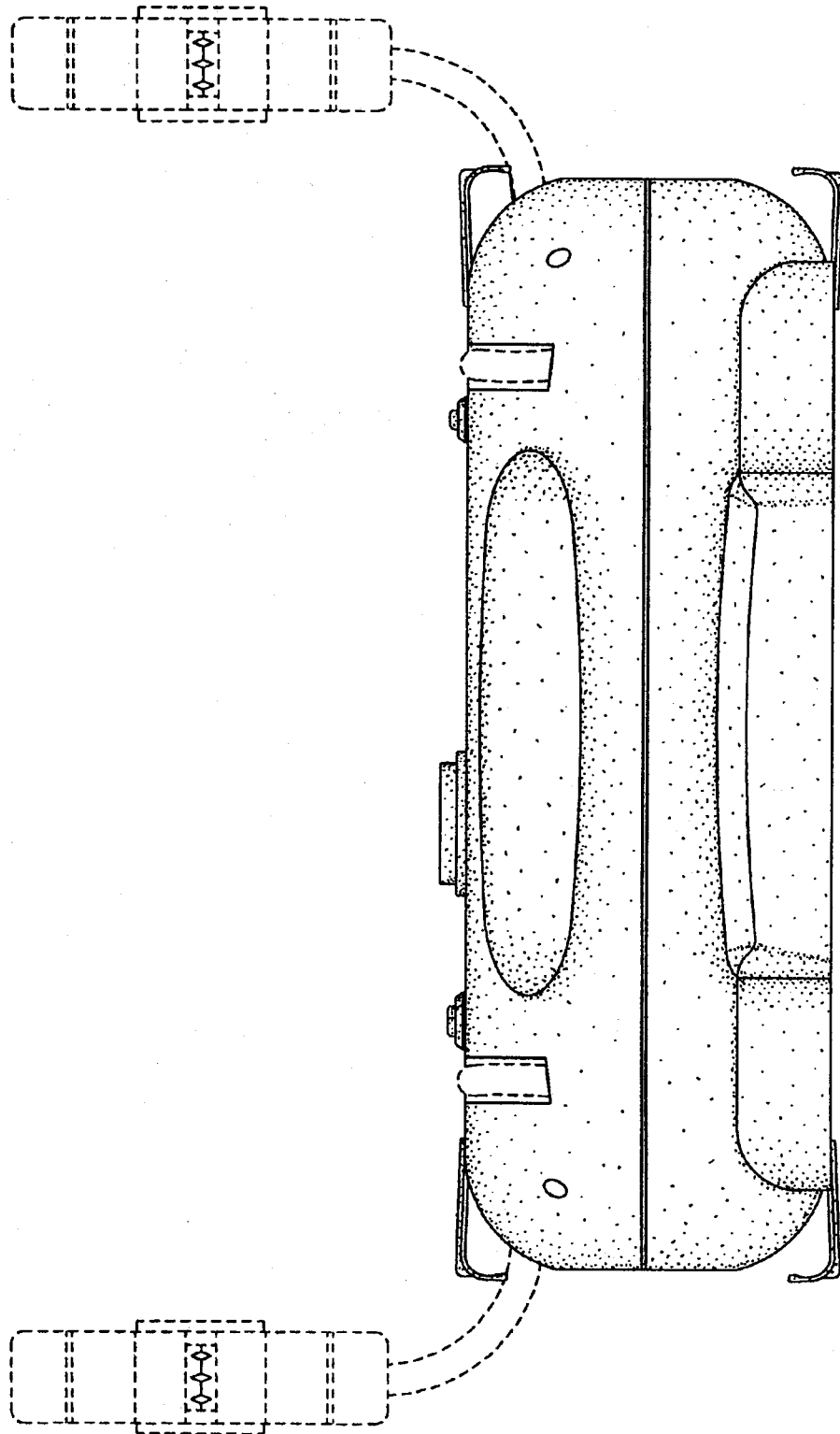


Fig. 6

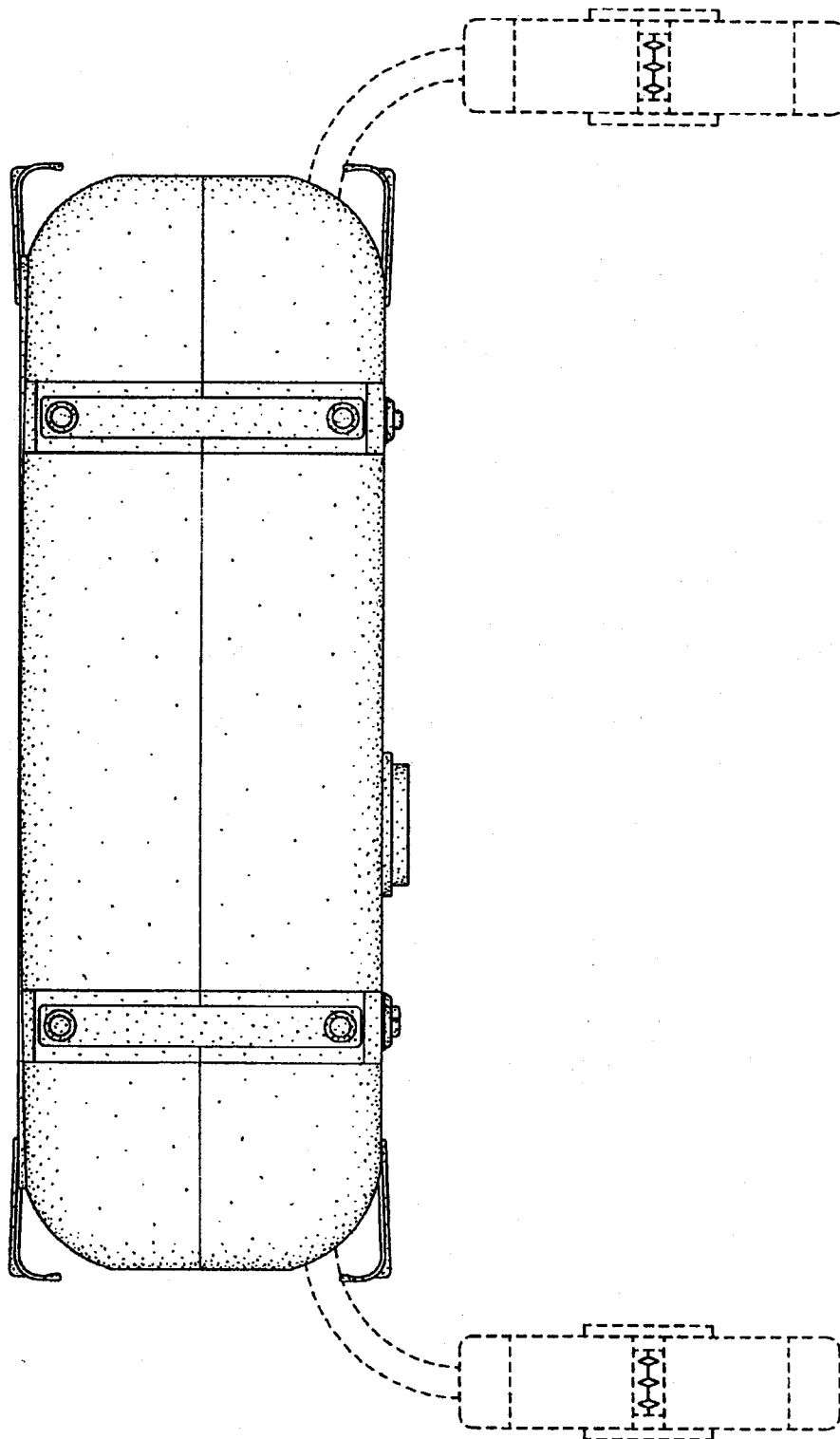
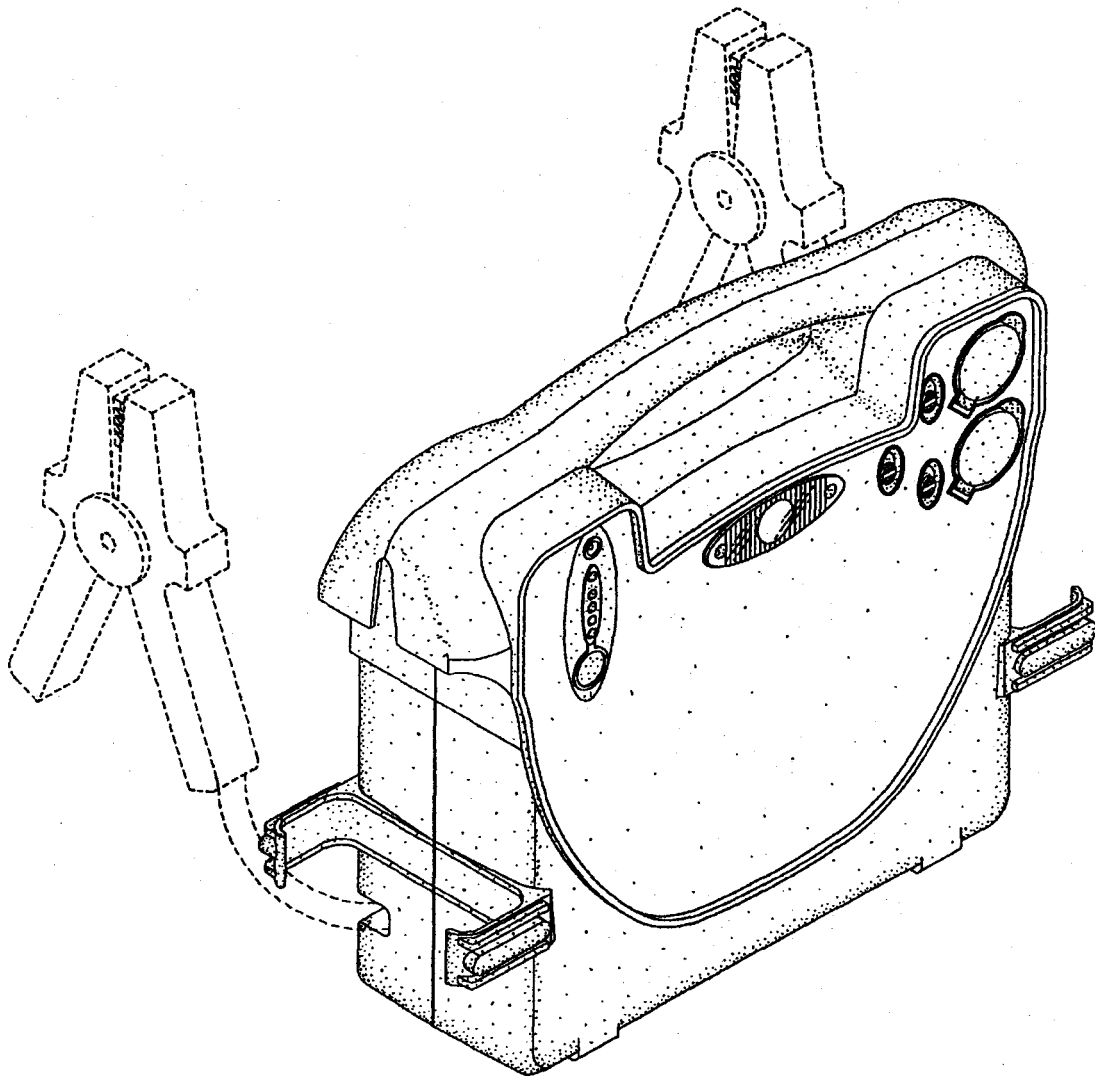


Fig. 7





US006356050B1

(12) **United States Patent**
Hussaini

(10) **Patent No.:** **US 6,356,050 B1**
(45) **Date of Patent:** **Mar. 12, 2002**

(54) **PORTABLE BOOSTER SUPPLY WITH
WIRELESS REMOTE CONTROL
ACTIVATION**

(75) **Inventor:** Saled Hussaini, Miami, FL (US)

(73) **Assignee:** Rally Manufacturing, Inc., Miami, FL
(US)

(*) **Notice:** Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) **Appl. No.:** 09/544,139

(22) **Filed:** Apr. 6, 2000

(51) **Int. Cl.⁷** H01M 10/44; H01M 10/46

(52) **U.S. Cl.** 320/105

(58) **Field of Search** 320/103, 104,
320/105, 114, 115, 165; 307/44, 46, 48

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,341,762 A	9/1967	Rockoff
3,343,057 A	9/1967	Smith
4,039,903 A	8/1977	Russell
4,163,134 A	7/1979	Budrose
5,094,635 A	3/1992	Thompson et al.
5,401,924 A	3/1995	Armanno, Sr.
5,635,818 A	6/1997	Quintero

* cited by examiner

Primary Examiner—Edward H. Tso

(74) *Attorney, Agent, or Firm*—Liniak, Berenato, Longacre
& White

(57) **ABSTRACT**

A portable battery booster supply having a wireless remote control activation device. The hand held battery booster has connecting jumpers adapted to be connected to the battery posts of a vehicle battery. The booster includes a rechargeable battery contained within the housing and a means to connect to an adapter charger. The portable booster also contains a receiver/controller for selectively switching/controlling voltage supplied to the positive jumper. A small hand held transmitter produces radio signals, or other electromagnetic signals that communicates with the controller. The transmitter has an on switch and an off switch for selectively sending a transmitting signal to the controller in the portable booster. The controller has a preset code to compare the signals received from the hand held transmitter to selectively activate the positive jumper lead. When an off signal is received the controller operates a switch to open the circuit between the rechargeable batteries and the positive jumper. Similarly, when an on signal is received, the controller operates the switch to close the circuit between the battery and the positive jumper cable to provide voltage thereto. This arrangement allows a user to make the connections between the booster battery and the battery to be charges and activate the voltage supply from a safe remote location.

6 Claims, 1 Drawing Sheet

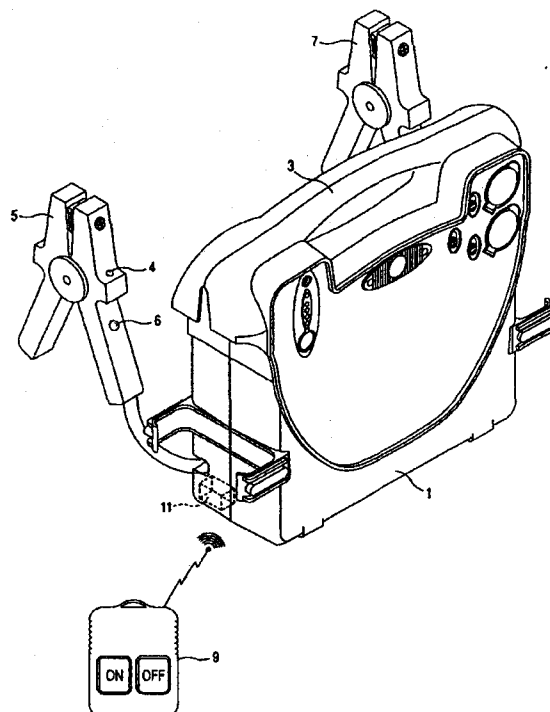
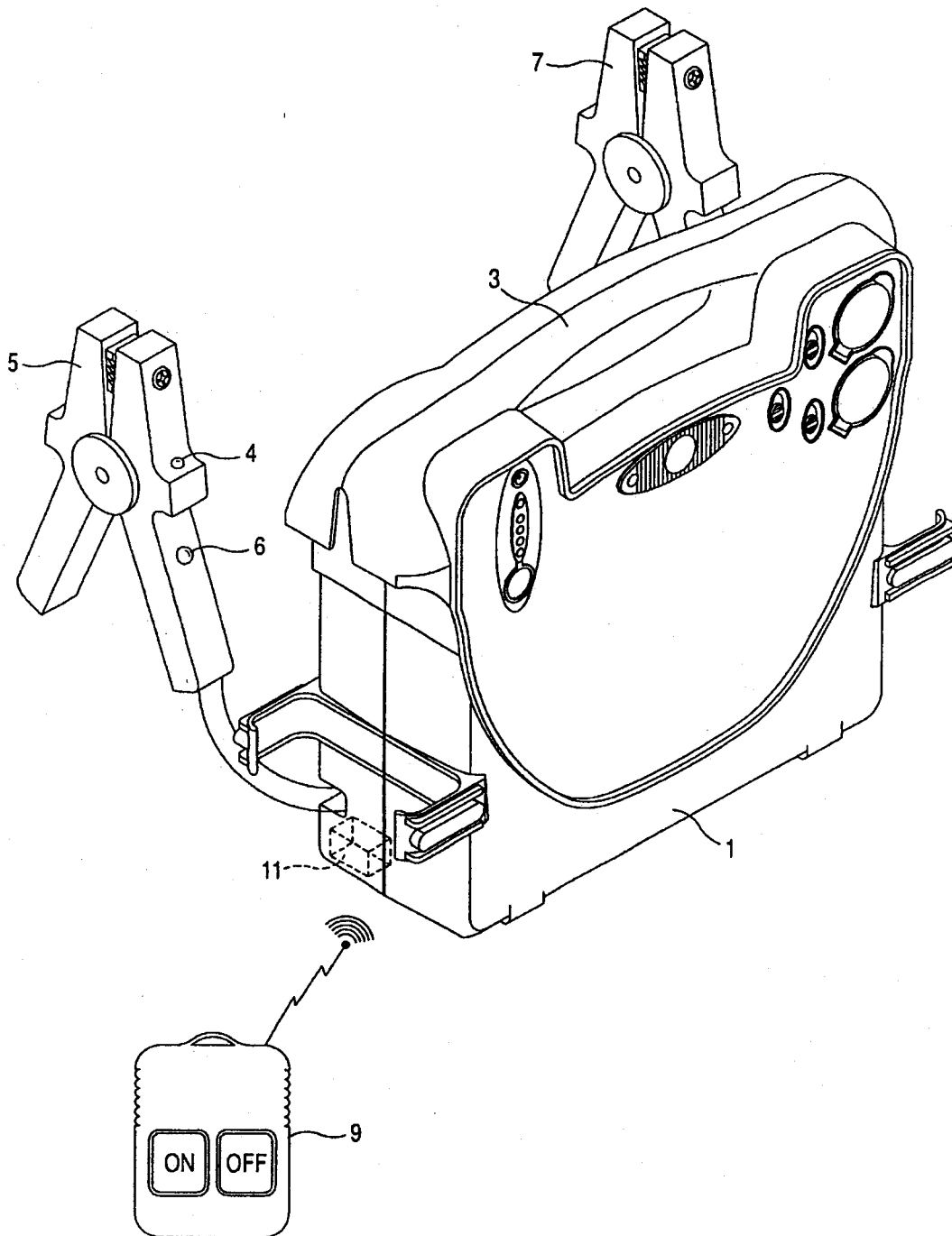


Fig. 1



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PORTABLE BOOSTER SUPPLY WITH WIRELESS REMOTE CONTROL ACTIVATION

BACKGROUND OF THE INVENTION

1. Field of the Invention

This invention is directed to a booster supply for boosting a run down vehicle battery and particularly is directed to a portable booster having a wireless remote control to remotely activate the supply of power from a safe distance.

2. Description of the Prior Art

Vehicular battery booster supplies are well known in the art. Historically, when a vehicle battery is run down to the point where there is insufficient voltage to start the vehicle, the battery must be jumped. This is often accomplished by jumping from a running vehicle or from a service vehicle. In such a case a live voltage source is connected to the run down battery to provide sufficient voltage to start the disabled vehicle. This operation often leads to unsafe conditions. Should the operator improperly make the electrical connection, or should a voltage surge occur, sparks can be generated. Vehicle batteries being jumped may produce hydrogen gas which can cause a fire or battery explosion causing severe injury to the operator. In an effort to increase safety in the boosting operation, it has been suggested to run wires from the voltage booster source to remotely activate the booster. Two such booster supplies are disclosed in U.S. Pat. Nos. 3,343,057 and 3,341,762 and are hereby incorporated herein by reference. However, these devices require a hard wire connection from the remote device to the service mounted booster supply. The use of a physical hard wire connection limits the range of use, reduces mobility, is expensive, bulky, and not easily adapted to be used with a portable hand carried device. The drawbacks of the hard wire remote activating switch are readily apparent.

It is also well known in the art to use wireless switching devices to remotely activate accessories within an automotive vehicle. Such devices are incorporated into keyless entry systems whereby an operator carries a small portable transmitter that selectively sends a radio signal to a control device in the vehicle to selectively operate vehicular accessories such as door locks, a power switch for the vehicle ignition and starter, or to disable a starting system. Two such keyless entry systems are disclosed in U.S. Pat. Nos. 5,689,142 and 4,672,375 and are hereby incorporated herein by reference.

It is the object of the present invention to improve on the booster supply devices of the prior art. Specifically, it is the object of the present invention to incorporate a wireless remote control to safely activate a hand held portable booster from a remote location.

SUMMARY OF THE INVENTION

The present invention is directed to portable hand held booster supply having a wireless remote control activation device. The hand held battery booster has connecting jumpers adapted to be connected to the battery posts of a vehicle battery. The booster includes a rechargeable battery contained within the housing and a means to connect to an adapter charger. The portable booster also contains a receiver and controller module for selectively switching/controlling voltage supplied to the positive jumper. A small hand held transmitter produces radio signals, or other electromagnetic signals, that communicates with the receiver/controller. The transmitter has an on switch and an off switch

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for selectively sending a transmitting signal to the receiver/controller in the portable booster. The receiver/controller has a preset code to compare the signals received from the hand held transmitter to selectively activate the positive jumper lead. When an off signal is received the receiver/controller operates a switch to open the circuit between the rechargeable battery and the positive jumper. Similarly, when an on signal is received, the receiver/controller operates the switch to close the circuit between the battery and the positive jumper cable to provide voltage thereto. This arrangement allows a user to make the connections between the booster battery and the battery to be charged and activate the voltage supply from a safe remote location.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1. is a perspective view of the portable booster device and wireless transmitter according to the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENT

FIG. 1 depicts the portable battery booster having a case 1, a handle 3, a positive jumper clamp 5 and a negative jumper clamp 7 each connected to the portable booster via cables. The jumper clamps 5, 7 are conventionally formed and adapted to easily connect to a vehicle battery. Within the case 1 is a rechargeable internal battery providing a voltage source to the positive jumper 5 for jumping a low vehicle battery. The case further includes a connection means to connect the internal battery to an adapter charger or other charging device as is known in the art. An LED display is provided on the front of the booster case 1 in order to provide a signal representative of the charge capacity of the internal battery. A light is also provided to illuminate the area proximate the portable booster and vehicle battery while establishing a connection there between. A pair of auxiliary connecting ports, such as to accommodate a cigarette lighter adapter, are also provided to enable connection to auxiliary DC devices. Each of the light and auxiliary ports are provided with a switch to selectively provide voltage thereto.

A hand held wireless transmitter 9 is provided to selectively send a signal to a controller 11 disposed within the portable booster. The wireless transmitter 9 and controller/receiver 11 devices may be as conventionally found in the keyless entry art and preferably act on radio frequency signals. The controller 11 is provided to selectively establish electrical connection between the internal battery and the positive jumper 5. Preferably, the wireless transmitter 9 has an on and off button to respectively send a closed an open signal. When the on button is pushed, the wireless transmitter 9 sends a closed signal to close a circuit between the internal battery and the positive jumper to provide voltage thereto. Similarly, when the off button is depressed, the wireless transmitter 9 sends an open signal to open the circuit between the internal battery and the positive jumper. This arrangement provides the ability to selectively activate the portable booster from a safe remote location.

In order to selectively connect and disconnect the internal battery from the positive jumper 5, a switch is disposed there between. The controller 11 simply activates the switch according to the signal received from the wireless transmitter. Preferably a relay switch or PCB board is disposed between the internal battery and the positive jumper 5.

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The method of using the portable booster will now be explained. When attempting to boost a low vehicle battery, the wireless transmitter 9 is operated to disconnect the connection to the positive jumper 5. Specifically, the off button is pushed to disconnect the positive jumper 5 from the voltage source. Then the portably battery is positioned proximate the low vehicle battery. Once the positive jumper 5 is deactivated, the positive and negative jumpers 5,7 of the portable booster are properly connected to the low battery and vehicle as is known in the art. The operator then moves a safe distance from the portable booster and low battery before activating the voltage source. Preferably, the operator retreats to within the vehicle and assumes a position to start the vehicle. Such a position provides protection to the operator in the event the connections were improperly made or a fire explosion occurs from a faulty battery. Once the operator assumes a safe position, the on button of the transmitter 9 is depressed to send the closed signal. This establishes voltage to the positive jumper 5 and low battery. After sufficient time has elapsed to initially provide some charging to the low battery, the operator starts the vehicle. Once the vehicle is running properly the operator depressed the off button on the transmitter before leaving the vehicle. Once the voltage source is disconnected from the positive jumper, the portable booster may be disconnected from the low battery in the vehicle. The portable booster may then be stored within the vehicle trunk or other storage compartments.

A pushbutton switch 2 is disposed on an assessable portion of the connector.

The positive jumper clamp 5 may be provided with an additional safety feature. A light 4, preferably an LED, is disposed on the jumper clamp 5 to provide light when connecting to a battery terminal post. An electrical connection is established between a push button switch 6 and the cable of opposite polarity. When the switch 6 is activated, the booster voltage source activates the light 4 independent of a connection to the battery to be charged. Alternatively, a rear portion of the jumper clamp 5 may contained an independent battery source, such as an AA battery, for providing an independent voltage source for the light 4. In either circumstance, light 4 may be selectively illuminated both prior to and after connection is established with a battery to be charged.

While the foregoing invention has been shown and described with reference to a preferred embodiment, it will be understood by those possessing skill in the art that various changes and modifications may be made without departing from the spirit and scope of the invention.

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What is claimed is:

1. A portable battery booster comprising:

- a case containing a voltage source and a positive and negative jumper provided for connection to a vehicle battery, said case containing a receiver/controller provided to selectively control a connection between said voltage source and said positive jumper; and
- a wireless transmitter adapted to transmit a signal to said controller, wherein said controller selectively connects said positive jumper to said voltage source in response to said signal.

2. The portable battery booster according to claim 1, further comprising:

- a switch disposed between said voltage source and said positive jumper, wherein said transmitter can selectively transmit one of an on and an off signal, said controller closing said switch in response to said on signal to provide said voltage source to said positive jumper and opening said switch to eliminate said connection between said voltage source and said positive jumper.

3. The portable battery booster according to claim 2, wherein said switch is one of a relay switch and a PCB board.

4. The portable battery booster according to claim 1, wherein said portable booster comprises a light to illuminate a working area proximate said vehicle battery.

5. The portable battery booster according to claim 1, wherein said booster has at least one auxiliary connection to connect said voltage source to an auxiliary device.

6. A method of boosting a vehicle having a low battery with a portable booster device, said method comprising the steps of:

- providing said portable booster device proximate said low battery;
- activating a wireless transmitter to send an open signal to said portable booster to disconnect a voltage source within said portable booster device;
- connecting said wireless booster device to said low battery;
- activating said wireless transmitter to send a closed signal to said portable booster to reconnect said voltage source thereby providing voltage to said low battery;
- starting said vehicle;
- activating said wireless transmitter to send said open signal thereby disconnecting said voltage source; and
- disconnecting said portable booster from said vehicle battery.

* * * * *



US006222342B1

(12) **United States Patent**
Eggert et al.

(10) **Patent No.:** **US 6,222,342 B1**
(45) Date of Patent: **Apr. 24, 2001**

(54) **JUMP START BATTERY PACK AND ENCLOSURE THEREFOR**

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(*) **Notice:** Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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Primary Examiner—Shawn Riley

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(74) *Attorney, Agent, or Firm*—Seyfarth Shaw

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(22) **Filed:** **Jul. 28, 2000**

(51) **Int. Cl.⁷** **H02J 7/00; H02M 2/02;**
H01R 3/00

(52) **U.S. Cl.** **320/105; 320/112; 307/66;**
D13/103; D13/107

(58) **Field of Search** **320/103, 104,**
320/105, 112; D13/DIG. 34, 103, 107,
109; 307/64, 66; 429/170; 439/500; 60/721

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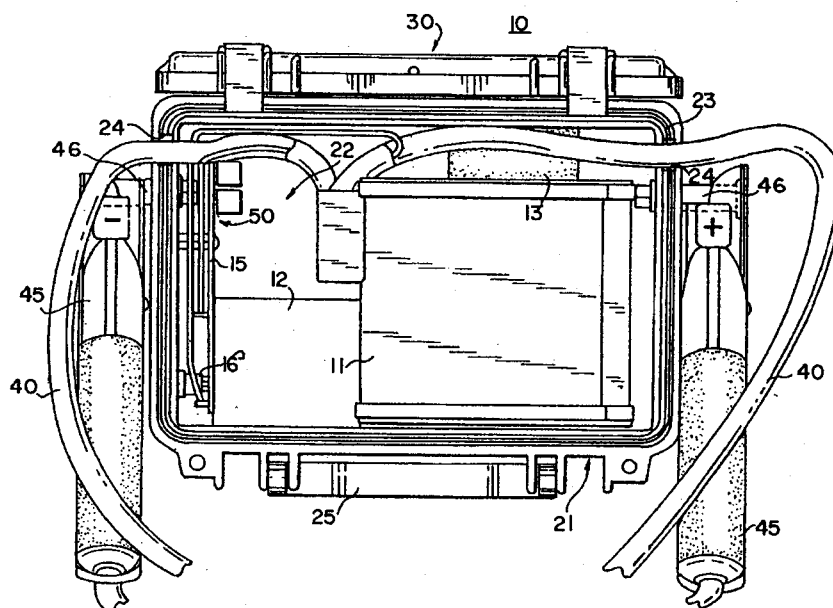
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(57) **ABSTRACT**

A portable battery jump start unit includes a portable, plastic, clamshell-type housing including a box-like open-top base and a box-like, open-bottom cover hingedly connected together for movement between open and closed conditions. A lead-acid battery is disposed in the housing, along with a state-of-charge monitoring circuit connected to the battery terminals and including a visible indicator viewable through an aperture in the housing. Cables are connected to the battery terminals and respectively extend through openings formed by cooperating notches in the mating edges of the base and cover, the cables respectively having clamp connectors at their distal ends which can be clamped onto posts projecting laterally outwardly from the sides of the base. A charger jack in the base is connected to the battery terminals for receiving a charger connector.

21 Claims, 5 Drawing Sheets



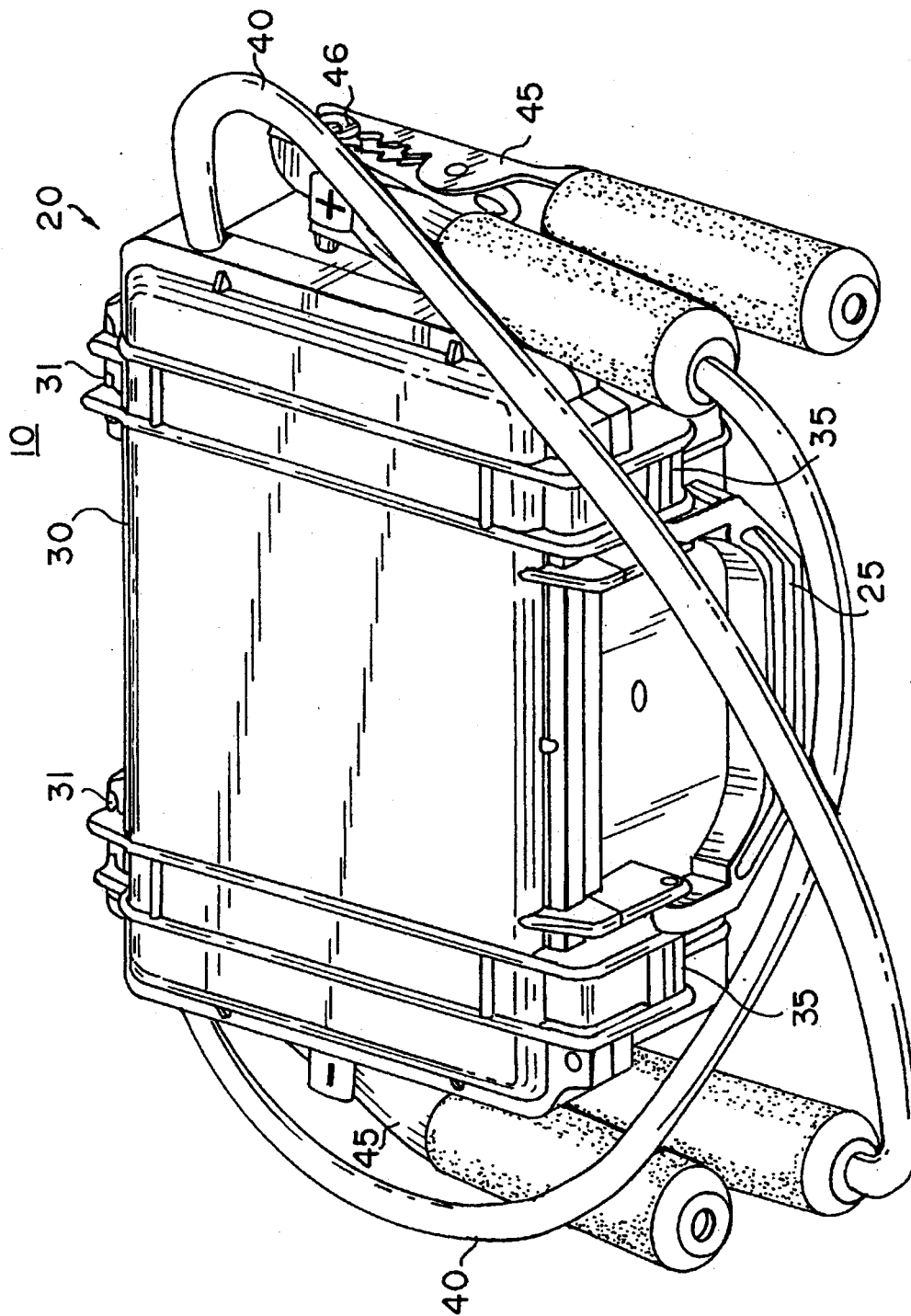


FIG. 1

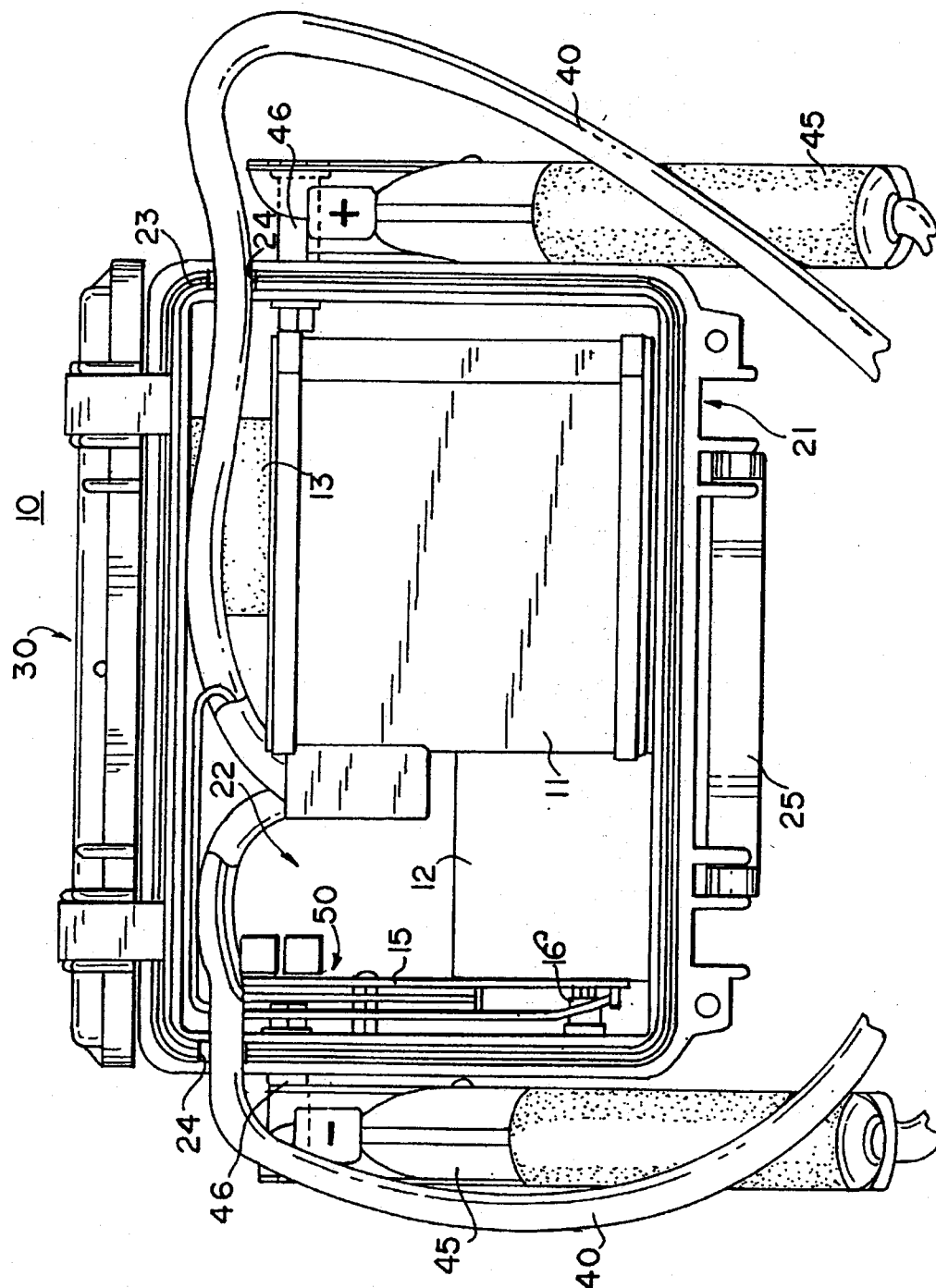


Fig. 2

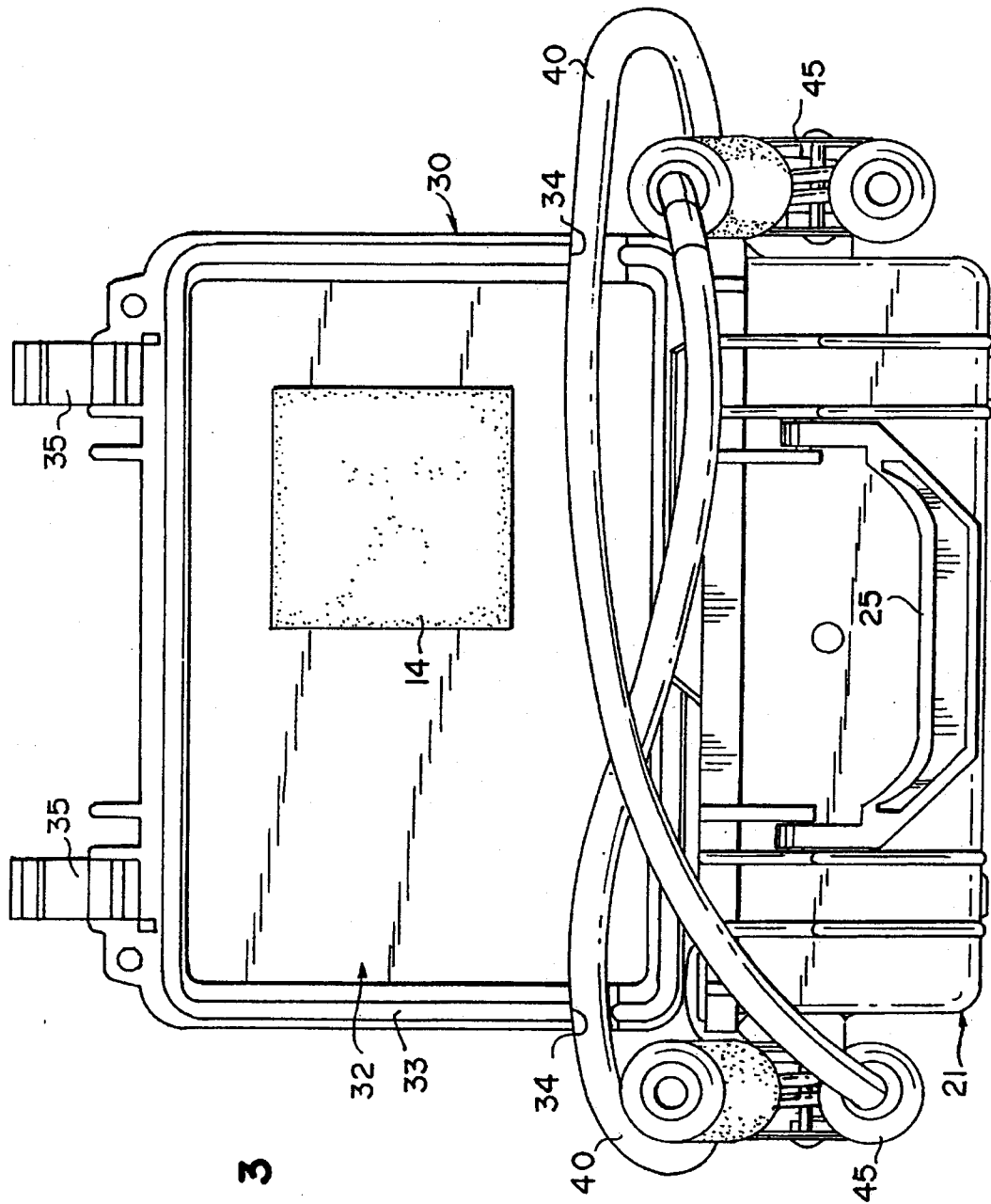


FIG. 3

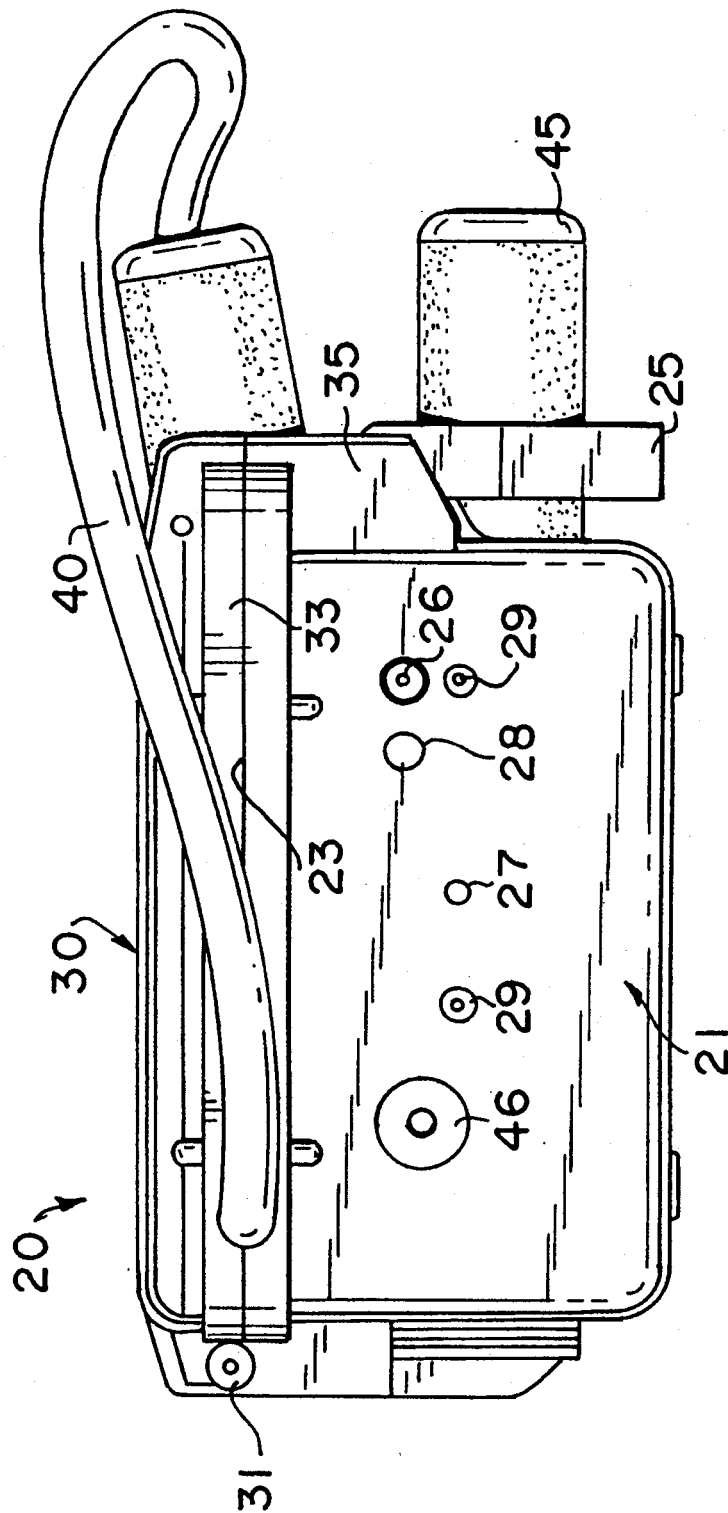
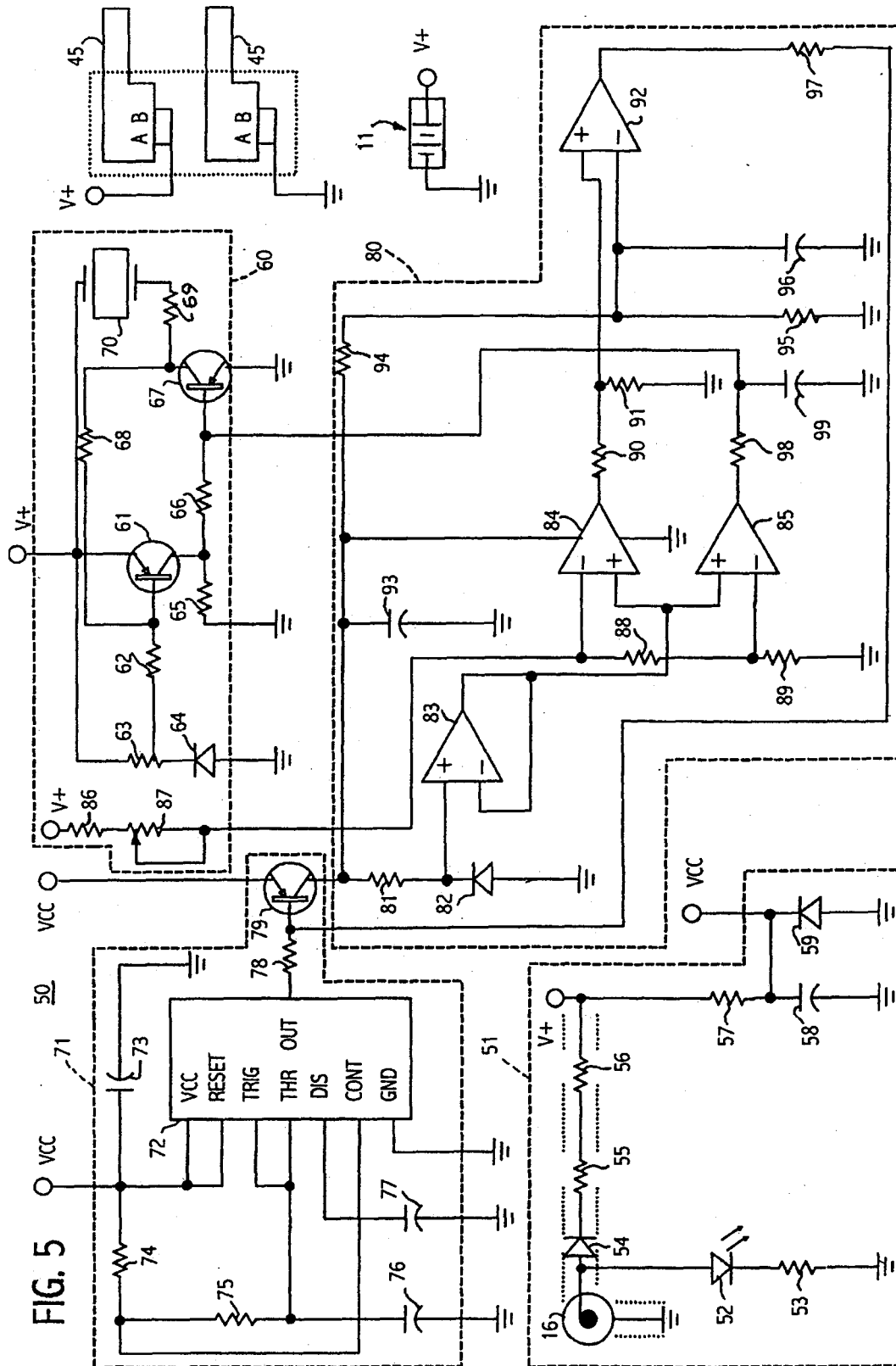


FIG. 4



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JUMP START BATTERY PACK AND ENCLOSURE THEREFOR

BACKGROUND

This application relates to portable battery pack units, and particularly to battery packs of the type used for jump starting automotive vehicles.

Various types of portable battery packs have heretofore been provided, including batteries disposed in portable carrying cases. These cases are typically specifically designed for the particular battery pack and often are provided with access panels or hatches for accessing the interior of the housing. One such battery pack is disclosed in U.S. Pat. No. 6,002,235, and includes cables projecting outwardly through openings in the housing and terminated in connector clamps which, when not in use, can be clamped onto posts projecting laterally outwardly from the case. Access to the inside of the housing requires the removal of a panel by removing a number of screws.

It is also known to provide clamshell-type or valise-type cases for various other types of applications, and at least one battery pack for use in starting airplane engines has been provided in a clamshell-type case.

SUMMARY

The present application is directed to a battery pack which is housed in a case which provides simple and easy access to the contents, but which is fully operable in the closed condition.

An aspect of the battery pack unit is that it is provided with monitoring circuitry which permits viewing of the state of charge of the battery from outside the case and/or viewing of the verification that the unit is charging. The circuitry also permits audible verification of the state of charge of the battery as well as certain overcharge conditions.

Another aspect is the provision of cables carried by the housing and with connector clamps mountable on the outside of the housing when not in use.

Another aspect is the provision of a charger port accessible when the housing is closed.

Still another aspect is the provision of a housing of the type set forth for use with a battery pack.

Certain ones of these and other aspects may be provided in a portable battery jump start unit comprising: a portable clamshell-type housing having a base and a cover hingedly connected together for movement between open and closed conditions and cooperating to define an interior space, a battery disposed in the interior space and having terminals, monitoring circuitry including a printed-circuit board disposed in the interior space and connected to the battery terminals and including a visible indicator, an aperture in the housing disposed for viewing the visible indicator therethrough, two cable openings in the housing, two cables respectively connected to the battery terminals and extending outwardly through the openings, two clamp connectors respectively connected to distal ends of the cables outside the housing, and a charger jack in the housing connected to the battery terminals and adapted for connection to an associated charger connector while the housing is in its closed condition.

BRIEF DESCRIPTION OF THE DRAWINGS

For the purpose of facilitating an understanding of the subject matter sought to be protected, there are illustrated in

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the accompanying drawings preferred embodiments thereof, from an inspection of which, when considered in connection with the following description, the subject matter sought to be protected, its construction and operation, and many of its advantages should be readily understood and appreciated.

FIG. 1 is a perspective view of an embodiment of a portable jumpstart battery pack unit with the enclosure closed;

FIG. 2 is a top plan view of the unit of FIG. 1, with the enclosure open;

FIG. 3 is a front elevational view of the unit of FIG. 2;

FIG. 4 is a side elevational view of the left-hand side of the unit of FIG. 1, with the associated battery clamp removed; and

FIG. 5 is an electrical schematic diagram of the circuitry of the unit of FIG. 1.

DETAILED DESCRIPTION

Referring to FIGS. 1-4, there is illustrated a battery pack unit, generally designated by the numeral 10, including a battery 11 disposed in a portable, clamshell-type housing 20. The battery 11 may be a lead-acid battery of the thin metal film-type, and may be of a type sold by Johnson Controls, Inc. under the trade designation "INSPIRA." Foam spacer blocks 12, 13 and 14 may be provided to snugly position the battery 11 in the housing 20 without shifting, rattling or vibration. Also disposed in the housing 20 is a printed circuit board 15 (FIG. 2), on which is mounted a monitoring circuit 50, to be described more fully below, which includes a coaxial charger jack 16 for coupling to an associated battery charging apparatus (not shown).

The housing 20 may be a modified form of a type of case sold by Chemplex Controls under the trade designation "PELICAN," and may be molded of a suitable plastic material. The housing 20 includes an open-top base 21 and an open-bottom cover 30 hingedly interconnected for movement between a closed condition, illustrated in FIGS. 1 and 4, and an open condition, illustrated in FIGS. 2 and 3. In its closed condition, the housing 20 is generally in the shape of a rectangular parallelepiped. The base 21 defines therein a receptacle 22 (see FIG. 2) and has a generally rectangular upper edge 23 having formed therein at opposite sides thereof adjacent to the rear end thereof, two generally semi-cylindrical cable notches 24. A handle 25 is pivotally mounted on the front wall of the base 21. Formed through the left-hand side wall of the base 21, as viewed in FIG. 4, is a charger opening 26 for providing access to the charger jack 16 (FIG. 2), as well as apertures 27 and 28, respectively, for an audible annunciator and for viewing an LED on the PC board 15. Formed through the same side wall are openings 29 for suitable fasteners to fixedly secure the printed circuit board 15 in place.

The cover 30 is hingedly connected to the base 21 by hinges 31, and defines an open-bottom receptacle 32 (see FIG. 3) which is much shallower than the receptacle 22 in the base 21, and cooperates therewith in the closed condition of the housing 20 to define an interior space or compartment. The cover 30 has a generally rectangular lower edge 33 in which are formed, respectively along the opposite sides thereof adjacent to the rear edge thereof, two semi-cylindrical cable notches 34. Latches 35 are provided on the front wall of the cover 30 for engagement with associated keepers on the base 21 for latching the housing 20 in its closed position. As can be seen in FIG. 2, the spacers 12 and 13 are disposed in the base receptacle 22, respectively along top and side edges of the battery 11, while the spacer 14

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(FIG. 3) is disposed in the cover receptacle 32, being fixedly secured therein by suitable means, and being of a thickness so as to engage the upper surface of the battery 11 when the housing 20 is in its closed condition.

Respectively connected to positive and negative terminals of the battery 11 are two cables 40, which respectively are seated in the cable notches 24 and extend outwardly of the housing 20, respectively terminating in cable clamps 45 of known construction, for clamping respectively onto the positive and negative terminals of an associated battery of an automotive vehicle to be jump started. It will be appreciated that, when the housing 20 is in its closed condition, the cable notches 24 in the base respectively cooperate with the cable notches 34 in the cover to form cable openings for snugly accommodating the cables 40. When not in use, the cable clamps 45 are respectively clamped onto posts 46, which respectively project laterally outwardly from the opposite side walls of the base 21, being fixedly secured thereon, as by suitable fasteners.

Referring now to FIG. 5, the monitoring circuit 50 includes a power supply and charger input circuit 51, which includes the charger jack 16. More specifically, the center or ungrounded terminal of the charger jack 16 is connected to the anode of an LED 52, the cathode of which is connected through a resistor 53 to ground, i.e., the negative terminal of the battery 11. The center terminal of the charger jack 16 is also connected to the anode of a diode 54, the cathode of which is connected through resistors 55 and 56 to the positive terminal of the battery 11, designated V+. A resistor 57 and a capacitor 58 are connected in a series across the terminals of the battery 11. The junction between the resistor 57 and the capacitor 58 is connected to the cathode of a Zener diode 59, the anode of which is grounded. A VCC supply voltage is provided at the cathode of the Zener diode 59.

The monitoring circuit 50 also includes an over-voltage alarm circuit 60, which includes a transistor 61 having its emitter connected to the positive terminal of the battery 11, and its base connected through a resistor 62 to the wiper of a potentiometer 63, connected to the cathode of a Zener diode 64, the anode of which is connected to ground. The collector of the transistor 61 is connected to ground through a resistor 65 and is connected through a resistor 66 to the base of a transistor 67, the emitter of which is grounded and the collector of which is connected through a resistor 68 to the base of the transistor 61. The collector of the transistor 67 is also connected through a resistor 69 to one terminal of a transducer 70, the other terminal of which is connected to the resistor 63 and the positive terminal of the battery. The transducer 70 may be any of a number of suitable audible annunciators.

The monitoring circuit 50 also includes a timer circuit 71, which includes an integrated circuit ("IC") timer 72, which may be an LM555CNB. The VCC supply is connected to the VCC and RESET terminals of the IC72, these terminals also being connected through a capacitor 73 to ground, and through the series connection of resistors 74 and 75 and a capacitor 76 to ground. The junction between the resistor 75 and the capacitor 76 is connected to the trigger and threshold terminals of the IC72, while the junction between the resistors 74 and 75 is connected to the discharge terminal of the IC72. The control terminal of the IC72 is connected to ground through a capacitor 77. The output terminal of the IC72 is connected through a resistor 78 to the base of a transistor 79, the emitter of which is connected to the VCC supply, and the collector of which is connected to a low-voltage comparator circuit 80.

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More particularly, the circuit 80 includes a resistor 81 and a Zener diode 82 connected in series between the collector of the transistor 79 and ground. The cathode of the diode 82 is connected to the non-inverting input terminal of an operational amplifier ("OP AMP") 83, the output of which is connected to its inverting input terminal and to the non-inverting input terminals of OP AMPS 84 and 85. Connected across the terminals of the battery 11 is a voltage divider including a resistor 86, a potentiometer or variable resistor 87, a resistor 88 and a resistor 89. The junction between the potentiometer 87 and the resistor 88 is connected to the inverting input terminal of the OP AMP 84, while the junction between the resistors 88 and 89 is connected to the inverting input terminal of the OP AMP 85. The output of the OP AMP 84 is connected to ground through resistors 90 and 91, the junction therebetween being connected to the inverting input terminal of an OP AMP 92. The collector of the transistor 79 is connected to ground through a capacitor 93 and through a voltage divider including resistors 94 and 95, the junction therebetween being connected to the non-inverting input terminal of the OP AMP 92, which is also connected to ground through a capacitor 96. The output of the OP AMP 92 is connected through a resistor 97 to the base of the transistor 79. The output of the OP AMP 85 is connected through a resistor 98 to the base of the transistor 67, which base is also connected to ground through a capacitor 99. The OP AMPS 83-85 and 92 may all be portions of a common integrated circuit, such as a TLC274, the supply terminal of which is connected to the VCC supply. The circuits 60 and 80 cooperate to form a state-of-charge monitor.

The power supply and charger input circuit 51 provides a means of safely charging the battery 11, and protects sensitive circuitry from voltage spikes. Jack 16 receives charging voltage from either a wall-mounted charger or a plug charger connected to the cigarette lighter of the associated vehicle. The rest voltage of the battery 11 may be higher than the rest voltage of the associated automotive battery so that, if left connected, the battery 11 would tend to discharge into the automotive battery through a cigarette lighter plug. The diode 54 prevents this from happening. Resistors 55 and 56 prevent high current inrushes, which could blow fuses in a vehicle used to charge the battery 11 through a cigarette lighter plug. The LED 52 lights in the presence of a charge voltage from any source, with current limiting by the resistor 53, the LED being visible through aperture 28 (FIG. 4). The combination of the resistor 57, the capacitor 58 and the Zener diode 59 form a protective network that suppresses high voltage spikes, which might otherwise damage the integrated circuits.

Under normal, non-charging conditions, the voltage of the battery pack battery 11 will be between 12.8 volts and 13.1 volts, well below the voltage rating of the Zener diode 64, which is preferably 15.0 volts $\pm 5\%$. In this normal, at rest condition, the total current draw of the circuit is approximately 190 μ A. When the battery 11 is being charged, its voltage will rise above 13.1 volts. If the charging method is applying an inappropriately high charging voltage, the battery's voltage could rise above 16.5 volts, which could lead to battery overheating and failure. As the battery voltage exceeds the threshold voltage of the Zener diode 64, the excess voltage will develop across the potentiometer 63. This potentiometer is adjusted such that transistor 61 will turn ON if the battery voltage exceeds 16.5 volts. When the voltage at the wiper of the potentiometer 63 exceeds the threshold value of the transistor 61, typically about 0.7 volts, the base-emitter junction of the transistor 61 conducts

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through the resistor 62 as a current limiter, causing the collector-emitter junction to conduct.

With transistor 61 ON, current flows through the resistor 65, bringing the collector of the transistor 61 from zero volts towards V+. When the voltage across the resistor 65 exceeds the threshold value of the transistor 67, its base-emitter junction conducts through the resistor 66 as a current limiter, turning ON the transistor 67. With the transistor 67 ON, current flows through the audible annunciator 70 and the current-limiting resistor 69, causing the audible annunciator 70 to emit a warning tone through aperture 27 (FIG. 4). Conduction of the base-emitter junction of the transistor 67 also provides positive feedback to the base of the transistor 61 through the resistor 68, forcing the transistor 61 into full conduction or saturation. This prevents a condition wherein the volume of the audible annunciator 70 transitions through a gradually increasing volume as the battery voltage transfers from below 16.5 volts to above 16.5 volts.

Once the voltage at the wiper of the potentiometer 63 falls below the threshold voltage of the transistor 61, the feedback through the resistor 68 cannot sustain the transistor 61 ON, and the circuit 60 turns OFF.

The timer circuit 71 administers a power pulse of approximately 400 ms duration, repeated every 35 seconds. This is done to conserve battery power by powering the IC of the low-voltage comparator circuit 80 only briefly. The momentary power causes, in effect, the comparator circuit to "check" for low battery voltage every 35 seconds. The capacitor 76 and the resistors 74 and 75 establish the timer's astable operation, and set the ON time at 400 ms and OFF time at 35 seconds. Capacitor 73 provides noise immunity to the power supply pins VCC and ground. Capacitor 77 stabilizes the control terminal of the timer IC 72. The output pin drives the transistor 79 through the current-limiting resistor 78.

The low-voltage comparator circuit 80 determines if the voltage of the battery 11 is below a safe storage level, and alerts the user if it is. The circuit includes two set points: at 12.4 V, where charging of the battery is needed, and again at 12.0 V, where immediate attention is required.

The admittance of current through transistor 79 powers the OP AMP integrated circuit, as well as the voltage reference diode 82, through current limiting resistor 81. The 1.250 V reference is buffered through OP AMP 83, configured as a unity voltage gain current amplifier, for stability. This reference voltage is applied to the positive inputs OP AMPS 84 and 85, configured as two comparators. The voltage divider circuit of resistors 86-89 causes the negative input of OP AMP 85 to exceed the reference voltage when the battery voltage is above 12.40 V, and causes the negative input of OP AMP 84 to exceed the reference voltage when the battery voltage is above 12.00 V. As a result, the output of OP AMP 85 is only HI if the battery voltage is less than 12.40 V, and the output of OP AMP 84 is only HI if the battery voltage is less than 12.00 V.

AHI at the output of OPAMP 85, through current limiting resistor 98, turns ON transistor 67 (of the over-voltage alarm circuit 60), causing the audible annunciator 70 to sound. A HI at the output of OP AMP 84 is halved at the voltage divider of resistors 90 and 91, and this voltage (about 5.5 V), is applied to the negative input of OP AMP 92, causing a LO at its output. This LO, through current limiting resistor 97, keeps transistor 79 ON, even after the timer output goes HI. This LO is transient, however, because resistors 94 and 95 and capacitor 96 form an R-C network that begins charging when transistor 79 turns ON. After about 1.5 seconds, the

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voltage at the positive input to OP AMP 92 exceeds that of the negative input, and the output goes HI.

The result of all this is that if the battery voltage falls below 12.4 V, the circuit emits an audible tone of about 400 ms duration, each 35 seconds. If the battery voltage falls below 12.0 V, the circuit emits an audible tone of about 1500 ms duration, each 35 seconds.

Capacitor 99 slows the response of the transistor 67 to transient changes at the output of OP AMP 85, reducing "click" sounds at voltages above alert levels. Capacitor 93 filters noise from the power supply line for the OP AMP IC.

The matter set forth in the foregoing description and accompanying drawings is offered by way of illustration only and not as a limitation. While a particular embodiment has been shown and described, it will be obvious to those skilled in the art that changes and modifications may be made without departing from the broader aspects of applicants' contribution. The actual scope of the protection sought is intended to be defined in the following claims when viewed in their proper perspective based on the prior art.

We claim:

1. A portable battery jump start unit comprising:

a portable clamshell-type housing having a base and a cover hingedly connected together for movement between open and closed conditions and cooperating to define an interior space,

a battery disposed in the interior space and having terminals,

two cable openings in the housing,

two cables respectively connected to the battery terminals and extending outwardly through the openings,

two clamp connectors respectively connected to distal ends of the cables outside the housing, and

a charger jack in the housing connected to the battery terminals and adapted for connection to an associated charger connector while the housing is in its closed condition.

2. The unit of claim 1, wherein the housing is formed of plastic.

3. The unit of claim 1, and further comprising two posts respectively projecting laterally outwardly from opposite sides of the housing.

4. The unit of claim 1, wherein the base has an upper edge and the cover has a lower edge, the upper edge having two spaced-apart substantially semi-cylindrical first recesses formed therein and the lower edge having two spaced-apart substantially semi-cylindrical second recesses formed therein, the second recesses respectively cooperating with the first recesses for forming the openings when the housing is in its closed condition.

5. The unit of claim 1, wherein one of the base and the cover includes keeper structure and the other includes latch structure releasably engageable with the keeper structure for retaining the housing in its closed condition.

6. The unit of claim 1, wherein the jack is disposed in the base.

7. The unit of claim 1, wherein the battery is a lead-acid battery.

8. The unit of claim 1, and further comprising monitoring circuitry including a printed-circuit board disposed in the interior space and connected to the battery terminals and including a visible indicator, and a first aperture in the housing disposed for viewing the indicator therethrough.

9. The unit of claim 8, wherein the monitoring circuitry includes a state-of-charge monitor.

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10. The unit of claim 9, wherein the state-of-charge monitor includes a low voltage detection circuit and an over-voltage detection circuit.

11. The unit of claim 10, wherein the monitoring circuitry, includes a timer for periodically activating the state-of-charge monitor. 5

12. The unit of claim 8, wherein the visible indicator is a light-emitting diode.

13. The unit of claim 8, wherein the monitoring circuitry includes an audible indicator, and a second aperture in the housing disposed to facilitate hearing the audible alert. 10

14. A housing for a portable jump start unit comprising: a box-like open-top base,

a box-like open-bottom cover hingedly connected to the base for movement between open and closed conditions in a clamshell-type fashion for cooperation with the base to define an interior space, 15

two openings in the housing dimensioned to permit cables to be disposed therethrough, and

two posts carried by the housing and projecting outwardly therefrom. 20

15. The housing of claim 14, wherein each of the base and the cover is formed of plastic.

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16. The housing of claim 14, wherein the posts are respectively disposed at opposite ends of the base.

17. The housing of claim 14, wherein the openings are respectively disposed at opposite ends of the housing.

18. The housing of claim 17, wherein the base has an upper edge and the cover has a lower edge, the upper edge having two spaced-apart substantially semi-cylindrical first recesses formed therein and the lower edge having two spaced-apart substantially semi-cylindrical second recesses formed therein, the second recesses respectively cooperating with the first recesses for forming the openings when the housing is in its closed condition.

19. The housing of claim 14, wherein one of the base and the cover includes keeper structure and the other includes latch structure releasably engageable with the keeper structure for retaining the housing in its closed condition.

20. The housing of claim 14, wherein the jack is disposed in a forward portion of the base.

21. The housing of claim 14, and further comprising a handle disposed on the base and projecting forwardly therefrom.

* * * * *



US006130519A

United States Patent [19]

Whiting et al.

[11] **Patent Number:** 6,130,519[45] **Date of Patent:** Oct. 10, 2000[54] **PORTABLE BATTERY CHARGER
INCLUDING AUTO-POLARITY SWITCH**[75] Inventors: **John S. Whiting**, Plymouth; **Branden
M. Reid**, Minneapolis, both of Minn.[73] Assignee: **Century Mfg. Co.**, Minneapolis, Minn.[21] Appl. No.: **09/400,423**[22] Filed: **Sep. 21, 1999****Related U.S. Application Data**

[60] Provisional application No. 60/104,651, Oct. 16, 1998.

[51] **Int. Cl.**⁷ **H02J 7/00**[52] **U.S. Cl.** **320/105**[58] **Field of Search** 320/103, 104,
320/105[56] **References Cited****U.S. PATENT DOCUMENTS**

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Primary Examiner—Peter S. Wong*Assistant Examiner*—Pia Tibbits*Attorney, Agent, or Firm*—Merchant & Gould P.C.[57] **ABSTRACT**

A portable hand-carrying auto-polarity battery booster apparatus includes a power storage device which is capable of providing power to an electrical system such as a battery. For example, the portable apparatus can be used to jump start a vehicle with a dead battery. The apparatus is compact in size which encloses the power storage device, the at least four connecting cables, and many other electrical circuits in a compact housing. The portable apparatus also includes an auto-polarity switch in the housing which ensures a proper polarity connection between the power storage device of the apparatus and the electrical system, as well as an ON/OFF switch located between the battery and the auto-polarity switch.

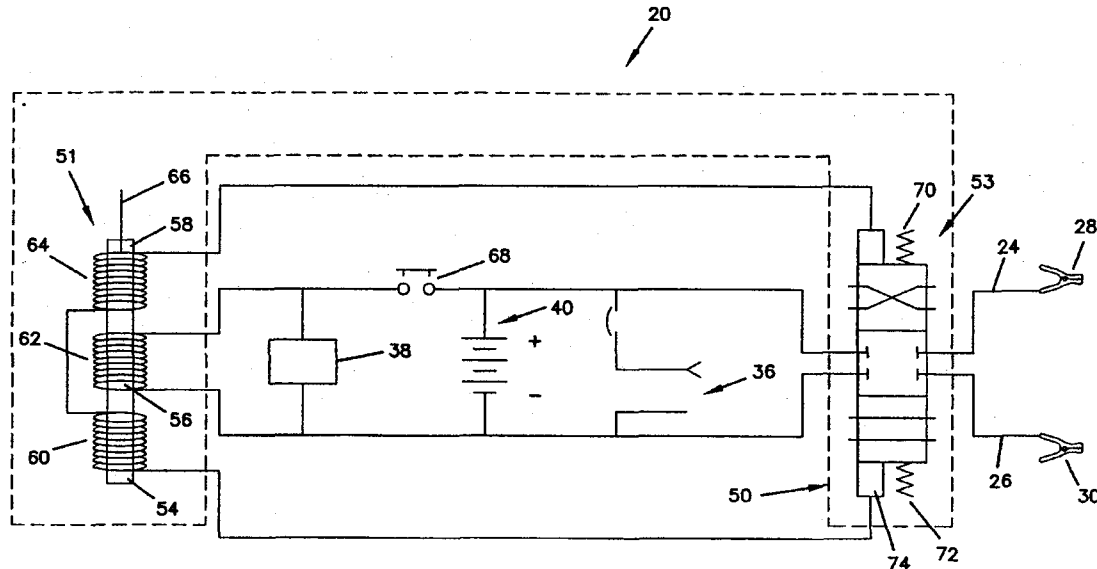
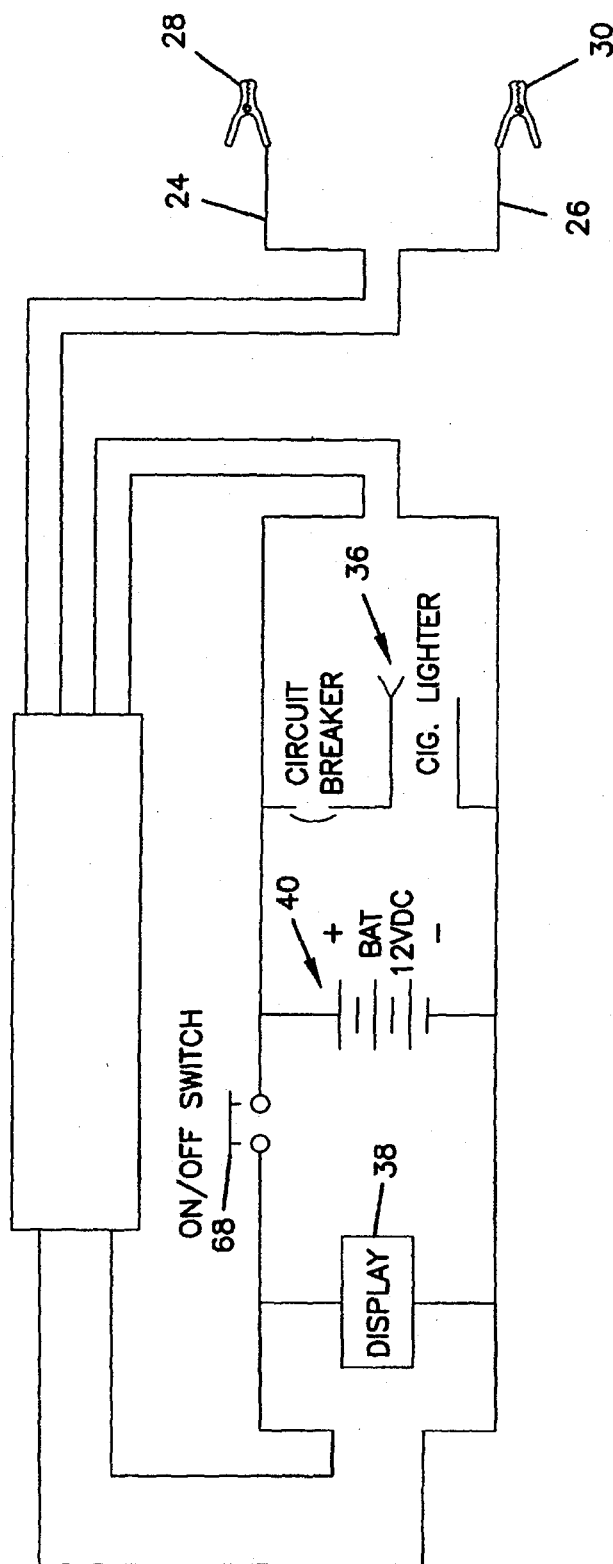
21 Claims, 12 Drawing Sheets

FIG. 1



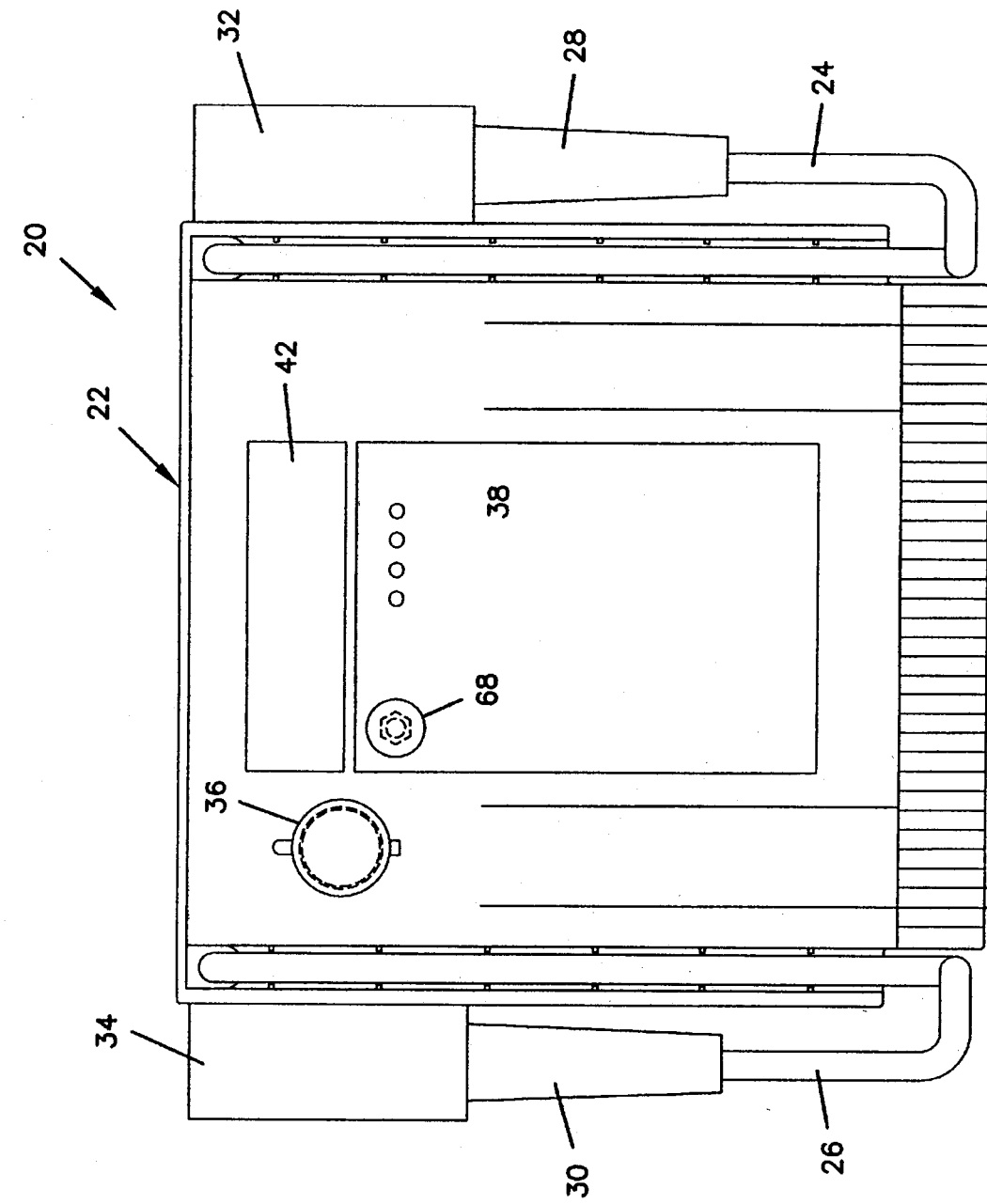


FIG. 2

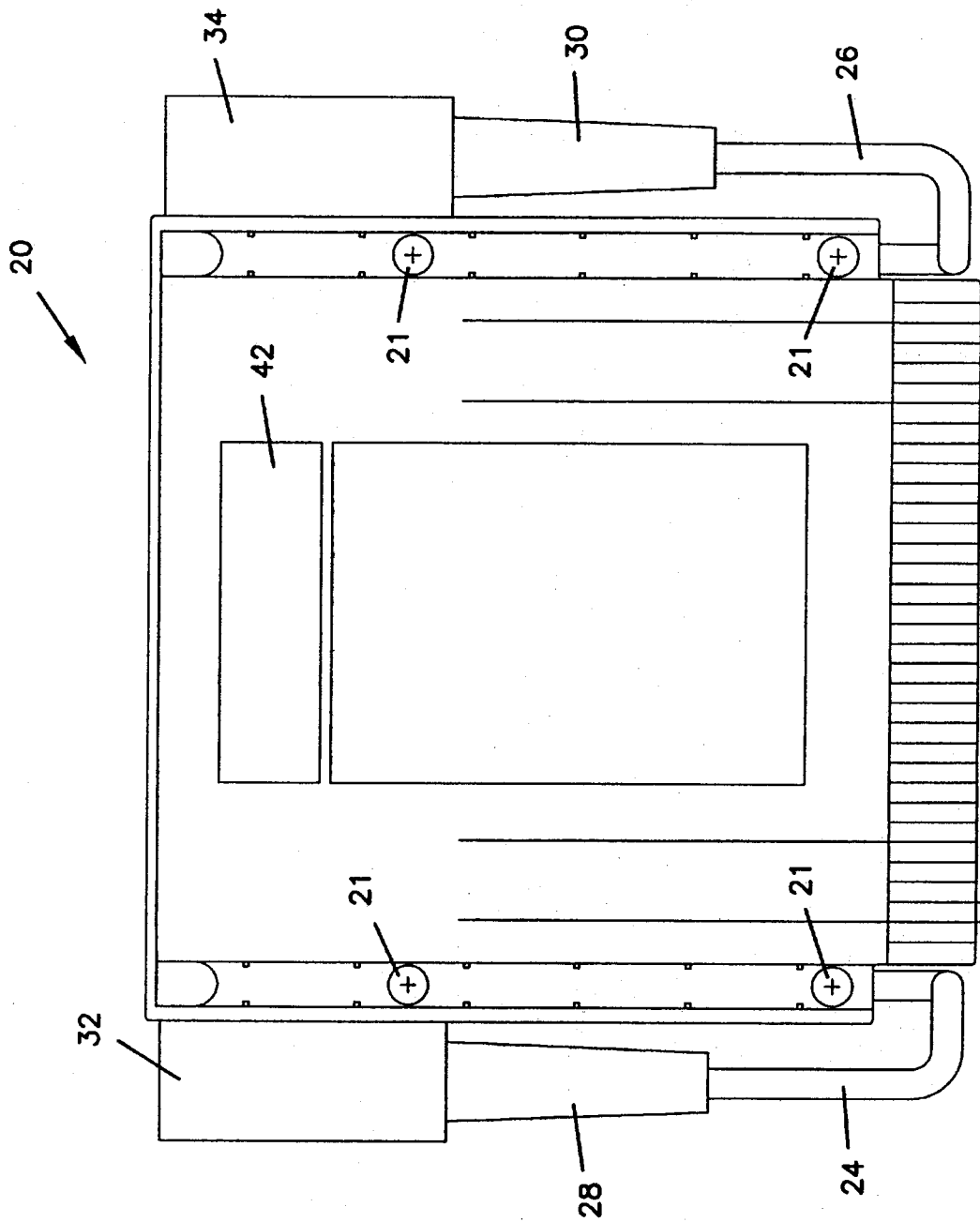


FIG. 3

FIG. 4

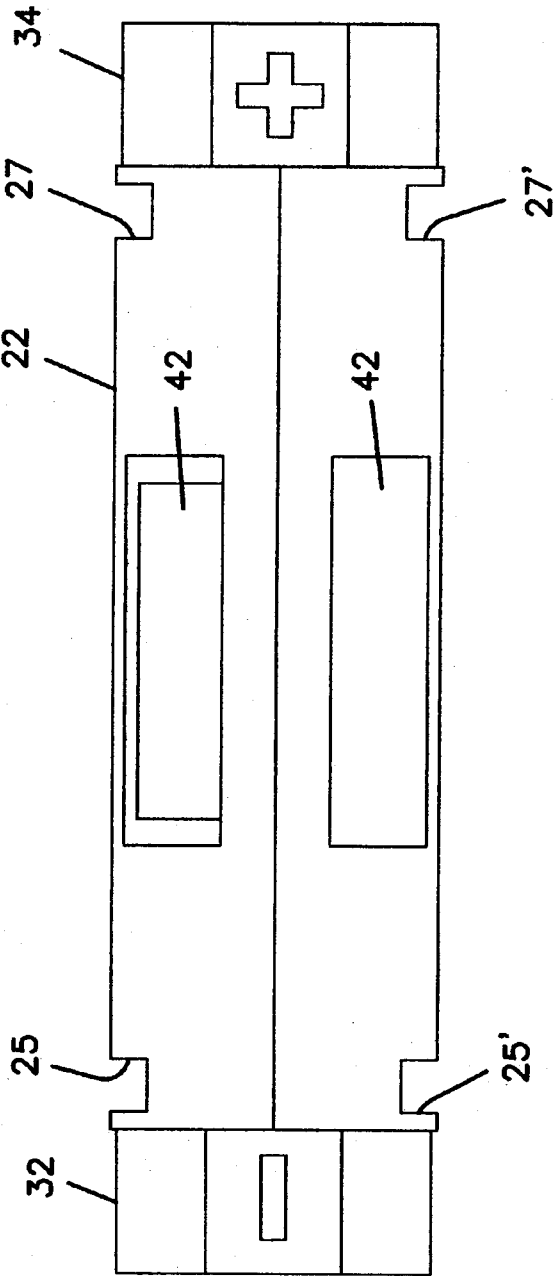
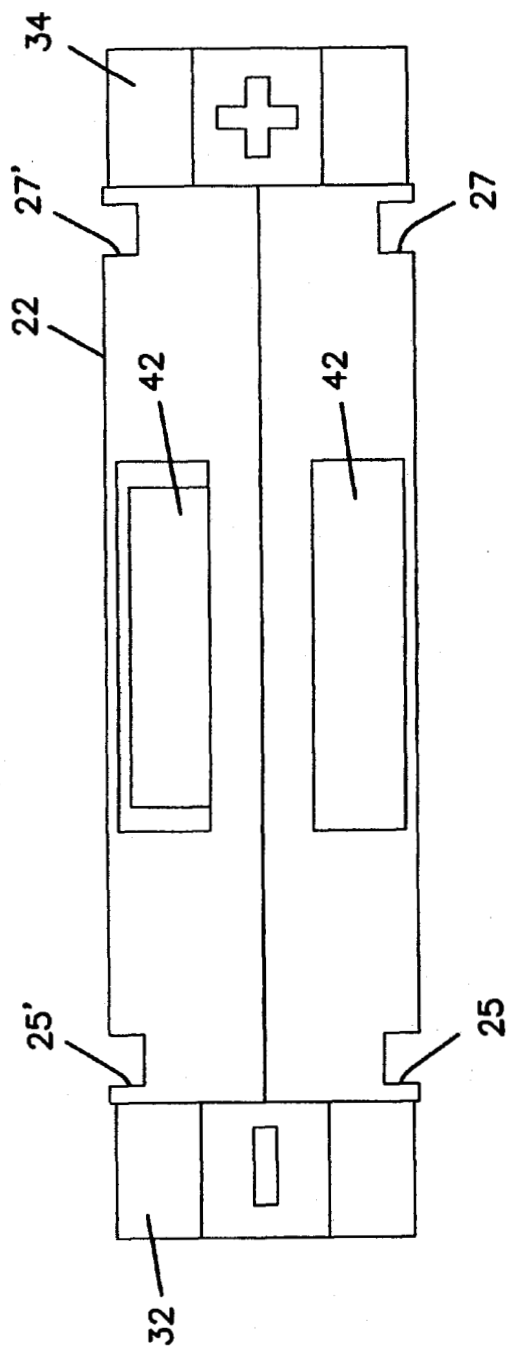


FIG. 5



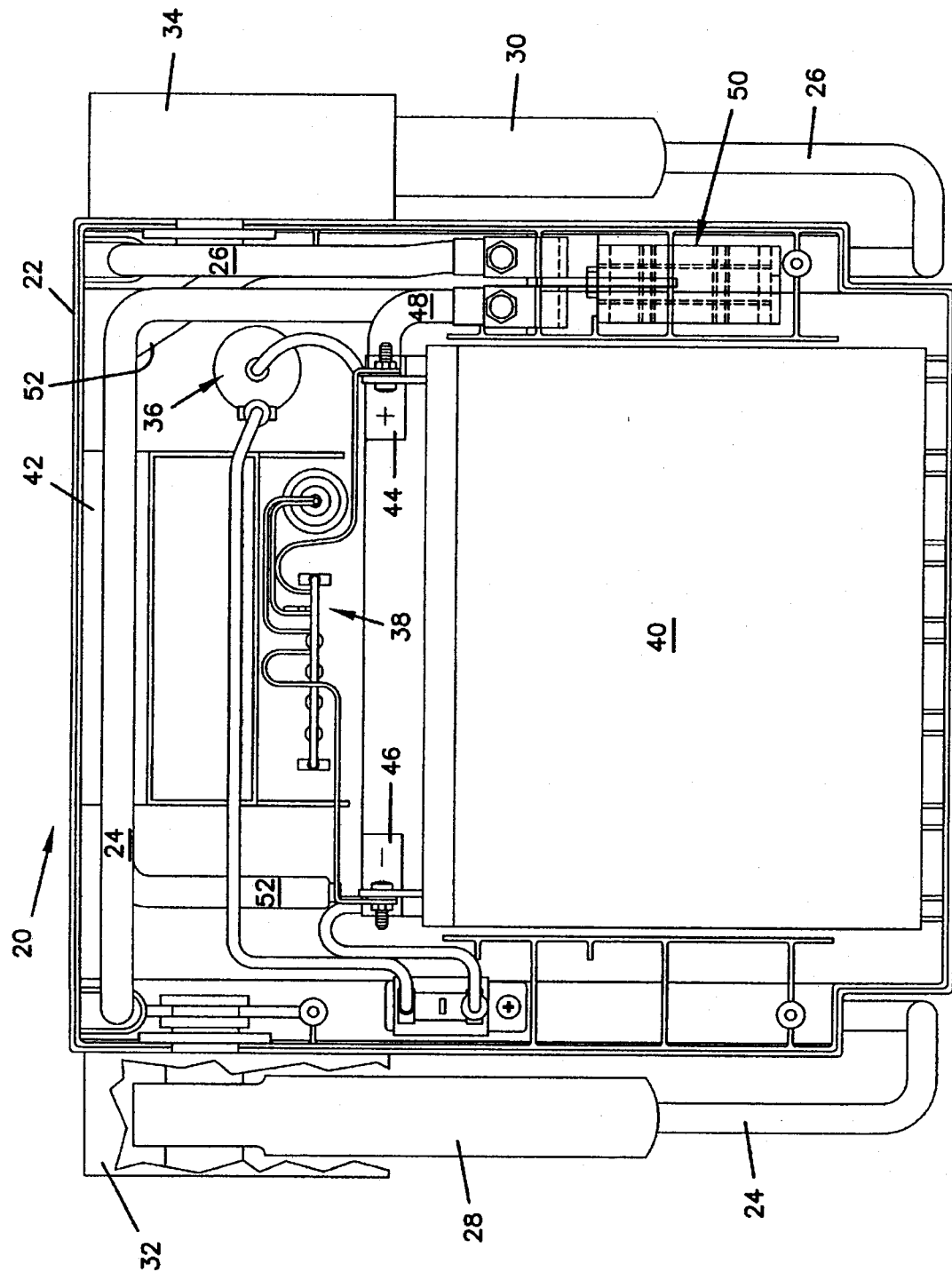


FIG. 6

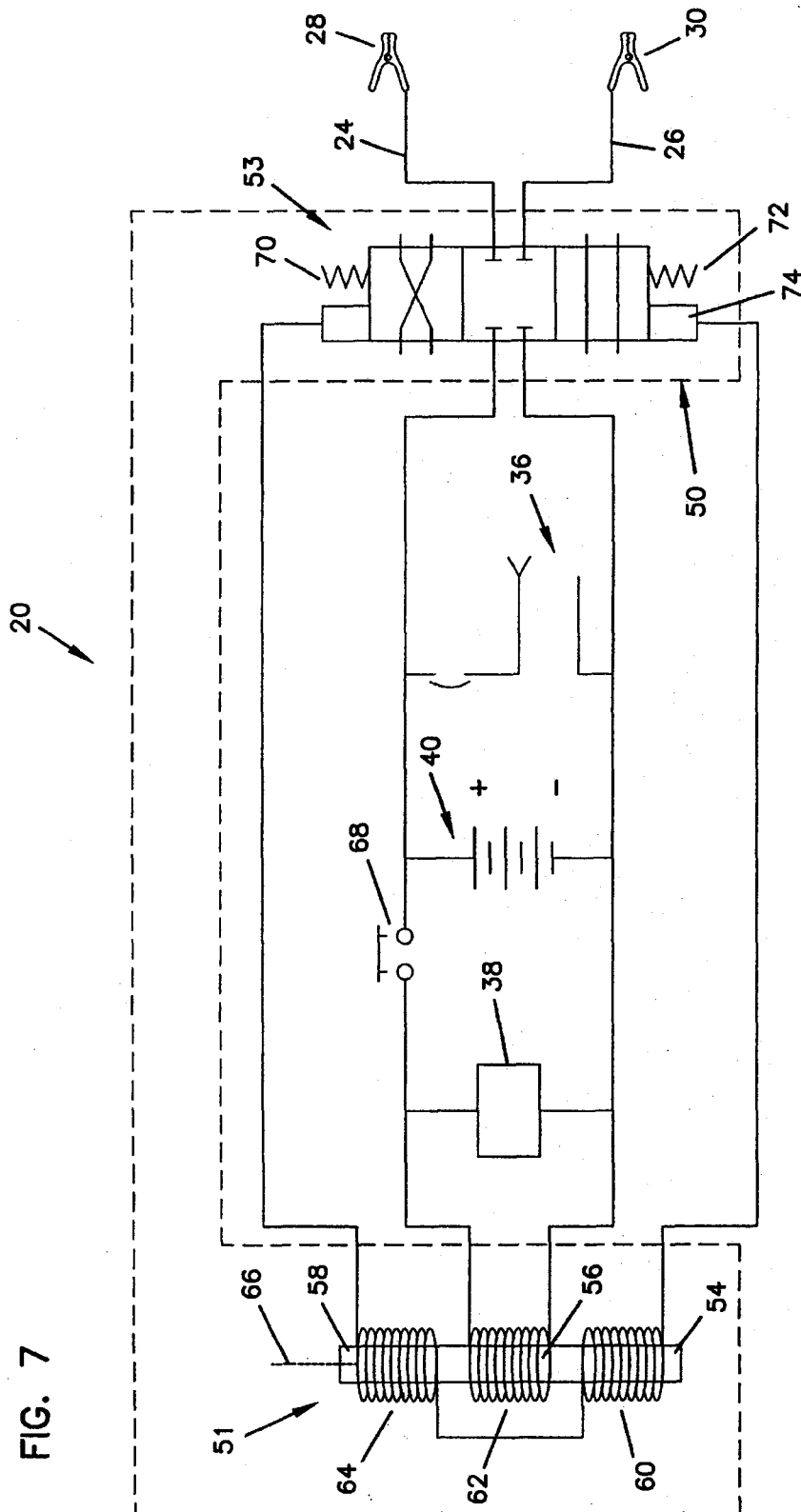
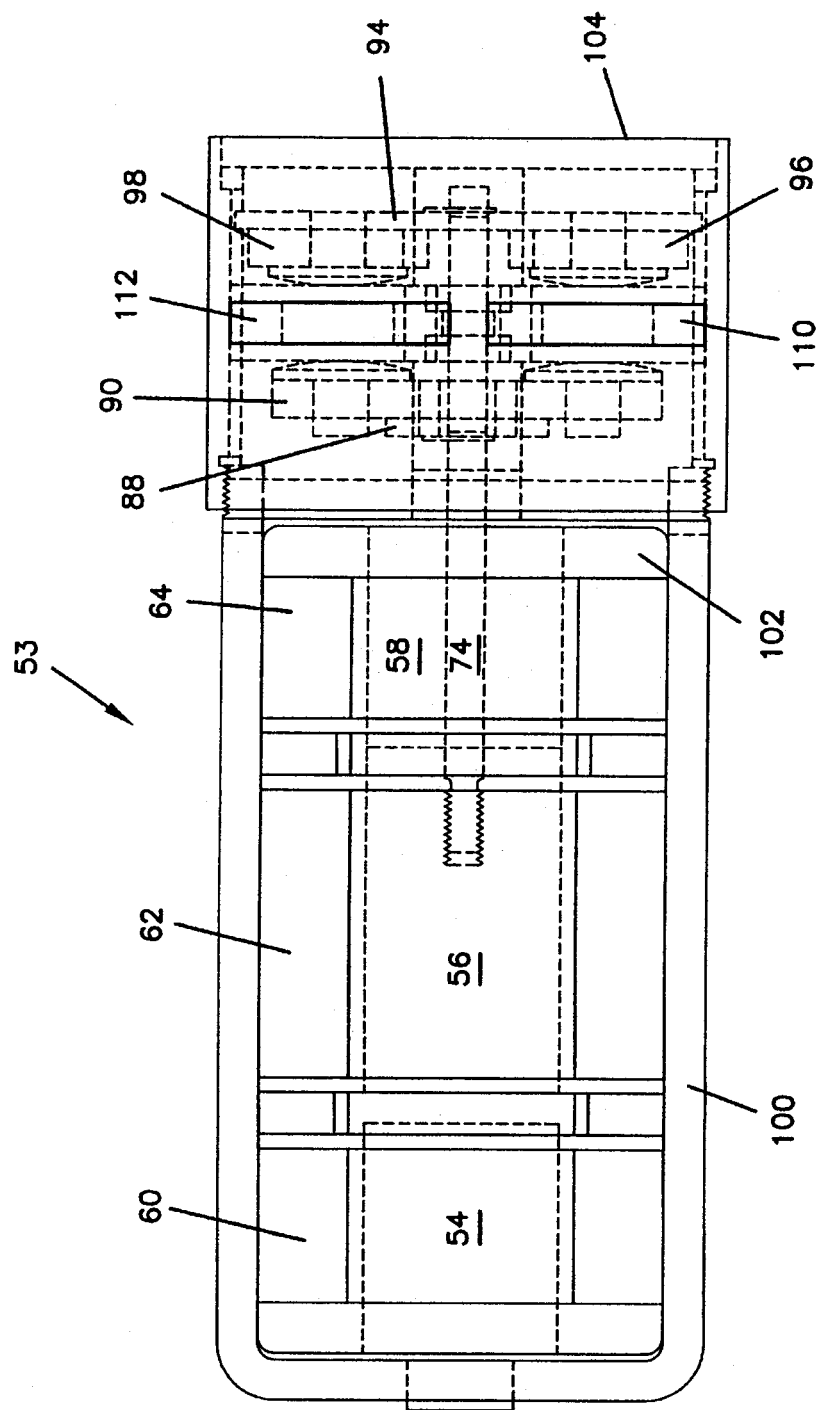


FIG. 8



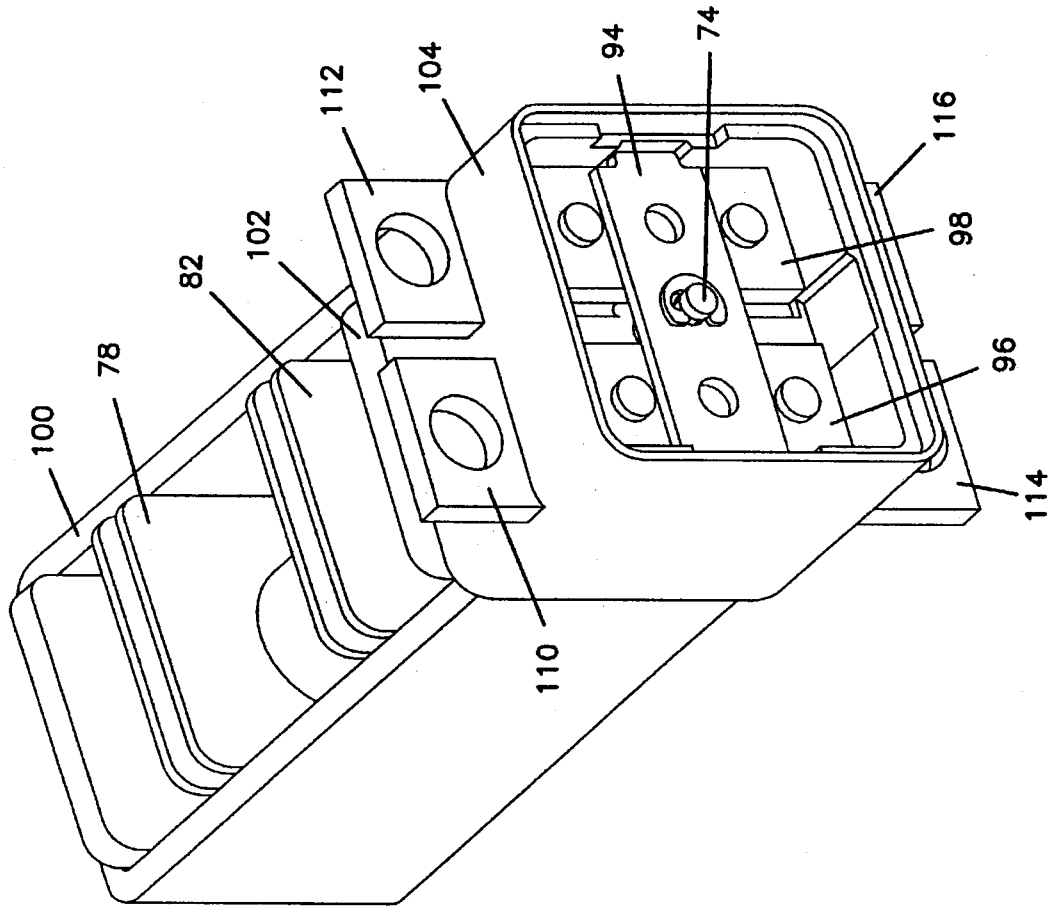


FIG. 9

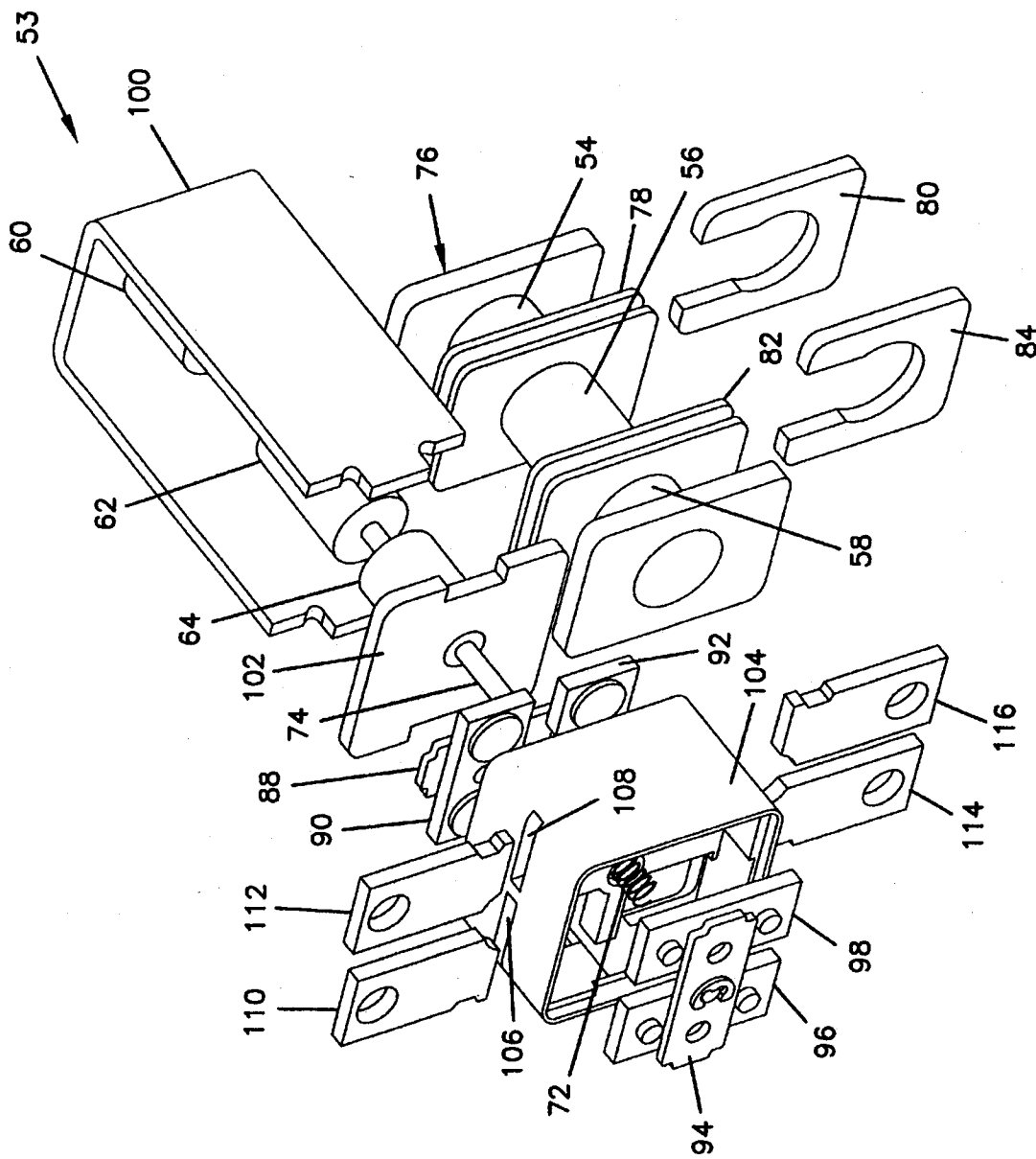


FIG. 10

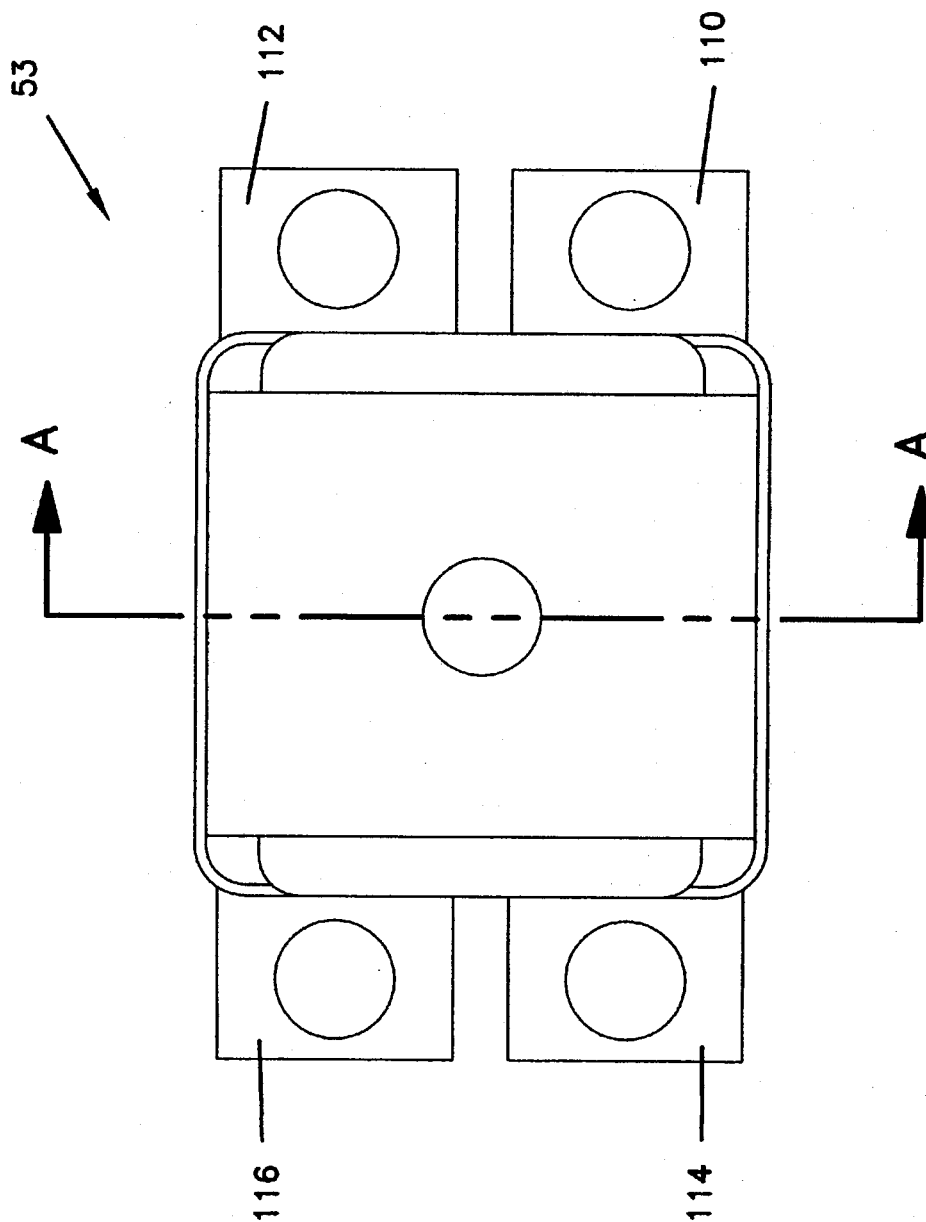
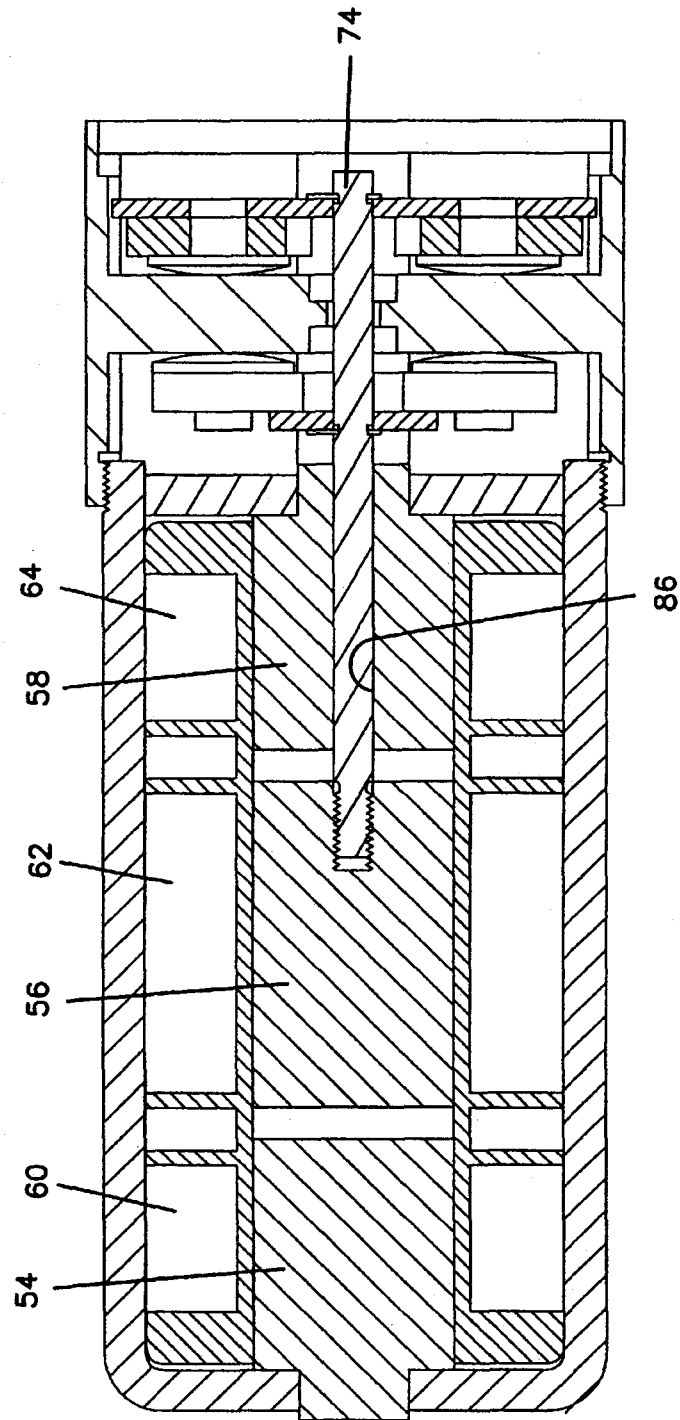


FIG. 11

FIG. 12



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PORTABLE BATTERY CHARGER INCLUDING AUTO-POLARITY SWITCH

This application claims the benefit of Provisional application Ser. No. 60/104,651 filed Oct. 16, 1998.

FIELD OF THE INVENTION

This invention relates generally to portable charging systems, and more specifically to portable charging systems including an auto-polarity switch for providing a proper polarity connection in charging batteries or jump starting an automobile which has a discharged battery.

BACKGROUND OF THE INVENTION

Most automobiles or other devices powered by an internal combustion engine, such as watercraft, etc., rely at least in part on a battery as part of its power source. For example, an automobile needs a battery to start the engine operating (the engine must be spun up to speed externally before they can operate). Once the engine is operating or running, the battery is charged by a generator connected to the engine. The battery is usually able to store enough charge for the periods that the engine is not running in order to start the engine when needed.

There are many things that will cause a battery to lose its charge, such as leaving the lights on or playing the radio after the engine is turned off, or a short circuit in the battery which drains the battery. When there is not enough charge in the battery, the engine cannot be started, i.e. the battery is discharged. As a result, the battery must be recharged or boosted by another power source. Very often another vehicle is brought to the vicinity of the vehicle with the discharged battery to jump start the vehicle. Another solution was proposed by U.S. Pat. No. 5,589,292 which provides a portable booster battery to recharge the discharged battery.

One concern in using a portable battery such as the portable booster battery disclosed U.S. Pat. No. 5,589,292 to charge batteries that have been discharged is making sure that proper polarity between the portable battery and the battery to be charged is proper. It is imperative to connect the portable battery and the battery to be charged with proper polarity, i.e. positive to positive terminal connections and negative to negative terminal connections. Failure to do so will rapidly drain the portable battery and further cause serious damage to either or both electrical systems that the two batteries are used in. However, it is often difficult to determine which battery terminal is positive and which is negative because of the accumulation of dirt or lack of lighting.

The polarity problem has been solved by a variety of methods in the battery jumper cable arrangement environment, such as the technique disclosed in U.S. Pat. No. 4,400,658. There is a need for a portable battery charging system that has the polarity problem solved. Such a portable charging system would need to provide a charge or boost to another battery regardless of the charging terminals of the battery to be charged to which the positive or negative terminals of the portable charging system are connected. Such a system would eliminate the problems and dangers associated with incorrect charging battery to discharged battery terminal connections.

SUMMARY OF THE INVENTION

The present invention relates to a portable battery with an auto-polarity switch for providing a proper polarity connection

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in charging a battery of a device, such as an automobile, i.e. jump starting an automobile, etc.

The present invention provides for a portable, auto-polarity battery booster apparatus. The portable apparatus in accordance with the present invention includes a power storage device, such as a battery, which is capable of providing power to an electrical system such as another battery. The portable apparatus can also be used in many other power supply applications or battery charging applications. For example, it can be used to jump start a vehicle with a dead battery. The apparatus can also be recharged by an outside power source. Further, the portable apparatus ensures a proper polarity connection between the power storage device of the apparatus and the electrical system.

BRIEF DESCRIPTION OF THE DRAWINGS

Referring to the drawings wherein like numerals represent like parts throughout the several views:

FIG. 1 is a schematic view of a general embodiment of the auto-polarity battery booster apparatus;

FIG. 2 is a front view of a portable, auto-polarity battery booster apparatus in accordance with the present invention;

FIG. 3 is a back view of the battery booster apparatus of FIG. 2;

FIG. 4 is a top view of the battery booster apparatus of FIG. 2;

FIG. 5 is a bottom view of the battery booster apparatus of FIG. 2;

FIG. 6 is a plane view of the inside of the battery booster apparatus of FIG. 2;

FIG. 7 is a schematic view of a specific embodiment of the auto-polarity battery booster apparatus;

FIG. 8 is a plane view of one embodiment of an auto-polarity switch of the battery booster apparatus;

FIG. 9 is a perspective view of the auto-polarity switch;

FIG. 10 is an exploded view of the auto-polarity switch;

FIG. 11 is a front view of the auto-polarity switch; and

FIG. 12 is a cross-sectional view of the auto-polarity switch along line A—A of FIG. 11.

GENERAL DESCRIPTION

In one embodiment constructed according to the principles of the present invention, a portable battery booster apparatus for boosting an electrical system with a first terminal and a second terminal, comprises a battery having a first terminal and a second terminal, a housing enclosing the battery and the first and second terminals of the battery, an automatic switch having a first pair of contacts and a second pair of contacts, the switch being enclosed in the housing, a first cable with a first end connected to the first terminal of the battery and a second end selectively connected to a first contact of the first pair of contacts of the switch, the first cable being enclosed in the housing, a second cable with a first end connected to the second terminal of the battery and a second end selectively connected to a second contact of the first pair of contacts of the switch, the second cable being enclosed in the housing, a third cable with a first end connected to a first contact of the second pair of contacts of the switch, the first end of the third cable being partially enclosed in the housing, a fourth cable with a first end connected to a second contact of the second pair of contacts of the switch, the first end of the fourth cable being partially enclosed in the housing, a first clamp connecting to a second end of the third cable, the first clamp

being detachably mounted onto a first side of outside of the housing, a second clamp connecting to a second end of the fourth cable, the second clamp being detachably mounted onto a second side of the outside of the housing, and wherein when the electrical system is boosted by the battery booster apparatus, the first clamp and the second clamp connect to the first terminal and the second terminal of the electrical system, respectively, such that the first and second terminals of the battery are electrically connected to the first and second terminals of the electrical system, the switch switches connections between the first pair of contacts and the second pair of the contacts to ensure that the first terminal of the battery and the first terminal of the electrical system have the same polarity, and the second terminal of the battery and the second terminal of the electrical system have the same polarity.

In one aspect of the invention, the apparatus may also include an ON/OFF switch. The ON/OFF switch is connected between the battery and the switch. When the ON/OFF switch is off, the switch is in a neutral, non-contacting position, and the first pair of contacts are not electrically connected with the second pair of contacts. When the ON/OFF switch is on, the electrical system is boosted by the battery.

In another aspect of the invention, in a first embodiment the auto-polarity switch comprises three coils which are wound around two cores and a plunger, respectively. The plunger is moved to a first contacting position when the first terminals of the battery and the electrical system have the same polarity and the second terminals of the battery and the electrical system have the same polarity. The plunger is moved to a second contacting position when the first terminals of the battery and the electrical system have the opposite polarity and the second terminals of the battery and the electrical system have the opposite polarity. In one embodiment, the switch is a double pole double throw switch.

In the first embodiment of the switch in accordance with the invention, the coil assembly has a bobbin shape with two dividers for separating the first, second, and third coils from each other. In addition, the switch may include an enclosure for enclosing the coil assembly. The plunger extends from the enclosure and connects to a double pole double throw switch.

In the first embodiment of the switch in accordance with the present invention, the first pair of contacts are arranged and configured to be a pair of bottom switch contacts and a pair of top switch contacts. When the plunger is in the first contacting position, the bottom switch contacts are connected to the second pair of contacts, and when the plunger is in the second contacting position, the top switch contacts are connected to the second pair of contacts.

In a second embodiment, the auto-polarity switch includes a polarity sensing circuit, and a reversing relay operated by the sensing circuit. The sensing circuit includes a full wave rectifier electrically connected to a differential amplifier circuit which is electrically connected to external cables having clamps electrically connected thereto, which are to be removably attached to the battery to be charged. The full wave rectifier provides a positive voltage to the polarity sensing circuit regardless of the polarity connection of the clamps to the source to be charged. The differential amplifier circuit is electrically connected to a first relay coil and a second relay coil which are electrically connected to the relay control circuit and cause selective activation of movable contacts in the relay to so that the reversing relay

acts as a switch in switching the connections between a first pair of contacts and a second pair of the contacts to ensure that a first terminal of the battery and a first terminal of the electrical system have the same polarity, and a second terminal of the battery and a second terminal of the electrical system have the same polarity.

In other embodiments, such as the embodiment shown in FIG. 1, the auto-polarity switch may be an electromechanical type, or an electronic type comprised of solid state devices or microprocessor type, all of which would include a type of change-over switch. In such other embodiments the auto-polarity switch would include a polarity detector comprised of a electric differential type, a processing circuit and two sets of pairs of electrical transmission conducting wires terminals. The switch would be installed between the portable charger battery 40 and the external charging source connection cables 24 and 26 to provide for switching between two pair of contacts connected to a pair of electrical transmission conducting wires terminals 28 and 30.

It will be appreciated that other types of suitable auto-polarity switches can be used without departing from the present invention. The examples of the other types of switches, but not limited to these examples, are solid-state switches or other electro-mechanical switching devices. It will be also appreciated that other conventional polarity detection/protection devices, or automatic polarity sensing battery charger devices can be used without departing from the present invention.

In another aspect of the invention, the housing includes switch retention grooves for retaining the switch in one side of the housing. The housing also includes cable A retention grooves for retaining the first, second, third, and fourth cables to the housing. Further, the housing includes battery retention grooves for retaining the battery to the housing.

In yet another aspect of the invention, the apparatus includes a first holster for receiving and covering the first clamp and a second holster for receiving and covering the second clamp. In one embodiment, the first and second holsters are attached to the outside of the housing. Further, the first and second holsters are pivotably connected to the housing.

In an additional aspect of the present invention, a plurality of accessories can be included to provide additional features and advantages of the invention. For example, a display can be included to display a level of charge remaining in the battery. A cigarette lighter female receptacle member can also be added to provide power to an electrical system which is equipped with a male cigarette lighter plug. Further, a male cigarette lighter plug member can be added to the present invention to provide it with the ability to connect the battery in the housing to a power source outside the housing to recharge the battery. Furthermore, a handle can be included to carry the battery booster apparatus such that the housing and the handle form a portable box unit.

While the invention will be described with respect to a preferred embodiment configuration and with respect to particular components, it will be understood that the invention is not to be construed as limited by such configurations or components. Further, while the preferred embodiment of the invention will be described in relation to boosting or recharging a vehicle or automobile battery, it will be understood that the scope of the invention is not to be limited by this environment in which the preferred embodiment is described herein.

These and various other advantages and features which characterize the invention are pointed out with particularity

in the claims annexed hereto and forming a part hereof. However, for a better understanding of the invention, its advantages and objectives obtained by its use, reference should be had to the drawings which form a further part hereof and to the accompanying descriptive matter, in which there is illustrated and described a preferred embodiment to the invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

The present invention provides for a portable, auto-polarity battery booster apparatus. The portable apparatus in accordance with the present invention includes a power storage device which is capable of providing power to an electrical system such as a battery. The portable apparatus can also be used in many other power supply applications or battery charging applications. For example, it can be used to jump start a vehicle with a dead battery. The apparatus can also be recharged by an outside power source. Further, the portable apparatus ensures a proper polarity connection between the power storage device of the apparatus and the electrical system.

In the following description of the exemplary embodiment, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration the specific embodiment in which the invention may be practiced. It is to be understood that other embodiments may be utilized when structural and other changes may be made without departing from the scope of the present invention.

Turning now to FIGS. 1-4, there is illustrated front, back, top, and bottom views of a preferred battery booster apparatus designated by the reference numeral 20. The apparatus 20 includes a housing 22 preferably made out of plastic. Electrical cables 24, 26 extend out of the housing 22 and are retained along outside grooves 25, 27 of the housing 22. Extra outside grooves 25', 27' can be arranged to wrap long cables and/or for aesthetic purposes. One end of each cable 24, 26 is connected to a clamp 28, 30 (best seen in FIG. 6), respectively. Each of the cables 24, 26 is connected to either the positive terminal or the negative terminal of a power storage device, such as a battery 40 (best seen in FIG. 6), in the housing 22. Accordingly, one of the clamps can be called positive clamp (marked as "+" in FIGS. 3 and 4), and the other one can be called negative clamp (marked as "-" in FIGS. 3 and 4). As it will be discussed later, since the apparatus 20 has an auto-polarity switch 50 (best seen in FIG. 7), it does not matter which clamp is positive and which is negative when the clamps are in use.

The clamps 28, 30 are preferably received and covered by holsters 32, 34, respectively, and are removably clamped to a bit within the holster. The holsters 32, 34 help avoiding accidental electrical contacts between the clamps and adjacent items. A cigarette lighter receptacle member 36 can also be included in the apparatus 20 to provide power to electrical systems such as cell-phones, lighting, etc. A charge status display 38 can also be included in the apparatus 20 to display the level of charge remaining in the power storage device, such as the battery 40 (best seen in FIG. 5), in the housing 22. The display 38 can be arranged and configured preferably in a plurality of LED lights to show a series of charge status. It will be appreciated that other suitable display means can be used to indicate the charge status within the scope of the invention. Further, a handle 42 can be molded into the housing 22 to allow easy carrying of the apparatus 20 to the vicinity of the dead battery. It will be appreciated

that other types of handles can be used within the principles of the invention. For example, an extra handle piece can be attached to the housing, etc.

FIG. 6 shows the internal details of the housing 22 (opened from the back side by removing the mounting screws 21). The battery 40 has a positive terminal 44 and a negative terminal 46. An electrical cable 48 is connected to the positive terminal 44 at one end of the cable 48. The other end of the cable 48 is connected to the auto-polarity switch 50. An electrical cable 52 is connected to the negative terminal 46 at one end of the cable 52. The other end of the cable 52 is connected to the switch 50. Also, it can be seen that the cable 24 extends from outside of the housing 22 to inside of the housing and is retained in the grooves of the housing. The other end of the cable 24 connects to the switch 50. Similarly, the cable 26 extends from outside of the housing 22 to inside of the housing and is retained in the grooves of the housing. The other end of the cable 26 connects to the switch 50. The details of the cigarette lighter receptacle member 36 and the display 38 are preferably shown in FIG. 5. It will be appreciated that other suitable arrangements of the cigarette lighter receptacle member 36 and the display 38 can be used without departing from the principles of the invention.

FIG. 7 illustrates a schematic, functional view of the battery booster apparatus 20. The apparatus 20 has the auto-polarity switch 50 interfacing with the battery 40. The switch 50 preferably includes a coil assembly 51 and a switching assembly 53. The coil assembly 51 has three coils 60, 62, 64 wound around three core members 54, 56, 58, wherein the central core member 56 is a plunger, respectively. The three coils 60, 62, 64, the outside cores 54, 58 and the central core (plunger) 56 are disposed along a longitudinal axis 66. The plunger 56 is placed between the outside cores 54, 58. The coil windings and three core assembly makes up two solenoids, a center solenoid comprised of core member 56 and windings 62 wound around the central core member 56 and an outside solenoid 65, 67, comprised of outside core members 54 and 58 and windings 60 and 64 wound around the outside core members 54, 56. The outside coil windings 60 and 64 of the outside solenoid are each wound with an equivalent number of turns. The center coil windings 62 of the central solenoid are wound with two times as many turns as either of outside coil windings 60, 64. Further, the windings of the first outside coil member 60 are opposite to the windings of the second outside coil member 64. The windings of the central coil 62 can be in the same direction as the windings of the first outside coil member 60 or in the same as the winding of the second outside coil member 64.

Accordingly, when the internal battery energizes the center coil windings 62, an electrical current flows through the center coil windings 62 (i.e. an ON/OFF switch 68 is closed) and the plunger 56 which is preferably comprised of a ferrous metal becomes magnetized. When there is no current flowing through the external coil windings 60 and 64 and the battery has energized the center coil windings 62, plunger 56 and switch contacts are held at a neutral position. This is because the center coil windings 62 is centered over the plunger 56 and there is no magnetic force being applied to the plunger 56 by either member of the outside solenoid. Outside coil windings 60 and 64 are connected in series and wound in opposite directions. The ends of outside coil windings 60 and 64 that are not connected in series are electrically connected to cables 24 and 26 and clamps 28 and 30. When the external clamps 28 and 30 are connected to the power source to be charged, current flows through the

outside coil windings 60 and 64, causing outside core members 54 and 58 to be magnetized. The magnetization of outside core members 54 and 58 creates a force pulling or pushing central core member 56, depending on the polarity connection of clamps 28 and 30 to the power source to be charged, and the direction of the turns in coil windings 60, 62 and 64. When the outside coil windings 60 and 62 and the central coil winding 64 are all energized, current flowing through the outside coil windings 64 can either flow in the same direction as the current through the central coil windings 62 or in the opposite direction and current flowing through the outside coil windings 60 can either flow in the same direction as the current through central coil windings 62 or in the opposite direction. When current is flowing through the central coil windings 62 in a clockwise direction and through the outside coil windings 64 in a clockwise direction, it is flowing through outside coil windings 60 in a counter clockwise direction. Or, when current is flowing through central coil windings 62 in a clockwise direction and through the outside coil windings 64 in a counter clockwise direction, it is flowing through the outside coil windings 60 in a clockwise direction. The direction of current flowing through outside coil windings 60 and 64 depends on the polarity connection of clamps 28 and 30 to the power source to be charged.

Assuming that current is flowing through central coil windings 62 in a clockwise direction and through the outside coil windings 64 in a clockwise direction, it is flowing through outside coil windings 60 in a counter clockwise direction. Accordingly, the magnetic field in core 54 is opposite the magnetic field within central core 56. The opposing magnetic forces causes the magnetic field in core 54 to negate half the magnetic force in the central core 56. The magnetic field generated by core 58 is in the same direction as that within central core 56 and adds its magnetic energy to core 54, causing full coil strength to be centered in the direction of core 58. This causes the central core 56 to be pulled in the direction of outside core 58.

When the central core 56 is pulled in the direction of outside core 58, the two cores 56 and 58 are very close but do not touch. This is because if either the outside coils 60 and 64 or central coil 62 are no longer energized, by way of removing clamps 28 and 30 from the power source to be charged or opening switch 68, the air space provides the central core 56 with the positioning necessary to return back to its neutral position. The same would also be true when the central core 56 is pulled in the direction of core 54. The two cores 54 and 56 are very close, but do not touch. The air space between the two cores 54 and 56 provides the central core 56 with the positioning necessary to return back to its neutral position.

A switch rod 74 is mounted on one end of the plunger 56 (best seen in FIG. 10). The plunger 56 is held between the two cores 60,64 by two springs 70,72. The springs 70,72 are shown schematically in FIG. 7. In one preferred embodiment, the springs 70,72 are attached to the switch rod 74, and the switch rod 74 is disposed proximate the center of the switch assembly 53. In this case, since the pulling forces are equal, the springs keep the plunger centered, i.e. the switch 50 is in a neutral, non-contacting position. To preserve the charges stored in the battery 40, the ON/OFF switch 68 is normally open.

FIGS. 8-12 illustrate details of the switch assembly 53. The coils 60, 62, 64 are wound around outside core members 54 and 58 and central core member 56. The coils 60 and 62 are separated by a divider 78. An insulating member 80 is mounted on the divider 78 to help insulate the magnetic

fields generated by each of the coils 60, 62. Similarly, the coils 62 and 64 are separated by a divider 82. An insulating member 84 is mounted on the divider 82 to help insulate the magnetic fields generated by each of the coils 62, 64.

As shown in FIGS. 8 and 12, the switch rod 74 is mounted in the plunger 56 at one end, extends through a center bore 86 of the core 58, is mounted on a bottom switch plate 88 which links a pair of bottom switch contacts 90,92, is attached to the two springs 70,72 (only spring 72 is shown in FIG. 7), and is mounted on a top switch plate 94 which links a pair of top switch contacts 96,98. The coils, the plunger, and the cores are preferably disposed in an enclosure 100 with a cover 102. The bottom switch plate 88, the bottom switch contacts 90,92, the springs 70,72, the top switch plate 94, and the top switch contacts 96,98 are retained in an enclosure 104. The enclosure 104 has a pair of top slots 106,108 and a pair of bottom slots (not shown). A pair of top bus bars or terminals 110,112 are inserted in the top slots 106,108 and retained by the enclosure 104. A pair of bottom bus bars or terminals 114,116 are inserted in the bottom slots and retained by the enclosure 104.

In one embodiment, the terminal 110 is electrically connected to the positive terminal of the internal power storage device or the battery 40 via the cable 48 as shown in FIG. 6. The terminal 116 is electrically connected to the negative terminal of the internal power storage device or the battery 40 via the cable 52 as shown in FIG. 6. The terminals 112 and 114 are electrically connected to the terminals of the external power source, e.g. a dead battery, via the cable 24 and the cable 26 and via the clamps 28,30, respectively. It is generally required that a proper polarity be established between the external power source and the internal power source, i.e. the positive terminals are electrically connected to each other, and the negative terminals are electrically connected to each other. In the present invention, the switch 50 connects the terminals with proper polarity in an automatic fashion. First, it is assumed that the positive terminal of the external power source is connected to the bus bar or terminal 112 and the negative terminal of the external power source is connected to the bus bar or terminal 114. When the ON/OFF switch 68 is closed, the plunger 62 is attracted toward the core 64 and repelled from the core 60. This action translates the movement of the switch rod 74 toward the enclosure 104. Accordingly, the spring 70 is compressed (not shown), and the spring 72 is expanded. Thus, the bottom contact 90 is connected between the top bus bar 110 and the top bus bar 112, i.e. the positive terminals of the internal and external power source devices are electrically connected. Meanwhile, the bottom contact 92 is connected between the bottom bus bar 114 and the bottom bus bar 116, i.e. the negative terminals of the internal and external power source devices are electrically connected.

Similarly, if the negative terminal of the external power source is connected to the bus bar or terminal 112 and the positive terminal of the external power source is connected to the bus bar or terminal 114, when the ON/OFF switch 68 is closed, the plunger 62 is attracted toward the core 60 and repelled from the core 64. This action translates the movement of the switch rod 74 toward the enclosure 100. Accordingly, the spring 70 is expanded (not shown), and the spring 72 is compressed. Thus, the top contact 96 is connected between the top bus bar 110 and the bottom bus bar 114, i.e. still the positive terminals of the internal and external power source devices are electrically connected. Meanwhile, the top contact 98 is connected between the top bus bar 112 and the bottom bus bar 116, i.e. still the negative terminals of the internal and external power source devices are electrically connected.

It will be appreciated that the arrangement of the bottom switch contacts 90,92, the top switch contacts 96,98, the top bus bars 110,112, and the bottom bus bars 114,116 can be varied within the scope and spirit of the invention. It will also be appreciated that connection arrangement of the positive and negative terminals of the internal power storage source or battery 40 can be varied within the scope and spirit of the invention. For example, the positive terminal of the internal battery 40 can be connected to the top bus bar 112, and the negative terminal of the internal battery 40 can be connected to the bottom bar 114, etc. It will be appreciated by a person skilled in the art that the connection arrangement can be varied as long as the positive and negative terminals of the internal power source are disposed in a diagonal position.

Therefore, even if a user applies the clamps to the wrong terminal connections, i.e. negative to positive or positive to negative between the external and the internal power sources, the switch 50 automatically switches to a proper polarity.

Further, as shown in FIGS. 8-12, the switch 50 is arranged and configured in such a manner that the switch 50 is able to fit in the housing 22. The housing 22 provides inside grooves to retain the switch 50 as well as many other components, such as cables, cigarette lighter wires, and the display wires, etc.

It will be appreciated that other types of suitable switches can be used without departing from the present invention. The examples of the other types of switches, but not limited to these examples, are solid-state switches, other electro-mechanical switching devices. It will be also appreciated that other conventional polarity detection/protection devices, or automatic polarity sensing battery charger devices can be used without departing from the present invention.

While a particular embodiment of the invention has been described with respect to its application for providing power to a battery, such as jump starting a vehicle, etc., it will be understood by those of skill in the art that the invention is not limited by such application or embodiment for the particular components disclosed and described herein. It will be appreciated by those skilled in the art that other circuit configurations that embody the principles of this invention and other applications therefor can be configured within the spirit and intent of this invention. The circuit configuration described herein is provided as only one example of an embodiment that incorporates and practices the principles of this invention. Other modifications and alterations are well within the knowledge of those skilled in the art and are to be included within the broad scope of the appended claims.

We claim:

1. A portable battery booster apparatus for boosting an electrical system with a first terminal and a second terminal, comprising:

- a) a battery having a first terminal and a second terminal;
- b) a housing enclosing the battery and the first and second terminals of the battery;
- c) a switch having a first pair of contacts and a second pair of contacts, the switch being enclosed in the housing;
- d) a first cable with a first end connected to the first terminal of the battery and a second end selectively connected to a first contact of the first pair of contacts of the switch, the first cable being enclosed in the housing;
- e) a second cable with a first end connected to the second terminal of the battery and a second end selectively

connected to a second contact of the first pair of contacts of the switch, the second cable being enclosed in the housing;

- f) a third cable with a first end connected to a first contact of the second pair of contacts of the switch, the first end of the third cable being partially enclosed in the housing;
- g) a fourth cable with a first end connected to a second contact of the second pair of contacts of the switch, the first end of the fourth cable being partially enclosed in the housing;
- h) a first clamp connecting to a second end of the third cable, the first clamp being detachably mounted onto a first side of outside of the housing;
- i) a second clamp connecting to a second end of the fourth cable, the second clamp being detachably mounted onto a second side of the outside of the housing; and
- j) wherein when the electrical system is boosted by the battery booster apparatus, the first clamp and the second clamp connect to the first terminal and the second terminal of the electrical system, respectively, such that the first and second terminals of the battery are electrically connected to the first and second terminals of the electrical system, the switch switches connections between the first pair of contacts and the second pair of the contacts to ensure that the first terminal of the battery and the first terminal of the electrical system have the same polarity, and the second terminal of the battery and the second terminal of the electrical system have the same polarity.

2. The apparatus of claim 1, further comprising an ON/OFF switch connected between the battery and the switch, wherein when the ON/OFF switch is off, the switch is in a neutral, non-contacting position, the first pair of contacts are not electrically communicated with the second pair of contacts, when the ON/OFF switch is on, the electrical system is boosted by the battery.

3. The apparatus of claim 1, wherein the electrical system is another battery.

4. The apparatus of claim 1, wherein the switch comprises:

- a coil assembly having a longitudinal axis, the coil assembly including:
 - a first coil, disposed around outside of the coil assembly, winding around a first core which is disposed inside of the coil assembly;
 - a second coil, disposed around outside of the coil assembly, winding around a second core which is disposed inside the coil assembly; and
 - a third coil, disposed around outside of the coil assembly and between the first coil and the second coil, winding around a plunger, the plunger being movable along the longitudinal axis between the first and second cores;

wherein the plunger is moved to a first contacting position when the first terminals of the battery and the electrical system have the same polarity and the second terminals of the battery and the electrical system have the same polarity, the plunger is moved to a second contacting position when the first terminals of the battery and the electrical system have the opposite polarity and the second terminals of the battery and the electrical system have the opposite polarity.

5. The apparatus of claim 4, wherein the switch is arranged and configured by a double pole double throw switch.

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6. The apparatus of claim 4, further comprising an ON/OFF switch connected between the battery and the switch, wherein when the ON/OFF switch is off, the switch is in a neutral, non-contacting position, the first pair of contacts are not electrically communicated with the second pair of contacts, when the ON/OFF switch is on, the electrical system is boosted by the battery, the ON/OFF switch is operated outside of the housing.

7. The apparatus of claim 6, wherein the switch is arranged and configured by a double pole double throw switch.

8. The apparatus of claim 7, wherein the first pair of contacts are arranged and configured to be a pair of bottom switch contacts and a pair of top switch contacts, wherein when the plunger is in the first contacting position, the bottom switch contacts are connected to the second pair of contacts, and when the plunger is in the second contacting position, the top switch contacts are connected to the second pair of contacts.

9. The apparatus of claim 4, wherein the coil assembly has a bobbin shape with two dividers for separating the first, second, and third coils from each other.

10. The apparatus of claim 9, further comprising an enclosure for enclosing the coil assembly, wherein the plunger extends from the enclosure and connects to a double pole double throw switch.

11. The apparatus of claim 1, further comprising switch retention grooves for retaining the switch in one side of the housing.

12. The apparatus of claim 11, further comprising cable retention grooves for retaining the first, second, third, and fourth cables to the housing.

13. The apparatus of claim 12, further comprising battery retention grooves for retaining the battery to the housing.

14. The apparatus of claim 1, further comprising a first holster for receiving and covering the first clamp and a second holster for receiving and covering the second clamp, wherein the first and second holsters are attached to the outside of the housing.

15. The apparatus of claim 14, wherein the first and second holsters are pivotably connected to the housing.

16. The apparatus of claim 1, further comprising a display for displaying a level of charge remaining in the battery.

17. The apparatus of claim 1, further comprising a cigarette lighter receptacle member disposed on the housing and electrically connected to the battery, the receptacle member providing power to an electrical system which equips with a cigarette lighter plug.

18. The apparatus of claim 1, further comprising a cigarette lighter plug member extending from the housing and electrically connected to the battery, the plug member being connectable to a power source outside the housing to recharge the battery inside the housing.

19. The apparatus of claim 1, further comprising a handle for carrying the battery booster apparatus, wherein the housing and the handle form a portable box unit.

20. A portable, auto-polarity battery booster apparatus for boosting an electrical system with a positive terminal and a negative terminal, comprising:

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a) a power storage device having a positive terminal and a negative terminal;

b) a housing enclosing the power storage device and the positive and negative terminals of the power storage device;

c) an auto-polarity switch enclosed in the housing for connecting between the power storage device and the electrical system;

d) a first cable with one end connected to the positive terminal of the power storage device and another end selectively connected to the switch, the first cable being enclosed in the housing;

e) a second cable with one end connected to the negative terminal of the power storage device and another end selectively connected to the switch, the second cable being enclosed in the housing;

f) a third cable with one end connected to the switch;

g) a fourth cable with one end connected to the switch;

h) a first clamp connecting to another end of the third cable and disposed outside of the housing;

i) a second clamp connecting to another end of the fourth cable and disposed outside of the housing; and

j) wherein when the positive and negative terminals of the electrical system are electrically connected to the first and second clamps, respectively, the switch switches to a first contacting position to ensure the positive terminal of the battery is electrically connected to the positive terminal of the electrical system, and the negative terminal of the battery is electrically connected to the negative terminal of the electrical system.

21. A portable, auto-polarity battery booster apparatus for boosting an electrical system with a positive terminal and a negative terminal, comprising:

a) a housing;

b) a battery, the battery having charges to provide power to the electrical system, the battery having a positive terminal and a negative terminal, the battery being enclosed in the housing;

c) a plurality of connecting members, the connecting members electrically connecting between the terminals of the battery and the terminals of the electrical system, the plurality of connecting members being enclosed in the housing, two of the plurality of connecting members being partially extended from the housing; and

d) a switch, the switch being arranged and configured to electrically connect between the battery and the electrical system, the switch being enclosed in the housing, the switch automatically switching between a first connecting position and a second connecting position to ensure that the positive terminal of the electrical system is electrically connected to the positive terminal of the battery and that the negative terminal of the electrical system is electrically connected to the negative terminal of the battery.

* * * * *



US006002235A

United States Patent [19]
Clore

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[45] **Date of Patent:** Dec. 14, 1999

[54] **BATTERY JUMP STARTER WITH JAW
SECURING MEANS**

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[73] **Assignee:** Bonnet Enterprises LLC, Kansas City,
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[22] **Filed:** Feb. 17, 1999

[51] **Int. Cl.⁶** H01M 2/02; H01R 3/00;
H02J 7/00

[52] **U.S. Cl.** 320/105; 429/170; 439/500

[58] **Field of Search** 320/105; 429/170;
439/500

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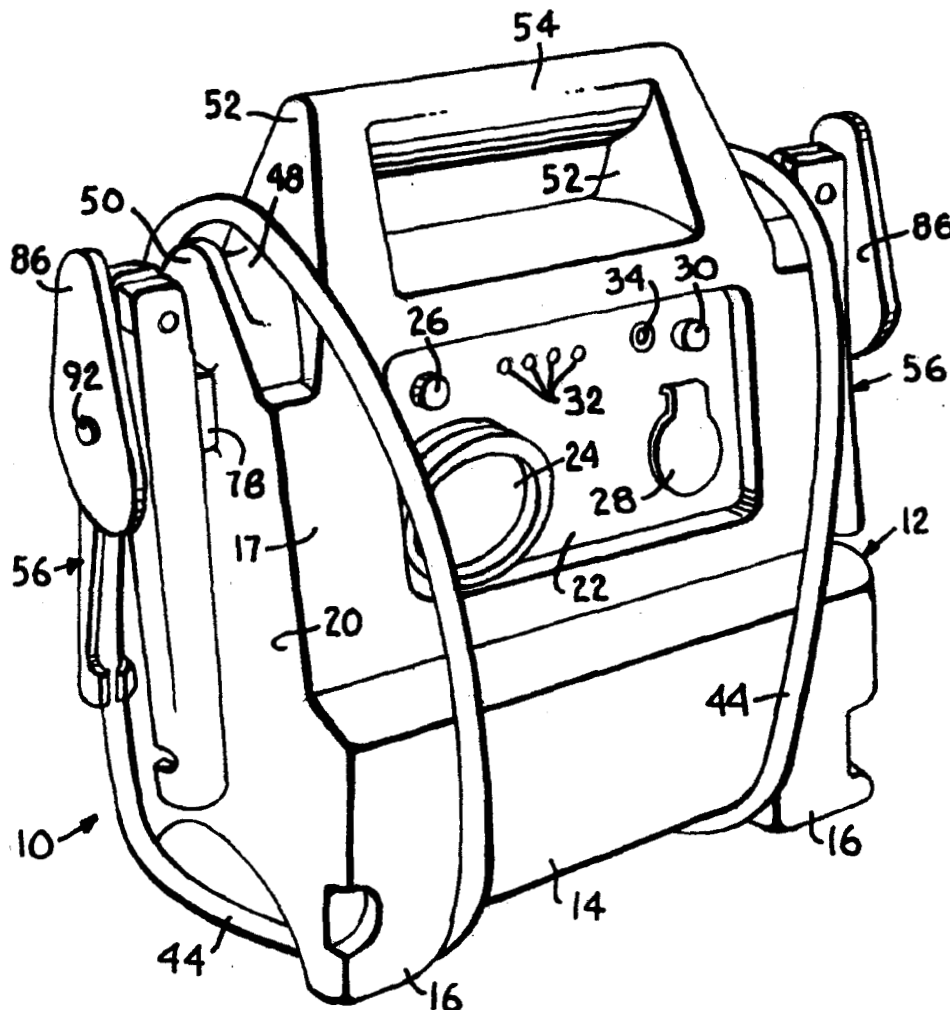
Primary Examiner—Gregory J. Toatley, Jr.

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[57] **ABSTRACT**

A portable battery jump start unit of the type having a portable housing containing a battery to which cables carrying battery clamps are permanently connected. The clamps are stored on rectangular posts on the sides of the housing, with flat areas on the jaw ends of the clamps held against the flat sides of the posts to prevent rotation. An end plate on each post prevents the clamps from slipping off. A storage compartment for a battery charger has slots in its bottom panel to receive electrical prongs of the charger body to hold the charger body in place when stored.

20 Claims, 2 Drawing Sheets



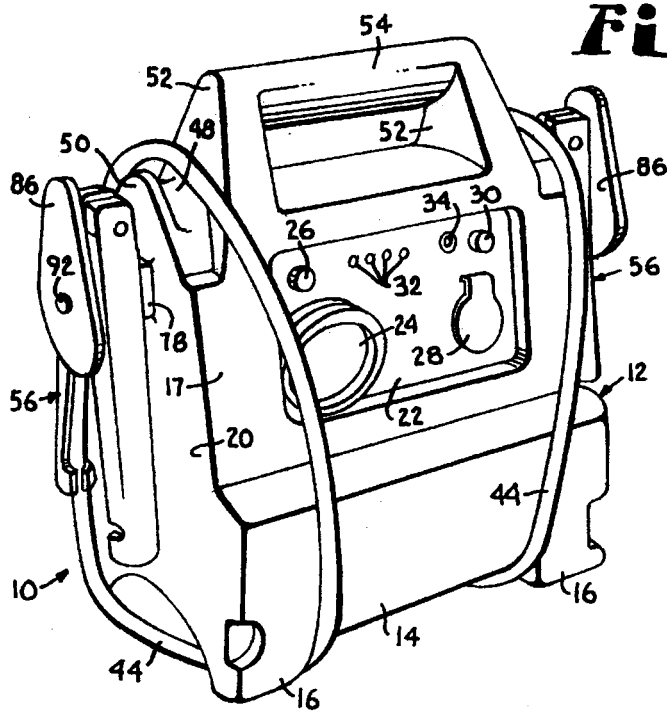


Fig. 1.

Fig. 4.

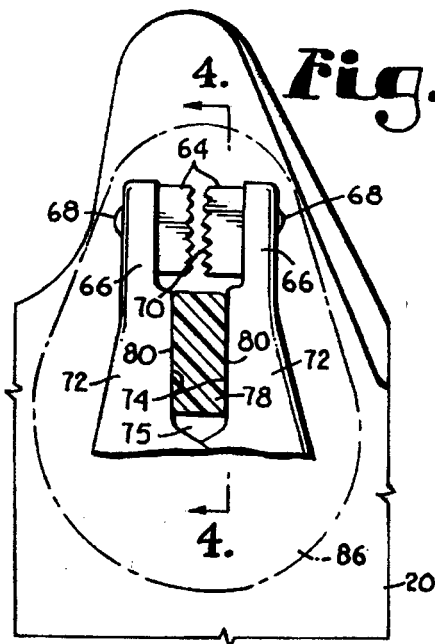
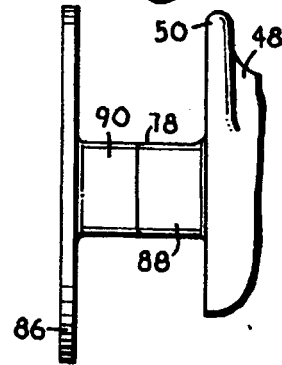


Fig. 3.

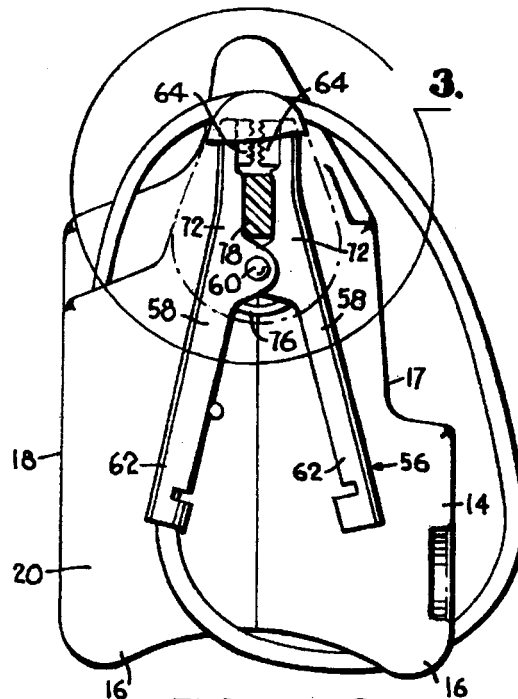
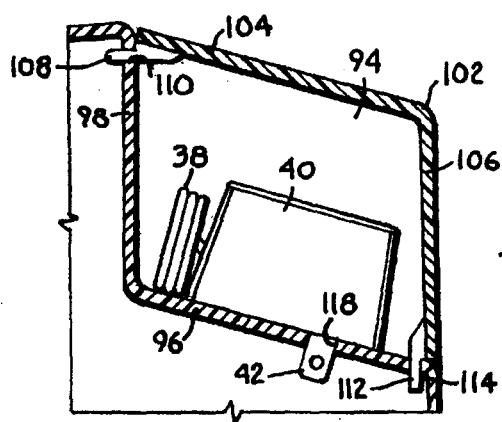
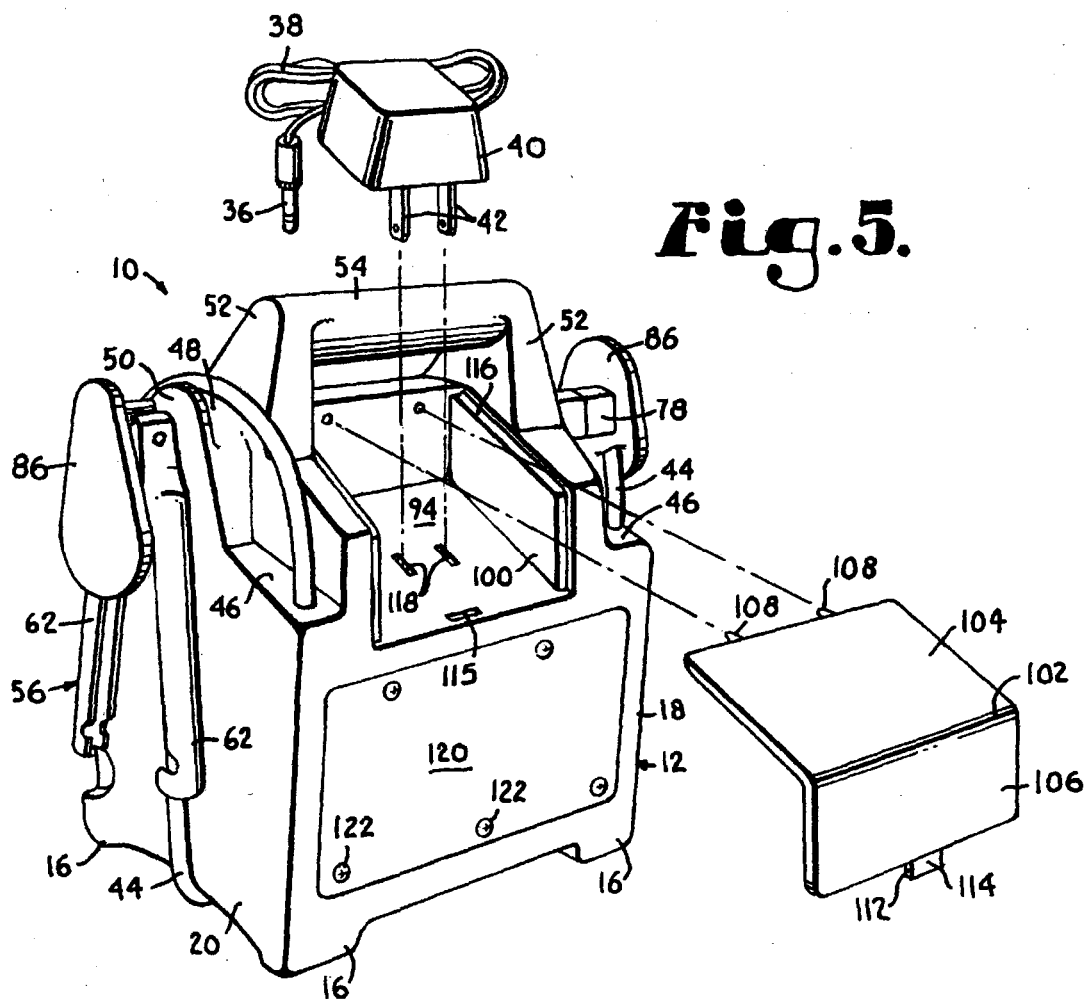


Fig. 2.



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BATTERY JUMP STARTER WITH JAW SECURING MEANS

FIELD OF THE INVENTION

This invention relates generally to the jump starting of automotive type batteries and more particularly to a self-contained jump start unit having battery clamps that are securely retained in place in a storage position when not in use. Additionally, the invention relates to a jump start unit that provides for improved storage of a battery charger when it is not being used.

BACKGROUND OF THE INVENTION

Battery jumper cables are commonly used to jump start automobiles and other vehicles having batteries that are discharged. A booster battery, typically one on another vehicle, is positioned in the vicinity of the discharged battery to allow use of the jumper cables to bypass the discharged battery and start the disabled vehicle. Once the vehicle has been started, its battery is charged to its normal voltage level by a generator or alternator carried on the vehicle.

Among the disadvantages of jump starting vehicles in this manner is the need for another vehicle having a properly charged battery. If the disabled vehicle is in an isolated location or if another vehicle is not readily available for some other reason, this procedure can involve considerable delay. Even then, it is necessary for space to be available close enough to the discharged battery for the second vehicle to be positioned where the cables can be extended between the two batteries. It is not unusual for the cables to be attached improperly to the battery terminals, and such improper connections can cause damage to the battery or electrical system of one or both vehicles.

In recognition of these and other problems, self-contained jump start units have been developed. On these devices, a booster battery is contained inside of a housing and is equipped with cables and battery clamps that are connected to the battery terminals, either permanently, removably, or through a switch. A vehicle with a discharged battery can be started by connecting the clamps of the jump start unit to the battery terminals of the disabled vehicle and using the booster battery to bypass the discharged battery.

In recent years, portable jump start units that can be carried by hand from place to place have been available in addition to larger wheel mounted units. The portable device is only slightly larger than a modern automotive battery and can be conveniently carried on a vehicle so that it is available in the event of a problem with the normal vehicle battery. It is also easy to position the jump start unit close to the discharged battery in situations where limited space is available. An example of a portable jump start unit is disclosed in U.S. Pat. No. 5,589,292.

A problem with the portable jump start units that have been proposed in the past is the way in which the battery clamps are stored when they are not being used. The clamps are often connected permanently to the battery terminals, and they are always "live" (although units with switches have been available). Therefore, the clamps should be stored so that they are protected from being inadvertently contacted by other objects or by the other clamp of the jump start unit. Units such as that described in U.S. Pat. No. 5,589,292 not only fail to secure the clamps in a secure and stable position when they are not being used, but they actually allow the clamps to turn about a round pin against which the jaws of the clamp are closed.

Another problem has been that the clamp handles or other parts of the clamps can project (due to rotation or otherwise)

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beyond the plane of the front or back of the housing. Consequently, if the unit is lying on a conductive surface, the only thing preventing a short circuit is insulation on the clamps, which can be worn away or otherwise lose its insulative effect.

Battery chargers are usually provided with the jump starter so that the booster battery can be maintained in a fully charged condition. The battery charger normally has a body containing a transformer that can be plugged into an electrical outlet. A plug carried on a cord extending from the charger body can be inserted into a jack on the jump starter housing in order to make charging current available to the battery contained in the housing.

If the battery charger is to be available when needed, it must be stored with the jump start unit. Portable units that have been available in the past and have provided for storage of the battery charger have stored the charger either in a simple cavity, where it can fall out, or loosely in a storage compartment, where it can rattle around unduly and fall out when the compartment is opened.

SUMMARY OF THE INVENTION

The present invention is directed to a portable battery jump start unit which is constructed to securely retain the battery clamps in a fixed storage position when they are not being used and which has an improved storage compartment for holding a battery charger.

In accordance with the invention, a jump start unit has a housing provided on its opposite sides with sidewardly projecting posts. The posts are generally rectangular in section so that they mate with similarly shaped openings on the jaw ends of the clamps. Each clamp opening has straight opposite edges that fit closely against the flat sides of the posts, with the spring which urges the jaws to close acting to retain the clamps in place against the posts.

This construction prevents the clamps from turning on the post or otherwise moving from the intended storage position. An enlarged end plate is located on the outer end of each post to shield the clamp jaws and prevent them from slipping off of the posts.

Another feature of the invention is that the housing is shaped to prevent any part of the clamps from projecting beyond the front or back planes of the housing. Consequently, the unit is not subject to a short circuit when placed on a conductive surface such as the floor of an automobile trunk.

The invention is further characterized by a uniquely arranged storage compartment on the back of the housing for storing a battery charger. The storage compartment has a removable cover which can be applied to enclose the charger when it is being stored. It is a particular feature of the invention that the bottom of the storage compartment has a pair of slots (or a rectangular opening) for receiving the electrical prongs of the charger body in order to prevent the charger body from sliding out of position. At the same time, space for storage of the battery charger cord is provided between the charger body and the front wall of the storage compartment.

BRIEF DESCRIPTION OF THE DRAWINGS

In the accompanying drawings which form a part of the specification and are to be read in conjunction therewith and in which like reference numerals are used to indicate like parts in the various views:

FIG. 1 is a front perspective view of a portable jump start unit constructed according to a preferred embodiment of the

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present invention, with the battery clamps of the jump start unit secured in their storage positions;

FIG. 2 is a side elevational view of the jump start unit taken from the left side;

FIG. 3 is a fragmentary elevational view on an enlarged scale showing the portion identified by detail 3 in FIG. 2;

FIG. 4 is a fragmentary view taken generally along line 4—4 of FIG. 3 in the direction of the arrows;

FIG. 5 is a rear perspective view of the jump start unit shown in FIG. 1, with selected components exploded for purposes of illustration; and

FIG. 6 is a fragmentary sectional view taken through the battery charger storage compartment of the jump start unit and showing the battery charger stored therein.

DETAILED DESCRIPTION OF THE INVENTION

Referring now to the drawings in more detail and initially to FIG. 1 in particular, numeral 10 generally designates a portable battery jump start unit constructed in accordance with a preferred embodiment of the present invention. The jump start unit 10 includes a hollow shell or housing which is generally identified by numeral 12 and which may be constructed of molded plastic or another suitable material. The housing 12 includes a relatively broad base 14 equipped with feet 16 that allow the unit to rest in a stable position on a suitable supporting surface.

The housing 12 includes a front face 17, a back 18 (FIG. 5), and a pair of opposite sides 20. The housing 12 carries within it a sealed automotive type battery (not shown). Above the base 14, the front face 17 of the housing is provided with a panel 22. The panel 22 includes a light 24 which is powered by the battery and which is controlled by an on/off button 26 mounted on panel 22. The panel 22 is also provided with a DC outlet which is normally plugged by a removable cover 28 and which provides an outlet for DC power from the unit. A status button 30 is also provided on panel 22 and may be depressed to check the status of the battery which is then indicated on a plurality of indicator lights 32 on panel 22. Finally, panel 22 has a charging jack 34 for receiving a plug 36 (see FIG. 5) carried on the end of a cord 38 of a battery charger. The battery charger includes a charger body 40 from which the cord 38 extends. The charger body 40 contains a conventional transformer (not shown) and is provided with a pair of electrical prongs 42 which allow the body 40 to be connected with a conventional AC power outlet.

The jump start unit 10 includes a pair of flexible cables 44, one of which is a positive cable connected with the positive battery terminal at a location within the housing 12. The other cable 44 is a negative cable which connects with the negative battery terminal at a location within the housing 12. As best shown in FIG. 5, the cables extend to the exterior of the housing 12 through ledges 46 located adjacent to the back 18 of the housing. Near its upper end, the housing 12 presents a pair of saddles or shoulders 48 located near the opposite sides 20 of the housing. Each shoulder 48 is provided with an adjacent flange 50 so that when one of the cables 44 is strung along the shoulder 48, the flange 50 helps to retain the cable on the shoulder 48. Upwardly extending lugs 52 are formed on housing 12 adjacent to the shoulders 48, and a handle 54 extends between the lugs 52 to provide a hand grip for carrying and handling of the unit 10.

Each cable 44 is equipped on its free end with a battery clamp 56 which may have a conventional construction. As

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best shown in FIG. 2, each clamp 56 includes a pair of arms 58 which are pivotally connected at intermediate locations thereon by a pivot coupling 60 which extends through inwardly projecting lugs on the arms 58. One end of each arm 58 provides a hand grip 62. The opposite end of each arm 58 carries a jaw 64. With particular reference to FIG. 3, the jaws 64 are mounted on end tabs 66 by suitable fasteners 68. The jaws 64 of each clamp 56 oppose one another and are provided with a plurality of teeth 70 for gripping against a battery terminal. The jaws 64 are constructed of an electrically conductive metal.

With continued reference to FIG. 3 in particular, each arm 58 has a shank portion 72 which extends from the end tab 66 to the area of the lug that receives the pivot coupling 60. The shank portions 72 of each clamp have opposing edges 74 which are spaced apart from one another. The edges 74 are straight along most of their lengths and provide surfaces having flat areas that face one another. An open space 75 is presented between the opposing edges 74 of the shank portions 72.

Each clamp 56 is provided with a spring 76 (FIG. 2) which continuously urges the hand grips 62 apart. The jaws 64 are thus continuously urged toward one another to a closed position in which the jaws come together. Each cable 44 extends generally inside of one of the hand grips 62 of each clamp 56 and has its conductive wires suitably connected with at least one of the jaws 64 of the clamp.

When not in use, the clamps 56 are stored on a pair of posts 78. The posts 78 project outwardly from the opposite sides 20 of the housing near the top of the unit at locations slightly below the flanges 50. The posts 78 are preferably aligned with one another on the opposite sides of the unit. As best shown in FIG. 3, each post 78 has a rectangular configuration in section and includes flat opposite side surfaces 80 and top and bottom surfaces which extend between the side surfaces 80. The sides 80 of each post are located to be flatly engaged by the flat areas 74 of the arms 58 when the jaws 64 are nearly but not completely closed against one another, as shown in FIG. 3. Thus, when the clamps 56 are applied to the posts 78, each post is closely received within and extends through the space 75 which is presented between the shanks 72.

The outer end of each post 78 is provided with an enlarged plate 86 which may be a curved shape having a relatively large lower portion and a smaller upper portion to generally conform with the shape of the top part of each clamp 56 when the jaws are nearly closed. As best shown in FIG. 3, the plate 86 is large enough to cover the upper portion of the clamp 56 when it is applied to the post 78. Thus, the jaws 64 are shielded on the outside by the plate 86.

With reference to FIG. 4 in particular, each post 78 may be constructed with a stub portion 88 which may be molded as part of the side 20. An outer portion 90 of each post may be formed as part of the plate 86 and may be suitably connected with the outer end of the stub 88, as by means of a screw or other fastener (not shown). For the sake of appearance, the passage in which the fastener is applied may be closed by a plug 92 (FIG. 1). Each post 78 is long enough to accommodate the thickness of the upper portions of the arms 58 between the housing side 20 and the end plate 86.

It is a particular feature of the invention that the clamping posts 78 are arranged to cooperate with the clamps 56 in a manner to prevent the clamps, when stored on the posts, from projecting beyond the front face 17 or the back face 18 of the housing. As best shown in FIG. 2, when the clamps are applied to the posts, all parts of the clamps 56 are

maintained between the plane of the back 18 and the front face 17, with no part of either arm 58, or either grip 62 or either jaw 64 projecting beyond the front face 17 or the back face 18. Consequently, if the unit is placed on its front face or back face on a conductive surface such as a metal automobile trunk floor, the clamps 56 do not contact the conductive surface to possibly create a short circuit. This prevents potential problems that plague units and is due to the anti-rotation provided by the fit between the posts and clamps as well as the configuration of the housing to avoid projection of the clamps beyond the plane of the front or back housing surface.

Referring now to FIGS. 5 and 6 in particular, the back of the housing 12 is provided with a storage compartment 94 which provides for storage of the battery charger, including the body 40, the cord 38 and the plug 36. The storage compartment 94 is located above the battery which is contained in the housing 12 and generally between the ledges 46. When the storage compartment 94 is open, it is accessible from the back of the unit.

The storage compartment has a floor or bottom panel 96 which inclines downwardly from front to back (see FIG. 6). A generally vertical front wall 98 of the storage compartment connects with the front edge of the bottom panel 96. The opposite sides of the storage compartment are formed by opposite side walls 100. The bottom panel 96, wall 98 and sides 100 may all be molded as an integral part of the housing 12.

A removable (or hinged) cover panel 102 for the storage compartment 94 may be removed to open the top and back of the storage compartment and provide access to its contents. The cover 102 has a generally L-shaped configuration and include a top panel 104 and a back panel 106 which is joined with the rear edge of the top panel 104. Both of the panels 104 and 106 are generally rectangular. The front edge of panel 104 is provided with a pair of forwardly projecting pins 108 that fit closely in a pair of openings 110 when the cover panel 102 is applied to the storage compartment. The openings 110 are formed through the front panel 98 of the storage compartment 94. The bottom edge portion of panel 106 is provided with a downwardly projecting tab 112 which is provided with a locking lip 114 and which may be fitted through a slot 115 formed near the back edge of the inclined bottom panel 96 of the storage compartment. The locking lip 114 fits beneath panel 96 to lock the cover 102 in place when the tab 112 is inserted through slot 115 to apply the cover and enclose the storage compartment 94. The cover 106 may be removed by flexing the back panel 106 forwardly far enough to allow the locking lip 114 to be removed upwardly through the slot 115. The cover 102 may then be pulled rearwardly to remove pins 108 from openings 110 in order to remove the cover 102. Each of the sides 100 of the storage compartment may be provided with recesses 116 extending along its top and back edge in order to receive the side edges of the cover 102 for a close fit of the cover on the storage compartment 94.

The inclined bottom panel 96 of the storage compartment is provided with a pair of parallel slots 118 which are spaced apart and sized to receive the electrical prongs 42 of the battery charger body 40. When the prongs 42 are fitted through the slots 118 as shown in FIG. 6, the charger body 40 rests on the bottom panel 96 with sufficient space provided in front of the charger body to accommodate storage of the cord 38 and the plug 36 between the charger body and the front wall 98 of the storage compartment 94. Because of the extension of the prongs 42 through the slots 118, the charger body 40 is held in place in the storage

compartment and does not rattle around unduly or slide out of the storage compartment when it is opened by removal of the cover 102. At the same time, the cover 102 can be easily removed to provide access to the battery charger when it is needed for charging of the battery contained within the housing 12 of the unit. As an alternative to the slots 118, a single rectangular opening may be provided in the panel 96 to closely receive the prongs 42 and thus serve the same function as the slots.

Access to the battery and the other internal components of the housing 12 is provided through an opening in the back of the housing which is normally covered by a removable cover panel 120. The cover panel 120 may be secured to the back 18 of the housing by a plurality of screws 122 or other fasteners which may be removed to allow removal of the panel 120. When the panel 120 is removed, a relatively large opening in the back 18 of the housing is exposed so that the battery and other internal components are accessible through the opening.

When the jump start unit 10 is not in use, the battery clamps 56 may be stored in the manner shown in FIG. 1. The cables 44 may be drawn upwardly along and over the shoulders 48 and then downwardly beneath the front feet 16. The cables are long enough that the clamps 56 can be applied to the storage posts 78 with the cables drawn relatively tight.

With particular reference to FIG. 3, the clamp surfaces 74 lie flatly along the opposite sides 80 of each post 78 such that there is a relatively long area of flat contact between each clamp and each post. The action of the spring 76 tending to close the jaws 64 retains the surfaces 74 securely against the post sides 80. It is noted that the jaws 64 are not completely closed so that the spring action is applied through the clamp to the post. This arrangement provides a stable, fixed storage position for each clamp and prevents any rotation of the clamp about the axis of the post or any other movement of the clamp when it is in the stored position. The end plates 86 act against the clamps to assure that they cannot slip off the ends of the posts.

When the clamps 56 are in their storage position, the hand grips 62 extend downwardly along the housing sides 20 well below the end plates 86. Accordingly, the hand grips 56 are readily accessible and can be pressed toward one another to release the shanks 72 from the posts 78, thus freeing the clamps and allowing them to be pulled downwardly far enough to clear the plates 86. The cables 44 can then be extended as necessary to reach the terminals of the discharged battery.

As previously indicated, the presence of the slots 118 (or a rectangular opening) in the battery charger storage compartment 94 provide for a stable position of the charger body 40 when it is not in use. When the battery charger is needed, the cover 102 can be easily removed, and the battery charger can be removed so that its plug 36 can be inserted into the jack 34 and the prongs 42 can be applied to an electrical outlet in order to charge the battery contained within the housing 18.

From the foregoing it will be seen that this invention is one well adapted to attain all ends and objects hereinabove set forth together with the other advantages which are obvious and which are inherent to the structure.

It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations. This is contemplated by and is within the scope of the claims.

Since many possible embodiments may be made of the invention without departing from the scope thereof, it is to

be understood that all matter herein set forth or shown in the accompanying drawings is to be interpreted as illustrative, and not in a limiting sense.

Having thus described the invention, what is claimed is:

1. A portable battery jump start unit comprising:
 - a substantially enclosed housing containing a battery;
 - a cable connected with said battery and extending outside of said housing;
 - a clamp on said cable which includes a pair of pivotally connected arms each having one end portion providing a hand grip and an opposite end portion providing a clamping jaw, at least one of said jaws being electrically conductive and connected with the cable;
 - means for urging said arms to pivot in a direction to move said jaws together;
 - a surface adjacent said opposite end portion of each arm having a generally flat area, said surfaces on the respective arms being spaced apart and opposing one another; and
 - a clamping post on said housing for storing said clamp, said post having opposite sides for receiving and mating with said flat areas of the arms in a manner to substantially prevent rotation of the clamp on the post when applied thereto.
2. A jump start unit as set forth in claim 1, wherein said post has a substantially rectangular configuration in section.
3. A jump start unit as set forth in claim 2, wherein said surfaces on the respective arms comprise substantially straight edges on the arms providing said flat areas.
4. A jump start unit as set forth in claim 1, wherein said surfaces on the respective arms comprise substantially straight edges on the arms providing said flat areas.
5. A jump start unit as set forth in claim 1, including an end plate on said post for retaining said clamp thereon.
6. A jump start unit as set forth in claim 1, including:
 - a charging jack on said housing for accepting power to charge the battery;
 - a battery charger including a charger body having a pair of electrical prongs and a cord having a plug for insertion into said jack;
 - a storage compartment on the back of said housing for storing of said charger when not in use, said compartment having a bottom surface presenting an opening for receiving said prongs to maintain said charger body in place on the floor when stored in said compartment; and
 - a removable cover for said compartment, said compartment being substantially enclosed when said cover is applied thereto.
7. A jump start unit as set forth in claim 6, wherein said opening comprises a pair of slots for closely receiving the respective prongs.
8. A jump start unit as set forth in claim 1, wherein:
 - said housing has front and back surfaces; and
 - said clamping post is arranged to retain said clamp applied thereto at a location to prevent any part of the clamp from extending beyond said front or back surface.
9. A portable battery jump start unit comprising:
 - a substantially enclosed housing containing a battery and having opposite sides;

- a pair of cables extending from the battery outside of the housing;
- a pair of clamps on the respective cables each including a pair of pivotally connected arms, each of said arms having one end portion providing a hand grip and an opposite end portion carrying a clamping jaw and including a substantially flat area;
- means for urging said arms to pivot in a direction to move said jaws together, said flat areas of the jaws in each pair being spaced apart and opposing one another when the jaws are substantially closed on one another; and
- a pair of clamping posts extending from the respective sides of the housing for storing said clamps, each post having opposite sides for receiving and mating with said flat areas of the corresponding clamp in a manner to resist turning of the clamp on the post.
10. A jump start unit as set forth in claim 9, wherein each post has a substantially rectangular configuration in section.
11. A jump start unit as set forth in claim 10, wherein said flat areas of the jaws are arranged to engage the opposite sides of the posts along substantially the entire vertical extent of said sides.
12. A jump start unit as set forth in claim 11, wherein said jaws are spaced away from one another when said flat areas are applied to the opposite sides of said posts.
13. A jump start unit as set forth in claim 11, including an end plate on each post for retaining the corresponding clamp thereon.
14. A jump start unit as set forth in claim 9, including an end plate on each post for retaining the corresponding clamp thereon.
15. A jump start unit as set forth in claim 9, wherein said jaws are spaced away from one another when said flat areas are applied to the opposite sides of said posts.
16. A jump start unit as set forth in claim 9, including:
 - a charging jack on said housing for accepting power to charge the battery;
 - a battery charger including a charger body having a pair of electrical prongs and a cord having a plug for insertion into said jack;
 - a storage compartment on the back of said housing for storing of said charger when not in use, said compartment having a bottom surface presenting an opening for receiving said prongs to maintain said charger body in place on the floor when stored in said compartment; and
 - a removable cover for said compartment, said compartment being substantially enclosed when said cover is applied thereto.
17. A jump start unit as set forth in claim 16, wherein said opening comprises a pair of slots for closely receiving the respective prongs therein.
18. A jump start unit as set forth in claim 9, wherein:
 - said housing includes front and back surfaces; and
 - said posts are arranged to retain the clamps applied thereto with no part of either clamp projecting beyond said front surface or said back surface.
19. A portable battery jump start unit comprising:
 - a substantially enclosed housing containing a rechargeable battery, said housing having a front and a back and

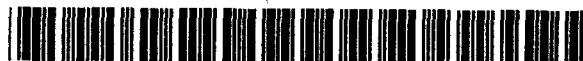
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- a charging jack for accepting power to charge the battery;
- a pair of cables connected with the battery each carrying a battery clamp;
- a battery charger including a charger body having a pair of electrical prongs and a cord having a plug for insertion into said jack;
- a storage compartment on the back of said housing for storing of said charger when not in use, said compartment having a bottom surface presenting an opening for

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- receiving said prongs to maintain said charger body in place on the floor when stored in said compartment; and
 - a removable cover for said compartment, said compartment being substantially enclosed when said cover is applied thereto.
20. A jump start unit as set forth in claim 19, wherein said opening comprises a pair of slots for closely receiving the respective prongs therein.

* * * * *



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United States Patent [19]
Krieger

[11] **Patent Number:** **5,982,138**

[45] **Date of Patent:** **Nov. 9, 1999**

[54] **PORTABLE ELECTRICAL ENERGY SOURCE**

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[52] **U.S. Cl.** **320/105; 320/104**

[58] **Field of Search** **320/104, 105,
320/115**

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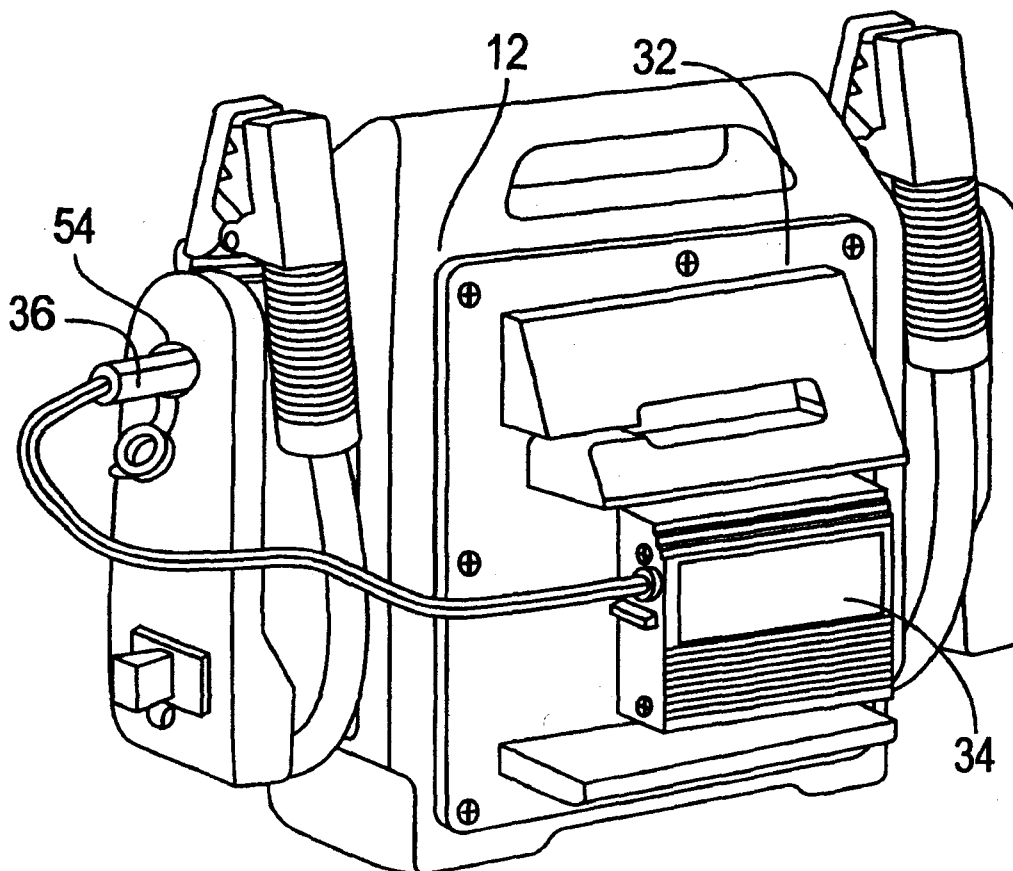
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[57] **ABSTRACT**

A portable electrical energy source includes a portable housing having an outer surface and defining an interior space. A direct current battery is disposed in the interior space. A direct current socket is supported on the housing and electrically coupled to the battery for supplying direct current from the battery to a device outside the housing. An inverter is provided for converting direct current into alternating current and includes an electrical lead have a male plug adapted for being electrically connected to the battery via the direct current socket and an electrical outlet for supplying alternating electrical current to a device outside the housing. The inverter is removably attached to the outer surface of the housing so that the inverter can be ported together with the battery inside the housing and selectively separated from the housing and used independently of the battery in the housing.

14 Claims, 4 Drawing Sheets



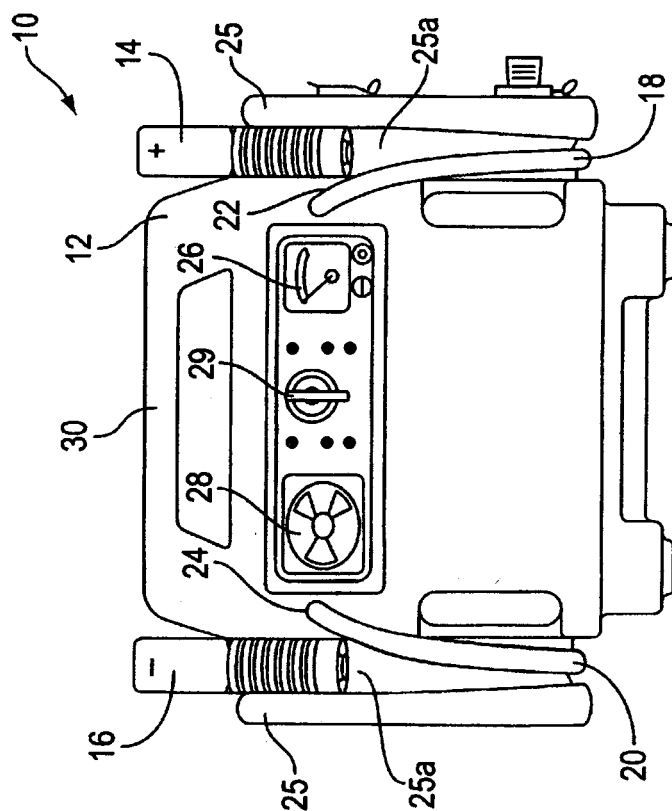


FIG. 1

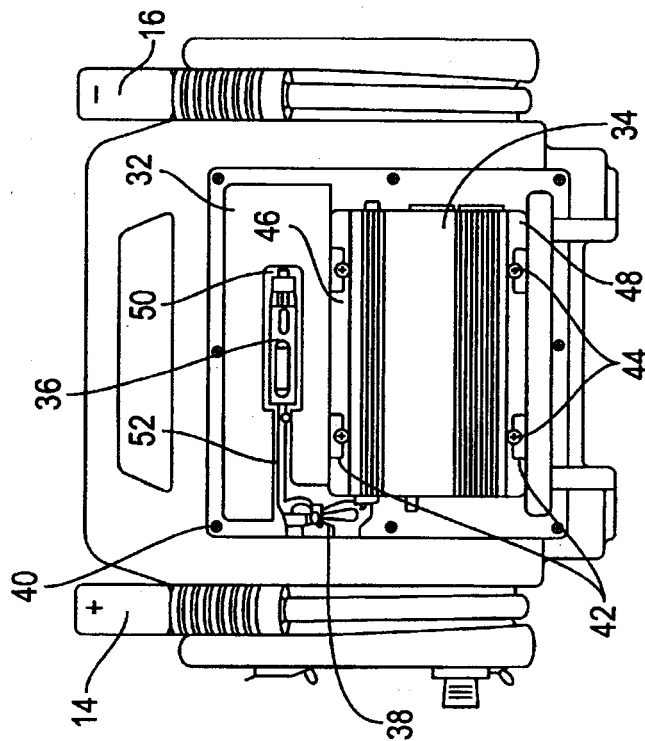


FIG. 2

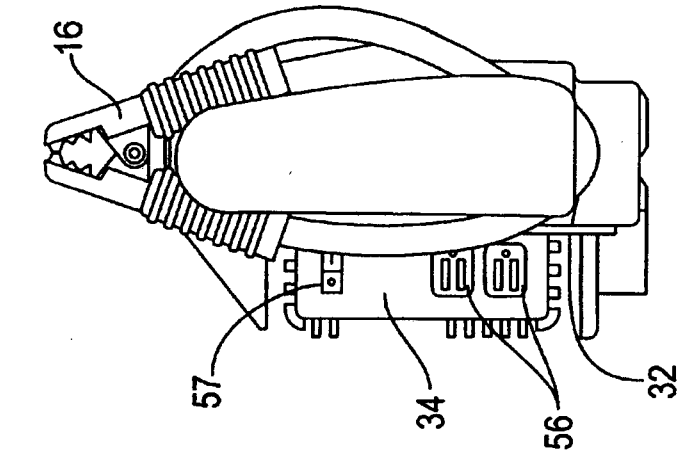


FIG. 4

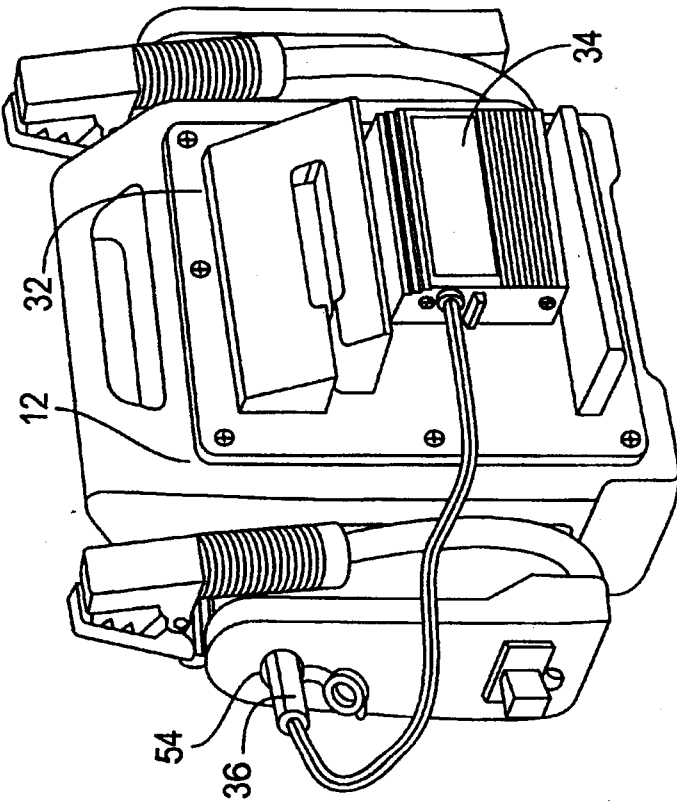


FIG. 3

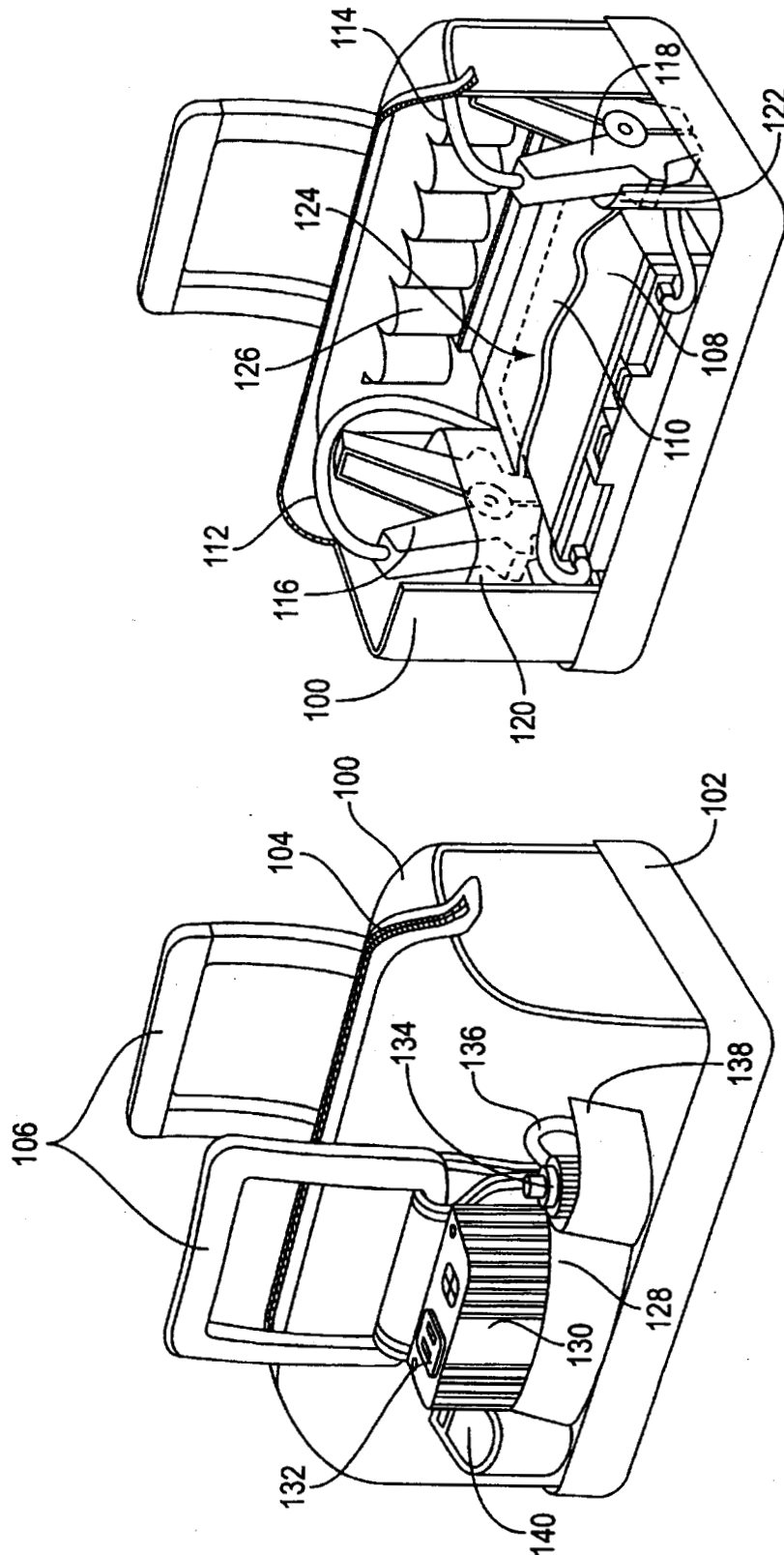


FIG. 6

FIG. 5

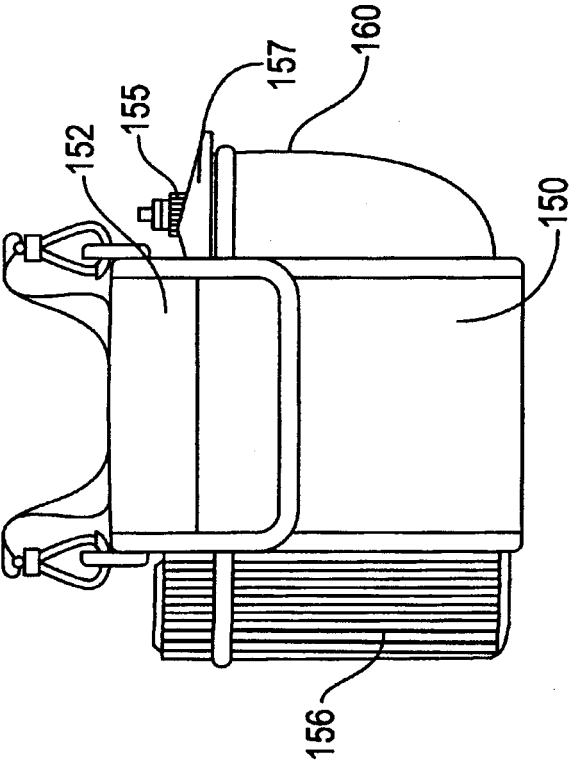


FIG. 7

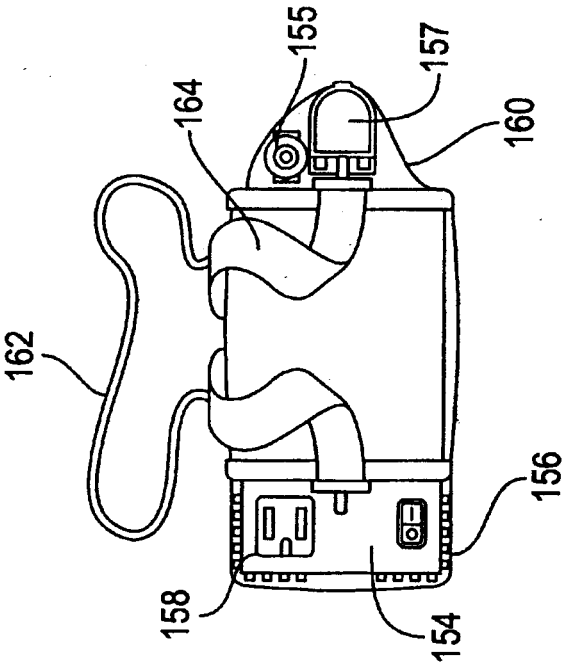


FIG. 8

PORTABLE ELECTRICAL ENERGY SOURCE

BACKGROUND OF THE INVENTION

The invention relates to a portable electrical energy source.

Portable electrical energy sources are known for providing direct current for jump starting an automobile, boat or other vehicle as well as for providing direct current for energizing various appliances. For example, a jump start system providing a 12-volt, 400 ampere power supply is manufactured and distributed by Vector Manufacturing, Ltd., of Hollywood, Fla. under the trade name PRO-CHARGE-900. In this jump start system, a heavy-duty 12-volt battery is arranged inside an impact-resistant polyethylene case. Heavy duty No. 4 industrial booster cables attached to the battery extend through the case and are attached to heavy duty 400 amp booster clamps which, when not in use, can rest on shoulders molded on the case. Dc appliances may be plugged into a DC outlet socket mounted in the casing and coupled to the battery. The jump start battery can be charged by a car battery via a DC/DC power cord provided with a male plug adapted to be inserted into the cigarette lighter of an automobile. The jump start battery may also be charged via an appropriate AC adapter/charger.

Power inverters are known which convert direct current into alternating current. For example, Vector Manufacturing, Ltd. also manufactures and distributes a self-contained power inverter under the trade name POWER FORCE 140 which plugs directly into a cigarette lighter socket or other 12-volt power source for converting 12-volt direct current energy to 110-volt alternating current. The PRO-CHARGE-900 jump start system also includes a cigarette lighter socket so that a power inverter such as the POWER FORCE 140 may be plugged into the jump start system. In such a configuration, the jump start system can provide either 12-volt direct current or 110-volt alternating current. A disadvantage of this configuration is that by having two separate components, they are separately portable and both may not be readily at hand when it is desired to use them together to furnish DC power to the inverter to produce alternating electrical power.

Jump start systems are known wherein a power inverter is included within the impact resistant casing along with the 12-volt battery to provide a portable jump start system which provides both a 12-volt direct current power supply and a 110-volt alternating current supply. While the integration of the power inverter along with the 12-volt battery inside the impact resistant casing has the advantage of providing a portable electrical power source that provides both AC and DC electrical energy, it is a disadvantage that the power inverter cannot be separated from the jump start system and used independently. For example, it would be desirable if the inverter of the portable jump start system could be plugged into the cigarette lighter socket of an automobile to produce alternating current off the automobile battery.

SUMMARY OF THE INVENTION

It is an object of the invention to eliminate the foregoing disadvantages by providing a combined portable electrical energy source for providing both AC and DC electrical energy in which the power inverter may be used independently of the DC battery in the portable system.

The above and other objects are accomplished according to the invention by the provision of a portable electrical

energy source which includes a portable housing having an outer surface and defining an interior space; a direct current battery disposed in the interior space; a direct current socket supported on the housing and electrically coupled to the battery for supplying direct current from the battery to a device outside the housing; an inverter for converting direct current into alternating current, the inverter including an electrical lead having a male plug adapted for being electrically connected to the battery via the direct current socket and an electrical outlet for supplying alternating electrical current to an alternating current device outside the housing; and means for removably attaching the inverter to the outer surface of the housing so that the inverter can be ported together with the battery and selectively separated from the housing and used independently of the battery within the portable housing.

In one embodiment of the invention, the battery is a 12-volt DC battery and booster cables are electrically coupled to the battery for jump starting a weak vehicle battery.

According to various aspects of the invention, the attaching means may slidably receive the inverter on the outer surface of the housing, or the attaching means may comprise a snap-on connection, or a pocket attached to the outer surface of the housing for removably receiving the inverter. Other mechanisms for removably attaching the inverter to the outer surface of the portable housing will be apparent to those skilled in the art. As another example, the inverter could be attached to the outer surface of the housing via VELCRO or other type of releaseable connection.

According to further embodiments of the invention, the mechanism for attaching the inverter to the housing may be molded or otherwise provided directly on the outer surface of the housing, or be retrofitted onto the housing by way of a panel which incorporates the attaching mechanism and which is fixable to the outer surface of the housing.

The housing may be rigid, for example, made of an impact resistant polyethylene plastic. Alternatively, the housing may be of a flexible material which, in effect, forms a bag for enclosing the DC battery. In this case, the mechanism for attaching the inverter to the exterior surface of the housing (bag) could be in the form of a pocket sewed onto the exterior surface of the bag.

It is also advantageous to provide the housing with a holder for the plug of the power inverter so that the plug can be properly stowed when not in use. A form fitting recess for the plug can be molded directly into the outer surface of the rigid housing or into a panel which is retrofitted onto the housing. When the housing is in the form of a bag comprised of flexible material, the holder for the plug may take the form of a pocket sewn or otherwise adhered onto the outer surface of the bag. In the case of the housing in the form of a flexible bag, it is preferable for the bottom of the bag to be fitted with a rigid support for supporting the battery. Furthermore, in a preferred form of this embodiment, an interior space is provided below a closeable opening at the top of the bag and above the battery to provide a space for stowing tools. This space can also be utilized for stowing battery booster cables. Preferably, holsters are also provided within the interior space for stowing battery cable clamps.

According to yet another embodiment of the invention, the housing, battery and inverter are of such size and weight as to constitute a belt pack or shoulder bag, and a belt or shoulder strap is attached to the bag to assist a person in porting the bag. In this embodiment of the invention, the battery may comprise, for example, a 12-volt, 3 amp-hour

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(up to 7 amp) battery and the inverter, a 50 watt (up to 100 watt) inverter, the combination preferably weighing not more than about 5 pounds, for supplying electrical energy to appliances having a relatively low energy requirement.

Further features and advantages of the invention will become apparent from the following detailed description with reference being made to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a front elevational view of a known portable electric power source.

FIG. 2 is a back elevational view of the portable electric power source shown in FIG. 1 retrofitted with a panel for removably holding a power inverter according to the principles of the invention.

FIG. 3 is a perspective view of a side and back of the power inverter in an operative condition.

FIG. 4 is a right side elevational view of FIG. 2.

FIG. 5 is a perspective view of a portable electric power source according to another embodiment of the invention.

FIG. 6 is a perspective view partially cut away of the portable electric power source of FIG. 5.

FIG. 7 is a front elevational view of a portably electric power source according to a further embodiment of the invention.

FIG. 8 is a plan view of the embodiment shown in FIG. 7.

DETAILED DESCRIPTION OF THE INVENTION

Referring to FIG. 1, there is shown a known jump start system 10 which includes an impact resistant polyethylene case 12 enclosing a heavy-duty 12-volt power supply (not shown) which has its positive and negative poles coupled, respectively to positive and negative booster clamps 14 and 16 via respective battery cables 18 and 20 which extend through openings 22 and 24, respectively in casing 12. Casing 12 includes molded on shoulders 25 provided with a groove 25a in which booster clamps 14 and 16 are intended to rest when not in use. The front elevational view of FIG. 1 additionally shows a battery condition indicator gage 26, a built-in work emergency light 28 and a safety on/off switch 29 for electrically connecting/disconnecting the battery cables to the battery. Jump start system 10 is portable and for this purpose includes a handle 30 so that the system can be easily ported from place to place.

FIG. 2 shows a back elevational view of the jump start system shown in FIG. 1 which incorporates, according to the invention, a panel 32, which may be a plastic injection molded product and which is shaped and adapted to removably hold a power inverter 34 and a power inverter plug 36 which is attached to the power inverter via an electrical lead 38. Panel 32 is fixed to casing 10, for example, by screws 40 spaced around an edge of panel 32. Other mechanisms for attaching panel 32 to the outer surface of casing 12 will be apparent to those skilled in the art, for an adhesive could be used. As shown in FIG. 2, a plurality of upper and lower strips 42 are attached to panel 32 by screws 44. Power inverter 34 has upper and lower lips 46 and 48, respectively, which can be slid between strips 42 and panel 32 when screws 44 are loosened. Once in place, the power inverter may be secured in place by tightening screws 44. Panel 32 is appropriately molded to provide a recess 50 for plug 36 and a connecting recess 52 to accommodate a region of lead 38 adjacent plug 36.

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It should be obvious to a person skilled in the art that power inverter 34 together with its plug 36 and lead 38 may be removably secured to the outer surface of casing 10 via a retrofitted panel 32 such as illustrated in the embodiment of FIG. 2, or alternatively, the outer surface of casing 12 may itself be specially molded and adapted to removably hold the power inverter together with its plug and lead.

FIG. 3 shows a perspective view of the side and back of jump start system 10 adapted with a panel 32 in accordance with the invention, wherein the removably secured power inverter 34 has its plug 36 inserted into a female receptacle 54 which is electrically coupled to the battery (not shown) within casing 12, for supplying the DC electrical energy of the battery to power inverter 34.

FIG. 4 shows a side view of the jump start system 10 with the attached power inverter where standard alternating current receptacles 56 and an inverter on/off switch 57 are illustrated. The power inverter converts the DC electrical energy of the battery to alternating electrical energy which is coupled to the receptacles 56, so that, for example, a 12 volt DC battery can be used to supply 110 volt alternating current wherever it is needed due to the portable nature of the system.

The principles of the invention may be implemented in connection with any portable DC power supply but is particularly useful in connection with a portable jump start system for jump starting cars, trucks, boats and other vehicles. A variety of jump start systems are commercially available which can be retrofitted with a panel of the type described above. Once such system is the above-mentioned PRO-CHARGE-900 manufactured and distributed by Vector Manufacturing, Ltd., Hollywood, Fla. Additionally, a variety of commercially available power inverters are known which can be utilized in implementing the invention, such as the above mentioned POWER FORCE-140, also manufactured and distributed by Vector Manufacturing, Ltd. In the illustrated embodiment, the power inverter is provided with a lip for being slidably received between strips 42 and panel 32. However, other mechanisms for removably attaching the power inverter are possible within the principles of the invention which would not require such lips. For example, the panel 32 or alternatively, the outer surface of casing 12, could be molded to provide a recess similar to that of recess 50 for receiving the power inverter with a friction fit. Another possibility is to provide a snap connection between the power inverter and panel 32 or the outer surface of casing 12. A VELCRO connection is also conceivable. Any other mechanism for removably securing the power inverter to the outer surface of casing 12 is conceivable within the principles of the invention and will be apparent to those skilled in the art.

The invention provides the advantage of easily porting the power inverter together with the DC battery supply to provide a combined AC/DC power source. When desirable, the power inverter can be easily removed from the casing 12 and utilized separately from the jump start system, for example, by inserting plug 36 into the cigarette lighter receptacle in an automobile to provide alternating current.

FIGS. 5 and 6 show another embodiment of the invention. In this embodiment, the housing for the portable electric power source is in the form of a bag 100 comprised of a flexible, yet heavy duty material, such as a reinforced fabric or leather. A rigid tray 102, made for example, of heavy-duty plastic or metal, is attached to the bottom of bag 100 to provide a reinforcing support for the load to be carried by the bag. The top of bag 100 is provided with a zipper 104 for

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opening and closing the bag. Handles 106 are attached to the bag so that the bag can be carried by hand. The partially cutaway view of FIG. 6 shows the interior of bag 100 wherein there is provided a DC battery 108 which sits on the bottom of the bag, preferably in a pocket 110 to hold the battery in place. Battery 108 has battery cables 112 and 114 attached to its terminals. The battery cables 112 and 114 are attached at the remote ends to booster clamps 116 and 118, respectively, which are stowed in holsters or pockets 120 and 122, respectively, which are sewn or glued to the interior surface of the bag. As clearly visible in FIG. 6, there is an interior space 124 provided above the battery and below the zippered top of the bag for the purpose of storing and transporting tools. Additionally, loops 126 may be sewn, glued or otherwise fixed to the interior surface of the bag to provide fixed storage spaces for elongated tools such as screw drivers, wrenches, pliers, etc.

Referring again to FIG. 5, there is shown attached to the exterior surface of bag 100 a first pocket 128 which is adapted to removably receive a power inverter 130 which has an exposed conventional AC outlet 132. A male plug 134, attached to power inverter 130 via a lead 136 is stored in a second pocket 138 and in operation can be plugged into a corresponding female receptacle 140 which is electrically coupled to battery 108 to furnish DC electrical energy to the inverter.

As in the embodiment of FIGS. 1-3, power inverter 130 together with its plug and cord can be ported and used with the battery stored within bag 100 or, alternatively, can be removed from the pocket and used separate and apart from the battery within the bag. Obviously, other types of mechanisms will be apparent to those skilled in the art for removably holding or securing the power inverter, including its cord and plug, to the outer surface of the bag in accordance with the principles of the invention.

FIGS. 7 and 8 show yet another embodiment of a portable electric power source employing the principles of the invention. In this embodiment, the battery and inverter are both smaller in size and produce lower levels of energy and thus can be placed in a belt pack or shoulder bag which can either be worn around the waist of a person or carried by way of a shoulder strap. In this embodiment, a battery such as 12-volt, 3 amp-hour battery (not shown) is placed in a flexible pouch 150 made for example of plastic, leather or cloth, which is covered by a flap 152 to secure the battery in place. An inverter, such as a 50 watt inverter 154 commercially available from Vector Manufacturing, Ltd., under the trade name POWER FORCE-50, is removably inserted into a pocket 156 sewn or otherwise attached to a side of pouch 150. Inverter 154 is electrically coupled to the battery within pouch 150 to provide alternating current via a standard AC outlet 158. A male plug 155 connected to inverter 154 by a lead (not shown) may be plugged into a 12-volt DC female socket which in FIGS. 7 and 8 is shown covered by a protective lid 157. The plug and socket are stored in yet another pocket 160 sewn or otherwise attached to an opposite side of pouch 150. A belt strap 162 and/or a shoulder strap 164 may be attached to pouch 150 in a known manner so that the pouch, together with the battery therein and the inverter and 12-volt outlet stored in the associated pockets may be worn around the waist of a person or slung over the shoulder for easy portability. Preferably, the combined weight of the portable electric power source in FIGS. 7 and 8 is no more than 5 pounds.

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The invention has been described in detail with respect to preferred embodiments, and it will now be apparent from the foregoing to those skilled in the art, that changes and modifications may be made without departing from the invention in its broader aspects, and the invention, therefore, as defined in the appended claims, is intended to cover all such changes and modifications that fall within the true spirit of the invention.

What is claimed:

1. A portable electrical energy source, comprising:

a portable housing having an outer surface and defining an interior space;

a direct current battery disposed in the interior space;

a direct current socket supported on the housing and electrically coupled to the battery for supplying direct current from the battery to a device outside the housing;

an inverter for converting direct current into alternating current, the inverter including an electrical lead having a male plug adapted for being electrically connected to the battery via the direct current socket and an electrical outlet for supplying alternating electrical current to a device outside the housing; and

attaching means for removably attaching the inverter to the outer surface of the housing so that the inverter can be ported together with the battery inside the housing and selectively separated from the housing and used independently of the battery in the housing.

2. The portable electrical energy source according to claim 1, wherein the battery is a 12 volt DC and further including battery booster cables attached to the battery for extending outside the housing for jump starting a weak battery of a vehicle.

3. The portable electrical energy source according to claim 1, wherein the attaching means slidably receives the inverter on the outer surface of the housing.

4. The portable electrical energy source according to claim 1, wherein the attaching means comprises a snap-on connection between the inverter and the outer surface of the housing.

5. The portable electrical energy source according to claim 1, wherein the attaching means comprises a pocket attached to the outer surface of the housing for removably receiving the inverter.

6. The portable electrical energy source according to claim 1, further including a holder arranged on the outer surface of the housing for stowing the male plug when not in use.

7. The portable electrical energy source according to claim 6, wherein the holder comprises a pocket.

8. The portable electrical energy source according to claim 6, wherein the holder comprises means defining a recess sized to have a friction fit with the male plug for holding the male plug in the recess.

9. The portable electrical energy source according to claim 1, wherein the housing comprises a rigid casing and further comprising a separate panel incorporating the attaching means, and means for fixing the separate panel to the rigid casing.

10. The portable electrical energy source according to claim 1, wherein the housing comprises a bag including a bottom having a rigid support for supporting the battery, the bag further including a top presenting a closable opening and defining an interior space above the battery and below the closable opening for stowing tools.

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11. The portable electrical energy source according to claim 10, wherein the bag includes a flexible side wall and the holder comprises a pocket attached to the flexible side wall for removably receiving the inverter.

12. The portable electrical energy source according to claim 10, and further comprising two battery cables connected to the battery and each having an end including a battery clamp, and the bag includes two holsters arranged in the interior space for stowing the clamps of the battery cables.

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13. The portable electrical energy source according to claim 1, wherein the housing, battery and inverter are sized to constitute a belt pack or a belt or shoulder strap, respectively, attached to the housing shoulder bag, and further comprising to assist a person in porting the portable electrical energy source.

14. The portable electrical energy source according to claim 1, and further including a handle attached to the housing to assist in porting the housing by hand.

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(US)

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F28D 7/10

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(58) Field of search

B3A

F4S

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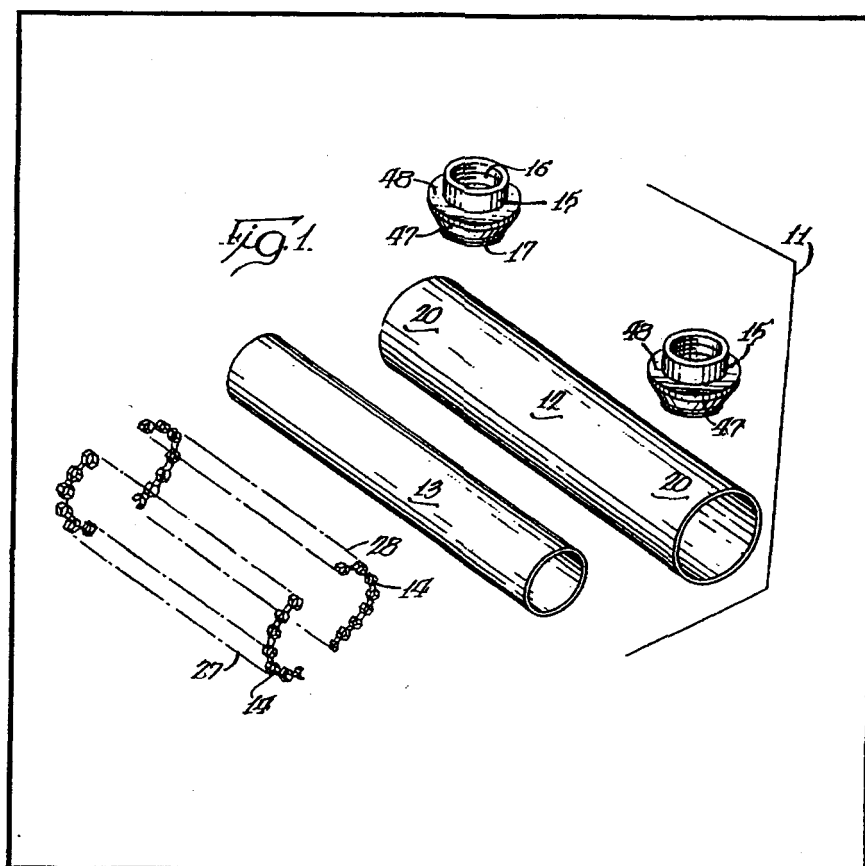
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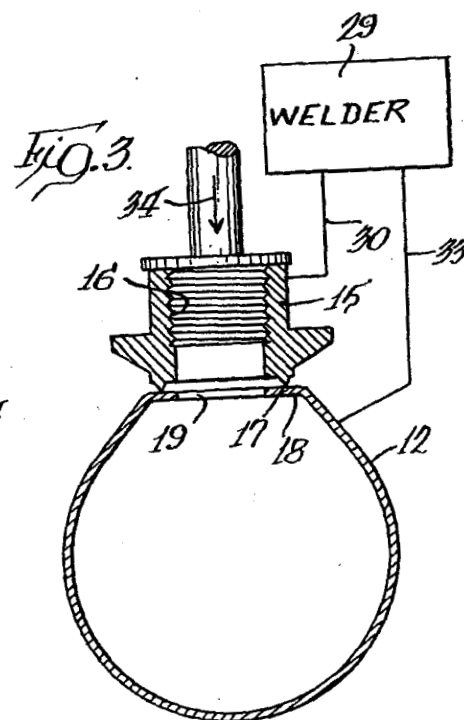
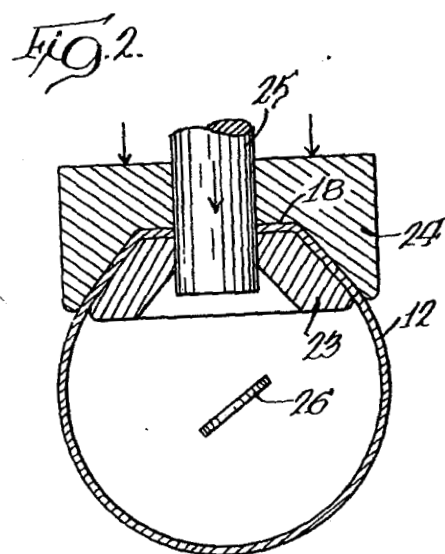
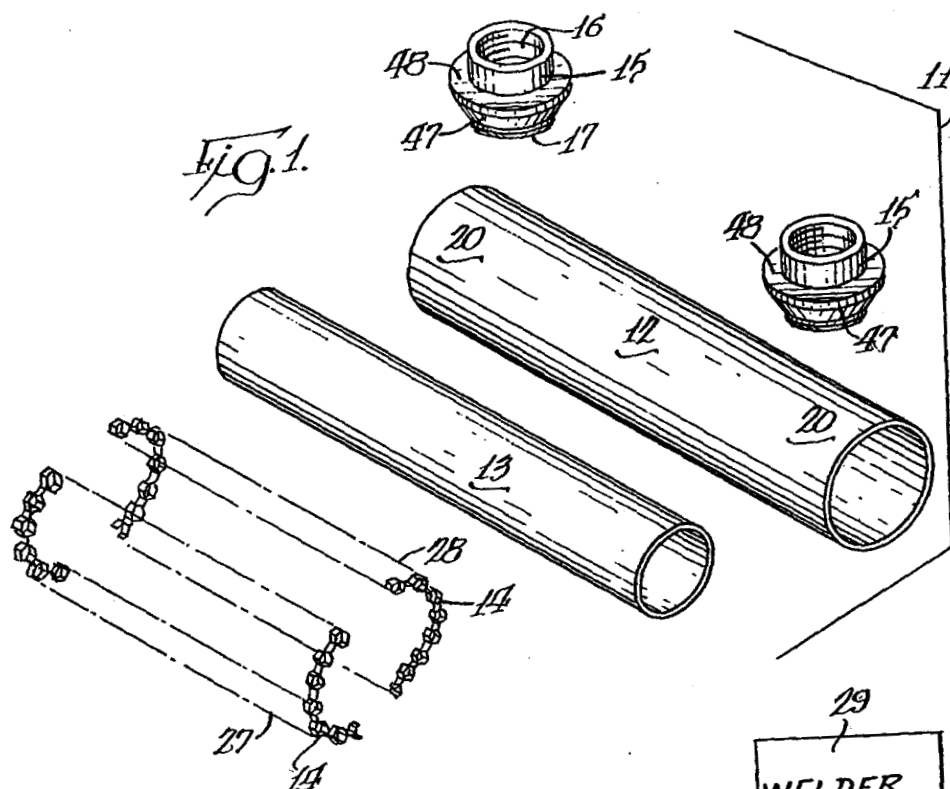
(54) Heat exchanger and method
of making

(57) A method of making a rigid stainless steel, corrosion resistant, welded, tubular heat exchanger and the resulting heat exchanger for exchanging heat between fluids in which a pair of steel tubes 12,13 with stainless surfaces are arranged in internested spaced relationship, providing a pair of cylindrical steel fittings 15 with stainless steel surfaces at each end 20 of the outer tube and welded to the outer tube on a shoulder of the outer tube by projection welding in which the contacting surfaces of the fitting and the tube are electric resistance welded, the ends of the inner and outer tubes being similarly welded together. Turbulators 14 are fitted between the tubes.

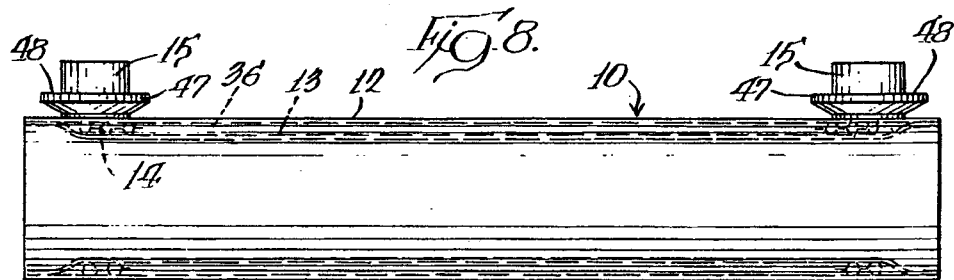
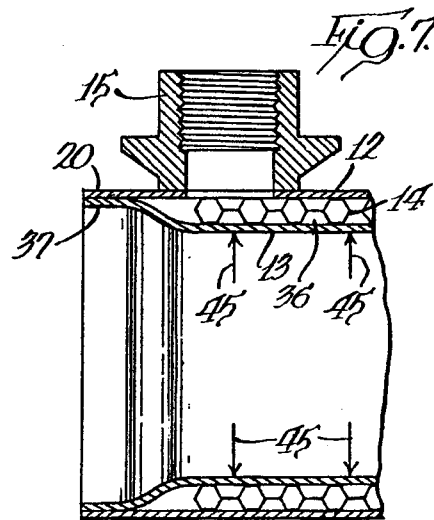
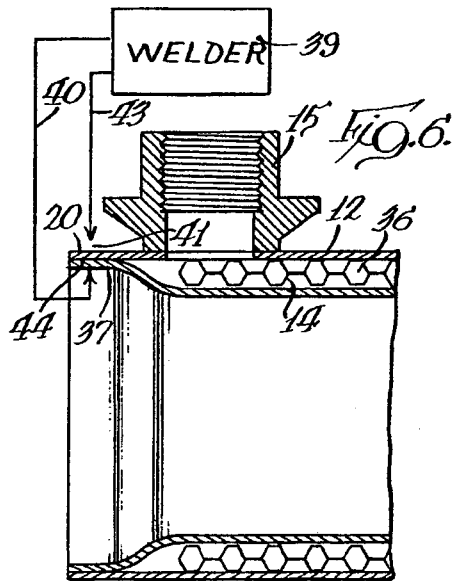
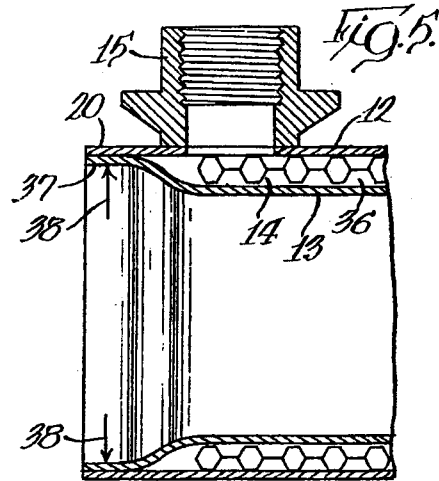
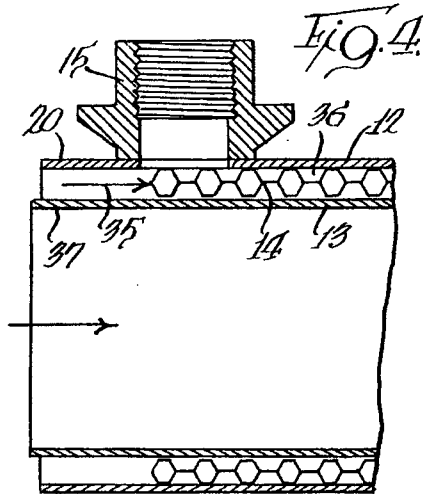


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SPECIFICATION

Heat exchanger and method of making

5 BACKGROUND OF THE INVENTION

10 Tubular heat exchangers, which are widely used as oil coolers for internal combustion engines, are customarily made of brass or copper parts joined together to provide one fluid space between the tubes and an internal core through which the heat exchange fluid is passed with the space between the tubes ordinarily having a turbulator therein for turbulizing flow of the liquid through this space to improve the heat exchange relationship between the fluids.

15 A fitting is provided at each end of this space between the tubes to pass the first fluid in one fitting through the space and out the other fitting. A heat exchanger of this type is illustrated in patents 2,752,125 and 3,959,867, both assigned to the assignee hereof, and patent 3,001,767. Such heat exchangers have also been made of aluminum as well as copper and brass, all of which are relatively lightweight and weak metals and alloys.

20 The heat exchanger of this invention is made of strong, corrosion resistant steel parts at least the surfaces of which are stainless steel. The invention also comprises a novel method of making such a tubular heat exchanger that is rigid, corrosion resistant and is a weldment construction of the stainless steel parts.

35 SUMMARY OF THE INVENTION

The parts of the tubular heat exchanger of the present invention are either solid stainless steel or steel with surface coatings that are stainless and these may be produced by procedures well known in the art such as those described in U.S. patents 3,093,556 and 3,184,331 and many others of similar nature.

40 Although the heat exchanger of this invention is preferably made of chromium containing stainless steel, other types of strong yet corrosion resistant steels may be used including Monel and Inconel. These steels are all very strong and, in addition, the heat exchanger of this invention is a weldment structure in that all joints between the internested tubes and between the external fittings and the outer one of the internested tubes are welded. This welding is by projection welding in which an electric current is passed between the contacting surfaces of the stainless steel parts, thereby resistance heating the parts to a molten condition while they are pressed together to produce the welded joint. Then the parts are cooled to provide a weldment heat exchanger.

45 A stainless steel heat exchanger of a different type is disclosed in U.S. patent 4,159,034, issued June 26, 1979, and as-

signed to the assignee hereof.

BRIEF DESCRIPTION OF THE DRAWINGS

Figure 1 is a collection perspective view of each of the parts comprising the specific embodiment of the tubular heat exchanger of this invention.

Figures 2-7 are successive sectional views illustrating successive steps of a method embodying the invention and producing a heat exchanger embodying the invention.

Figure 8 is a side elevational view of the completed heat exchanger embodiment of the invention.

80 DESCRIPTION OF THE PREFERRED EMBODIMENT

As illustrated in the collective view of Fig. 1, the heat exchanger 10 parts 11 comprise a pair of tubes consisting of an outer tube 12 and an inner tube 13 of different diameters and a pair of semicylindrical expanded metal turbulator 14 halves. The parts 11 also include a pair of cylindrical fittings 15. Each fitting 15 is generally cylindrical and has an axial, fluid flow passage 16 through it and a circular contact area ring 17 having a small, sharp surface for contact with a shoulder 18 at each end of the large or outer tube 12.

95 Fig. 2 illustrates the step in the formation of the shoulders 18 on each end of the tube 12 and, simultaneously, an opening 19 surrounded by a shoulder 18. In the forming step of Fig. 2 and end 20 of the tube is clamped between forming die members 23 and 24 to form each shoulder 18 and a cylindrical punch 25 is forced through the center of the shoulder 18 while so clamped to displace a disc 26 of metal from the tube 12 and thereby form the fluid flow opening 19. A similar step is performed at each end 20 of the larger or outer tube 12 to provide a shoulder 18 and corresponding opening 19 in each of these ends.

110 There is also provided a turbulator 14 which in this embodiment is in the form of two semi-cylindrical halves 27 and 28. Turbulators of this displaced metal construction are well known in the art and are illustrated in the following patents, all of which are assigned to the assignee hereof: 3,688,372; 3,732,921; 3,734,135; 3,734,177 and 3,959,867.

The tubes 12 and 13 are of the above described stainless steel while the turbulator 14 and the fittings 15 are of carbon steel.

120 In producing the heat exchanger after the end shoulders 18 and openings 19 are formed, as described and as illustrated in Fig. 2, each fitting 15 is a projection welded to its shoulder 18 by applying an electric current to the fitting as illustrated in Fig. 3. This current is supplied from a welder 29 through electric leads 30 and 33, one of which 30 is connected to the fitting 15 while the other 33 is connected to the tube 12. As is customary, a

current is thereby passed from the welder 29 through the very small circular ring contact area 17. Because the sharp contact area 17 is so small, the resistance to current flow is high with the result that the steel in the area of the circular area 17 becomes molten, so that pressure 34 applied to this area by way of the fitting 15, as illustrated schematically in Fig. 3, causes the molten metal to bond on cooling to form the projection weld. This technique, of course, is well understood in the art.

In the next step in the method of making the turbulator of this invention the smaller tube 13 is inserted as indicated by the arrow 35 into the larger tube 13 in spaced relationship so as to provide a fluid flow space 36. The two halves 27 and 28 of the cylindrical turbulator 14 are arranged in this space and both ends of the inner tube 13, as illustrated at 37, are flared outwardly by internal pressure 38 into contact with the inner surface of the outer tube end 20 as illustrated in Fig. 5. Each pair of contacting ends 20 and 37 are then welded together in either a GTAW (gas tungsten arc welding) or TIG welding with the electric current supplied from a welder 39 through electric leads 40 and 43. The arc space is illustrated in Fig. 6 at 41.

After the welding of the pairs of opposite ends 20 and 37 together in the manner described, the inner tube 13 is expanded outwardly by internal pressure illustrated schematically at 45 into tight contact with the turbulator 14 and thereby tight contact of the turbulator with the outer cylinder 12.

As can be seen in Fig. 8, the joined ends 20 and 37 of the outer 12 and inner 13 tubes are in areas outwardly, or longitudinally, beyond the fittings 15.

As stated, the parts including the pair of tubes 12 and 13 are of stainless steel. The pair of fittings 15 as well as the turbulator 14 may be of stainless steel but are preferably low carbon steel. Some or all of these parts can either be solid stainless steel or may have only their surfaces of stainless such as occurs when the surface of ordinary steel is impregnated with a steel alloy ingredient comprising chromium.

The preferred stainless steel used in making at least the tubes 12 and 13 of the parts 11 of the exchanger of this invention is one that contains chromium in an amount of at least 12%, such as from 12-32%, although other types of stainless alloys may be used. A steel that contains chromium is preferred because the chromium is a strong promoter of hardenability as it decreases the critical cooling rate of steel, and the steel alloy containing the chromium has good creep particularly at high temperatures and pressures.

The completed oil cooler as shown in side elevation in Fig. 8 with certain internal parts shown in broken lines is usually mounted in position as on a tank by attaching means at

the projecting ledge 47 which is an integral part of each fitting 15. This may customarily be done by providing a copper coating (not shown) to the radial surface of the ledge 47 as by welding, plating or spraying the copper onto the top surface 48 of each ledge 47. This copper coating, for example, would be essential if the unit 10 were soldered or the like to a brass tank. However, if the tank is plastic or stainless steel no such copper coating would be required and, in that event, the structure would be cheaper to build.

Having described our invention as related to the embodiment shown in the accompanying drawings, it is our intention that the invention be not limited by any of the details of description, unless otherwise specified, but rather be construed broadly within its spirit and scope as set out in the appended claims.

CLAIMS

1. The method of making a rigid stainless steel, corrosion resistant, weldment, tubular heat exchanger for exchanging heat between fluids, comprising the following steps:

(a) providing a pair of steel tubes of different diameters each having stainless steel surfaces;

(b) forming a fluid flow opening with a surrounding shoulder adjacent each end of the larger tube;

(c) arranging said pair of tubes as spaced inner and outer tubes;

(d) providing a pair of cylindrical steel fittings each having a fluid flow passage through the fitting and a circular contact area ring on one end of the fitting surrounding the flow passage, said ring having a contact surface of small area for contact with said shoulder;

(e) projection welding each fitting to a said shoulder by applying an electric current to the fitting and the outer tube for current flow therebetween by way of said contact ring while pressing a said fitting against its said shoulder, thereby melting said fitting and said shoulder at said contact area to a molten condition and then cooling to provide a resistance electric weld of each said fitting to said outer tube; and

(f) fusion welding the ends of said inner and outer tubes together.

2. The method of claim 1 wherein said pair of tubes are solid stainless steel.

3. The method of claim 2 wherein said stainless steel is No. 409 stainless.

4. The method of claim 1 wherein said spaced inner and outer tubes define a fluid flow space therebetween and a steel liquid turbulator is inserted within said space.

5. The method of claim 1 wherein said spaced inner and outer tubes define a fluid flow space therebetween and a steel liquid turbulator is inserted within said space after which said inner tube is expanded outwardly to force said tubes and turbulator into perma-

nent pressure contact.

6. The method of claim 1 wherein the ends of said inner tube are expanded outwardly into contact with the inner surface of said outer tube in areas outwardly of said fittings, the welding of the tubes to each other being at these contacting ends.

7. The method of claim 1 wherein said spaced inner and outer tubes define a fluid flow space therebetween and a steel liquid turbulator is inserted within said space and the ends of said inner tube are expanded outwardly into contact with the inner surface of said outer tube in areas outwardly of said fittings, the welding of the tubes to each other being at these contacting ends.

8. The method of claim 1 wherein said pair of tubes have steel surfaces impregnated with a steel alloying ingredient comprising chromium.

9. A rigid stainless steel, corrosion resistant, weldment, tubular heat exchanger for exchanging heat between fluids, comprising:
(a) a pair of spaced inner and outer tubes having a fluid flow opening with a surrounding shoulder adjacent each end of the larger outer tube;

(b) a pair of cylindrical steel fittings each having a fluid flow passage through the fitting, a circular contact area ring on one end of the fitting surrounding said passage and a continuous projection weld at said contact ring joining each fitting to a said shoulder; and

(c) a continuous weld joining the ends of said inner and outer cylinders together, each said weld joining a said fitting to said outer tube and joining said cylinders together having as weld metal essentially only solid, previously molten metal from adjacent portions of said tubes.

10. The heat exchanger of claim 9 wherein said pair of tubes are solid stainless steel.

11. The heat exchanger of claim 10 wherein said stainless steel is No. 409 stainless.

12. The heat exchanger of claim 9 wherein said spaced inner and outer tubes define a fluid flow space therebetween and there is provided a steel liquid turbulator within said space.

13. The heat exchanger of claim 9 wherein said spaced inner and outer tubes define a fluid flow space therebetween and there is provided a steel liquid turbulator within said space in permanent pressure contact with said tubes.

14. The heat exchanger of claim 9 wherein the ends of said inner tube are enlarged and in contact with the inner surface of said outer tube in areas outwardly of said fittings, the welding of the tubes of each other being at these contacting ends.

15. The heat exchanger of claim 9

wherein said spaced inner and outer tubes define a fluid flow space therebetween and there is provided a steel liquid turbulator within said space and the ends of said inner tube are enlarged and in contact with the inner surface of said outer tube in areas outwardly of said fittings, the welding of the tubes to each other being at these contacting ends.

16. The heat exchanger of claim 9 wherein said pair of tubes have steel surfaces impregnated with a steel alloying ingredient comprising chromium.

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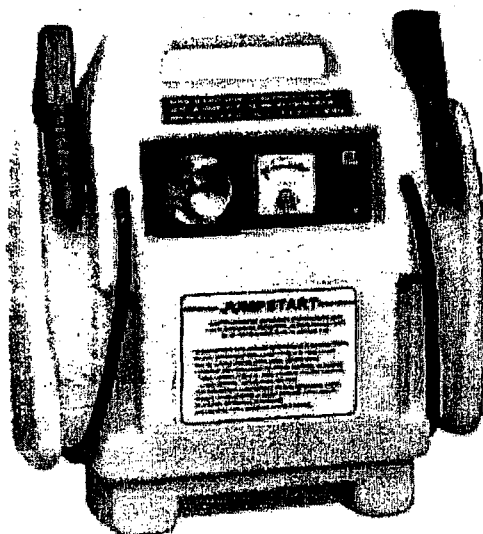
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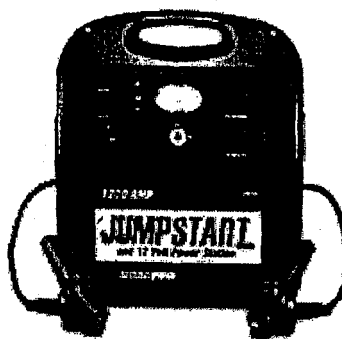
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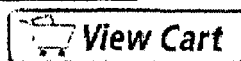
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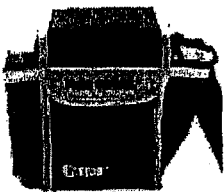


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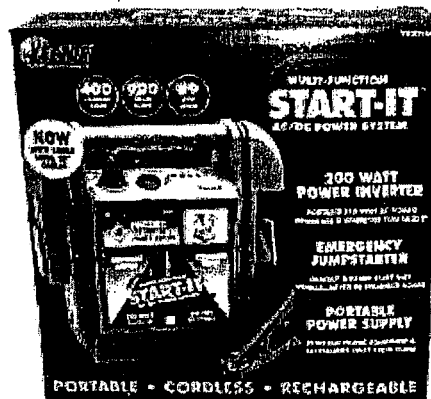
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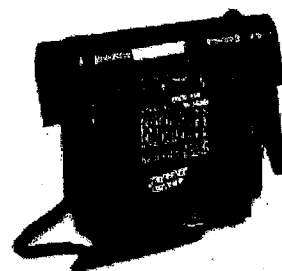
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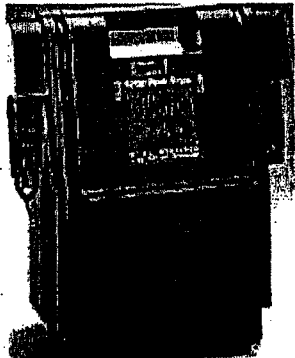
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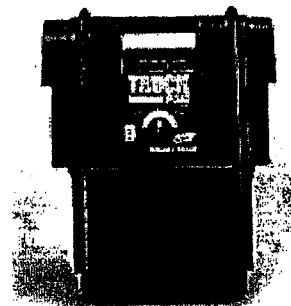
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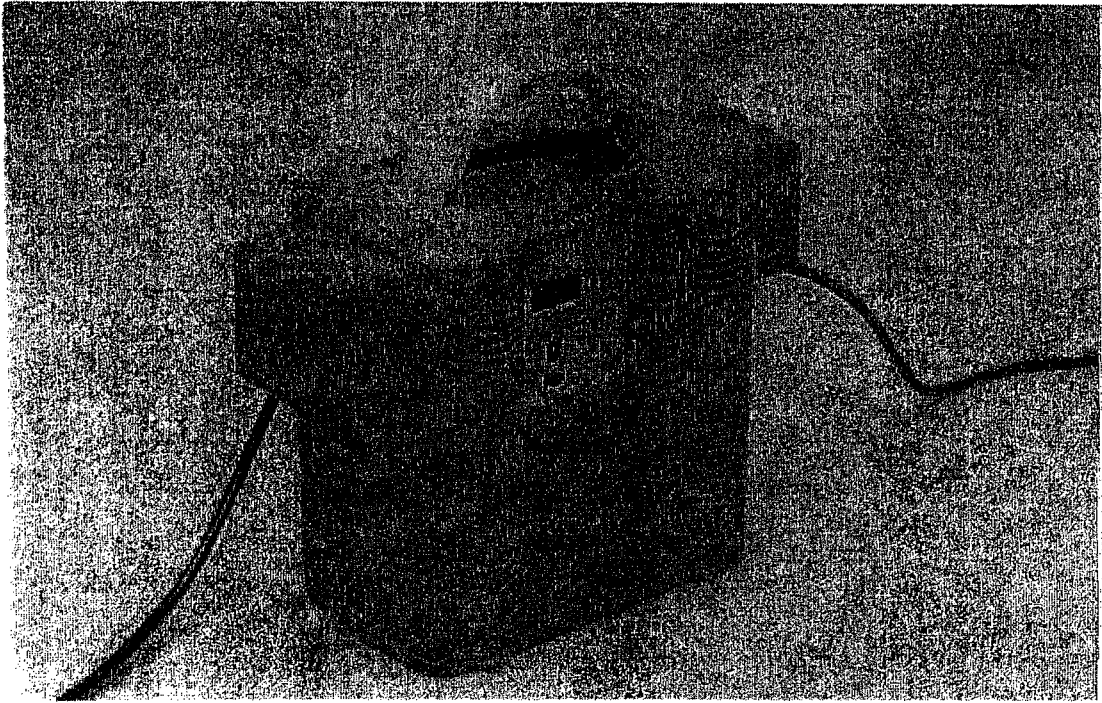
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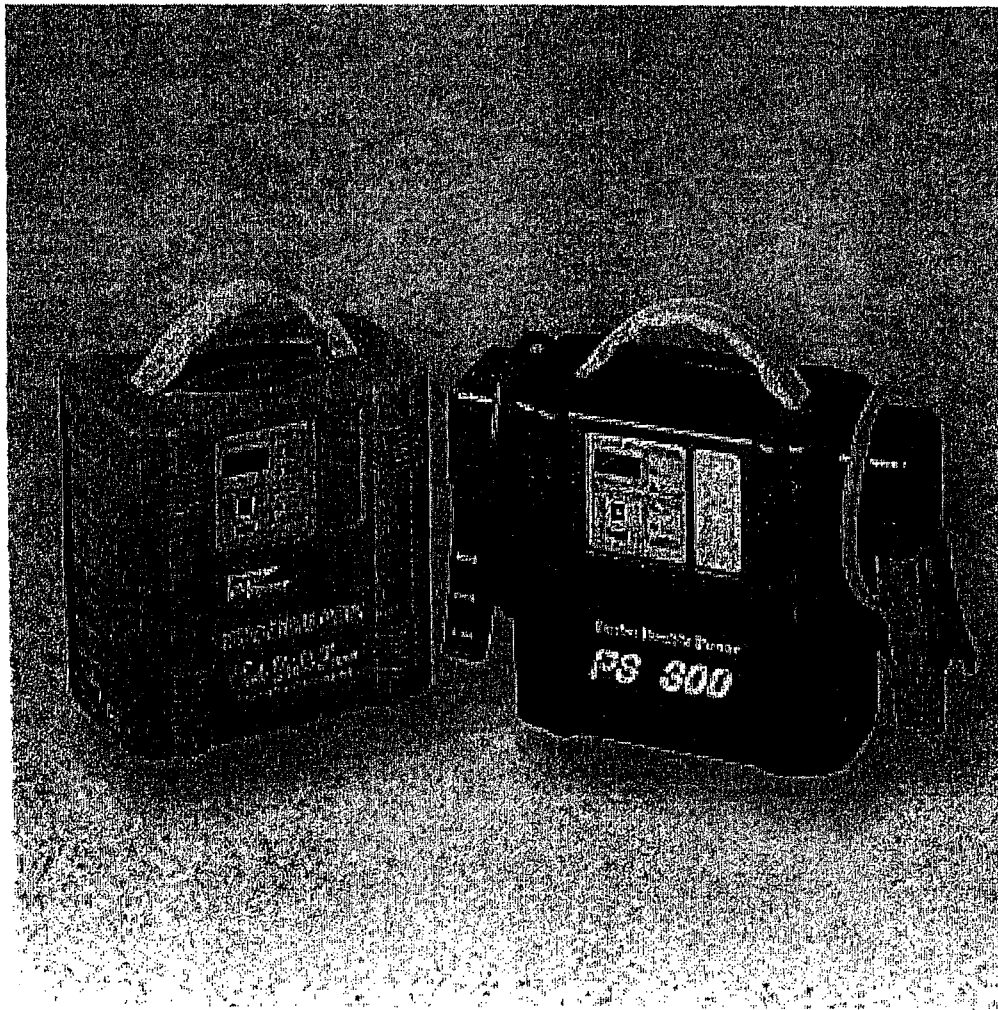
UK Registration **Design #2055060**

Design Number	2055060	Design Number	N/A
Relevant Dates	25/07/2016	Application Date	23/07/2016
Current Date	25/07/2016	Registration Date	23/07/2016
Applicant Name	Please Specify		
Priority Date	23/07/2016	Design Name	Unknown
Inventor Name	Starwood (Proprietor)	Salmon H4 HC 0013	N/A
	Hamilton (Service Agent)		
Priority Document Class	13/02/07	Relation	05



Smart Technologies & Investment Ltd

PS 200/300 Portable Power Station/Power Bank with Cigarette Lighter Socket Output



http://www.asia.globalsources.com/GeneralManager?&catalog_id=2000000003862&design=clean&language=en&action=GetProduct&page=ProductDetail&product_id=8812988668&action=GetPoint&point_id=3000000151145&big=y

Accessed Sunday, October 21, 2001

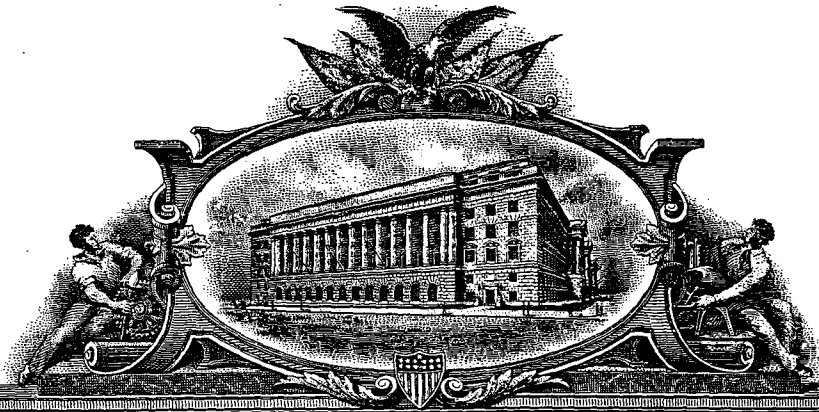
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Exhibit 4

A 1340547



THE UNITED STATES OF AMERICA

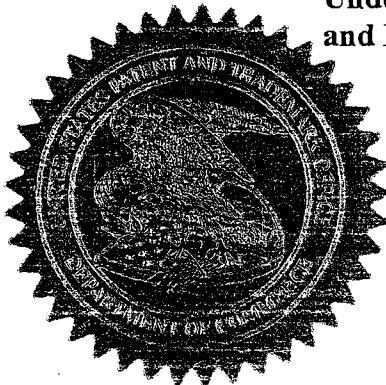
TO ALL TO WHOM THESE PRESENTS SHALL COME:

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office

July 06, 2005

THIS IS TO CERTIFY THAT ANNEXED IS A TRUE COPY FROM THE
RECORDS OF THIS OFFICE OF A DOCUMENT RECORDED ON
February 08, 2002

By Authority of the
Under Secretary of Commerce for Intellectual Property
and Director of the United States Patent and Trademark Office



H. L. Jackson
H. L. JACKSON
Certifying Officer

02-20-2002

101987516
PATENTS ONLYForm PTO-1595
(Rev. 03/01)

OMB No. 0651-0027 (exp. 5/31/2002)

U.S. DEPARTMENT OF COMMERCE
U.S. Patent and Trademark Office

To the Honorable Commissioner of Patents and Trademarks: Please record the attached original documents or copy thereof

1. Name of conveying party(ies):

Mervin A. Dayan

2-8-02

Additional name(s) of conveying party(ies)
attached? ☐ Yes ☒ No

3. Nature of Conveyance:

☒ Assignment☐ Merger☐ Security Agreement☐ Change of Name☐ Other

Execution Date: January 10, 2002

2. Name and address of receiving party(ies)

Name: Roadmaster (USA) Corp.

Internal Address:

Street Address:
6 Industrial Way West
Building G

City: Eatontown

State: NJ

Zip: 07724

Additional name(s) &
address(es) attached: ☐ Yes ☒ No

4. Application number(s) or patent number(s):

If this document is being filed together with a new application, the execution date of the new application is:

A. Patent Application No.(s):

29/150,972

B. Patent No.(s):

F.B. - 8 0000

Additional numbers attached? ☐ Yes ☒ No5. Name and address of party to whom correspondence
concerning document should be mailed:Name: Thomas M. Palisi
LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP

Internal Address:

Street Address: 600 South Avenue West

City:
WestfieldState: NJ Zip:
070906. Total number of applications and
patents involved: 1

7. Total fee (37 CFR 3.41) \$ 40.00

☐ Enclosed☒ Authorized to be charged to deposit account

8. Deposit account number:

12-1095

(Attach duplicate copy of this page if paying by deposit account)

DO NOT USE THIS SPACE

9. Statement and signature.

To the best of my knowledge and belief, the foregoing information is true and correct and any attached copy
is a true copy of the original document.

Thomas M. Palisi

Name of Person Signing

Signature

January 15, 2002

Date

Total number of pages including cover sheet, attachments, and documents: 2

I hereby certify that this correspondence is being deposited with the U.S. Postal Service with sufficient postage as First Class Mail, in
an envelope addressed to: Commissioner for Patents, Washington, DC 20231, on the date shown below.

Dated: January 15, 2002

Signature:

(Thomas M. Palisi)

02/20/2002 RMHED1 00000040 121095 29150972
01 FC:501 40.00 CHPATENT
REEL: 012575 FRAME: 0481

ASSIGNMENT OF DESIGN APPLICATION

former PTO/SB/15 (R-00) Modified LDI K&M
Docket Number (Optional)

ROADM 3.1-002

Whereas, I, Mervin A. Dayan of
251 Park Avenue, Oakhurst, NJ 07755

hereafter referred to as assignor, have invented certain new and useful improvements in

PORTABLE POWER STATION

☒ for which an application for a United States Design Patent was filed on October 30, 2001

Application Number 29/150,972

☐ for which an application for a United States Design Patent was executed by me on _____
And

Whereas, Roadmaster (USA) Corp.

a corporation of New Jersey herein referred to as "assignee" whose mailing address is

6 Industrial Way West, Building G; Eatontown, New Jersey 07724

is desirous of acquiring the entire right, title and interest in the same;

NOW, THEREFORE, in consideration of the sum of one dollar (\$ 1.00), the receipt whereof is acknowledged, and other good and valuable consideration, I as assignor hereby sell, assign and set over to said assignee the entire right, title and interest for the United States of America and all other countries in and to said invention and the aforesaid design patent application and all original, divisional, continuation, substitute or reissue applications and patents applied for or granted therefor in the United States of America and all other countries, for said invention, including without limitation all applications and patents for said invention claiming priority or benefit of the aforesaid design application pursuant to any law or treaty, and including the right to claim such priority or benefit and the Commissioner of Patents and Trademarks is hereby authorized and requested to issue all patents on said improvements or resulting therefrom to said assignee herein, as assignee of the entire interest therein; and the undersigned for me and my legal representatives, heirs and assigns do hereby agree and covenant without further remuneration, to execute and deliver all original, divisional, continuation, reissue and other applications for Design Patent on said improvements and all assignments thereof to said assignee or its assigns, to communicate to said assignee or its representatives all facts known to the undersigned respecting said improvements, whenever requested, to testify in any interferences or other legal proceedings in which any of said applications or patents may become involved, to sign all lawful papers, make all rightful oaths, and to do generally everything necessary to aid assignee, its successors, assigns and nominees to obtain patent protection for said improvements in all countries, the expenses incident to said applications to be borne and paid by said assignee.

(Date)

(Signature)

State of NEW JERSEY) SS:

County of MONMOUTH)

On this 10 day of JANUARY, 2002 before me THOMAS J. MANN personally came the above-named Mervin A. Dayan, to me personally known as the individual who executed the foregoing instrument, who acknowledged to me that the same was executed by him/her of his/her own free will for the purposes therein set forth.

(Notary Public)

RECORDED: 02/08/2002

PATENT
REEL: 012575 FRAME: 0482

Exh-5

Certificate of Registration



This Certificate issued under the seal of the Copyright Office in accordance with title 17, United States Code, attests that registration has been made for the work identified below. The information on this certificate has been made a part of the Copyright Office records.

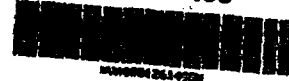
Marybeth Peters

Register of Copyrights, United States of America

FORM VA

— Work of the Visual Arts —

VA 1-261-495



EFFECTIVE DATE OF REGISTRATION
JULY 21 2004
 Month Day Year

DO NOT WRITE ABOVE THIS LINE IF YOU NEED MORE SPACE USE A SEPARATE CONTINUATION SHEET

1

TITLE OF THIS WORK ▼

JNS1880 Portable Power & Air Station Product Packaging

NATURE OF THIS WORK ▼ See instructions

Product Packaging

PREVIOUS OR ALTERNATIVE TITLES ▼

Publication as a Contribution If this work was published as a contribution to a periodical, serial, or collection, give information about the collective work in which the contribution appeared. Title of Collective Work ▼

If published in a periodical or serial give Volume ▼

Number ▼

Issue Date ▼

On Pages ▼

2

NOTE

Under the law the author of a "work made for hire" is generally the employer not the employee (see instructions). For any part of this work that was made for hire check Yes in the space provided give the employer (or other person for whom the work was prepared) as Author of that part and leave the space for dates of birth and death blank.

NAME OF AUTHOR ▼

a Roadmaster (USA) Corp

DATES OF BIRTH AND DEATH

Year Born ▼

Year Died ▼

Was this contribution to the work a "work made for hire"?
☐ Yes
☐ No

Author's Nationality or Domicile
 Name of Country

OR { Citizen of ► USA
 Domiciled in ► USA

Was This Author's Contribution to the Work

Anonymous? ☐ Yes ☒ No

Pseudonymous? ☐ Yes ☒ No

If the answer to either of these questions is "Yes," see detailed instructions.

NATURE OF AUTHORSHIP Check appropriate box(es) See instructions

☒ 2 Dimensional artwork

☐ Map

☐ Technical drawing

☐ Reproduction of work of art

☐ Photograph

☐ Text

☐ Jewelry design

☐ Architectural work

NAME OF AUTHOR ▼

b

DATES OF BIRTH AND DEATH

Year Born ▼

Year Died ▼

Was this contribution to the work a "work made for hire"?
☐ Yes
☐ No

Author's Nationality or Domicile
 Name of Country

OR { Citizen of ►
 Domiciled in ►

Was This Author's Contribution to the Work

Anonymous? ☐ Yes ☐ No

Pseudonymous? ☐ Yes ☐ No

If the answer to either of these questions is "Yes," see detailed instructions.

NATURE OF AUTHORSHIP Check appropriate box(es) See instructions

☐ 3 Dimensional sculpture

☐ Map

☐ Technical drawing

☐ 2 Dimensional artwork

☐ Photograph

☐ Text

☐ Reproduction of work of art

☐ Jewelry design

☐ Architectural work

3

Year in Which Creation of This Work Was Completed

2001

This information must be given in all cases.

Date and Nation of First Publication of This Particular Work

Complete this information ONLY if this work has been published.

Month ► June

Day ► 15

Year ► 2001

Nation ► USA

4

See instructions before completing this space

COPYRIGHT CLAIMANT(S) Name and address must be given even if the claimant is the same as the author given in space 2 ▼

Roadmaster (USA) Corp

6 Industrial Way West, Building G, Eatontown, NJ 07724

Transfer If the claimant(s) named here in space 4 is (are) different from the author(s) named in space 2 give a brief statement of how the claimant(s) obtained ownership of the copyright. ▼

APPLICATION RECEIVED

JUL 21 2004

ONE DEPOSIT RECEIVED

JUL 21 2004

TWO DEPOSITS RECEIVED

FUNDS RECEIVED

MORE ON BACK ►

Complete all applicable spaces (numbers 5-8) on the reverse side of this page. See detailed instructions. Sign the form at line 8.

DO NOT WRITE HERE

Page 1 of _____ pages

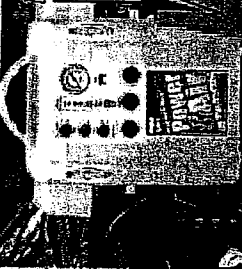
Roadmaster.

JNS1880

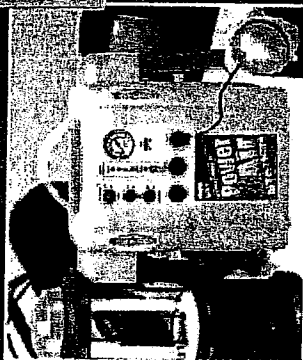
**Jump
Start**

PORTABLE RECHARGEABLE

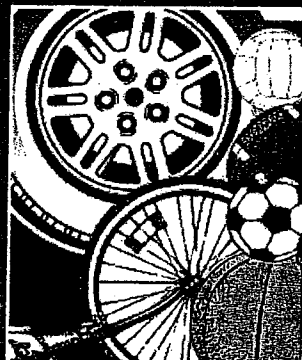
**Power Station
Air Compressor**



Built-in booster cables help jump-start your car when there's no help around!



Provides three 12V DC outputs to provide power to any 12Volt DC product.



Built-in air compressor inflates tires, air mattresses, water rafts, inflatable toys, sports equipment. Includes air pressure gauge!

Recharge at home
or in the car!



**Jump
Start
PORTABLE
RECHARGEABLE
POWER
STATION
AIR
COMPRESSOR**

**JUMP
START
YOUR CAR
IN THE MIDDLE
OF NOWHERE
WHEN THERE ARE
NO OTHER CARS AROUND!**

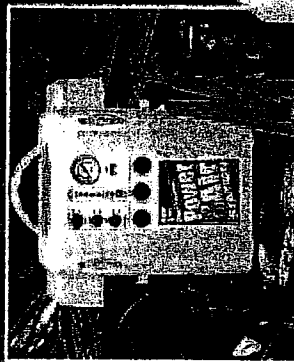
JNS1880 ENGLISH FRONT PANEL
VERIFY IN FIELD - SIZE TO FIT

Roadmaster.

JNS 23 90

**Jump
n
Start**

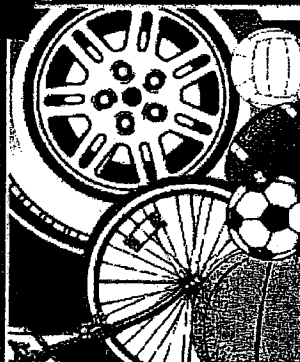
JNS1880
SPANISH BACK PANEL
VERIFY IN FIELD
SIZE TO FIT



Con Cables Para El Arranque De Su Carro.

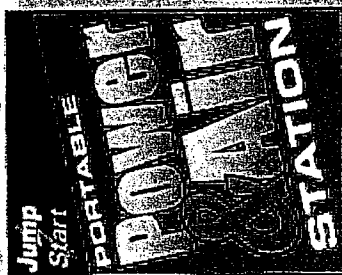


Contiene Dos Salidas De 12 Voltios De Corriente Directa Para Usar Con Productos De 12 Voltios DC.



Incluye un compresor de aire, para inflar llantas, colchón de aire, balsa, juguetes y artículos desmontables.
Incluye medidor de presión!

ESTACION DE Energia & COMPLETOR DE ANGE PORTATIL



Rebaja en su
precio el auto!

ARRANQUE
SU CARRO EN
LUGARES
DESCARROS
SIN OTROS
CARRROS ALBEREDON

JNS1880 ENGLISH SIDE PANEL
VERIFY IN FIELD - SIZE TO FIT

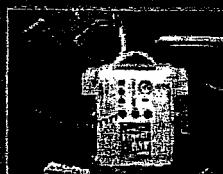
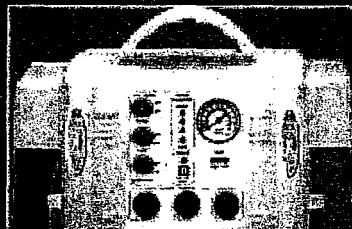
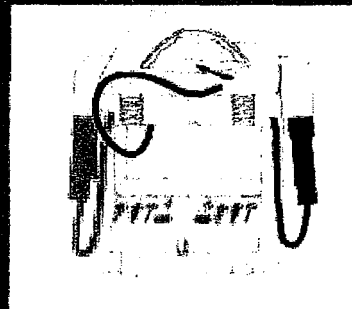
Roadmaster.

JNS1800

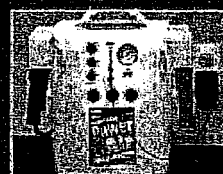
PORTABLE
RECHARGEABLE
**POWER
STATION**
**& AIR
COMPRESSOR**
Jump
Start

FEATURES:

- Battery booster cables.
- **250 PSI AIR COMPRESSOR** with pressure gauge and auto shut-off feature.
- **TWO** powerful work lights.
- **THREE 12 Volt DC** outputs.
- LED warning lights for battery condition and charging status.
- In-line fuse protection.
- **SIX** inflating needles.
- Heavy duty compressor hose with tire valve fitting
- AC and DC adapter stored in back compartment.
- **FOUR** extra fuses.



• Recharge unit in a 110 volt wall outlet or with the 12 volt DC cigarette lighter adapter in your car.



JNS1880 SPANISH SIDE PANEL
VERIFY IN FIELD - SIZE TO FIT

Roadmaster.

JNS1800

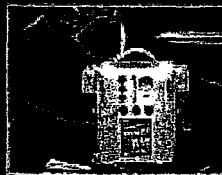
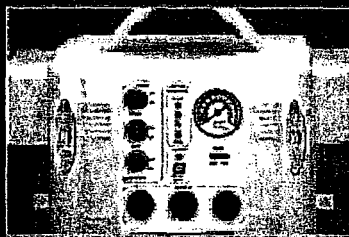
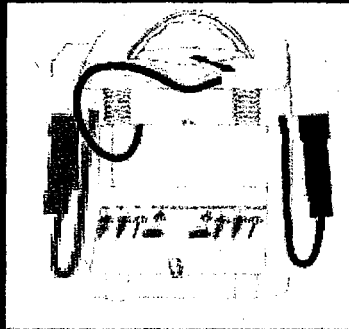
ESTACION DE Energia COMPRESOR de Aire PORTATIL

Recarga en su
casa o en el auto!

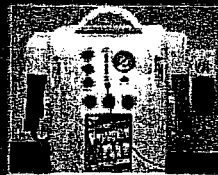
Jump
n
Start

FUNCIONES:

- Cables para arranque de su carro.
- COMPRESOR de Aire con 250 psi con medidor de presión.
- DOS lamparas de trabajo de alto alumbrado.
- TRES salidas de 12 Volts.
- Luces de aviso para condiciones de la bateria.
- Fusible en linea para proteger su unidad.
- SEIS adaptadores para inflar.
- Compresor fuerte con valvula de cierre.
- Compartimiento para los adaptadores de corriente de alterna y directa.
- CUATRO fosibles de repuesto.



• Recarga su unidad con 110V en su casa y con adaptador de 12V en el encendedor de cigarrillo en su auto.



JNS1880 TOP PANEL
VERIFY IN FIELD - SIZE TO FIT

Roadmaster.

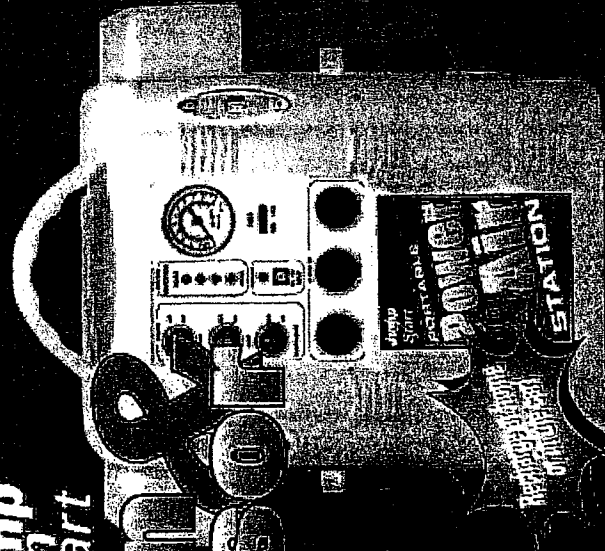
JNS1880

**Jump
n'
Start**

PORTABLE • RECHARGEABLE

**Power Station®
Air Compressor**

- Built-in booster cables to jump your car.
- 12 volt 17.2Ah sealed lead-acid **RECHARGEABLE** battery.
- LED displays for **FULL/CHARGE/LOW/GOOD** indications.
- Provides **THREE** 12 volt DC outputs to provide power to any 12 volt DC product.
- Recharge unit with 12 volt adapter using 110 volt wall outlet or with 12 volt DC cigarette lighter adapter in your car.
- Built-in **250 PSI AIR COMPRESSOR** with air pressure gauge for inflating tires, inflatable toys, air mattresses, water rafts, sports equipment, etc.



JN5188U Bottom Panel

As part of our commitment to product excellence, ROADMASTER routinely improves the designs, materials or production methods of its existing products. Because it is impractical to publicize all changes in every product, we reserve the right to make such changes without notice.	© Copyright 2001 ROADMASTER (USA) Corp Six G Industrial Way West Edgewater, New Jersey 07724 Made in China 00008A	JN5188U  0 12 31 54 0 1796 1
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exh-6

Int. Cl.: 9

Prior U.S. Cls.: 21, 23, 26, 36, and 38

United States Patent and Trademark Office

Reg. No. 2,594,538

Registered July 16, 2002

**TRADEMARK
PRINCIPAL REGISTER**

JUMP N START

ROADMASTER USA CORP. (NEW JERSEY CORPORATION)
6 INDUSTRIAL WAY WEST, BLDG. G
EATONTOWN, NJ 07724

FOR: BATTERY-CHARGING SYSTEM, NAMELY,
A PORTABLE BATTERY AND BATTERY CABLES,
IN CLASS 9 (U.S. CLS. 21, 23, 26, 36 AND 38).

FIRST USE 3-0-1998; IN COMMERCE 3-0-1998.

SN 75-412,359, FILED 12-30-1997.

JAMES GRIFFIN, EXAMINING ATTORNEY